DISCRETE BEURLING NUMBER SYSTEMS WITH INTEGER ERROR $O(x^\theta)$, $\theta < 1/2$, WHOSE ZETA FUNCTION HAS NO ZERO-FREE HALF-PLANE

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Abstract. We construct a discrete Beurling generalized number system whose integer counting function satisfies $N(x) = x + O(x^\theta \log^* x)$, where $\theta \in [\frac{1}{4}, \frac{1}{2} - 10^{-4}]$ is fixed and $\log^*$ is the iterated logarithm, and whose zeta function $\zeta_{\mathcal{P}}$ has zeros ρ_k with $\operatorname{Re}\rho_k \to 1$. In particular $\zeta_{\mathcal{P}}$ is zero-free in no half-plane $\operatorname{Re} s > 1 - \varepsilon$. Equivalently, there is a discrete $[1, \beta]$-system with $\beta \le \theta < \frac{1}{2}$. So the hypothesis $N(x) = Ax + O(x^\theta)$ with $\theta < \frac{1}{2}$ does not by itself force a zero-free half-plane.

The system is built by a randomised greedy rule with feedback from the integer count. Generalized primes are placed one slot at a time, against a target, adapted to the construction, whose Mellin transform carries designed zeros, and a prime is placed only when the queue of composite elements is empty. The heart of the proof is a positive drift for this queue at every scale. It comes from a sharp resolvent bound of Beurling–Saias type for the realised system, together with a counting theorem for clusters of composites in short windows.

Several steps are computer-assisted, with certificates in interval or exact arithmetic. The construction, the proofs and the certificate code were developed with extensive use of AI agents, and so far they have been reviewed only by AI agents; no human expert has yet verified the proof (see Appendix C).

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1. Introduction

1.1. **Beurling systems and the question.** A *discrete Beurling generalized number system* is a non-decreasing sequence $\mathcal{P} = (p_j)_{j\geq 1}$ of real numbers $p_j > 1$ with $p_j \to \infty$, the *generalized*

primes, together with the multiplicative semigroup $\mathcal{N}$ they generate. Its elements are the finite products $\prod_j p_j^{e_j}$, counted with multiplicity: different exponent vectors are different elements, even when their values coincide. Write

$$N_{\mathcal{P}}(x) := \#\{n \in \mathcal{N} : \ n \le x\}, \qquad \psi_{\mathcal{P}}(x) := \sum_{p_j^k \le x} \log p_j, \qquad \zeta_{\mathcal{P}}(s) := \sum_{n \in \mathcal{N}} n^{-s} = \prod_j (1 - p_j^{-s})^{-1}.$$

The rational primes give $(\mathbb{N}, \zeta)$. Following [4], $\mathcal{P}$ is an $[\alpha, \beta]$-*system* if $\psi_{\mathcal{P}}(x) = x + O(x^{\alpha+\varepsilon})$ and $N_{\mathcal{P}}(x) = Ax + O(x^{\beta+\varepsilon})$ for every $\varepsilon > 0$, with $A > 0$ and both exponents optimal. Hilberdink [10] proved that always $\max(\alpha, \beta) \ge \frac{1}{2}$.

This paper answers the following question.

Definition 1.1 (Question Q)**.** Is it true that every discrete Beurling system with $N_{\mathcal{P}}(x) = Ax + O(x^\theta)$ for some $A > 0$ and some $\theta < \frac{1}{2}$ has $\zeta_{\mathcal{P}}$ zero-free on some half-plane $\operatorname{Re} s > 1 - \varepsilon$, $\varepsilon > 0$?

Applied to $\mathbb{N}$, where $N(x) = \lfloor x \rfloor = x + O(1)$, a positive answer would give a zero-free half-plane for the Riemann zeta function. Our result shows that the integer-counting hypothesis alone cannot do this. It says nothing about ζ itself.

The answer to Q is equivalent to the existence question for $[1, \beta]$-systems with $\beta < \frac{1}{2}$ (Remark 1.2).

1.2. **The result.**

Theorem A (informal form; see Theorem 2.1). Let $\theta \in [\frac{1}{4}, \frac{1}{2} - 10^{-4}]$. There is a discrete Beurling system $\mathcal{P}$ with

$$N_{\mathcal{P}}(x) = x + O(x^\theta \log^* x),$$

whose zeta function has zeros ρ_k, $k \ge 1$, with

$$\operatorname{Re} \rho_k = 1 - \frac{3 \log\log x_k + O(1)}{\log x_k} \longrightarrow 1$$

along an explicit sequence $x_k \to \infty$. In particular $\zeta_{\mathcal{P}}$ is zero-free in no half-plane $\operatorname{Re} s > 1 - \varepsilon$, the answer to Q is negative, and $\mathcal{P}$ is a discrete $[1, \beta]$-system with $\beta \le \theta < \frac{1}{2}$.

The system is random, and Theorem 2.1 holds on an event of probability at least $1 - 1.5/\log X_0 > 0$, where X_0 is the scale at which the construction starts (Section 2). Every realisation in that event is a fixed discrete system with the stated properties. Since Q quantifies over deterministic systems, one realisation suffices. The value $\theta = \frac{1}{4}$ already gives a negative answer to Q.

Remark 1.2 (Q and $[1, \beta]$-systems). For a system with $N_{\mathcal{P}}(x) = Ax + O(x^\beta)$, $A > 0$, $\beta < \frac{1}{2}$, the following are equivalent:

(a) $\zeta_{\mathcal{P}}$ is zero-free on some half-plane $\operatorname{Re} s > 1 - \varepsilon$;
(b) $\psi_{\mathcal{P}}(x) = x + O(x^{1-\varepsilon'})$ for some $\varepsilon' > 0$.

(b) implies (a). For $\sigma > 1$, partial summation gives

$$\zeta_{\mathcal{P}}(s) - \frac{As}{s-1} = s \int_1^\infty (N_{\mathcal{P}}(x) - Ax)\, x^{-s-1}\, dx, \qquad -\frac{\zeta_{\mathcal{P}}'(s)}{\zeta_{\mathcal{P}}(s)} - \frac{s}{s-1} = s \int_1^\infty (\psi_{\mathcal{P}}(x) - x)\, x^{-s-1}\, dx.$$

The first integral converges absolutely and locally uniformly on $\sigma > \beta$, so $\zeta_{\mathcal{P}}$ is meromorphic there, with only the simple pole at $s = 1$. By (b) the second converges likewise on $\sigma > 1 - \varepsilon'$, so $\zeta_{\mathcal{P}}'/\zeta_{\mathcal{P}}$ has no pole other than $s = 1$ on $\sigma > \max(\beta, 1 - \varepsilon')$. A zero of $\zeta_{\mathcal{P}}$ there would be such a pole. Hence (a) holds with $\varepsilon = \min(\varepsilon', 1 - \beta)$.

(a) implies (b). This is the standard converse. $\zeta_{\mathcal{P}}$ grows at most polynomially on $\sigma \geq \beta + \delta$ away from $s = 1$, so the Borel–Carathéodory lemma bounds $\zeta'_{\mathcal{P}}/\zeta_{\mathcal{P}}$ polynomially on $\sigma \geq 1-\varepsilon+\delta$. Perron inversion for a smoothed $\psi_{\mathcal{P}}$, followed by an elementary Tauberian step (since $\psi_{\mathcal{P}}$ is non-decreasing), then gives (b). The same argument for $N_{\mathcal{P}}$ is [4, Remark 4.1]; see [7] for the general theory.

So a negative answer to Q is the same as the existence of a discrete $[1, \beta]$-system with $\beta < \frac{1}{2}$. Only the direction (b)⇒(a) is needed for Corollary 2.2.

1.3. **Background.** *Well-behaved systems.* The notion of an $[\alpha, \beta]$-system is due to Hilberdink [10]. If the integers are β-well-behaved for some $\beta < 1$, then $\psi_{\mathcal{P}}(x) = x + O(x \exp(-c\sqrt{\log x}))$ (see [4, §1]). Diamond, Montgomery and Vorhauer [6] showed that this de la Vallée Poussin-type error term cannot be improved in general. Zhang [19] gave the first unconditional well-behaved example, with $\psi_{\mathcal{P}}(x) = x + O(x^{1/2+\varepsilon})$ and $N_{\mathcal{P}}(x) = ax + O(x^{1/2+\varepsilon})$, using the probabilistic method of [6]. Broucke and Vindas [5] refined that method and obtained $[0, \frac{1}{2}]$-systems. Broucke, Debruyne and Révész [4] constructed $[\alpha, \beta]$-systems for every $\alpha \in [0, 1)$ and $\beta \in [\frac{1}{2}, 1)$, building on Révész's construction of systems with prescribed zeta zeros [15]. Assuming the Riemann hypothesis, they also constructed $[\frac{1}{2}, \beta]$-systems for $0 \leq \beta < \frac{1}{2}$, and $[\alpha, \beta]$-systems for $\frac{1}{2} < \alpha < \frac{2}{3}$ and $2\alpha/(\alpha + 2) \leq \beta < \frac{1}{2}$ [4, Thm. 1.3]. A $[1, \beta_0]$-system with some $\beta_0 \leq \frac{1}{2}$ is in [7, Ch. 17], as recorded in [4, p. 3], and [4, footnote 4] remarks: "Most likely the value of β_0 equals $1/2$, but in principle it is still possible that β_0 could be smaller."

The barrier at $x^{1/2}$. The random approximation procedures discretise a continuous system, and they lose a square root. As [4, §5] put it, "a direct application of Theorem 1.2 will always yield an error term of at least $O(x^{1/2})$ on the integers". To get below $x^{1/2}$ "one has to construct a discrete example right from the start", and "it seems natural to define a Beurling system through the sequence of the integers rather than the primes. Yet [...] it is often extremely difficult to show which behavior the primes must admit."

The closest prior result. Broucke [2, Thm. 1.6] proves the following. Let f be eventually C^1, decreasing and strictly convex, with $1/u = o(f(u))$, and regularly varying of index $-a \in [-1, 0)$, or slowly varying and satisfying [2, (1.10)]. Then there is a discrete Beurling system with:

(1) $N_{\mathcal{P}}(x) = Ax + O_{\varepsilon}(x^{1/2+\varepsilon})$;
(2) $\psi_{\mathcal{P}}(x) - x = \Omega_{\pm}\big(x \exp(-\omega(x) + (1 - \delta) f(u_0(x)) u_0(x))\big)$;
(3) infinitely many zeros on the curve $\sigma = 1 - f(\log |t|)$, and none to its right.

The exponent $\frac{1}{2} + \varepsilon$ comes from randomly discretising a continuous system of Diamond–Montgomery–Vorhauer type.

Landau's region. If $N_{\mathcal{P}}(x) = Ax + O(x^{\theta})$ with $\theta < 1$, then $\zeta_{\mathcal{P}}(s) \neq 0$ for $\sigma > 1 - c(1-\theta)/\log |t|$, $|t| \geq t_0(\mathcal{P})$, with c absolute. This is [2, (1.2)], derived there from [2, Thm. 6.1, (1.4)]. The zeros of Theorem 2.1 are consistent with it (Remark 1.3).

1.4. **What is new, and what is not shown.** Theorem 2.1 makes the step from $\theta = \frac{1}{2} + \varepsilon$ to $\theta < \frac{1}{2}$. It differs from [2, Thm. 1.6] in three ways:

(a) $\theta < \frac{1}{2}$, which the existing random approximation procedures [6, 19, 5, 4] do not reach;
(b) the construction is discrete from the start and integer-first: generalized primes are placed by a greedy rule with feedback from the integer count, which is the kind of construction [4, §5] suggests;
(c) the zero information is weaker: the zeros are designed, but there is no zero contour and no Ω-result.

Remark 1.3 (What Theorem 2.1 does not show).

- It does not show that Landau's region is optimal for $\theta < \frac{1}{2}$. Here $1 - \operatorname{Re}\rho_k \asymp 3(1-\theta)\log\log\gamma_k/\log\gamma_k$, where $\gamma_k = \operatorname{Im}\rho_k$. That is outside Landau's region by a factor $\log\log\gamma_k$. In this design the loss comes from the amplitude condition (2), which gives the factor $3\log\log$ rather than $1\log\log$. Positivity of the target alone would allow larger amplitudes; the smaller ones are needed to keep the drift margin of Section 5 at scales where γ_k is not yet resolvable.
- It gives no system with prime error x^α, $\frac{1}{2} < \alpha < 1$, and integer error $O(x^\theta)$, $\theta < \frac{1}{2}$. Infinitely many designed zeros make the prime error $x^{1-o(1)}$.
- It gives no $[\alpha,\beta]$-system with $\alpha < 1$ and $\beta < \frac{1}{2}$: the greedy construction has $\alpha = 1$.
- It prescribes no zero contour and gives no Ω-result for $\psi_{\mathcal{P}}$ (compare [2, Thm. 1.6]).
- It says nothing about the Riemann zeta function.

1.5. **Outline of the proof.** *The construction* (Section 3). Below a large scale X_0 the system coincides with $\mathbb{N}$, apart from the few generalized primes that land within $H(X_0)/2$ below X_0. Above X_0 we follow a slightly modified target $T(x) \approx x$, built along with the system (its absorbed part and the heights γ_k depend on it). Its continuous part removes the density of the X_0-smooth integers, averaged over windows $[u,(1+\kappa)u]$. It also carries oscillations R_k that switch on at very sparse scales x_k and plant zeros near $1+i\gamma_k$. The level sets of T cut $[X_0,\infty)$ into unit *slots*. Slots are processed in order: a slot receives a new generalized prime exactly when the queue of composite elements is empty and no composite lands in the slot, and the prime lands uniformly in a box of width $H(s_j)$ around the slot start. So $N_{\mathcal{P}} - T$ is controlled by the queue length. Bounding the queue is the whole difficulty.

Zeros (Section 4). If $|N_{\mathcal{P}} - T| \le x^\eta$ (property (12)), a Rouché argument places a zero of $\zeta_{\mathcal{P}}$ near each designed point. The heights γ_k are chosen from the system below x_k, so the target stays adapted to the construction.

The queue (Sections 5 and 6). The queue follows a Lindley recursion. In a window W of slots, the excess of composite arrivals over slots has predictable compensator $-\int_W(\text{margin})$, plus martingale noise. Two facts drive it:

- a *margin*: at every scale the compensator of jittered composites falls short of the number of continuous slots by a relative amount $\ge c'/\log x$ (Section 5). This rests on a sharp bound for the resolvent coefficients of the realised system, of Beurling–Saias type, proved with a computer-assisted certificate; on a local prime number theorem; and on a careful treatment of the jump of the target at X_0;
- a *cluster bound*: outside an event of small probability, no generalized prime is the latest prime of more than $2M_1$ composites in a slot window, where M_1 is an explicit parameter (Section 6 and Appendix A).

A first-failure (stopping time) argument orders all the events by time and closes the induction. The probability that some event fails is $\le 1.5/\log X_0$.

The base range (Section 7). The rigid part of the target averages the X_0-smooth integers over linear windows $[u,(1+\kappa)u]$. This makes the rational identity behind the margin exact, up to the error of the short-interval prime number theorem of Huxley and Heath-Brown, and it removes a family of edge effects.

Section 2 states Theorem 2.1 precisely, fixes all parameters, and records the logical structure, including why the induction is not circular. Appendix B describes the computer-assisted parts, and Appendix C is a statement on the use of AI.

2. Statement, parameters and logical structure

Throughout, $L := \log x$ and $\ell := \log\log x$ at the scale x under discussion, and $\Lambda := \log X_0$. The exponent $\theta \in [\frac14, \theta_1]$, $\theta_1 := \frac12 - 10^{-4}$, is fixed. Numerical values are quoted for $\theta = \frac14$, which is the worst case for the absorption estimates of Section 6.

2.1. **The main theorem.**

Theorem 2.1 (Theorem A)**.** *Fix $\theta \in [\frac14, \theta_1]$, $\theta_1 := \frac12 - 10^{-4}$, and the parameters of Section 2.2; in particular $X_0 \geq \max(e^{10^8}, X_{\mathrm{Hux}})$ is an even integer, where X_{Hux} is defined in (80). Let $\mathcal{P}$ be the random discrete Beurling system of Section 3: the hard greedy rule with box-centred jitter, for the target T of (5). Then on the event $\mathcal{G} = \{\tau = \infty\}$ of Definition 6.8, which has probability at least $1 - 1.5/\log X_0 > 0$:*

(i) $|N_{\mathcal{P}}(x) - x| \leq C_{X_0} x^\theta + 3x^\theta \log^* x + x^\eta$ *for all $x \geq 1$, where $\log^*$ is the iterated logarithm, $\eta := \min(\theta, \frac12 - \theta)/2$ and $C_{X_0} < \infty$;*

(ii) $\zeta_{\mathcal{P}}$ *has zeros ρ_k $(k \geq 1)$ with* $\operatorname{Re}\rho_k = 1 - \big(3\log\log x_k + O_{X_0}(1)\big)/\log x_k \to 1$.

The threshold X_{Hux} is finite by Heath-Brown's theorem [9] and depends on no parameter of the construction. Its value is never used: every quantitative estimate in the proof holds for all $\log X_0 \geq 10^8$, with the estimates that rest on (80) (Lemma 7.3(a) and its consumers) needing only $X_0 \geq X_{\mathrm{Hux}}$ in addition.

Corollary 2.2. *The answer to Question 1.1 is negative. Equivalently, a discrete $[1,\beta]$-system with $\beta < \frac12$ exists.*

Proof. Fix any $\omega \in \mathcal{G}$. The realised prime sequence $\mathcal{P}(\omega)$ is a fixed discrete Beurling system: it has at most one prime per slot, hence finitely many below any bound. By Theorem 2.1(i) it has $N = x + O(x^\theta \log^* x)$, and by Theorem 2.1(ii) it has no zero-free half-plane. That T was constructed adaptively plays no role in this conclusion. By Remark 1.2, direction (b)⇒(a), $\psi_{\mathcal{P}}(x) - x$ has no power saving, so $\mathcal{P}(\omega)$ is a $[1,\beta]$-system with $\beta \leq \theta < \frac12$. □

Proof of Theorem 2.1, assuming the results of Sections 3–7. Part (i) follows from the target error (7), the property (12) and $|R| \leq 2x^\theta \log^* x$. Property (12) follows from the two-sided lag (11) and the queue bound, Theorem 6.6, on $\mathcal{G}$. The queue bound needs two inputs.

- The drift of Section 5: the noise response, the margin, and R1′ (Corollary 5.40). The jump of T' at X_0 is Lemma 5.27, with Lemma 5.32 in its largest range.
- The probabilistic inputs of Section 6: Freedman's inequality, the cluster event (Proposition 6.10) via the rank-full counting of Appendix A, the local count (Corollary 6.3), and the sparsity of the absorbed set (Lemma 6.25).

Part (ii) is Theorem 4.9, from (12) and Lemma 4.4. Finally, $\mathbb{P}(\mathcal{G}) \geq 1 - 1.5/\log X_0$ is Lemma 6.9. □

2.2. **Parameters.**

All parameters are deterministic, and they are fixed here, before the construction starts. The jitter width H, the capacity threshold c and the smooth majorants $\bar r_0$, $\bar M_1$ are functions of the height itself; the integers r_0, M_1 and the quantities L_0, y, B are frozen per scale.

parameter	value
θ, θ'	$\theta \in [\frac14, \theta_1]$, $\theta_1 := \frac12 - 10^{-4}$, fixed ($\theta = \frac14$ suffices for Corollary 2.2); the local prime number theorem uses the exponent $\theta' := \theta + 0.01$.
X_0	$X_0 \geq \max(e^{10^8}, X_{\mathrm{Hux}})$, an even integer (for instance the least one): the single regime for the whole proof. It is even so that the jump of T at X_0 is the jump of $\lfloor x \rfloor$ at the rigid integer X_0 (Section 3.2). X_{Hux} is defined in (80), depends on no parameter, and gives $\mathrm{MA}[\mathbb{P}](q) \log q \geq 0.99$ for every $q \geq X_0$ (Lemma 7.3(a)). The local prime number theorem is used only above $X_{\mathrm{LP}}(X_0)$, with $\log X_{\mathrm{LP}} = O(\Lambda^2 \log^2 \Lambda)$ (84). X_{Hux} is effective but has not been computed, so X_0 is not fixed at e^{10^8}; every quantitative estimate therefore holds uniformly in $\Lambda = \log X_0 \geq 10^8$, in particular Theorem 5.6, Theorem 6.22 and (83), which use computer-assisted steps (Appendix B).
κ	$\kappa := X_0^{-5/12+2\varepsilon_\kappa}$ with $\varepsilon_\kappa := 0.01$, i.e. $\kappa = X_0^{-0.3967} \approx e^{-3.97 \cdot 10^7}$ (the window of the rigid density; every use needs only $0 < \varepsilon_\kappa \leq 0.04$).
$H(x)$	$H(v) := \max(H_{\mathrm{std}}(v), \bar{M}_1(v)^2)$ for $v \geq X_0$ and $H(v) := H_{\mathrm{std}}(v)$ for $v < X_0$, with $H_{\mathrm{std}}(v) := \exp(8(\log\log v)^2)$, where $\bar{M}_1(v) := \bar{r}_0 + \varphi_v(\bar{r}_0) + 1$ and $\bar{r}_0(v) := \log v/(3\log\log v) + 4$ are smooth majorants of M_1 and r_0 (φ_v is φ below with $c = c(v)$). So H is a function of the height, continuous and non-decreasing, with $d\log H/d\log z \leq 4 \cdot 10^{-6}$ (Theorem 6.22(c)) and $H \geq M_1^2$ (Section 6.1); the definition below X_0 makes H continuous at X_0 and equal to H_{std} on $[X_0^-/2, X_0^2]$; $H(X_0) \approx e^{2715}$. The raise $\bar{M}_1^2$ can exceed H_{std} only on the raised range of Theorem 6.22, which ends at $\ell_{\mathrm{std}} = \sqrt{(7.5\Lambda + \log 4)/4}$ ($1.37 \cdot 10^4$ at $\Lambda = 10^8$). Always $\log H - \log H_{\mathrm{std}} \leq 10^{-6} L$, so $H = x^{o(1)}$; also $H \leq 16 X_0^{15}$ for $\ell \leq \ell_{\mathrm{std}}$, and $H = H_{\mathrm{std}}$ beyond.
λ	$\lambda = 4L$ (Theorem 6.6); $\lambda_u = 3\log u + 2\log\log u$ (per-scale noise); $\lambda'_u = \lambda_u + 2 + \log H(u)$ (short windows); $\lambda_P = 4\log P$ (Lemma 6.1).
c'	$c' = 0.15$ (the drift margin).
r_0, c, M_1	$r_0 = \lceil L/(3\log L) \rceil + 3$, the rank threshold. Capacity threshold $\log c = \max(4\ell^2 - \log 4, \gamma(s)\Lambda)$, $s := \ell - 2\log\Lambda$, with the explicit ramp $\gamma \leq 7.5$ of Theorem 6.22; so $c = \frac14 H_{\mathrm{std}}^{1/2}$ for $\ell \geq \ell_{\mathrm{std}}$, and $c \leq X_0^{7.5}$ for $\ell \leq \ell_{\mathrm{std}}$. $M_1 = r_0 + \varphi(r_0) + 1$ with $\varphi(u) = c + uc^{1/2} + u(u+1)/2$ is the cluster bound, and $M_1 \leq \bar{M}_1 \leq 2c$. Tolerance: $M_1 \leq H/L^{14}$.
L_0, $q_{\min}$, B	$L_0 = \max(L^5, 48 M_1 L^3/c'^2)$; $q_{\min} = L_0 L^6$; queue bound $B(x) \leq 5 M_1 L^2/c'$ (71); $K_{\mathrm{loc}} = 2B(4P) + 7\lambda_P/3 + 23$.
absorption	$L_* = 10^7$ (inactive, since $\log X_0 \geq 10^8$); $y(n) := y(x) = H(x) L_0(x) (\log x)^6$ for $n \in [x, 2x)$, x dyadic (frozen per scale), with $y \leq X_0^{22.6} L^9$ for $\ell \leq \ell_{\mathrm{std}}$ and $\log 2y \leq 12.1\ell^2$ beyond; $\mathcal{A} = \emptyset$ wherever $y < X_0^- = X_0 - H(X_0)/2$, which holds for $\ell \leq 2\log\Lambda + s_H$, $s_H = 5.9745$ (Theorem 6.22(b)); $h(u) = u^{\theta/2}$; Lemma 6.25 uses $\sup_{[u,2u]} y$.
cluster window	$\delta_1 = 2L^{-6}$.
(12)	$\lvert D \rvert \leq x^\eta$ with $\eta = \min(\theta, \frac12 - \theta)/2$, for all $x \geq X_0^-$; $D = 0$ below X_0^-.
amplitudes	$\kappa'_k = \kappa_0/\log^2 x_k$, $\kappa_0 \in [X_0^{-1}, \frac14]$; $\Gamma_k = \kappa_0 x_k^{1-\theta}/(2\log^3 x_k)$; $\gamma_k \in [\Gamma_k, 2\Gamma_k]$; $\alpha_k \in [1, 2]$; $\kappa_+ = 0.8$.
switch-on	$x_1 \geq \exp(X_0^2)$, $x_{k+1} \geq \exp(x_k)$; $R_k \equiv 0$ on $[1, x_k + 2H(x_k)]$.
budget	$\mathbb{P}(\tau < \infty) \leq \sum_m (\Lambda + m\log 2)^{-2} + O(X_0^{-1}) \leq 1.5/\log X_0$ (Lemma 6.9).

This regime is far beyond anything that can be simulated. Experiments at small scales (with $X_0 \leq 10^7$ and $x \leq 5 \cdot 10^8$) reproduce the qualitative behaviour (an observed queue length of order $\log^2 x$, far below the proved bound $B = e^{O(\ell^2)}$, and designed zeros near the predicted points), but they test only the mechanism, not any inequality used in the proof.

2.3. **Logical structure.** The table lists the main results with their inputs. An input marked $\downarrow$ is used only at a strictly earlier time or a strictly smaller height; every other input is used at the same time. The same-time dependencies form an acyclic graph; this was checked mechanically on the full dependency graph (79 nodes, 193 edges, 27 of them $\downarrow$; Appendix B.8). The table is an abridgement of that graph. The time-ordering of the $\downarrow$ edges is Section 2.4.

result	inputs	used in
(80)	Heath-Brown, de la Vallée Poussin	(6), Lemmas 7.1, 7.3
Lemma 3.2 (adaptedness)	Lemma 3.5 (causality), Section 2.2	Section 2.4, Lemma 6.9
(6) (positivity)	(80), the cap, (2)	slots, (F1)
(7) (target error)	Rankin; Lemma 6.25↓	Theorem 2.1(i), Section 5, Lemma 4.4
Lemmas 3.6–3.9	the greedy rule	Theorem 6.6, Corollary 6.3, (12)
Lemmas 4.2, 4.3	(3); partial summation	Section 4; Theorem 5.24
Lemma 4.4	Lemma 6.25↓, Theorem 6.22(b)↓	Lemma 4.8, Theorem 4.9
Lemmas 4.6–4.8	Lemma 4.2; mean values; (12)↓ on $[1, x_k)$, Lemma 4.4↓	Theorem 4.9, choice of γ_k
(12)	(11), Theorem 6.6, Lemma 6.9	Theorem 2.1, Theorem 4.9
Theorem 4.9	(12), Lemmas 4.2, 4.4, 4.6–4.8; Rouché	Theorem 2.1(ii)
Lemmas 7.1, 7.3	(80); Lemma 5.1↓, (7)↓, (12)↓ on cofactors	Proposition 5.35
Lemma 5.1 (Chebyshev–Mertens)	(21) from Theorem 6.6↓	Section 5, (F3′), (F4)
Theorem 5.6, Lemma 5.8	Lemmas 5.1, 5.3; certificate	Corollary 5.9, Lemma 5.32
Theorems 5.14, 5.15, Corollary 5.17, Proposition 5.18	Lemmas 5.5, 5.20, Corollary 5.9, (18); for the absorbed count: Corollary 3.8, Lemma 6.25, Proposition 6.10	Proposition 5.26, Lemma 5.27
Lemma 5.23 (threshold mass)	Lemma 5.1; Corollary 6.3↓	Proposition 5.35
Lemma 5.22 (per-scale noise)	Freedman; Theorem 6.6↓, Proposition 6.10↓, Lemma 6.25↓	Proposition 5.38
Theorem 5.24 (local PNT)	Lemma 4.3, (17), Lemma 3.4 ((7), (8)), Lemma 3.3, Lemma 5.1; $\lvert N - T_0\rvert$ bound↓	Proposition 5.35, Lemma 5.32
(63)	(3), (5); Lemma 6.26(c) via Lemma 6.25↓	Propositions 5.26, 5.38
Lemmas 5.27, 5.32 (jump at X_0)	Theorem 5.15, Corollary 5.17, Propositions 5.18, 5.26, Lemma 5.20, Theorem 5.24, Lemma 5.8; Corollary 6.3↓, Lemma 6.25↓	Propositions 5.35, 5.38, Theorem 5.37
Proposition 5.35, Theorem 5.37, Proposition 5.38	Lemma 7.3, Proposition 5.26, Theorem 5.24, Lemmas 5.27, 5.22, (63)	Corollary 5.40
Corollary 5.40 (R1′, every window)	Proposition 5.38	Theorem 6.6
Lemma 6.1, Corollary 6.3	(F1), (F2), Freedman, Lemma 3.6, Theorem 6.22(c); Theorem 6.6↓	Section 5, Lemma 6.25 (↓ uses)
Theorem 6.6 (queue bound)	Corollary 5.40, (F1), (F2), Lemmas 3.7, 3.9, Theorem 6.22(b),(c), Freedman; Proposition 6.10↓, Corollary 6.11↓, (F3′)↓, Theorem 6.6↓	(12), Lemma 6.9
Definition 6.8, Lemma 6.9	Theorem 6.6, Proposition 6.10, Lemmas 6.1, 5.22	Theorem 2.1
Proposition 6.10, Corollary 6.11	Theorem A.12, Lemma 6.13, Theorem 6.22	Theorem 6.6, Lemmas 5.22, 6.25, Proposition 5.18
Theorem 6.22(a) (capacity)	combinatorial covering lemmas, Rankin, Rosser–Schoenfeld	Proposition 6.10
Theorem 6.22(b),(c)	none (definitions of the parameters)	(F3′), Lemmas 6.1, 6.25, 4.4, Theorem 6.6
Lemmas 6.25, 6.26 (absorption)	Lemma 5.1, Theorem 6.22(b), Corollary 6.11; Proposition 6.10↓, Corollary 6.3↓	(5), (7), Lemma 4.4, (63), Proposition 5.18
Theorem A.12 (Appendix A)	(F1)–(F4)	Proposition 6.10

The three loops of the proof close only through ↓ edges:

- queue → R1′ → per-scale noise → queue: Lemma 5.22 reads the queue at earlier slots;

- queue → cluster event → Lemma 5.1 → (21) → queue: the cluster event reads Mertens bounds at heights $\le s - H$;
- local count → queue → local count: Corollary 6.3 at height z is read only by steps that run after $z + 7H(z)$ (Remark 6.5).

2.4. **Why the induction is not circular.** The induction runs over *time* (slots), not over scales. Every bad event is frozen at a time before any consumer reads it.

(N0) Deterministic parameters. H, c, M_1, L_0, y, r_0 and the switch-on points are deterministic functions of the height or of the scale, fixed in Section 2.2 before the construction starts. The only random objects are the decisions P_j and the jitters U_j.

(N1) First failure (Definition 6.8, Lemma 6.9).

- Every bad event E carries a time $t(E)$, a slot, and is $\mathcal{F}_{t(E)}$-measurable. Put

$$\mathrm{Good}(<t) := \{Q_j \le B(s_j) \text{ for all } j \text{ with } t_j < t\} \cap \bigcap_{t(E)<t} E,$$

  where the queue constraint at j carries the time $t_j := \lceil s_{j+1} \rceil + 1$, the reach of its outer deterministic window. Slots can be longer than 1, but $s_{j+1} - s_j \le 11.3L$ by (6). The time t_j is predictable, since s_{j+1} is determined by T on $[1, s_{j+1}]$ (Lemma 3.2). Then $\tau := \min\{t : \mathrm{Good}(\le t) \text{ fails}\}$ is an $(\mathcal{F}_t)$-stopping time.
- Each event's bound holds on $\mathrm{Good}(< t(E))$. Freedman's inequality is used in the form $\mathbb{P}(S \ge a,\ V \le v)$, or peeled, so no conditioning on the future enters.
- (D) The queue bound at s_j is deterministic on $\mathrm{Good}(< t_j) \cap \bigcap_{t(E)\le t_j} E$ (Theorem 6.6(ii)). The relevant window event has time $\max I \le t_j$. Events with time in $(s_j, t_j]$ read only lower heights, so none of them uses Q_j.
- Hence $\mathbb{P}(\tau < \infty) \le \sum_E \mathbb{P}(E^c \cap \mathrm{Good}(< t(E))) \le 1.5/\log X_0$, and on $\mathcal{G} = \{\tau = \infty\}$ every event holds at every scale.

(N2) The inputs of the queue bound at $s_j \in [x, 2x)$ all have time $\le t_j$. R1′ (Corollary 5.40) on a window I with $\max I \le t_j$ holds on $\mathrm{Good}(< \max I)$, and it reads:

- the local prime number theorem (Theorem 5.24, against the target without the absorbed part) on $[1, u(1-2\delta)]$, which is pathwise in the realised count below, so no event is needed;
- local counts at heights $\le u/2$ or $\le \sqrt{u}$, and one local count at height $u(1-\delta)$ (the proof of Proposition 5.35(B), Step 1), read before $u - \delta u + 7H < u$;
- the per-scale noise event for the noise on $[1, u/2]$, at time $2v \le u$;
- the short-window events, internally ordered by window start, each window's variance bound coming from strictly earlier windows (Proposition 5.38);
- the cluster event at heights $\le 8u/L^{11} + H$.

(N3) The cluster event follows from Theorem A.12 at lower heights, independently of the queue.

- The moment bounds of Section 6 and Appendix A carry the good-event indicator progressively, $\mathbf{1}(\mathrm{Good}(< s_{p_1})) \ge \mathbf{1}(\mathrm{Good}(< s_{p_2})) \ge \cdots$ at successive pivots.
- They use it only through: the harmonic sums (F3′) over elements whose primes are all decided before s; Mertens bounds at heights $\le s - H$, which hold surely on $\mathrm{Good}(< s)$ (Lemma 5.1, local form); and the sure slot facts (F1), (F2).
- No queue bound at heights in the window is used quantitatively, and the window's time is at most the cluster event's time.

(N4) The local count uses the queue bound strictly below the current scale.

- The local count at height z (Corollary 6.3: Lemma 6.1 plus the count identity) reads queue bounds at slots $\le z + H(z)$ and the landing event at height z.

- Its uses (Remark 6.5) are: step (ii) of Lemma 5.23 and Lemma 5.20, at heights $\le \sqrt{x}$; Lemma 5.29, at heights $\le y/2$; Step 1 of the proof of Proposition 5.35(B), for a window that ends $\delta x \gg H$ before x; and optionally step (iv) of Lemma 6.25, at heights $u/d \le u^{1-\theta/2}$. Each is read at a strictly earlier time than its consumer.
- The per-prime increment in Theorem 6.6 uses the cluster event (Corollary 6.11), never the local count, because a local count at height x/q would force $B(x) \ge (3/c')L^2 B(x/q)$.

(N5) Absorption and zeros.

- Membership in $\mathcal{A}$ is decided at the latest prime $r < y(n) \ll n$, and $\rho_{\mathcal{A}}$ at u reads only data decided before $u - H$ (Lemma 3.2).
- Lemma 6.25 at u reads the cluster event and local counts at heights $u/d < u$: an induction on height.
- γ_k is chosen from Z_∞ (deterministic) and the system on $[1, x_k)$ (Lemma 4.8), and $R_k \equiv 0$ on $[1, x_k + 2H(x_k)]$. So T is adapted.
- (12) and Section 4 are pathwise consequences on $\mathcal{G}$.

3. The construction

The parameters X_0, κ, H, M_1, L_0, y, x_k, γ_k, α_k, κ_0 are those of Section 2.2.

3.1. The base, the rigid set and the semigroup.

- *Base.* Below $X_0^- := X_0 - H(X_0)/2$ the system is exactly $\mathbb{N}$. Its primes are the rational primes $p < X_0$, placed at their own values with no jitter. On $[X_0^-, X_0)$ it is $\mathbb{N}$ together with the at most $H(X_0)/2+1$ jittered primes that are decided at slots with $s_j - H(s_j)/2 < X_0$ and land there.
- *Jittered primes.* Every prime placed by the construction (Section 3.4) has value $\ge X_0 - H(X_0)/2$ and is called *jittered*.
- *Elements.* $\mathcal{N}$ is the free commutative semigroup on all primes, rational and jittered. An element is an exponent vector, and its value is $n = \prod p^{e_p}$. $N(x)$ counts elements of value $\le x$, with multiplicity. An element is *composite* if $\sum e_p \ge 2$. Its *latest prime* is the prime factor decided last (Section 3.4).
- *The rigid set* $\mathcal{D}$ is the set of X_0-smooth integers, i.e. integers all of whose prime factors are $< X_0$. Write $\Psi(x, X_0) := \#\mathcal{D} \cap [1, x]$ and $\Psi(I, X_0) := \#\mathcal{D} \cap I$. Every $d \in \mathcal{D}$ with $d \ge X_0$ is composite.

3.2. The target.

Oscillations. For $k \ge 1$ put $x_k^+ := x_k + 2H(x_k)$ and

$$R_k(x) := \alpha_k x^\theta \cos(\gamma_k \log x)\, \phi\big(\log(x/x_k^+)\big), \qquad R := \sum_k R_k, \tag{1}$$

where:

- $\phi \in C^\infty$, $0 \le \phi \le 1$, $\phi = 0$ on $(-\infty, 0]$, $\phi = 1$ on $[1, \infty)$ and $\phi' \ge 0$. So $R_k \equiv 0$ on $[1, x_k + 2H(x_k)]$. The term $2H$ makes Lemma 3.2 exact: γ_k reads primes decided at slots up to $x_k + H(x_k)/2$, and T on $[1, u]$ is read by slots $\le u - H(u)$.
- The *amplitude condition*:

$$\alpha_k \gamma_k x_k^{\theta-1} = \kappa_k' / \log x_k, \qquad \kappa_k' := \kappa_0 / \log^2 x_k, \quad \kappa_0 \in [X_0^{-1}, \tfrac{1}{4}]. \tag{2}$$

  It gives $|R'(x)| \le \kappa_+ / \log x$ with $\kappa_+ = 0.8$ (see (6)).
- $\gamma_k \in [\Gamma_k, 2\Gamma_k]$ is chosen in Lemma 4.8 from data on $[1, x_k)$ only, and $\alpha_k \in [1, 2]$.

In Section 4, x_k means x_k^+. The difference is a relative $O(H(x_k)/x_k)$, which is invisible in every estimate.

The rigid density. With $\kappa := X_0^{-5/12+2\varepsilon_\kappa}$, $\varepsilon_\kappa := 0.01$, put

$$\rho_{\mathrm{det}}(u) := \frac{\Psi([u,(1+\kappa)u],X_0)}{\kappa u} \qquad (u \geq X_0). \tag{3}$$

This is a linear-window average with linear normalisation. Each rigid $n \geq X_0(1+\kappa)$ contributes a ramp of total mass $\log(1+\kappa)/\kappa = 1-\kappa/2+O(\kappa^2)$ on $[n/(1+\kappa), n]$.

The absorbed set (random and adapted). Put $L_* := 10^7$, $h(u) := u^{\theta/2}$ and $y(n) := y(x) = H(x)L_0(x)(\log x)^6$, where x is the dyadic scale with $n \in [x, 2x)$ and H, L_0 are the parameters of Section 2.2; so y is frozen per scale. Then

$$\mathcal{A} := \{n \in \mathcal{N} : \ \log n \geq L_*, \text{ the latest prime } r \text{ of } n \text{ is jittered, and } r < y(n)\}, \tag{4}$$

with any cofactor. Also put $\rho_{\mathcal{A}}(u) := \#(\mathcal{A} \cap [u, u+h(u)])/h(u)$.

The target. Put $T(x) := \lfloor x \rfloor$ for $x < X_0$, and for $x \geq X_0$

$$\begin{aligned} T(x) &:= X_0 + \big[\Psi(x,X_0) - \Psi(X_0,X_0)\big] + \#(\mathcal{A} \cap [X_0,x]) + \int_{X_0}^{x} T'_{\mathrm{cont}}(u)\,du, \\ T'_{\mathrm{cont}} &:= 1 - \rho_{\mathrm{det}} - \bar{\rho}_{\mathcal{A}} + R', \qquad \bar{\rho}_{\mathcal{A}}(u) := \min\big(\rho_{\mathcal{A}}(u), 0.1/\log u\big). \end{aligned} \tag{5}$$

So on $[X_0, \infty)$, T has a unit jump at every $d \in \mathcal{D} \cap [X_0, \infty)$ and at every $n \in \mathcal{A}$, and no other jump. At $d = X_0$ this is the step from $\lfloor x \rfloor = X_0 - 1$ to $T(X_0) = X_0$: as X_0 is even, it lies in $\mathcal{D}$, and the term $\Psi(x, X_0) - \Psi(X_0, X_0)$ counts the $d \in \mathcal{D} \cap (X_0, x]$.

Remark 3.1 (The stopped process). Every probabilistic statement in Sections 4–6 concerns the process stopped at the first failure time τ of Definition 6.8.

- Before τ, i.e. on Good$(< u)$, the event of Lemma 6.25 for the window $[u, u+h(u)]$ holds, since its time is $\leq u - H(u)$ (Lemma 3.2). So $\rho_{\mathcal{A}}(u) \leq (\log u)^{-4} < 0.1/\log u$ and $\bar{\rho}_{\mathcal{A}}(u) = \rho_{\mathcal{A}}(u)$.
- The cap can bind only after τ, where nothing is claimed. Its role is to keep T, the slots and the construction defined on every ω.
- The cap is adapted, being a deterministic function of the adapted $\rho_{\mathcal{A}}$.

Lemma 3.2 (Adaptedness)**.** *T restricted to $[1,u]$ is determined by the decisions and jitters at slots $s_j \leq u - H(u)$.*

Proof. An element $n \leq u + h(u)$ of $\mathcal{A}$ has latest prime $r < y(n) \leq n^{o(1)} \ll u - H(u)$. So r, and every other prime of n, is decided before slot $u - H(u)$ (by (10) below). The set $\mathcal{D}$ and the density ρ_{det} are deterministic, and the parameters are deterministic functions of the height or of the scale. The oscillation R on $[1,u]$ involves only those R_k with $x_k + 2H(x_k) < u$, and each such γ_k reads primes decided at slots up to $x_k + H(x_k)/2 \leq u - H(u)$. Hence $\rho_{\mathcal{A}}$ on $[X_0, u]$, R on $[1,u]$ and the jumps of T in $[1,u]$ are known before slot u. □

3.3. **Positivity of the target and the target error.**

Lemma 3.3 (Positivity margin)**.** *For every $u \geq X_0$ and every ω,*

$$T'_{\mathrm{cont}}(u) \geq \frac{0.99 - 0.1 - \kappa_+}{\log u} - \frac{1}{\kappa u} = \frac{0.09}{\log u} - \frac{1}{\kappa u} > 0, \qquad \kappa_+ = 0.8. \tag{6}$$

In particular T is non-decreasing on every ω.

Proof. There are three inputs.

- *Rigid part.* $1 - \rho_{\mathrm{det}}(u) \geq 0.99/\log u - 1/(\kappa u)$. Every window $[u, (1+\kappa)u]$ with $u \geq X_0$ lies in the range of the short-interval prime number theorem ((80), Lemma 7.3(a)), so it contains at least $0.99\kappa u/\log u$ rational primes among its at most $\kappa u + 1$ integers.

With the linear normalisation (3) there is no loss of $\kappa/2$; such a loss belongs to the logarithmically normalised kernel, and a bound of the form $0.99/\log u - \kappa/2$ would be negative once $\log u > 2/\kappa \approx e^{3.97\cdot 10^7}$, which is still below x_1.

- *Absorbed part.* $\bar\rho_{\mathcal A} \le 0.1/\log u$, by the cap.
- *Oscillations.* $R \equiv 0$ below $x_1 + 2H(x_1)$. For $x \ge x_k$, by (2),

$$|R_k'(x)| \le \alpha_k x^{\theta-1}(\gamma_k + \theta + \|\phi'\|_\infty) \le 2\alpha_k\gamma_k x^{\theta-1} = \frac{2\kappa_0 (x/x_k)^{\theta-1}}{\log^3 x_k}.$$

For the last switched-on k: if $x \le x_k^2$ then $\log x \le 2\log x_k$; otherwise $(x/x_k)^{\theta-1} \le x^{-(1-\theta)/2}$. Earlier k contribute less, by the spacing $x_{k+1} \ge \exp(x_k)$. So $|R'(x)| \le 16\kappa_0/\log^3 x + 4\kappa_0 x^{-(1-\theta)/2} \le \kappa_+/\log x$, since $\log x \ge X_0^2$.

Finally $1/(\kappa u) \le X_0^{-7/12} \ll 0.09/\log u$ for $u \ge X_0$. □

Lemma 3.4 (Target error). *Write $E(x) := T(x) - x - R(x)$. On the event of Lemma 6.25,*

$$|E(x)| \le C_{X_0} x^\theta + 2x^{\theta/2} \qquad \text{for all } x \ge X_0, \quad C_{X_0} < \infty. \tag{7}$$

Moreover $|R(x)| \le 2x^\theta \log^* x$, and the rigid part $E_{\mathrm{rig}}(x) := \Psi(x, X_0) - \Psi(X_0, X_0) - \int_{X_0}^x \rho_{\det}$ of E satisfies, for every $x \ge X_0$,

$$|E_{\mathrm{rig}}(x)| \le \min\Big(\Psi(2x, X_0),\ \kappa X_0 + 1 + \min\big(\kappa x + 1, \Psi(2x, X_0)\big)\Big). \tag{8}$$

Proof. Rigid part. Each rigid $d > X_0$ contributes a unit jump at d and the ramp mass $\kappa^{-1}\log\big(\min(x,d)/\max(X_0, d/(1+\kappa))\big)^+$ to $\int_{X_0}^x \rho_{\det}$. So $E_{\mathrm{rig}}(x) = E_0 + E_{\mathrm{dr}} - E_{\mathrm{fl}}$, where:

- $E_0 \in [0, \kappa X_0 + 1]$ comes from the $d \in (X_0, X_0(1+\kappa))$, whose ramps are truncated at X_0 (Lemma 7.1(4));
- $E_{\mathrm{dr}} \in [0, \frac{\kappa}{2}\Psi(x, X_0)]$ is the drift: the $d \in [X_0(1+\kappa), x]$ each carry ramp mass $\log(1+\kappa)/\kappa = 1 - \kappa/2 + O(\kappa^2)$ (Lemma 7.1(5));
- $E_{\mathrm{fl}} \in [0, \#(\mathcal D \cap (x, (1+\kappa)x))]$ is the mass in flight.

Each rigid $d \le x$ contributes at most 1 to $E_0 + E_{\mathrm{dr}}$, so $E_0 + E_{\mathrm{dr}} \le \Psi(x, X_0)$, and $E_{\mathrm{fl}} \le \Psi(2x, X_0) - \Psi(x, X_0)$. Since $E_0 + E_{\mathrm{dr}}$ and E_{fl} are non-negative, $|E_{\mathrm{rig}}| \le \max(E_0 + E_{\mathrm{dr}}, E_{\mathrm{fl}})$. Both E_{dr} and E_{fl} are $\le \min(\kappa x + 1, \Psi(2x, X_0))$ (as $\Psi(x, X_0) \le x$), which gives (8).

Rankin's method gives $\Psi(2x, X_0) \le (2x)^\theta C'_{X_0}$, where $C'_{X_0} := \prod_{p<X_0}(1 - p^{-\theta})^{-1} < \infty$, since $\Psi(y, X_0) \le \sum_{d\in\mathcal D}(y/d)^\theta$. So $|E_{\mathrm{rig}}(x)| \le C_{X_0} x^\theta$ with $C_{X_0} := 2C'_{X_0}$. At the scales $\log x \ge X_0^2$ of the zeros the drift E_{dr} is $\le \kappa x^{0.011}$, since by Rankin $\Psi(x, X_0) \le x^{0.01}\prod_{p<X_0}(1 - p^{-0.01})^{-1} \le x^{0.011}$ there: each factor is at most $(1 - 2^{-0.01})^{-1} < 145 < e^5$, so the product is $\le e^{5\pi(X_0)} \le e^{0.001X_0^2} \le x^{0.001}$.

Absorbed part. Put $b := \theta/2$ and, for $a \in \mathcal A$, let $u_a + h(u_a) = a$ and $M_a := \int_{u_a}^a du/h(u)$, the mass of the smoothed element a. An element a contributes to $\rho_{\mathcal A}(u)$ exactly for $u \in [u_a, a]$, so

$$\#\mathcal A(\le x) - \int^x \rho_{\mathcal A} = \sum_{a\in\mathcal A,\, a\le x}(1 - M_a) - \sum_{a\in\mathcal A,\, u_a < x < a}\int_{u_a}^x \frac{du}{h(u)}.$$

The second sum runs over $a \in (x, x + h(x))$, and is at most $\#(\mathcal A \cap [x-h, x+h]) \le 2h(x)(\log x)^{-4}$ by Lemma 6.25. In the first, h is increasing, so $(u_a/a)^b = h(u_a)/h(a) \le M_a \le 1$; with $a - u_a = h(u_a) \le a^b$ this gives $0 \le 1 - M_a \le -b\log(u_a/a) \le b\, u_a^{b-1} \le 1.01\, b\, a^{b-1}$. To sum this, note that $a^{b-1} \le M_a^{-1}\int_{u_a}^a u^{b-1}du/h(u)$ with $M_a^{-1} \le 1.001$, and that $\sum_{a\in\mathcal A}\mathbf 1[u_a \le u \le a]/h(u) = \rho_{\mathcal A}(u) \le$

$(\log u)^{-4}$ by Lemma 6.26(a), where $\log u \geq 10^7 - 1$ wherever $\rho_{\mathcal{A}}(u) \neq 0$ (since $L_* = 10^7$). Hence

$$\sum_{a \in \mathcal{A},\, a \leq x} a^{b-1} \leq 1.001 \int_{e^{10^7-1}}^{x} u^{b-1} (\log u)^{-4}\, du \leq \frac{1.01}{b}\, x^b (\log x)^{-4},$$

because $u^{b-1}(\log u)^{-4} \leq (1 - 4/(b \log u))^{-1} \frac{d}{du}\big(u^b (\log u)^{-4}/b\big)$ and $4/(b \log u) \leq 4 \cdot 10^{-6}$. So the first sum is at most $1.01\, b \cdot 1.01\, b^{-1} x^{\theta/2} (\log x)^{-4} \leq 1.03\, x^{\theta/2} (\log x)^{-4}$, and $|\#\mathcal{A}(\leq x) - \int^x \rho_{\mathcal{A}}| \leq 3.1\, x^{\theta/2} (\log x)^{-4} + O(1)$, well within (7).

Oscillations. $|R(x)| \leq 2x^\theta \cdot \#\{k : x_k \leq x\} \leq 2x^\theta \log^* x$. □

3.4. **Slots and the greedy rule.** *Slots.* Fix $\vartheta_{\mathrm{s}} \in (0, \frac{1}{2}]$ such that almost surely no jump of T starts on a level $j + \vartheta_{\mathrm{s}}$. This is possible, since only countably many ϑ_{s} are excluded: the deterministic jump levels form a countable set, and for $n \in \mathcal{A}$ the level $T(n-)$ is a continuous strictly increasing function of the landing position of n, which has a continuous law. For integers $j \geq X_0$, *slot* j is $J_j := T^{-1}([j + \vartheta_{\mathrm{s}}, j + 1 + \vartheta_{\mathrm{s}}))$, with $s_j := \inf J_j$. Each unit jump of T (at $d \in \mathcal{D}$ or $n \in \mathcal{A}$) crosses exactly one level and lies in its own *jump slot*. Every other slot is a *continuous slot*. Consecutive continuous-slot starts are at least $1/(1 + 1/(\kappa u) + \kappa_+/\log u)$ apart at height u, since $T'_{\mathrm{cont}} \leq 1 + 1/(\kappa u) + \kappa_+/\log u$. A single continuous slot can be shorter than 1 where rigid integers are dense.

The hard greedy rule with box-centred jitter. Process $j = X_0, X_0 + 1, \ldots$ in order, keeping $Q_{j-1} \geq 0$.

- A_j is the number of composite elements whose value lies in J_j.
- $P_j := \max(0, 1 - Q_{j-1} - A_j) \in \{0, 1\}$ and $Q_j := Q_{j-1} + A_j + P_j - 1 \geq 0$. No prime is placed while the queue is positive, and a jump slot ($A_j \geq 1$) never receives a prime.
- If $P_j = 1$, place one new prime at $s_j + H(s_j)(U_j - \frac{1}{2})$, where the U_j are i.i.d. uniform on $[0, 1]$ and independent of the past. This holds for every slot, including those near X_0, so the landing density matches the slot count up to $O(1/H)$; Section 5 uses this.
- The *jitter width* is

$$H(x) := \max\big(H_{\mathrm{std}}(x), \bar{M}_1(x)^2\big) \quad (x \geq X_0), \qquad H(x) := H_{\mathrm{std}}(x) \quad (x < X_0), \tag{9}$$
$$H_{\mathrm{std}}(x) := \exp\big(8(\log\log x)^2\big),$$

  with the smooth majorant $\bar{M}_1 \geq M_1$ of Section 2.2. So H is a function of the height itself, not frozen per scale; it is continuous and non-decreasing. The definition below X_0 makes H continuous at X_0 and equal to H_{std} on $[X_0^-/2, X_0^2]$. *Box-centred* means that the landing law of a prime decided at s_j is uniform on $[s_j - H(s_j)/2, s_j + H(s_j)/2)$.
- *Decision order.* $t(p)$ is the slot at which p is decided, with $t(p) := p$ for the rational primes $p < X_0$. For an element w, $\tau_w := \max_{p|w} t(p)$ is the decision time of its latest prime.
- $\mathcal{F}_t$ is the σ-field of all decisions and offsets at slots $\leq t$.

Lemma 3.5 (Causality)**.** *A_j is $\mathcal{F}_{s_j-}$-measurable:*

$$A_j \in \mathcal{F}_{s_j-}. \tag{10}$$

Proof. Let $n \in J_j$ be composite. It has a prime factor $p \leq n/2 < s_{j+1}/2$, which was decided at a slot j' with $s_{j'} \leq p + H/2 \leq s_{j+1}/2 + H/2 < s_j$, once $s_j > 2H + 4$. Conversely, a prime decided at a slot $\geq j$ lands at $\geq s_j - H/2 > s_{j+1}/2$, so it is not a factor of any composite in J_j. □

3.5. **Bookkeeping.**

Lemma 3.6 (Count identity). *Let $j_0 := X_0$ be the first slot (recall that X_0 is an even integer) and $Q_{j_0-1} := 0$. For $j \geq j_0 - 1$ put*

$$V_j := \#(\text{composites of value } < s_{j+1}) + \#(\text{primes decided at slots } \leq j, \text{ including the rational primes } < X_0) + 1.$$

Then $V_j = j + 1 + Q_j$.

Proof. V increases by $A_j + P_j$ at step j, and $Q_j = Q_{j-1} + A_j + P_j - 1$, so $V_j - j - Q_j$ does not depend on j. It remains to compute it at $j = j_0 - 1$, where no slot has been processed and V_{j_0-1} counts 1, the composites of value $< s_{j_0}$ and the rational primes $< X_0$. By (6), T rises by 2 within $30 \log X_0$ of X_0, so $s_{j_0} < X_0 + 30 \log X_0 < 2X_0^-$. Hence every composite of value $< s_{j_0}$ is an integer (an element with a jittered prime factor has value $\geq 2X_0^-$), and jittered primes do not enter V_{j_0-1}. The composite integers in $[X_0, s_{j_0})$ are exactly X_0 itself: $T(X_0) = X_0 < X_0 + \vartheta_s$, so $X_0 < s_{j_0}$, and X_0 is composite (it is even); a composite $n \in (X_0, s_{j_0})$ would be rigid (as $n < 2X_0$), and its unit jump would give $T(n) \geq X_0 + 1 > j_0 + \vartheta_s$, which is impossible for $n < s_{j_0}$. So $V_{j_0-1} = \#(\mathbb{N} \cap [1, X_0)) + 1 = j_0$. □

Lemma 3.7 (Lindley). $Q_j = \max_{j_0-1 \leq i \leq j} \big[\sum_{l=i+1}^{j} A_l - (j - i)\big]$, *with j_0 and $Q_{j_0-1} = 0$ as in Lemma 3.6. In words, the queue is the largest excess of composite arrivals over slots, taken over slot windows ending at j.*

Proof. This is the standard solution of the recursion $Q_j = \max(0, Q_{j-1} + A_j - 1)$, which is equivalent to the rule above. □

Corollary 3.8 (Two-sided lag). *With $D := N - T$, for $x \geq X_0 + 2H(x)$,*

$$-\tfrac{H(x)}{2}\Big(1 + \frac{\kappa_+}{\log x}\Big) - 4 \leq D(x) \leq \max_{s_j \leq x} Q_j + \tfrac{H(x)}{2}\Big(1 + \frac{\kappa_+}{\log x}\Big) + 3. \tag{11}$$

Proof. Let j be the slot with $x \in J_j$, so $s_j \leq x < s_{j+1}$ and $T(x) \in [j + \vartheta_s, j + 1 + \vartheta_s)$. Let E_+ be the number of primes decided at slots $> j$ that land in $[1, x]$, let E_- be the number decided at slots $\leq j$ that land in (x, ∞), and let $C((x, s_{j+1}))$ be the number of composites of value in (x, s_{j+1}). The rational primes are decided before slot j_0 and lie below x. So $N(x) = V_j - C((x, s_{j+1})) + E_+ - E_-$, and Lemma 3.6 gives

$$D(x) = \big(j - T(x)\big) + Q_j + 1 - C((x, s_{j+1})) + E_+ - E_-.$$

The terms are bounded as follows.

- $j - T(x) \in (-1 - \vartheta_s, -\vartheta_s]$.
- $C((x, s_{j+1})) \leq A_j$, and $Q_j - A_j = Q_{j-1} + P_j - 1 \geq -1$.
- A prime decided at slot j' lands within $H(s_{j'})/2$ of $s_{j'}$, and only continuous slots receive primes. The slots counted by E_- therefore start in $(x - H(x)/2, x]$, since H is non-decreasing. Those counted by E_+ start in $(x, x + H(s_{j'})/2]$, and $H(s_{j'}) \leq H(x) + 10^{-6}$ by Theorem 6.22(c), since $s_{j'} - x \leq x^{0.1}$. Each unit jump of T crosses exactly one level, so T gains exactly one unit of continuous mass between consecutive continuous-slot starts; an interval I therefore contains at most $\int_I T'_{\mathrm{cont}} + 1$ of them. With $T'_{\mathrm{cont}} \leq 1 + 1/(\kappa u) + \kappa_+/\log u$ this gives $E_\pm \leq \frac{H(x)}{2}(1 + \kappa_+/\log x) + 1 + \epsilon$, where $\epsilon < 10^{-6}$ covers the term $H/(\kappa x)$ and the variation of H and $\log u$ across the window. (Where $R' > 0$ the slot density exceeds 1 by up to $\kappa_+/\log x$, so $H/2 + O(1)$ would not suffice.)

Hence $D(x) \leq Q_j + (1 - \vartheta_s) + E_+ < \max_{s_j \leq x} Q_j + \frac{H(x)}{2}(1 + \kappa_+/\log x) + 2 + \epsilon$, and $D(x) > -(1 + \vartheta_s) - 1 + 1 - E_- \geq -\frac{H(x)}{2}(1 + \kappa_+/\log x) - 2 - \vartheta_s - \epsilon$, which is $\geq -\frac{H(x)}{2}(1 + \kappa_+/\log x) - 3$ because $\vartheta_s \leq \frac{1}{2}$; in particular the lower bound in (11) holds. □

Lemma 3.9 (Excess identity)**.** *Let $C_{\mathrm{jit}}(W)$ be the number of composites in W whose latest prime is jittered and which are not in $\mathcal{A}$. For every slot window W,*

$$C(W) - \#\mathrm{slots}(W) = C_{\mathrm{jit}}(W) - \int_W T'_{\mathrm{cont}} \pm 2.$$

Proof. The rigid set and $\mathcal{A}$ drop out identically, because each of their elements sits on its own jump slot; the continuous slots in W number $\int_W T'_{\mathrm{cont}} \pm 2$. No smooth-number or Tijdeman-type input enters. □

Reduction. If $C(W) - \#\mathrm{slots}(W) \le B$ for every slot window $W \subset [X_0, x]$, then $D \le B + \frac{H}{2}(1 + \kappa_+/\log x) + 3$ on $[X_0 + 2H, x]$. Everything therefore reduces to the queue bound of Section 6 (Theorem 6.6), whose drift input is Section 5.

4. Zeros: everything reduces to a bound on the lag

Proposition 4.1 (The lag bound)**.** *On the event $\mathcal{G} = \{\tau = \infty\}$ of Definition 6.8,*

$$|D(x)| \le x^{\eta} \qquad \text{for all } x \ge X_0^-, \qquad \eta := \min(\theta, \tfrac{1}{2} - \theta)/2, \tag{12}$$

and $D = 0$ below X_0^-.

Proof. By (11) and Theorem 6.6, $|D| \le B(x) + \frac{H(x)}{2}(1 + \kappa_+/\log x) + 4 \le B(x) + H(x) + 3$ for $x \ge X_0 + 2H$. On $[X_0^-, X_0 + 2H]$ we have $|D| \le B(X_0) + 3H(X_0) + 3$. Indeed, for $x \in [X_0, X_0 + 2H]$ the accounting in the proof of Corollary 3.8 applies verbatim (with $j := j_0 - 1$ if $x < s_{j_0}$, where $V_{j_0 - 1} = j_0$ and $T(x) \in [j_0, j_0 + \vartheta_{\mathrm{s}})$), and gives $|D| \le B(X_0) + \frac{H(X_0)}{2}(1 + \kappa_+/\log X_0) + 5 \le B(X_0) + H(X_0) + 5$. For $x \in [X_0^-, X_0)$, $D(x)$ is the number of jittered primes that land in $[X_0^-, x]$, which is at most $H(X_0)/2 + 1$.

At every scale, $\log(B + 3H + 3) \le 8\ell^2 + 10^{-6}L + 2$. Indeed $\log H - \log H_{\mathrm{std}} \le 10^{-6}L$ by Theorem 6.22(c), and $B \le H$ by (71), since $\log M_1 \le \frac{1}{2}\log H$ and $4\ell^2 \ge 2\ell + 3.6$. Since $(8\ell^2 + 2)/L$ decreases for $L \ge \Lambda \ge 10^8$, this is $\le 2.82 \cdot 10^{-5} L$ for all $L \ge \Lambda - 1$. For $\theta \le \theta_1 = \frac{1}{2} - 10^{-4}$ we have $\eta \ge 5 \cdot 10^{-5}$, so the right side is $\le x^{\eta}$ for all $x \ge X_0^-$. □

This is where $\theta \le \theta_1$ is used. At $X_0 = e^{10^8}$ one has $\log H(X_0) = 2715$, so (12) fails near X_0 for $\eta < 2.8 \cdot 10^{-5}$; going beyond θ_1 would need a θ-dependent X_0, and one value of θ suffices for Corollary 2.2. A polylogarithmic form of (12) is false: the landing lag alone is $H/2 = x^{o(1)}$.

4.1. Decomposition of the zeta function.

For $\sigma > 1$,

$$\zeta_{\mathcal{P}}(s) = \int x^{-s}\, dN = Z_B(s) + \widehat{E}_{\mathcal{A}}(s) + \widehat{R}(s) + \widehat{D}(s), \tag{13}$$

where:

- $Z_B(s) := \sum_{n < X_0} n^{-s} + \sum_{d \in \mathcal{D},\, d \ge X_0} d^{-s} + \int_{X_0}^{\infty} x^{-s}(1 - \rho_{\mathrm{det}})\, dx$;
- $\widehat{E}_{\mathcal{A}}(s) := \int x^{-s}\, d\big(\#\mathcal{A} - \int \bar{\rho}_{\mathcal{A}}\big)$. On $\mathcal{G}$ the cap is inactive, and $\widehat{E}_{\mathcal{A}}(s) = \sum_{a \in \mathcal{A}} a^{-s}(1 - w_a(s))$, where

$$w_a(s) := a^s \int_{u_a}^{a} u^{-s} g(u)\, du, \qquad g := 1/h, \qquad u_a + h(u_a) = a,$$

is the weight of a in the average $\rho_{\mathcal{A}}(u) = \#(\mathcal{A} \cap [u, u + h(u)])/h(u)$ over the forward window $[u, u + h(u)]$. Put $b := \theta/2$, so that $h(u) = u^b$ and g is decreasing, and $M := \int_{u_a}^{a} g$. For $a \ge X_0/2$ (so for every $a \in \mathcal{A}$) and $0 \le \sigma \le 2$:

(i) $(u_a/a)^b \le M \le 1$ and $a/u_a \le (1 - a^{b-1})^{-1}$;

(ii) $|w_a(s)| \le (a/u_a)^{\sigma} M \le 1 + 1.02\, \sigma a^{b-1}$;

(iii) $|w_a(s)| \le 2.01\, a^{1-b}/|t|$ for $t \ne 0$;
(iv) $|1 - w_a(s)| \le 1.01(|s| + b)a^{b-1}$; hence $|1 - w_a(s)| \le 2|t|a^{b-1}$ for $|t| \ge 2$.
Indeed, put $\epsilon := a^{b-1} \le (X_0/2)^{-3/4}$. (i) Since g is decreasing, $h(u_a)/h(a) = (a - u_a)g(a) \le M \le (a - u_a)g(u_a) = 1$, and $h(u_a)/h(a) = (u_a/a)^b$; also $a - u_a = h(u_a) \le a^b$. (ii) $|u^{-s}| \le u_a^{-\sigma}$ on $[u_a, a]$, and $(1 - \epsilon)^{-\sigma} \le 1 + 1.02\,\sigma\epsilon$. (iii) Integrating, $w_a(s) = (a^{1-b} - a^s u_a^{1-s-b})/(1 - s - b)$, so $|w_a(s)| \le a^{1-b}\big(1 + (a/u_a)^{\sigma+b-1}\big)/|t|$, and $(a/u_a)^{\sigma+b-1} \le \max\big(1, (1-\epsilon)^{-2}\big) \le 1.01$. (iv) $1 - w_a(s) = (1 - M) + \int_{u_a}^{a} \big(1 - (a/u)^s\big) g(u)\, du$. Here $1 - M \le 1 - (1-\epsilon)^b \le 1.01\, b\epsilon$, and, by $|e^z - 1| \le |z| e^{\max(0, \operatorname{Re} z)}$, $|1 - (a/u)^s| \le |s| \log(a/u_a)(a/u_a)^\sigma \le |s|\epsilon(1 - \epsilon)^{-3} \le 1.01|s|\epsilon$; with $M \le 1$ this gives the first bound. For $|t| \ge 2 \ge \sigma$ we have $|s| \le \sqrt{2}|t|$, and $1.01(\sqrt{2}|t| + b) \le 2|t|$.

- $\widehat{R} := \sum_k \widehat{R}_k$ with $\widehat{R}_k(s) := \int x^{-s} R_k'(x)\, dx$;
- $\widehat{D}(s) := \int_{X_0^-}^{\infty} x^{-s}\, dD = s \int_{X_0^-}^{\infty} D(x) x^{-s-1}\, dx$, since $D = 0$ below X_0^-. Under (12) this converges absolutely for $\sigma > \eta$.

Lemma 4.2 (The base factor)**.** *Let $Z_\infty(s) := \prod_{p < X_0} (1 - p^{-s})^{-1} = \sum_{n \in \mathcal{D}} n^{-s}$. For $|t| \ge \Gamma_1$ and $\sigma \ge \frac{1}{2}$,*

$$|Z_B(s) - Z_\infty(s)| \le \exp(-X_0^2/4).$$

On $\sigma \ge 1 - \delta$ with $\delta \log X_0 \to 0$: $|Z_\infty| \le 2 \log X_0 =: Z_{\max}$ and $|Z_\infty'| \le 4 \log^2 X_0$.

Proof. Each rigid $n \ge X_0(1 + \kappa)$ contributes a unit jump at n and the ramp $-du/(\kappa u)$ on $[n/(1+\kappa), n]$. The at most $\kappa X_0 + 1$ rigid $n \in [X_0, X_0(1+\kappa))$ have ramps truncated at X_0; each truncation changes the transform by $(X_0^{-s} - n^{-s})/(s\kappa)$, of modulus $\le 2X_0^{-\sigma}/(\kappa|s|)$, so these boundary terms total $\le 3X_0^{1-\sigma}/|t|$. Up to them,

(14)
$$Z_B = Z_\infty - W(s) S_{\ge X_0}(s) + \frac{X_0^{1-s}}{s-1}, \qquad W(s) := \frac{(1+\kappa)^s - 1}{s\kappa}, \qquad S_{\ge X_0}(s) := \sum_{d \in \mathcal{D},\, d \ge X_0} d^{-s}.$$

For $\sigma \le 2$, $|W| \le 6/(\kappa|t|)$ and $|S_{\ge X_0}(s)| \le Z_\infty(\sigma)$. Also $\log Z_\infty(\frac{1}{2} + \delta) \lesssim 2X_0^{1/2}/\log X_0$, while $\log(\kappa\Gamma_1) \ge (1 - \theta)X_0^2 - O(\log X_0)$, since $\log x_1 \ge X_0^2$; the boundary terms and $X_0^{1-s}/(s-1)$ are $\le 4X_0^{1/2}/\Gamma_1$ for $\sigma \ge \frac{1}{2}$. On $\sigma \ge 1 - \delta$: $Z_\infty(\sigma) \le e^\gamma \log X_0 \cdot e^{\delta \log X_0}(1 + o(1))$, and $|Z_\infty'| \le Z_\infty(\sigma) \sum_{p < X_0} \log p/(p^\sigma - 1)$. □

Lemma 4.3 (The base factor at all heights)**.** *Let $0 < a \le 8$ and $\mathrm{Ein}(a) := \int_0^a (e^t - 1)\, dt/t$, so that $\mathrm{Ein}(1) = 1.32$ and $\mathrm{Ein}(8) = 437.7$. Put $\widehat{E}(s) := Z_B(s) - s/(s-1)$. Then for $\sigma \ge 1 - a/\Lambda$ and every t,*

$$\begin{aligned} |\widehat{E}(s)| &\le C_0 e^a \Lambda + 4.04 e^{\mathrm{Ein}(a)} \min\big(1, \kappa(1 + |s|)\big) e^\gamma \Lambda + 3\kappa e^a + 1 \\ &\le C_a (1 + \kappa|s|) e^\gamma \log X_0, \qquad C_a := C_0 e^a + 5 e^{\mathrm{Ein}(a)}, \end{aligned}$$

with $C_0 := 2.1$. Also $1 - W(1) = 0$, so Z_B has the pole and residue of $s/(s-1)$.

Proof. From (14) and $s/(s-1) = 1 + \int_1^{X_0} x^{-s}\, dx + X_0^{1-s}/(s-1)$,

$$\widehat{E}(s) = \Big[\sum_{n < X_0} n^{-s} - 1 - \int_1^{X_0} x^{-s}\, dx \Big] + (1 - W(s)) S_{\ge X_0}(s) + (\text{boundary terms}).$$

The bracket. For $\sigma > 2$ it is at most 2; let $\sigma \le 2$. For every integer $M \ge 2$, partial summation gives the exact identity

$$\zeta(s) - \frac{1}{s-1} = \sum_{n \le M} n^{-s} - \int_1^M x^{-s}\, dx - s \int_M^\infty \{x\} x^{-s-1}\, dx. \tag{15}$$

Let $|t| \le X_0$. With $M := \lceil X_0 \rceil - 1$, (15) shows that the bracket equals $\zeta(s) - 1/(s-1) - 1$ up to $s\int_M^\infty \{x\}x^{-s-1}\,dx - \int_M^{X_0} x^{-s}\,dx$, which has modulus $\le (|s|/\sigma + 1)M^{-\sigma} \le 1.1e^a$. (This makes explicit the error term $O(X_0^{-\sigma})$ of [18, Thm. 4.11].) With $M := \lfloor |t| \rfloor + 2$ instead, $|s| \le M+1$ and $\max(1, M^{1-\sigma}) \le (X_0+2)^{a/\Lambda} \le 1.001e^a$, so (15) gives $|\zeta(s) - 1/(s-1)| \le 1.001e^a(1 + 2\log M + 1.52) \le 2.01e^a\Lambda$. For $|t| > X_0$ the bracket is $\le 2 + 2\int_1^{X_0} x^{a/\Lambda - 1}\,dx \le 2 + 2e^a\Lambda$. So the bracket is $\le C_0 e^a \Lambda$ with $C_0 := 2.1$.

The rigid part. $|1 - W(s)| \le \kappa|s-1|/2 + O(\kappa^2(1+|s|)^2) \le \kappa(1+|s|)$ when $\kappa|s| \le 1$, and $|W| \le 3$ otherwise (for $\sigma \le 2$; for $\sigma > 2$ the term is $\le 2/X_0$). Next, $|S_{\ge X_0}(s)| \le Z_\infty(\sigma)$, and

$$\log Z_\infty(\sigma) - \log Z_\infty(1) \le \sum_{p < X_0} p^{-1}(p^{a/\Lambda} - 1) + O(a/\Lambda) \le \mathrm{Ein}(a) + O(ae^a/\Lambda).$$

So $Z_\infty(\sigma) \le 1.01 e^{\mathrm{Ein}(a)} e^\gamma \Lambda$ by Mertens' product.

Boundary. The at most $\kappa X_0 + 1$ rigid $n \in [X_0, X_0(1+\kappa))$ each contribute $\le 3n^{-\sigma} \le 3X_0^{a/\Lambda - 1}$. □

The local prime number theorem (Theorem 5.24) compares N with the target without its absorbed part, $T_0 := T - (\#\mathcal{A} - \int \bar\rho_{\mathcal{A}})$, whose transform is $\widehat{T_0} = Z_B + \widehat{R} = s/(s-1) + \widehat{R} + \widehat{E}$. The absorbed part $\widehat{E}_{\mathcal{A}}$ is *not* bounded on all of $\sigma \ge 1 - a/\Lambda$ at large $|t|$ (Remark 4.5), so it is not put into $\widehat{T_0}$. On Good($< y$), $|N - T_0| \le |N - T| + 3.1u^{\theta/2}(\log u)^{-4} + 2$ on $[1, y]$ by the absorbed part of (7). The constant of Lemma 4.3 enters only through $\log C_a$; with $C_0 = 2.1$, $\log C_8 < 439.4$.

Lemma 4.4 (The absorbed part)**.** *Put $u_* := \Lambda e^{s_H}/22.6$, with $s_H = 5.9745$ as in Theorem 6.22(b); so $u_* \ge 1.7 \cdot 10^9$.*

- *(a) On $\mathcal{G}$, $|\widehat{E}_{\mathcal{A}}(s)| \le 2\exp(-0.45\,u_* \log u_*) + 5.4(\log x_k)^9 \exp\big(-\log x_k/(22\log\log x_k)\big) \le e^{-10^{10}}$, where k is the index of the designed zero concerned. This holds on every disc $|s - (1 + i\gamma_k)| \le \delta_k$, on every Rouché circle of Lemma 4.6, and on $\sigma = 1$ at the designed heights.*
- *(b) On $\mathcal{G}$, the term $\int_{X_0}^{x_k} x^{-s}\rho_{\mathcal{A}}\,dx$ has mean square $\le x_k^{\theta - 1/2 - o(1)}$ over $t \in [\Gamma_k, 2\Gamma_k]$, $\sigma \in [\frac34, \frac54]$.*

Proof. The elements of $\mathcal{A}$ are $2y$-smooth, and Rankin's method for the realised system (Lemma 6.25, proof steps (i) and (v)) gives $\#\mathcal{A} \cap [1,x] \le x\exp(-0.9u\log u)$, with $u = \log x/\log(2y) \ge 100$. Moreover $\mathcal{A} = \emptyset$ wherever $y < X_0^-$, which holds for $\log L \le 2\log\Lambda + s_H$ (Theorem 6.22(b)). Where $\mathcal{A} \ne \emptyset$ and $\ell \le \ell_{\mathrm{std}}$, $\log 2y \le 22.6\Lambda$ (Section 2.2). In the standard regime $\ell \ge \ell_{\mathrm{std}}$ instead $H = H_{\mathrm{std}}$, $c = \frac14 H_{\mathrm{std}}^{1/2}$ and $\log 2y \le 12\ell^2 + 9\ell + 8 \le 12.1\ell^2$. So, in the variable $v = \log x$, wherever $\mathcal{A} \ne \emptyset$ we have $\log 2y \le \lambda(v) := \max\big(22.6\Lambda,\, 12(\log v)^2 + 9\log v + 8\big)$, and therefore

$$\#\mathcal{A} \cap [1,x] \le x\,e^{-\varsigma_*(\log x)}, \qquad \varsigma_*(v) := 0.9\,\tilde u(v)\log\tilde u(v), \quad \tilde u(v) := v/\lambda(v).$$

The minorant ς_* is continuous and increasing, since $v\lambda'(v) \le 24\log v + 9 < \lambda(v)$. For $\log v \ge \ell_{\mathrm{std}}$ we have $\lambda(v) \le 12.1(\log v)^2$, hence $\varsigma_*(v) \ge v/(14\log v)$. Since $\mathcal{A} = \emptyset$ for $\log v \le 2\log\Lambda + s_H$, the least v_0 with $e^{v_0} \in \mathcal{A}$ satisfies $\tilde u(v_0) \ge \Lambda^2 e^{s_H}/(22.6\Lambda) = u_*$, so $\varsigma_*(v_0)/2 \ge 0.45\,u_*\log u_* \ge 1.66 \cdot 10^{10}$ for every $\Lambda \ge 10^8$.

(a) Fix $|t| \in [\Gamma_k/2, 3\Gamma_k]$ and $1 - \delta_k \le \sigma \le 2$ (this covers all the sets in (a)), and put $Y := |t|^{1/(1-\theta/2)} \le x_k$. For the terms $a \le Y$ we use $|1 - w_a(s)| \le 1 + |w_a(s)| \le 2 + 2.1a^{\theta/2 - 1} \le 2.0001$, by (ii) after (13). By partial summation these terms are then at most 2.0001 times $\exp(\delta_k \log Y - \varsigma_*(\log Y)) + \int_{v_0}^{\log Y} \exp(\delta_k v - \varsigma_*(v))\,dv$. This is $\le 2\exp(-\varsigma_*(v_0)/2) \le 2\exp(-0.45\,u_*\log u_*)$,

uniformly in k. Indeed, ς_* is increasing, so $e^{-\varsigma_*(v)} \le e^{-\varsigma_*(v_0)/2} e^{-\varsigma_*(v)/2}$, and it suffices that $e^{\delta_k \log Y - \varsigma_*(\log Y)/2} + \int_{v_0}^{\log Y} e^{\delta_k v - \varsigma_*(v)/2}\,dv \le 10^{-6}$. Put $V := e^{\ell_{\mathrm{std}}}$. On $[v_0, V]$ we have $\delta_k v \le 18\Lambda e^{-2\Lambda} V \le 10^{-6}$, since $\delta_k \le 9 \log\log x_k / \log x_k \le 18\Lambda e^{-2\Lambda}$, and $\varsigma_*(v)/2 \ge \varsigma_*(v_0)/2 \ge \ell_{\mathrm{std}} + 20$, so that part is $\le 1.01\, V e^{-\ell_{\mathrm{std}} - 20} \le 10^{-8}$. On $[V, \log Y]$ we have $\log v \le \log\log x_k$, hence $\delta_k v \le \varsigma_*(v)/4$ (as $504(\log\log x_k)^2 \le \log x_k$), and $\varsigma_*(v) \ge v/(14 \log v)$; so that part, and the boundary term, are $\le e^{-V/(60 \log V)} \le 10^{-8}$. The terms $a > Y$ use $|1 - w_a| \le 2|t| a^{\theta/2 - 1}$, from (iv) after (13). With $b = \theta/2$, $|t| = Y^{1-b}$ and partial summation, they are at most

$$2|t| \sum_{a \in \mathcal{A},\, a > Y} a^{b-2+\delta_k} \le 2 Y^{1-b} \frac{2-b}{1-b-\delta_k} Y^{b-1+\delta_k} e^{-\varsigma_*(\log Y)} \le 5.4\, Y^{\delta_k} e^{-\varsigma_*(\log Y)}.$$

Here $Y \le x_k$, so $Y^{\delta_k} \le x_k^{\delta_k} \le (\log x_k)^9$. Also $\log Y \ge 0.66 \log x_k$, since $\log \Gamma_k = (1-\theta)\log x_k - O(\log\log x_k)$ and $(1-\theta)/(1-\theta/2) > \frac{2}{3}$. These a lie in the standard regime ($\log\log Y \ge 2\Lambda - 1 > \ell_{\mathrm{std}}$), where $\varsigma_*(\log Y) \ge \log Y / (14 \log\log Y) \ge \log x_k / (22 \log\log x_k)$. So these terms are $\le 5.4 (\log x_k)^9 \exp(-\log x_k / (22 \log\log x_k))$, which is $\le e^{-10^{11}}$ since $\log x_k \ge X_0^2$.

(b) Mellin–Plancherel gives, at $\sigma = \frac{3}{4}$, $(4\pi/\Gamma_k) \sum_{a \in \mathcal{A},\, a \le 2x_k} a^{1-2\sigma} \le x_k^{1/2 - 1/(14 \log\log x_k)}/\Gamma_k + 2\exp(e^{\ell_{\mathrm{std}}}/2)/\Gamma_k$, and $\Gamma_k \asymp x_k^{1-\theta}/\log^3 x_k$. □

Remark 4.5. $\widehat{E}_{\mathcal{A}}$ can be huge off these sets, for example at $\sigma = 1 - 1/\log X_0$. No step of this section evaluates it there. Lemma 4.4 needs only $y \le n^{o(1)}$ on the range of $\mathcal{A}$, $\log y \le 23 \log X_0$ for $\ell \le \ell_{\mathrm{std}}$ and $\log 2y \le 12.1 \ell^2$ beyond, so it holds for every absorption threshold of Section 2.2, uniformly in $\Lambda \ge 10^8$.

4.2. **The oscillations.** Put $\Phi(w) := \int_0^1 e^{-wv} \phi'(v)\,dv$. Then

$$\widehat{R}_k(s) = \frac{\alpha_k s}{2} \Big[x_k^{-w} \frac{\Phi(w)}{w} + x_k^{-w'} \frac{\Phi(w')}{w'} \Big], \qquad w = s - \theta - i\gamma_k, \quad w' = s - \theta + i\gamma_k. \tag{16}$$

- Φ is entire, with $\Phi(0) = 1$ and $|\Phi(w)| \le C_m |w|^{-m}$ for $\operatorname{Re} w \ge 0$.
- $\operatorname{Re}\Phi(w) \ge e^{-1}\cos 1 =: \operatorname{Re}\Phi_{\min}$ for $0 \le \operatorname{Re} w \le 1$ and $|\operatorname{Im} w| \le 1$.
- So $\zeta_{\mathcal{P}}$ continues to $\sigma > \max(\theta, \eta)$, except for the simple pole at 1.
- Write M_k for the resonant term (the one with w) and E_k for the conjugate one. Near $t = \gamma_k$: $|E_k| \le 2\alpha_k C_2 x_k^{\theta-\sigma}/\gamma_k^2$; $\sum_{j<k} |\widehat{R}_j| \ll \sum_j |\gamma_k - \gamma_j|^{-m}$; and $\sum_{j>k} |\widehat{R}_j| \ll x_j^{\theta-\sigma}\gamma_k$, since $x_j \ge \exp(x_k)$. All three are $o(1)$.
- On $\sigma \ge 1 - 8/\log X_0$, at every height,

$$|\widehat{R}(s)| \le |s|\, e^{-0.39 X_0^2}. \tag{17}$$

Indeed there $\operatorname{Re} w, \operatorname{Re} w' \ge \frac{1}{2} - 8/\log X_0 > 0.4$, $|\Phi| \le \int \phi' = 1$ and $|1/w| \le 2.5$, so $|\widehat{R}_k| \le 5|s| x_k^{-0.4}$. Theorem 5.24 uses this.

4.3. **The designed zeros.**

Lemma 4.6 (A designed zero)**.** *Suppose $|Z_B| \in [\frac{1}{2}, Z_{\max}]$ on $|s - (1 + i\gamma_k)| \le \delta_k$, where $\delta_k := 9 \log\log x_k / \log x_k$. Then $Z_B + M_k$ has a zero s^* with*

$$\operatorname{Re} s^* = 1 - \frac{3 \log\log x_k + O_{X_0}(1)}{\log x_k},$$

and $|Z_B + M_k| \ge \frac{1}{4}$ on the circle $|s - s^*| = 1/\log x_k$.

Proof. 1. The equation is $x_k^w = G(w) := -(\alpha_k s/2)\Phi(w)/(w Z_B(s))$.

2. Since $\alpha_k \gamma_k = \kappa_0 x_k^{1-\theta}/\log^3 x_k$, we have $\log|G| = (1-\theta)\log x_k - 3\log\log x_k + O_{X_0}(1)$ on the box.

3. The map $w \mapsto (\log G(w))/\log x_k$ has derivative $O(\log^2 X_0/\log x_k) = o(1)$ (as $x_1 \geq \exp(X_0^2)$), so it is a contraction with a unique fixed point w^*, and $\operatorname{Re} s^* = \theta + \operatorname{Re} w^*$.
4. On the circle, with $\xi := (w - w^*)\log x_k$: $M_k(s)/M_k(s^*) = e^{-\xi}(1 + o(1))$ and $Z_B(s)/Z_B(s^*) = 1 + o(1)$.
5. Since $Z_B(s^*) = -M_k(s^*)$, we get $|Z_B + M_k| \geq |Z_B(s^*)|(1 - e^{-1} - o(1)) \geq \frac{1}{4}$.
6. The circle lies in the disc. In step 2 the $O_{X_0}(1)$ term lies in $[\log\kappa_0 - \log(2\Lambda) - 1.5,\ \log\kappa_0 + 0.72]$, since $e^{-1} \leq |\Phi(w)| \leq 1$, $0.49 \leq |w| \leq 0.76$ and $\frac{1}{2} \leq |Z_B| \leq 2\Lambda$ there; it is negative, as $\kappa_0 \leq \frac{1}{4}$. Choosing the branch of $\log G$ with $|\operatorname{Im}\log G| \leq \pi$ gives $|\operatorname{Im} w^*| \leq \pi/\log x_k$. Hence $|s^* - (1 + i\gamma_k)| + 1/\log x_k \leq (3\log\log x_k + \log(1/\kappa_0) + \log(2\Lambda) + 5.7)/\log x_k \leq \delta_k$, because $\kappa_0 \geq X_0^{-1}$ and $\log\log x_k \geq 2\Lambda$. □

The discs $|s - (1 + i\gamma_k)| \leq \delta_k$ are small for every $k \geq 1$: $\delta_k \leq \delta_1 \leq 18\log X_0/X_0^2 \ll 1/\log^2 X_0$. So Z_B is evaluated only where Lemma 4.2 applies.

Lemma 4.7 (Late noise)**.** *On the circle of Lemma 4.6, with* $\widehat{D}_{\geq x_k}(s) := \int_{x_k}^{\infty} x^{-s}\, dD$,

$$|\widehat{D}_{\geq x_k}(s)| \leq 6e \cdot |w^*| \cdot x_k^{\eta-\theta} \cdot \frac{Z_{\max}}{\alpha_k \operatorname{Re}\Phi_{\min}} \longrightarrow 0 \qquad (|w^*| \leq 1).$$

Proof. By parts, $|\int_{x_k}^{\infty} x^{-s}\, dD| \leq 3|s| x_k^{\eta-\sigma}$, using (12). On the circle, $x_k^{-\sigma} \leq e\, x_k^{-\operatorname{Re} s^*} = e\, x_k^{-\theta}/|G(w^*)|$, and $|G(w^*)| \geq |s|\alpha_k \operatorname{Re}\Phi_{\min}/(2|w^*|Z_{\max})$. The factor $|s| \approx \gamma_k$ cancels, and the bound tends to 0 because $\eta < \theta$. □

Lemma 4.8 (Good heights exist)**.** *Let* $\Gamma_k := \kappa_0 x_k^{1-\theta}/(2\log^3 x_k)$. *There is* $\gamma_k \in [\Gamma_k, 2\Gamma_k]$ *with:*

(i) $|Z_B(1 + i\gamma_k)| \geq 0.6$*;*
(ii) $|\widehat{D}_{<x_k}(s)| \leq 1/16$ *on* $|s - (1 + i\gamma_k)| \leq \frac{1}{8}$, *where* $\widehat{D}_{<x_k}(s) := \int_{X_0^-}^{x_k} x^{-s}\, dD$.

The choice uses only the system on $[1, x_k)$, *which does not depend on* γ_k. *So* R_k, *and hence* T, *is adapted (Lemma 3.2).*

Proof. (i) We use the mean value theorem of Montgomery and Vaughan [13] for the absolutely convergent Z_∞ and Z_∞^2 on $\sigma = 1$. Over $t \in [\Gamma, 2\Gamma]$:

$$\mathbb{E}|Z_\infty(1 + it)|^2 = \prod_{p<X_0} (1 - p^{-2})^{-1} + O(\log X_0/\Gamma) \in [1, \zeta(2)],$$

$$\mathbb{E}|Z_\infty|^4 \leq \sum_{n\in\mathcal{D}} \tau(n)^2 n^{-2} + o(1) \leq \frac{\zeta(2)^4}{\zeta(4)} + o(1) \leq 6.8.$$

The Paley–Zygmund inequality gives $|Z_\infty(1+it)| \geq 2^{-1/2}$ on a set of t of density $\geq \frac{1}{4}/6.8 > 0.036$, independently of X_0. Lemma 4.2 moves this by at most $\exp(-X_0^2/4)$.

(ii) We have

$$\widehat{D}_{<x_k} = \sum_{n\in\mathcal{N}\setminus(\mathcal{D}\cup\mathcal{A}),\ X_0^- \leq n < x_k} n^{-s} - \int_{X_0}^{x_k} x^{-s}(1 - \rho_{\det} - \rho_{\mathcal{A}} + R')\, dx.$$

- The R' and 1 parts are $O(x_k^{1-\sigma}/\Gamma) + O(\sum_{j<k} |\widehat{R}_j|)$.
- The $\rho_{\det}$ part is a truncated $\sum_{d\in\mathcal{D}} d^{-s} W$, and it is $\leq 6Z_\infty(\frac{3}{4})/(\kappa\Gamma) + O(1/\Gamma) \leq \exp(-X_0^2/4)$ pointwise on $\sigma \geq \frac{3}{4}$.
- The $\rho_{\mathcal{A}}$ part is Lemma 4.4(b).

- For the Dirichlet sum, on each dyadic block $(N, 2N]$ use the mean value theorem for generalised Dirichlet polynomials [1, Thm. 2.1] (from [3]):
$$\int_\Gamma^{2\Gamma} \Big| \sum_{n\in(N,2N]} n^{-\sigma-it} \Big|^2 dt \ll (\Gamma + N)\, K(N)\, N^{1-2\sigma},$$
where $K(N)$ is the largest number of elements of $\mathcal{N} \setminus (\mathcal{D} \cup \mathcal{A})$ in a window of length 2 at scale $\le 2N$. Those elements are counted by $D + \int T'_{\mathrm{cont}}$, so (11) and (12) below x_k give $K(N) \le 2(2N)^\eta + 5$.
- Sum the blocks by Cauchy–Schwarz and average over $\sigma \in [\frac{3}{4}, \frac{5}{4}]$. The mean of $|\widehat{D}_{<x_k}|^2$ over the strip is $\ll K(X_0)X_0^{-1/2} + x_k^{\theta+\eta-1/2}\log^3 x_k + x_k^{\theta-1/2-o(1)}$.
- By subharmonicity, $|f(s)|^2 \le (64/\pi)\iint_{|z-s|\le 1/8} |f|^2$, so the bad set has density < 0.018 once X_0 is large.

Intersecting with (i) leaves a set of density ≥ 0.018. □

Theorem 4.9 (Zeros)**.** *Choose γ_k by Lemma 4.8, and $\alpha_k := \kappa_0 x_k^{1-\theta}/(\gamma_k \log^3 x_k) \in [1, 2]$. On $\mathcal{G}$, $\zeta_{\mathcal{P}}$ has a zero ρ_k with $|\rho_k - s_k^*| \le 1/\log x_k$ for every k. Hence*
$$\operatorname{Re}\rho_k = 1 - \frac{3\log\log x_k + O_{X_0}(1)}{\log x_k} \longrightarrow 1.$$

Proof. Apply Rouché's theorem on the circle of Lemma 4.6, comparing $\zeta_{\mathcal{P}}$ with $Z_B + M_k$. The perturbation is bounded by
$$|E_k| + \sum_{j\ne k} |\widehat{R}_j| + |\widehat{D}_{<x_k}| + |\widehat{D}_{\ge x_k}| + |\widehat{E}_{\mathcal{A}}| \le o(1) + \tfrac{1}{16} + o(1) + e^{-10^{10}} < \tfrac{1}{4},$$
by the bullets after (16) and Lemmas 4.8, 4.7 and 4.4. The hypothesis $|Z_B| \ge \frac{1}{2}$ on the disc follows from Lemma 4.8(i), Lemma 4.2 and $|Z'_\infty| \le 4\log^2 X_0$. □

Remark 4.10. $1 - \operatorname{Re}\rho_k \asymp 3(1-\theta)\log\log\gamma_k/\log\gamma_k$. This is outside Landau's region $1 - c(1-\theta)/\log|t|$ by a factor $\log\log\gamma_k$, so it does not contradict it. In this design the loss comes from the amplitude condition $\kappa'_k = \kappa_0/\log^2 x_k$, which gives $3\log\log$ rather than $1\log\log$. Positivity (6) alone would allow a constant $\kappa'_k = \kappa_0$. The margin needs a decaying amplitude: the response of the drift to the oscillation R_k must stay $o(1/\log x)$ on $x_k^2 < x < \exp(C'\log^2 x_k)$, $C' = (4(1-\theta)/c_{\mathrm{LP}})^2$, where γ_k is not yet resolvable (proof of Proposition 5.35, Step 4), and this requires $\kappa'_k \log x_k \to 0$. Any power $\log^{-1-\delta} x_k$ with $\delta > 0$ would do, giving $(2+\delta)\log\log$; the exponent 2 is a choice of the design.

5. The noise response and the margin of the compensator

This section proves that the predictable compensator of the jittered composites leaves a positive margin at every scale. Write $L := \log x$.

- Theorem 5.37 shows that the $H(x)$-window average M of the margin satisfies $\log x \cdot M(x) \ge 0.97 - o(1) - j_{X_0}(x)$ for $x \ge X_0^{1.01}$, and the same with 0.82 in place of 0.97 for $2X_0 \le x < X_0^{1.01}$.
- Proposition 5.38 extends this to the pointwise margin density m.
- Corollary 5.40 deduces the drift inequality R1′ on every window, with $c' = 0.15$. This is the input of Theorem 6.6.

The jump of the target at X_0 is handled exactly by Lemma 5.27, using the short-interval Saias bound of Lemma 5.32. The noise enters only through Saias-weighted dyadic sums (46), which depend on X_0 alone. This is what removes the divergence of the sup-norm noise response.

The section is organised as follows.

- §5.1: notation.
- §5.2: prime inputs (Chebyshev–Mertens and smooth elements).
- §5.3: the margin identity and its push-through form.
- §5.4: the sharp coefficients (Lemma M) and the Beurling–Saias bound.
- §5.5: the operator kernels.
- §5.6: the noise, per scale.
- §5.7: a local prime number theorem.
- §5.8: the bridge from the operator to the sharp reference, the X_0 jump, and positivity of the sharp reference.
- §5.9: the margin and R1′.

5.1. **Notation and standing conventions.** Throughout, $\mathcal{N}$ is the realised multiplicative semigroup and $\mathcal{P}$ the set of realised primes. Values are real numbers, and elements with equal value are counted as distinct. The variable x denotes the current scale, $L_u := \log u$, and dyadic scales are $u = X_0 2^j$. We write $\Lambda := \log X_0 \geq 10^8$ (§2.2).
Jitter and slots. (§3).

- A prime q decided at slot $s_q \geq X_0$ has, conditionally on $\mathcal{F}_{s_q-}$, its value uniformly distributed on the box J_q of width $H(s_q)$, centred at s_q: the offset is independent of the past, while the slot s_q and the decision itself depend on the history. Only this conditional law is used below (predictable compensators and martingales in decision order). Hence jittered values are $\geq X_0^- := X_0 - H(X_0)/2$, and $\mathcal{N} = \mathbb{N}$ below X_0^-.
- H is the jitter width (9) (§3.4, §6.1); it exceeds $H_{\mathrm{std}}(u) := \exp(8(\log\log u)^2)$ only on the raised range of Theorem 6.22.
- $\sigma := \sum_q \delta_{s_q}$ is the slot measure of the decided primes, and $\tilde{\rho} := J\sigma$, i.e. $\tilde{\rho}(y) = \sum_q \mathrm{box}_{H(s_q)}(y - s_q)$, is the jitter-smoothed slot density. Here $\mathrm{box}_h(x) := h^{-1}\mathbf{1}[-h/2 \leq x < h/2]$: the boxes are taken half-open, which fixes $\tilde{\rho}$ at the box edges and does not change the landing law.

Regularity of H.

$$\frac{H(vw)}{H(v)} \leq w^{\varepsilon} \qquad (v \geq X_0^-/2,\ w \geq 1), \qquad \varepsilon := 4 \cdot 10^{-6}\ (\leq 0.03), \tag{18}$$

and $H(v)/v$ is decreasing, for every $\Lambda \geq 10^8$. Indeed H is continuous, non-decreasing and piecewise C^1 ((9) and Theorem 6.22(c)). Where $H = H_{\mathrm{std}}$ (in particular on $[X_0^-/2, X_0)$), $d\log H/d\log v = 16\log\log v/\log v \leq 16\log\log(X_0^-/2)/\log(X_0^-/2) \leq 3\cdot 10^{-6}$; on the raised range $d\log H/d\log v \leq 4 \cdot 10^{-6}$ (Theorem 6.22(c)). Integrating gives (18), and $d\log(H/v)/d\log v \leq \varepsilon - 1 < 0$. Every use below needs only $\varepsilon \leq 0.03$, or $\varepsilon \leq 4 \cdot 10^{-6}$.
Decision order. (§3). $t(p)$ is the decision slot of p, so $|t(p) - p| \leq H(p)$; for rational primes $p < X_0$, $t(p) = p$. For $w \in \mathcal{N}$ put $\tau_w := \max_{p|w} t(p)$.
Operators on functions of $y \geq X_0^-$.

- The output-scale average: $(Bg)(y) := H(y)^{-1}\int_{W_y} g$, with $W_y := [y - H(y)/2,\ y + H(y)/2]$.
- The pointwise sharp latest-prime operator:

$$(K_{\mathrm{pt}}g)(y) := \sum_{w \in \mathcal{N},\ w \geq 2} \frac{1}{w}\,\mathbf{1}[\tau_w < y/w]\,\mathbf{1}[y/w \geq X_0^-]\,g(y/w).$$

  The cutoff X_0^- is the bottom of the support of $\tilde{\rho}$; every function to which K_{pt} is applied (T', F, $\tilde{\rho}$ below) vanishes below X_0^-.

The target. (§3).

- T is the target (5). Its continuous source is $T'_{\mathrm{cont}} = 1 - \rho_{\mathrm{det}} - \bar\rho_{\mathcal A} + R'$, with $\rho_{\mathrm{det}}(u) = \Psi([u,(1+\kappa)u], X_0)/(\kappa u)$ (3) and $\kappa = X_0^{-5/12+2\varepsilon_\kappa}$. On Good($< u$) the cap is inactive: $\bar\rho_{\mathcal A} = \rho_{\mathcal A}$ (Remark 3.1).
- $R = \sum_k R_k$ is the designed oscillation. It vanishes below $x_1 \ge \exp(X_0^2)$, has frequency $\gamma_k \in [\Gamma_k, 2\Gamma_k]$, and $x_{k+1} \ge \exp(x_k)$.
- The amplitude condition (2) reads $\alpha_k\gamma_k x_k^{\theta-1} = \kappa'_k/\log x_k$, with $\kappa'_k = \kappa_0/\log^2 x_k$ and $\kappa_0 \le \frac14$. Hence

$$|R'_k(v)| \le \kappa_k\,(v/x_k)^{\theta-1}, \qquad \kappa_k := 2\kappa'_k/\log x_k. \tag{19}$$

- $\mathcal A$ is the absorbed set (4). $K_{\mathcal A}$ denotes the part of K_{pt} whose latest prime is absorbed, and $\sigma_{\mathcal A} := K_{\mathcal A}\tilde\rho - \rho_{\mathcal A}$. By Lemma 6.26(b), $|\sigma_{\mathcal A}| \le 2L^{-4}$, and $\sigma_{\mathcal A}$ vanishes unless $\log\log u > 2\log\Lambda + 5.97$ ($= 42.8$ at $\Lambda = 10^8$), since $\mathcal A$ is empty below that height.
- *The effective source* is

$$T' := T'_{\mathrm{cont}} + K_{\mathcal A}\tilde\rho = 1 - \rho_{\mathrm{det}} + R' + \sigma_{\mathcal A}.$$

  The term $\sigma_{\mathcal A}$ is always kept in T' (§5.3).
- *The jump at X_0.* $T' = 0$ below X_0 (there $T = \lfloor\cdot\rfloor$), and $T'(X_0^+) = j_0 := 1 - \rho_{\mathrm{det}}(X_0) \le 1.0001/\log X_0$ by (80). Put

$$J := j_0\mathbf{1}_{[X_0,\infty)}, \qquad T'_{\mathrm c} := T' - J,$$

  so that $T'_{\mathrm c}$ is continuous at X_0. Every oscillation bound below refers to $T'_{\mathrm c}$. The jump is carried exactly, in chain form, by Lemma 5.27.
- $\tilde T'(y) := \#\{\text{continuous slots in } W_y\}/H(y) + (BK_{\mathcal A}\tilde\rho)(y)$. Then

$$\tilde T' = BT' + \vartheta/H, \qquad |\vartheta| \le 6, \tag{20}$$

  where $\vartheta = \vartheta(y)$ collects the granularity: slots against $\int T'_{\mathrm{cont}}$, the two slot boundaries, and $\tilde\rho$ against $\sigma(W_y)/H(y)$.

Sharp coefficients. Put

$$c_n(y) := \mu(n)\chi_y^t(n)\,\mathbf{1}[y/n \ge X_0^-] \quad \text{(Lemma 5.5)}, \qquad \text{and} \qquad \rho^{\mathrm{sh}}(y) := \sum_{n\in\mathcal N} c_n(y)T'(y/n)/n$$

is the pointwise sharp reference.

Constants.

- The margin target is $c' := 0.15$, the drift constant of R1′ (§6).
- $M_1(u)$, $L_0(u)$, $y(u)$ and $B(u)$ are the per-scale parameters (65), with the tolerance $M_1 \le H/L^{14}$. By (71), $B(u) \le 28M_1(u)L_u^2$ for $\log u \ge 10^8$.
- $K_M = 1.01$ and $K' = 2.20$ (Lemma 5.1); $C_0 = 0.70$ (Lemma 5.3).
- $B_1 := 1$, so that $\sum_{n\in\mathcal N,\,n\le y} 1/n \le \log y + B_1$ for every $y \ge 1$. For $\mathbb N$ the harmonic sum is $\le \log y + 1$; for $y \ge X_0^-$ the $\mathbb N$-excess is $\le \gamma + 1/y$, and by (21) the system $\mathcal N$ adds $\le \int_{X_0^-}^y \varepsilon_1(v)\,dv/v + \varepsilon_1(y) \le 2\eta_1 \le 0.002$.
- $C_{\mathrm{BS}} \le 38.3$ and $C_S \le 77$ (Theorem 5.6, Corollary 5.9); $C_N := C_S(2.04 + 4.71/\log X_0) \le 158$ (Corollary 5.17).

Regime and $o(1)$. $X_0 \ge \max(e^{10^8}, X_{\mathrm{Hux}})$ is fixed; it is never sent to infinity. In this section, $o(1)$ denotes a term bounded, at every $\Lambda \ge 10^8$ and uniformly in x, by $\exp(-10^{-3}X_0^{0.005}/\Lambda) + 10^5\Lambda^2/X_0^2 \le 10^{-3}$. These are the terms of Proposition 5.35(B), Steps 1–4: errors of the local prime number theorem at $\log x \ge X_0^{1/100}$, and κ_0-terms with $\log x_k \ge X_0^2$. So $0.97 - o(1) \ge 0.969$, $0.96 - o(1) \ge 0.959$ and $0.82 - o(1) \ge 0.819$.

5.2. **Prime inputs.**
The hypothesis ($\mathrm{H_{BS}}$). $N(v) = \lfloor v \rfloor$ for $1 \le v < X_0^-$, and $|N(v) - \lfloor v \rfloor| \le \varepsilon_1(v)\, v$ for $v \ge X_0^-$. Put

$$\eta_1 := \sup_x \sum_{k\ge 0} \sup\{\varepsilon_1(v) : v \in [x2^{-k-1}, x2^{-k}] \cap [X_0^-, \infty)\}, \qquad \eta_2 := \sup_{v \ge X_0^-} \varepsilon_1(v) \log v, \tag{21}$$

and require $\eta_1, \eta_2 \le 10^{-3}$. In the construction, write $N - \lfloor \cdot \rfloor = (N - T) + (T - \lfloor \cdot \rfloor)$.

- On Good$(< z)$ (Definition 6.8), $|N - T| \le B(u) + \frac{1}{2}H(u)(1 + \kappa_+/\log u) + 4$ at scales $u < z$, by Theorem 6.6 and Corollary 3.8.
- For $u \ge X_0$, $T - \lfloor \cdot \rfloor = E_{\mathrm{rig}} + (\#\mathcal{A} - \int \bar\rho_{\mathcal{A}}) + R + \{u\}$, with the rigid error E_{rig} of Lemma 3.4. By (8), $|E_{\mathrm{rig}}(u)| \le \kappa X_0 + 1 + \min(\kappa u + 1, \Psi(2u, X_0))$. Rankin's method with exponent $1 - 1/\Lambda$ gives $\Psi(y, X_0) \le y\, e^{-\log y/\Lambda} \prod_{p < X_0} (1 - p^{1/\Lambda - 1})^{-1} \le e^{7.2}\Lambda\, y\, e^{-\log y/\Lambda}$; the product is bounded as in the proof of Lemma 5.30 with $z = X_0$, using the rational primes and Mertens' estimates [16] instead of Lemma 5.1, so no use of (21) is made. Hence

$$\frac{|E_{\mathrm{rig}}(u)|}{u} \le \frac{\kappa X_0 + 1}{u} + \min\Big(1.001\kappa,\ 2700\,\Lambda e^{-\log u/\Lambda}\Big).$$

  The absorbed part is $\le 2u^{\theta/2}$ (Lemma 3.4), and $\mathcal{A} = \emptyset$ for $\log\log u \le 2\log\Lambda + s_H$; $R \equiv 0$ below $x_1 \ge \exp(X_0^2)$, and $|R(u)| \le 2u^{\theta} \log^* u$; and $\{u\} < 1$.
- On $[X_0^-, X_0)$, $N - \lfloor \cdot \rfloor$ counts the jittered primes landing below X_0. There are at most $\#\{\text{slots in } [X_0, X_0 + H]\} \le 2H(X_0)$ of them, so $\varepsilon_1 \le 3H(X_0)/X_0$ there.
- Hence $\eta_1, \eta_2 \le \kappa\Lambda^2 + X_0^{-1/2} \ll 10^{-3}$. Indeed, in the rigid part the term $(\kappa X_0 + 1)/u$ contributes $\le 4.01\kappa$ to η_1 (a geometric sum over the dyadic blocks above X_0^-). The minimum equals 1.001κ on at most $\Lambda \log(2700\Lambda/\kappa)/\log 2 + 1 \le 0.61\Lambda^2$ blocks (as $\log(1/\kappa) \le \frac{5}{12}\Lambda$ for every $\varepsilon_\kappa > 0$), and beyond them it decays geometrically with ratio $e^{-\log 2/\Lambda}$, so it contributes $\le 0.62\kappa\Lambda^2 + 1.45\kappa\Lambda$. For η_2, the maximum of $\min(\cdot)\log u$ is at most $1.001\kappa \cdot \Lambda \log(2700\Lambda/\kappa) \le 0.43\kappa\Lambda^2$, since $t \mapsto 2700\Lambda t e^{-t/\Lambda}$ decreases for $t \ge \Lambda$; and $(\kappa X_0 + 1)\log u/u \le 1.001\kappa\Lambda$. The other parts (the lag $|N - T|$, the absorbed part, R, $\{u\}$, and the band $[X_0^-, X_0)$) contribute $\le X_0^{-1/2}$ to each.

No sup-norm power saving is assumed; for the target (5) it would fail near $u \approx X_0^2$.

Lemma 5.1 (Chebyshev–Mertens)**.** *Assume* ($\mathrm{H_{BS}}$) *on* $[1, z]$, *and let* $\psi(x) := \sum_{p^k \le x} \log p$, *summed over the realised primes. Then for all* $1 \le x \le z$*:*

(i) $\psi(x) - \psi(x/2) \le 0.71x$, *and* $\psi(x) \le 1.42x$*;*
(ii) $S(x) := \sum_{p^k \le x} \log p/p^k$ *satisfies* $-1.01 \le S(x) - \log x \le 0.43$*; we put* $K_M := 1.01$*;*
(iii) $S_p(x) := \sum_{p \le x} \log p/p$ *satisfies* $-1.77 \le S_p(x) - \log x \le 0.43$*;*
(iv) *for* $2 \le y \le x \le z$,

$$\Big| \sum_{y < p \le x} \frac{1}{p} - \log \frac{\log x}{\log y} \Big| \le \frac{2.20}{\log y} =: \frac{K'}{\log y}.$$

The proof at x *uses* N *only on* $[1, x]$. *In particular* (i)–(iv) *hold surely on* Good$(< z)$.

Proof. Write $N = \lfloor \cdot \rfloor + E_1$, with E_1 supported on $[X_0^-, \infty)$ and $|E_1(v)| \le \varepsilon_1(v)v$. Since log is additive on $\mathcal{N}$, we have $\log \cdot\, dN = d\psi * dN$, that is, the Chebyshev identity $\int_{[1,x]} \log v\, dN(v) = \int_{[1,x]} N(x/a)\, d\psi(a)$. Inserting $N = \lfloor \cdot \rfloor + E_1$ on both sides and writing $\lfloor x/a \rfloor = x/a - \{x/a\}$ gives

$$x\, S(x) = \sum_{n \le x} \log n + \int \{x/a\}\, d\psi(a) + \int_{[1,x]} \log v\, dE_1(v) - \int E_1(x/a)\, d\psi(a). \tag{22}$$

Two error integrals. By partial summation, $\int_{[1,x]} \log v \, dE_1(v) = E_1(x)\log x - \int_{X_0^-}^x E_1(v)\, dv/v$, so its absolute value is at most $x(\eta_2 + \sup\varepsilon_1)$. Next, group a into dyadic blocks $[A, 2A)$ with $x/A \geq X_0^-$. On such a block, $|E_1(x/a)| \leq \varepsilon_1(x/a)\, x/a$, and $\int_{[A,2A)} (x/a)\, d\psi(a) \leq (x/A)(\psi(2A) - \psi(A)) \leq 1.42x$ by (i) at the smaller scale $2A \leq 2x/X_0^-$. Summing over the blocks,

$$\int |E_1(x/a)|\, d\psi(a) \leq 1.42\, \eta_1\, x. \tag{23}$$

The proof of (i) is by induction on x, so this use of (i) at scales $\leq 2x/X_0^-$ is legitimate.

(i). Put $\ell_N(x) := \int_{[1,x]} \log v \, dN$. The Chebyshev identity gives $\ell_N(x) - 2\ell_N(x/2) = \int [N(x/a) - 2N(x/2a)]\, d\psi(a)$. In the bracket, $\lfloor v \rfloor - 2\lfloor v/2 \rfloor \geq \mathbf{1}_{[1,2)}(v)$, so the $\lfloor\cdot\rfloor$ part contributes at least $\psi(x) - \psi(x/2)$. Hence

$$\psi(x) - \psi(x/2) \leq \big[S_0(x) - 2S_0(x/2)\big] + \Big|\int_{[1,x]} \log v \, dE_1\Big| + 2\Big|\int_{[1,x/2]} \log v \, dE_1\Big| + \int \big(|E_1(x/a)| + 2|E_1(x/2a)|\big)\, d\psi(a),$$

with $S_0(x) := \sum_{n\leq x} \log n$. By Stirling, $S_0(x) - 2S_0(x/2) \leq x\log 2 + 3\log x + 3$. The two log-integrals are at most $4x(\eta_2 + \sup\varepsilon_1) \leq 0.008x$, and by (23) the last integral is at most $2.84\,\eta_1 x \leq 0.0043x$. So $\psi(x) - \psi(x/2) \leq (\log 2 + 0.008 + 0.0043)x + 3\log x + 3 \leq 0.71x$ for $x \geq X_0^-$. Below X_0^- the system is $\mathbb{N}$, where $E_1 = 0$ and the same argument gives $\psi(x) - \psi(x/2) \leq x\log 2 + 3\log x + 3 \leq 0.71x$ for $x \geq 1500$; a direct computation for $x < 1500$ gives the maximum 0.6803 of $(\psi(x) - \psi(x/2))/x$, attained at $x = 5$ (see Appendix B, `finite-n`). Summing over dyadic blocks gives $\psi(x) \leq 0.71x \sum_{k\geq 0} 2^{-k} = 1.42x$.

(ii). In (22), $0 \leq \int \{x/a\}\, d\psi \leq \psi(x) \leq 1.42x$, and $|S_0(x) - (x\log x - x)| \leq \log(2\pi x)$. The two E_1-terms are at most $(\eta_2 + \sup\varepsilon_1 + 1.42\eta_1)x \leq 0.004x$. So $S(x) - \log x \in [-1.004,\, 0.424]$ for $x \geq X_0^-$. Below X_0^- it equals the value for $\mathbb{N}$, which lies in $[-0.753, 0]$: this is a direct computation for $x \leq 10^7$ (see Appendix B, `finite-n`). For $10^7 \leq x < X_0^-$, Rosser and Schoenfeld [16, (2.8) and Thm. 6, (3.21)–(3.22)] give $\sum_{p\leq x} (\log p)/p = \log x + E \pm 1/(2\log x)$, where the lower bound holds for $x > 1$ and the upper for $x \geq 319$. By their (2.8), adding the prime powers gives $S(x) - \log x = -\gamma \pm 0.04$ there.

(iii). $S - S_p = \sum_{p^k \leq x,\, k\geq 2} \log p/p^k \in [0, 0.7554]$: the rational primes contribute at most $\sum_p \log p/(p(p-1)) \leq 0.75537$ (Appendix B, `finite-n`), and primes $\geq X_0^-$ add at most $3/X_0$. Combine with (ii).

(iv). Put $E_p := S_p - \log \in [-1.77, 0.43]$. Partial summation on $(y, x]$ gives

$$\sum_{y<p\leq x} \frac{1}{p} - \log\frac{\log x}{\log y} = \frac{E_p(x)}{\log x} - \frac{E_p(y)}{\log y} + \int_y^x \frac{E_p(v)\, dv}{v\log^2 v}.$$

If $m \leq E_p \leq M'$, the first and third terms together lie in $[m/\log y,\, M'/\log y]$, and the second in $[-M'/\log y,\, -m/\log y]$. So the left side has absolute value at most $(M' - m)/\log y = (0.43 + 1.77)/\log y$. □

Remark 5.2. Part (iv) is the form used in the pivot chains of §6 and Appendix A. It applies once per chain, so nothing accumulates. A single-function version also exists: for σ_a with $e^{\sigma_a} \leq z$, the function $E(\sigma) := \sum_{p\leq e^\sigma} 1/p - \log\sigma - M_a$, with M_a chosen so that $E(\sigma_a) = 0$, satisfies $|E(\sigma)| \leq 2.20/\sigma$ for $\log 2 \leq \sigma \leq \sigma_a$, by (iv) with $y = e^\sigma$ and $x = e^{\sigma_a}$. It depends on the top point σ_a, so it is not adapted.

Facts on Dickman's function. Let ρ be Dickman's function, with $\rho(v) := 1$ for $v \le 1$, and put $r(v) := -\rho'(v)/\rho(v) = \rho(v-1)/(v\rho(v))$ for $v > 1$.

(R1) Exact identities. $v\rho(v) = \int_{v-1}^{v} \rho$ for $v \ge 1$, and $\int_a^\infty \rho(w-1)\,dw/w = \rho(a)$ for $a \ge 1$ (both sides vanish at ∞ and have derivative $-\rho(a-1)/a$). For $u \ge 2$,

$$\int_{u-1}^{u} \rho(w-1)(u-w)\frac{dw}{w} = \rho(u-1) - u\rho(u) = \rho(u-1)\big(1 - 1/r(u)\big),$$

which follows from the first two identities: the left side is $u(\rho(u-1)-\rho(u))-(u-1)\rho(u-1)$.

(R2) r is increasing on $(1,\infty)$, so ρ is log-concave on $[1,\infty)$. On $[1,2]$, $r = 1/(v(1-\log v))$, which is increasing. For $v > 2$, the first identity of (R1) gives $r(v) = 1/I(v)$ with $I(v) := \int_0^1 \exp(-\int_{v-1}^{v-1+s} r)\,ds$, and $I'(v) = -\int_0^1 e^{(\cdots)}\big(r(v-1+s) - r(v-1)\big)\,ds$. Let $v^* := \inf\{v > 2 : r'(v) \le 0\}$. Then r is strictly increasing on $[v^*-1, v^*]$, so $I'(v^*) < 0$ and $r'(v^*) > 0$, a contradiction.

(R3) $r(2) = 1/(2-2\log 2) = 1.6294$. Log-concavity gives $I(v) \le (1-e^{-r(v-1)})/r(v-1)$, hence $r(v) \ge r(v-1)/(1-e^{-r(v-1)})$. Iterating from $r(2)$ gives a computable lower bound r_{low}. Also $\rho(w-1)/\rho(u-1) \le e^{-r(u-1)(w-u)}$ for $w \ge u$.

(R4) For $\delta \in (0,1)$, (R1) and monotonicity give $v\rho(v) \ge \delta\rho(v-1+\delta)$, so $\exp(\int_{v-1+\delta}^{v} r) = \rho(v-1+\delta)/\rho(v) \le v/\delta$. Since r is increasing, $\int_{v-1+\delta}^{v} r \ge (1-\delta)r(v-1+\delta)$, and with $x := v-1+\delta$,

$$r(x) \le \frac{\log((x+1)/\delta)}{1-\delta} \qquad (0 < \delta < 1).$$

The minimum over δ is denoted $r_{\mathrm{up}}(x)$; for example $r_{\mathrm{up}}(100) = 7.66$. Taking $\delta = 1/\log x$ gives $r(x) \le \log x + \log\log x + 2$ for $x \ge 30$.

Lemma 5.3 (smooth elements, with Dickman quality)**.** *Assume* ($\mathrm{H_{BS}}$). *Let* $z > 1$, $L_z := \log z \ge 100$, *and* $K := 3.2 \ge K_M$. *For* $\ell := \log Y = uL_z$ *with* $1 \le u \le L_z$,

$$v(\ell, z) := \sum_{n \in \mathcal{N}\cap(Y,2Y],\ P^+(n)\le z} \frac{1}{n} \le C_0\,\rho(u)\,e^{a_V}, \qquad a_V := \frac{4Ku^2\log(u+2)}{\ell}, \qquad C_0 := 0.70.$$

Proof. Hildebrand's identity. Since $\log n = \sum_{p^k \mid n} \log p$ in the free semigroup $\mathcal{N}$, and writing $n = p^k m$,

$$\sum_{n\in(Y,2Y],\ P^+(n)\le z} \frac{\log n}{n} = \sum_{p\le z,\ k\ge 1} \frac{\log p}{p^k}\, v(\ell - k\log p, z).$$

The dyadic window shifts exactly, so this is an identity. Its left side is $\ge \ell\, v(\ell, z)$.

Small heights. For $0 \le \ell' \le L_z$ put $Y' := e^{\ell'}$; then $v(\ell', z) \le \sum_{n\in\mathcal{N}\cap(Y',2Y']} 1/n$.

- If $72 \le Y' < X_0^-/2$, the window lies below X_0^-, where $\mathcal{N} = \mathbb{N}$. There $\sum_{Y'<n\le 2Y'} 1/n \le \log 2 + 1/\lceil Y'\rceil \le 0.6933$ for $Y' \ge 10^4$, and a direct computation for $72 \le Y' < 10^4$ gives at most 0.69659 (attained for $Y' \in [72.5, 73)$; see Appendix B, `finite-n`).
- If $Y' \ge X_0^-/2$, write $N = \lfloor\cdot\rfloor + E_1$ and sum by parts:

$$\sum_{n\in\mathcal{N}\cap(Y',2Y']} 1/n = \sum_{Y'<n\le 2Y'} 1/n + \big[E_1(v)/v\big]_{Y'}^{2Y'} + \int_{Y'}^{2Y'} E_1(v)\,dv/v^2$$
$$\le \log 2 + 1/Y' + 3\sup\varepsilon_1 \le 0.6992.$$

- If $1 \le Y' < 72$, it is the value for $\mathbb{N}$, which is at most $5/6 \le 0.8334$ (attained for $Y' \in [1.5, 2)$, where the sum is $\frac12 + \frac13$). For $\ell' < 0$, $v(\ell', z) \le 1$.

So $v(\ell', z) \le C_0$ for $72 \le Y' \le z$.

Induction. We show $v(\ell', z) \le G_V(\ell') := C_0\rho(\ell'/L_z)e^{a_V(\ell')}$ for $\log 72 \le \ell' \le L_z^2$, by induction on ℓ' at fixed z; for $\ell' \le L_z$ this is the bound just proved. As a function of $u' = \ell'/L_z$, $a_V = 4Ku'\log(u'+2)/L_z$ is increasing. Let $\ell \in (L_z, L_z^2]$, so $u \in (1, L_z]$, and put $F(\sigma) := G_V(\ell - \sigma)$ for $0 \le \sigma \le L_z$. Since ρ is non-increasing and a_V is increasing,

$$\max_{0\le\sigma\le L_z} F(\sigma) \le C_0\,\rho(u-1)\,e^{a_V(u)}. \tag{24}$$

Moreover F is unimodal. As a function of u', G_V is increasing for $u' \le 1$, where $\rho(u') = 1$; and for $u' \ge 1$,

$$\frac{d\log G_V}{du'} = -r(u') + \frac{4K}{L_z}\Big(\log(u'+2) + \frac{u'}{u'+2}\Big) \le -1 + \frac{12.8\,(\log(L_z+2)+1)}{L_z} < 0,$$

since $r \ge 1$ on $(1,\infty)$ (R2), $u' \le L_z$ and $L_z \ge 100$. So $F(\sigma) = G_V(\ell-\sigma)$ first increases and then decreases in σ; it is increasing when $u \ge 2$, but for $1 < u < 2$ it is only unimodal.

- *Main terms, $p^k \le z$.* Partial summation against $S = \log + E$, where $E(0) = 0$ and $|E| \le K_M$ (Lemma 5.1(ii)), gives $\int F\,dE = F(L_z)E(L_z) - \int E\,dF \le K_M F(L_z) + K_M\ \mathrm{TV}(F) \le 2K_M \max F$, since F is unimodal. Hence

$$\sum_{p^k\le z} \frac{\log p}{p^k} F(k\log p) \le \int_0^{L_z} F(\sigma)\,d\sigma + 2K_M\,C_0\rho(u-1)e^{a_V(u)}.$$

- *Small heights.* Here $\ell - k\log p \ge \ell - L_z = (u-1)L_z$. If $e^{(u-1)L_z} < 72$, some of these terms have $e^{\ell - k\log p} < 72$, where v may exceed G_V by up to $0.8334 - C_0 \le 0.1334$. Their total excess is at most

$$0.1334 \sum_{p^k\in(Y/72,z]} \frac{\log p}{p^k} \le 0.1334\,(\log 72 + 1.44) \le 0.7627,$$

  by Lemma 5.1(ii). This can happen only for $1 < u < 1 + \log 72/L_z \le 1.043$.
- *Prime powers $p^k > z$ with $p \le z$* (so $k \ge 2$). Put $\ell' := \ell - k\log p$ and $t := k\log p/L_z$. Then $v(\ell', z) \le 1.001\,G_V(\ell')/C_0$ for every ℓ': by induction for $\ell' \ge \log 72$; because $v \le 1$ and $G_V(\ell')/C_0 = e^{a_V(\ell')} \ge e^{-7\cdot 10^{-4}}$ for $-\log 2 \le \ell' < \log 72$; and because $v = 0$ for $\ell' < -\log 2$. Since r is non-negative and non-decreasing, $\rho(u-t)/\rho(u) \le e^{t\,r(u)}$, and $a_V(\ell') \le a_V(\ell)$. Hence each term is at most $1.001\,(\log p)\,p^{-k\beta}\rho(u)e^{a_V(u)}$, with $\beta := 1 - r(u)/L_z \ge 1 - r_{\mathrm{up}}(100)/100 \ge 0.92$. For $p \le \sqrt{z}$ the k-sum is at most $z^{-\beta}/(1-2^{-\beta}) \le 2.12z^{-\beta}$, and $\sum_{p\le\sqrt{z}}\log p \le 1.42\sqrt{z}$ (Lemma 5.1(i)), so these p give at most $3.02z^{-0.42}$. For $p > \sqrt{z}$ every $k \ge 2$ occurs, the k-sum is at most $1.01p^{-2\beta}$, and partial summation with $\psi(t) \le 1.42t$ gives $\sum_{p>\sqrt{z}}(\log p)p^{-2\beta} \le 1.42\cdot\frac{2\beta}{2\beta-1}\,z^{1/2-\beta} \le 3.12z^{-0.42}$. Altogether, with $\rho(u)e^{a_V(u)} = \mathcal{N}(u)/(C_0u) \le \mathcal{N}(u)/C_0$ ($\mathcal{N}$ is the normaliser below), these terms contribute at most $1.001\,(3.02{+}1.01{\cdot}3.12)\,z^{-0.42}\mathcal{N}(u)/C_0 \le 9e^{-0.42L_z}\mathcal{N}(u) \le 22e^{-0.4L_z}\mathcal{N}(u)$.

The one-step inequality. Substitute $s = u - \sigma/L_z$. Then $\int_0^{L_z} F = L_zC_0\int_{u-1}^u \rho(s)e^{a_V(s)}ds$, where $a_V(s)$ is evaluated at $\ell' = sL_z$, and $\ell G_V(\ell) = uL_zC_0\rho(u)e^{a_V(u)}$. By the first identity of (R1), $u\rho(u) = \int_{u-1}^u \rho(s)\,ds$. Let $\alpha := 4K\big(\log(u+1) + (u-1)/(u+1)\big)/L_z$; then $a_V(u) - a_V(s) \ge \alpha(u-s)$ on $[u-1,u]$, and $e^{-\alpha t} \le 1 - (1-e^{-\alpha})t$ for $t \in [0,1]$. Let E_* be a lower bound for the mean of $u-s$ under the density proportional to $\rho(s)$ on $[u-1,u]$. With $\rho(u-1) = u\,r(u)\,\rho(u)$,

this gives

$$\ell G_V(\ell) - \int_0^{L_z} F - 2K_M C_0 \rho(u-1) e^{a_V(u)} \;\geq\; \mathcal{N}(u) \cdot \Big[L_z(1-e^{-\alpha}) E_* - 2K_M\, r(u) \Big],$$
$$\mathcal{N}(u) := C_0 e^{a_V(u)} \rho(u)\, u.$$

Both sides are in the absolute units of the inequality $\ell v(\ell) \leq \ell G_V(\ell)$. The bracket is of order 1, not of order L_z: $L_z(1-e^{-\alpha}) \approx 4K \log(u+1)$. Two choices of E_* are available:

- $E_* = \frac{1}{2}$ always, since that density is decreasing;
- $E_* = 1 - 1/r(u-1)$ for $u \geq 2$: by (R2) the density is dominated in likelihood ratio by the truncated exponential with rate $r(u-1)$, whose mean is at least $1 - 1/r(u-1)$.

Dividing the excesses by the normaliser $\mathcal{N}(u)$, the induction step closes if

$$L_z(1-e^{-\alpha}) \max\left(\tfrac{1}{2},\, 1 - 1/r_{\mathrm{low}}(u-1)\right) - 2K_M\, r_{\mathrm{up}}(u) - 22e^{-0.4L_z} - X(u) > 0, \tag{25}$$

where $X(u) := 0.7627/(C_0(1-\log u)u)$ for $u < 1 + \log 72/L_z$ (using $\mathcal{N}(u) \geq C_0\rho(u)u$ and $\rho(u) = 1 - \log u$ on $[1,2]$), and $X(u) := 0$ otherwise. The left side of (25) is increasing in L_z, and the range of X shrinks as L_z grows, so $L_z = \max(100, u)$ is the worst case.

- *Computer-assisted check* (see Appendix B). With r enclosed in $[r_{\mathrm{low}}, r_{\mathrm{up}}]$ (and exact on $[1,2]$), the left side of (25) without X is positive on $u \in [1, 10^6]$: 13,823 cells in ball arithmetic, no failures, minimum certified value 1.724. On $u \in [1,2]$, the left side *with* X and $22e^{-0.4L_z}$, and even with an additional allowance $2K_M r_{\mathrm{up}}(u)(e^{0.141}-1)$, is positive: 1,387 cells, no failures, minimum certified value 0.829.
- For $u > 10^6$, we have $L_z \geq u > 10^6$, so $\alpha \leq 4K(\log(L_z+1)+1)/L_z \leq 2 \cdot 10^{-4}$ and $L_z(1-e^{-\alpha}) \geq 0.9999\, L_z\alpha \geq 12.79 \log(u+1)$. Use $E_* = \frac{1}{2}$ and $r_{\mathrm{up}}(u) \leq \log u + \log\log u + 2$ (R4). The left side is then at least $6.39\log(u+1) - 2.02(\log u + \log\log u + 2) - 10^{-10^5} > 0$.

This completes the induction. The range $\ell' \leq L_z^2$ covers $u \leq L_z$. □

5.3. **The margin identity and the push-through form.**
Bookkeeping. (Lemmas 3.6, 3.9, Corollary 3.8). Rigid integers and elements of $\mathcal{A}$ sit on their own unit jumps of T (jump slots). Every level crossing of T is either a jump slot or a continuous slot, so on every window W

$$\#\{\text{continuous slots in } W\} = \int_W T'_{\mathrm{cont}} \pm 2.$$

By Lemma 3.6, on a slot window W the number of primes decided equals the number of continuous slots, minus the non-absorbed jittered composites landing in W, plus the change in the queue:

$$\sigma(W) = \#\{\text{continuous slots in } W\} - C_{\mathrm{jit}}(W) + \Delta Q(W) \pm 2, \tag{26}$$

where the ± 2 accounts for the two slot boundaries. Here C_{jit} counts the composites $n \notin \mathcal{A}$ whose latest prime q (in decision order) is jittered; the cofactor may be any element of $\mathcal{N}$. Write $C_{\mathrm{jit}} = C^{(1)} + C^{(\geq 2)}$, according to whether $q \,\|\, n$ or $q^2 \mid n$.

- Comp is the predictable compensator of $C^{(1)}$. Given $\mathcal{F}_{q-}$, the element $q \cdot w$ ($q \nmid w$) lands in W with conditional probability $|J_q \cap W/w|/H(s_q)$, and $q \cdot w$ is formed only if q is decided after every prime of w, i.e. $s_q > \tau_w$. Hence

$$\mathrm{Comp}(W) = \sum_w \int_{W/w} J(\Pi_{\tau_w}\sigma) \cdot \mathbf{1}[qw \notin \mathcal{A}],$$

where $\Pi_\tau\sigma$ keeps the slots $s_q > \tau$.

- $\mathrm{Comp}^{(2)}$ is the predictable compensator of $C^{(\geq 2)}$: $\mathrm{Comp}^{(2)}(W) := \sum_q \sum_{a\geq 2} \sum_m \mathbb{P}(q^a m \in W \mid \mathcal{F}_{q-})$.
- $\mathrm{Mart}' := (C^{(1)} - \mathrm{Comp}) + (C^{(\geq 2)} - \mathrm{Comp}^{(2)}) =: \mathrm{Mart} + \mathrm{Mart}^{(2)}$. Both parts are martingales in decision order. They are controlled by separate Freedman bounds (§5.6), because the increment of q enters Mart through W/q and $\mathrm{Mart}^{(2)}$ through W/q^a.

Since $\int_{W_x/w} g = \frac{1}{w}\int_{W_x} g(y/w)\,dy$,

$$\mathrm{Comp}(W_x)/H(x) = (BK_{\mathrm{pt}}\tilde{\rho})(x) - (BK_{\mathcal{A}}\tilde{\rho})(x) + \mathrm{NT}(x).$$

- $K_{\mathcal{A}}$ is the part of K_{pt} whose landing is absorbed ($y/w < y(y)$, $\log y \geq L_*$).
- The cutoff $\mathbf{1}[y/w \geq X_0^-]$ in K_{pt} costs nothing, since $\tilde{\rho}$ vanishes below X_0^-.
- NT replaces the slot filter $s_q > \tau_w$ by the landing filter $y/w > \tau_w$ on the non-absorbed part. It is supported on τ-*near-threshold* cofactors, i.e. those with $|\tau_w - y/w| \leq 2H(y/w)$, and

$$|\mathrm{NT}(y)| \leq 2\eta(y)\cdot \sup\tilde{\rho}, \qquad \eta(y) := \sum_{w\ \tau\text{-near at } y} \frac{1}{w}. \tag{27}$$

- $\mathrm{NT}_{\mathrm{pt}}$ denotes NT before the B-average, and $m := T' - K_{\mathrm{pt}}\tilde{\rho} - \mathrm{NT}_{\mathrm{pt}}$ is the *pointwise margin density*.

Proposition 5.4 (the margin identity)**.** *For every $y \geq X_0$,*

$$\tilde{\rho} + BK_{\mathrm{pt}}\tilde{\rho} = F, \qquad F := \tilde{T}' - \nu - \mathrm{NT}, \qquad \nu(y) := \frac{\mathrm{Mart}'(W_y) - \Delta Q(W_y) + \mathrm{Comp}^{(2)}(W_y)}{H(y)}. \tag{28}$$

Proof. Apply (26) to the window W_y and divide by $H(y)$. On the left, $\tilde{\rho}(y) = \sigma(W_y)/H(y) + O(2/H(y))$: every box is centred and has width $H(s_q) = H(y)(1+O(\varepsilon H(y)/y))$, so at most two slots near the ends of W_y are counted differently. On the right, write $C_{\mathrm{jit}} = \mathrm{Comp} + \mathrm{Comp}^{(2)} + \mathrm{Mart}'$, use the display for $\mathrm{Comp}(W_y)/H(y)$ above, and move $BK_{\mathrm{pt}}\tilde{\rho}$ to the left. The terms $BK_{\mathcal{A}}\tilde{\rho}$ and $\#\{\text{continuous slots in } W_y\}/H$ make up $\tilde{T}'$. The $O(1/H)$ terms are part of ϑ in (20). □

The compensator margin density at y is

$$M(y) := \tilde{T}'(y) - (BK_{\mathrm{pt}}\tilde{\rho})(y) = \tilde{\rho}(y) + \nu(y) + \mathrm{NT}(y). \tag{29}$$

The noise ν enters only at the current scale y, as an $H(y)$-window function (Lemma 5.22). Every other quantity in (28) at y involves σ only at scales $\leq y/2 + H$.

The push-through form. K_{pt} maps functions supported on scales $\geq v$ to functions supported on scales $\geq 2v$. So $1 + BK_{\mathrm{pt}}$ and $1 + K_{\mathrm{pt}}B$ are invertible, and their Neumann series terminate on bounded ranges. For any such operators B, K,

$$(1+BK)^{-1} = (1-B) + B(1+KB)^{-1}\big(1 - K(1-B)\big),$$

since $(1+BK)(1-B) + B(1-K(1-B)) = 1 - B + BK - BKB + B - BK + BKB = 1$. With $K = K_{\mathrm{pt}}$ and $\mathcal{R} := (1+K_{\mathrm{pt}}B)^{-1}$, (28) gives

$$\tilde{\rho} = (1-B)F + B\mathcal{R}F - B\mathcal{R}K_{\mathrm{pt}}(1-B)F. \tag{30}$$

- $\mathcal{R}$ is the *long* operator: at each step the sharp landing filter, followed by an $H(u)$-average at the landing scale u. Its kernels C_n are studied in §5.5.
- $(1-B)$ acts only on the source. The response $\tilde{\rho}$ is never differentiated, so no regularity lemma for it is needed.

- Applying the pointwise inverse $(1+K_{\mathrm{pt}})^{-1}$ to (28) instead would leave the response term $(1-B)K_{\mathrm{pt}}\tilde{\rho}$, which is of size $\asymp 1/\log y$ in sup norm. This is why the push-through form is used.

Note also the push-through identity $(1+BK_{\mathrm{pt}})^{-1}B = B\mathcal{R}$ and its consequence $K_{\mathrm{pt}}B\mathcal{R} = 1-\mathcal{R}$. The jump split. Split $T' = T'_{\mathrm{c}} + J$ with $J = j_0\mathbf{1}_{[X_0,\infty)}$ (§5.1). The J-part of F is BJ. By the push-through identity its contribution to $\tilde{\rho}$, and hence to the margin $M = \tilde{\rho}+\nu+\mathrm{NT}$, is exactly $B\mathcal{R}J$. For the rest put

$$F_c := F - BJ = T'_{\mathrm{c}} + F_r, \qquad F_r := (B-1)T'_{\mathrm{c}} + \vartheta/H - \nu - \mathrm{NT}, \qquad G := K_{\mathrm{pt}}(1-B)F_c. \tag{31}$$

By (30) applied to F_c, $M = B\mathcal{R}J + (1-B)F_c + B\mathcal{R}F_c - B\mathcal{R}G + \nu + \mathrm{NT}$.

Placement convention. The absorbed source $\sigma_{\mathcal{A}}$ is part of T'_{c}, never of F_r. Only its H-oscillation enters, through $(B-1)T'_{\mathrm{c}}$ and o_u (see (63)). The jump enters only through $B\mathcal{R}J$, that is, through Lemma 5.27.

5.4. **Sharp coefficients: Lemma M and the Beurling–Saias bound.** For a start y, the sharp resolvent is $((1+K_{\mathrm{pt}})^{-1}g)(y) = \sum_{n\in\mathcal{N}} c_n(y)\, g(y/n)/n$, where $c_1 = 1$ and

$$c_n = -\sum_{d\mid n,\ d<n} \mathbf{1}[\tau_{n/d} < y/n]\cdot \mathbf{1}[y/n \ge X_0^-]\cdot c_d.$$

Expanding the recursion, $c_n(y)$ is the signed count of ordered factorisations $n = w_1\cdots w_k$ into $w_i \in \mathcal{N}\setminus\{1\}$ whose cumulative filters $\tau_{w_i}\cdot w_1\cdots w_i < y$ and cutoffs $y/(w_1\cdots w_i) \ge X_0^-$ all pass, with sign $(-1)^k$. Since the landing points $y/(w_1\cdots w_i)$ decrease along a chain, the cutoffs reduce to the single condition $y/n \ge X_0^-$.

Lemma 5.5 (Lemma M)**.** *Let $t : \mathcal{P} \to (0,\infty)$ be any weight, with ties broken by a fixed order, and $\tau_w := \max_{p\mid w} t(p)$. Then $c_n(y) = \mu(n)\,\chi_y^t(n)\,\mathbf{1}[y/n \ge X_0^-]$, where $\chi_y^t(n) = 1$ if and only if n is squarefree and*

$$t(p)\cdot \prod_{q\mid n,\ t(q)\ge t(p)} q < y \qquad \text{for every prime } p \mid n.$$

In words: strip the primes of n in decreasing t-order; after removing p, the current point must still exceed $t(p)$.

Proof. We may assume $y/n \ge X_0^-$. Write $q \succ q'$ for the t-order.

n squarefree. Chains are ordered set partitions $(B_1,\dots,B_k)$ of the set of primes of n, with $w_i = \prod B_i$. Let $\Pi_i := w_1\cdots w_i$ and $F(B) := \max_{p\in B} t(p)$. A chain is valid if and only if $F(B_i)\,\Pi_i < y$ for all i.

- Let $q_1 \succ \cdots \succ q_r$ be the primes of n. The canonical chain $(\{q_1\},\dots,\{q_r\})$ is valid if and only if $\chi_y^t(n) = 1$, since the primes q with $t(q) \ge t(q_i)$ are exactly $q_1,\dots,q_i$. Its sign is $(-1)^r = \mu(n)$.
- For a non-canonical valid chain, let i be the first index with $B_i \ne \{q_i\}$, put $q := q_i$, and let $j \ge i$ be the index with $q \in B_j$. Every block from position i on consists of primes $\preceq q$, so $F(B_j) = t(q)$.
  - (S) If $|B_j| \ge 2$, split B_j into the two consecutive blocks $B_j\setminus\{q\}, \{q\}$. The first new condition is implied by the old condition of B_j, since both its F and its partial product are smaller. The second new condition is $t(q)\,\Pi_j < y$, which is the old condition of B_j.
  - (M) If $B_j = \{q\}$ (so $j > i$), merge B_{j-1} and $\{q\}$ into one block. Its F is $t(q)$ and its partial product is Π_j, so its condition is the old condition of $\{q\}$; the other conditions are unchanged.

The maps (S) and (M) are mutually inverse, preserve validity and the first deviation (i, q), and change k by 1. So the non-canonical valid chains cancel in pairs.

n not squarefree (induction on $\Omega(n)$). By induction, $c_d = \mu(d)\chi_y^t(d)$ for $d < n$, so $c_n = -\sum_{d\in\mathcal{F}_n,\ d \text{ squarefree}} \mu(d)\chi_y^t(d)$ with $\mathcal{F}_n := \{d \mid n,\ d < n :\ \tau_{n/d} \cdot n < y\}$. Let s be the t-largest prime with $s^2 \mid n$.

- $\mathcal{F}_n$ is stable under $d \mapsto d\triangle\{s\}$ (toggling s). Indeed $d \cdot s \mid n$ whenever d is squarefree, $d \mid n$ and $s \nmid d$; and n/d, $n/(ds)$ have the same prime support, because both still contain s. So $\tau_{n/d} = \tau_{n/(ds)}$. Moreover, if $\mathcal{F}_n \neq \emptyset$ then $t(s)\, n \le \tau_{n/d}\, n < y$.
- $\chi_y^t(d) = \chi_y^t(d\triangle\{s\})$. The conditions at primes $p \succ s$ do not involve s. The conditions at the other primes p hold for both, since $t(p)\prod_{t(q)\ge t(p)} q \le t(s)\,\mathrm{rad}(n) \le t(s)\, n < y$.
- μ changes sign under $d \mapsto d\triangle\{s\}$, so the terms cancel in pairs and $c_n = 0$. □

Consequences.

- $|c_n| \le 1$, and $\sum_n |c_n|/n \le \sum_{n\in\mathcal{N},\, n\le y/X_0^-} 1/n \le \log y + B_1$.
- *Density.* With the decision order t ($|t(p) - p| \le H(p)$, and $t(p) = p$ for rational primes), $\chi_y^t(n) = 1$ and $y/n \in [v, 2v)$ force n to be squarefree and *$4v$-dense*: $p_{j+1} \le 4v\, p_1 \cdots p_j$ for $0 \le j < r$, where $p_1 < \cdots < p_r$ are the primes of n sorted by value and the empty product is 1 (so $p_1 \le 4v$). To see this, fix $0 \le j < r$ and let p^* be the prime of smallest t among $p_{j+1}, \dots, p_r$. All of $p_{j+1}, \dots, p_r$ have $t \ge t(p^*)$, so the condition at p^* gives $t(p^*)\, p_{j+1} \cdots p_r < y < 2v\, n$, that is, $t(p^*) < 2v\, p_1 \cdots p_j$ (for $j = 0$: $t(p^*) < 2v$). Since p_{j+1} has the smallest value among $p_{j+1}, \dots, p_r$, and $p^* \le t(p^*) + H(p^*)$ with $H(p^*) \le p^*/10^{10}$ (and $t(p) = p$ for rational primes), $p_{j+1} \le p^* \le (1 - 10^{-10})^{-1} t(p^*) \le 4v\, p_1 \cdots p_j$. The passage from the decision order to the value order is made by the choice of p^*; the factor 4 in place of 2 only has to absorb $(1 - 10^{-10})^{-1}$.
- Lemma 5.5 concerns *sharp scalar* filters. For the true operator $\mathcal{R}$, whose filter is averaged across the threshold, the pointwise analogue is false: for $n = 4$ with void filter and constant H, $\mathrm{TV}(C_4) = \frac{1}{4}$ while $\mu(4) = 0$. The transfer to $\mathcal{R}$ is Theorem 5.14 and its random-width form Theorem 5.15, which use only the masses c_n.

Theorem 5.6 (Beurling–Saias, upper bound)**.** *Assume* ($\mathrm{H_{BS}}$) *and* $\log X_0 \ge 10^8$. *For $t \ge X_0$ put $\tau := \log t$. Call a squarefree $n \in \mathcal{N}$ t-*dense *if $p_{j+1} \le t\, p_1 \cdots p_j$ for $0 \le j < r$, where $p_1 < \cdots < p_r$ are its primes and the empty product is 1; in particular $p_1 \le t$. Then for every $Y \ge t$,*

$$W_t(Y) := \sum_{n\in\mathcal{N}\cap(Y,2Y],\ t\text{-dense}} \frac{1}{n} \le C_{\mathrm{BS}} \frac{\log t}{\log Y},$$

$$C_{\mathrm{BS}} := C_0 \max\left(e^{\kappa_\Phi/\tau},\, 30e^{\kappa_\Phi/(30\tau)}\right), \quad \kappa_\Phi := 4\tau + 1500.$$

In particular $C_{\mathrm{BS}} \le 0.70\, e^{4+1.5\cdot 10^{-5}} \le 38.3$. The same bound holds with t replaced by $4t$, at the cost of a factor $1 + \log 4/\log X_0$.

For $\mathcal{N} = \mathbb{N}$ the order of magnitude is due to Saias [17]. The point here is a proof that uses only Lemmas 5.1 and 5.3, so that it applies to the realised system $\mathcal{N}$. The exponent of $1/\log Y$ is exactly 1; this is what makes the noise sums of §5.6 independent of x.

Proof of Theorem 5.6. Step 1: recursion. Put $W_t(Y; z) := \sum_{n\in(Y,2Y],\ t\text{-dense},\ P^+(n)<z} 1/n$. If n is t-dense and $p = P^+(n)$, then n/p is t-dense with $P^+(n/p) < p$, and $p \le t \cdot n/p$ (for $n = p$ this is the case $j = 0$, $p \le t$), so $p^2 \le tn \le 2tY$. Summing over p first (for each p the cofactor range

$(Y/p, 2Y/p]$ is a single dyadic block, so the sum is exact),

$$W_t(Y;z) \le \sum_{p<z,\ p\le\sqrt{2tY}} \frac{1}{p}\, W_t(Y/p;p). \tag{32}$$

Step 2: logarithmic variables. Put $\ell := \log Y$, $u := \ell/\log z$, and $\Psi(\ell,u) := W_t(e^\ell; e^{\ell/u}) \cdot \max(1,\ell/\tau)$. For a prime p write

$$\sigma := \log p, \qquad m := \ell - \sigma, \qquad w := m/\sigma, \qquad \nu := \ell/m = (1+w)/w, \qquad w_1 := \frac{\ell - \tau - \log 2}{\ell + \tau + \log 2}.$$

Then $p < z$ if and only if $w > u-1$, and $p \le \sqrt{2tY}$ if and only if $w \ge w_1$. For $\ell \ge 3\tau + \log 2$ every term has $m \ge (\ell - \tau - \log 2)/2 \ge \tau$, so $\max(1,\ell/\tau)/\max(1,m/\tau) = \nu$, and (32) becomes

$$\Psi(\ell,u) \le \sum_{p:\ w\ge w_{\mathrm{lo}}} \frac{1}{p}\,\nu\,\Psi(m,w), \qquad w_{\mathrm{lo}} := \max(u-1, w_1). \tag{33}$$

In the continuum the kernel is dw/w. Its exact fixed point is $g(u) = \rho(u-1)$: by (R1), $\int_{u-1}^{\infty} g(w)\,dw/w = g(u)$ for $u \ge 2$, while for $u < 2$ $\int_{w_1}^{\infty} g(w)\,dw/w = 1 + \log(1/w_1)$. So the kernel is critical, and every error must be summable along chains. This is why the exponent of $1/\log Y$ is exactly 1.

Step 3: the profile. Fix the constants

$$B := 5K' = 11, \qquad \kappa_\Phi := 4\tau + 1500, \qquad \ell_1 := 30\tau,$$

and the functions

$$u_0(\ell) := \ell^{1/4}, \qquad U(\ell) := u_0 + \ell^{1/4}\sqrt{2\log(\ell/\tau) + 4},$$

$$Q(\ell,u) := A + a, \quad A := \frac{B(1+u)^3\log(2+u)}{\ell}, \quad a := \frac{(u-u_0)_+^2}{2\sqrt{\ell}},$$

$$\Phi(\ell) := \Phi_\infty e^{-\kappa_\Phi/\ell}, \qquad \Phi_\infty := C_0 \max\left(e^{\kappa_\Phi/\tau},\, 30 e^{\kappa_\Phi/\ell_1}\right).$$

We prove, by induction on ℓ, the claim

$$\Psi(\ell,u) \le G(\ell,u) := \Phi(\ell)\,\rho(u-1)\,e^{Q(\ell,u)} \qquad (\ell \ge \tau,\ 0 \le u \le U(\ell)). \tag{34}$$

At $u = 0$ (that is, $z = \infty$) this gives the theorem, since

$$W_t(Y) = \Psi(\ell,0)\tau/\ell \le \Phi_\infty e^{(B\log 2 - \kappa_\Phi)/\ell}\,\tau/\ell \le \Phi_\infty \tau/\ell$$

and $\Phi_\infty = C_{\mathrm{BS}}$. The three factors of G play the following roles:

- A absorbs the errors of the prime sums;
- a inflates the profile gradually, by $\log(\ell/\tau) + 2$ at $u = U$; this pays for the density factor ℓ/τ in the tail (Step 6);
- Φ absorbs the boundary term $\log(1/w_1) \approx 2\tau/\ell$ of the critical zone $u < 2$. The ratio is exact: $\Phi(m)/\Phi(\ell) = e^{-\kappa_\Phi/(w\ell)}$, since $1/m - 1/\ell = 1/(w\ell)$.

Step 4: base region $\tau \le \ell \le \ell_1$. Drop the density condition, which costs at most a factor $\max(1,\ell/\tau)$. First, $(\ell/\tau)C_0 \le \Phi_\infty e^{-\kappa_\Phi/\ell} = \Phi(\ell)$ on $[\tau,\ell_1]$, by the choice of Φ_∞: the function $(\ell/\tau)e^{\kappa_\Phi/\ell}$ is convex in ℓ, so its maximum on $[\tau,\ell_1]$ is at an endpoint.

- For $1 \le u \le U(\ell)$, apply Lemma 5.3. Its hypotheses hold: $u \le U(\ell) \le 4.3\ell^{1/4}$, so $L_z = \ell/u \ge \ell^{3/4}/4.3 \ge 100$ and $u \le L_z$. This gives $\Psi \le (\ell/\tau)C_0\rho(u)e^{a_V} \le G(\ell,u)$, because $\rho(u) \le \rho(u-1)$ and $a_V \le A$ (as $11(1+u)^3 \ge 12.8u^2$).
- For $0 \le u < 1$, Lemma 5.3 does not apply, and the trivial bound suffices: by the small-height bound in its proof (valid for every $Y' \ge 72$), $W_t(e^\ell; e^{\ell/u}) \le \sum_{n\in\mathcal{N}\cap(e^\ell, 2e^\ell]} 1/n \le C_0$. So $\Psi \le (\ell/\tau)C_0 \le \Phi(\ell) \le G(\ell,u)$, since $\rho(u-1) = 1$ and $Q \ge 0$.

Step 5: the induction step, discretised. Let $\ell > \ell_1$ and $0 \le u \le U(\ell)$, and assume (34) for all smaller ℓ. Split the primes in (33) into the range $w \le U(m)$, where the induction hypothesis applies, and the tail $w > U(m)$ (Step 6). The range $w_{\mathrm{lo}} \le w \le U(m)$ is an interval $\sigma \in [\sigma_J, \sigma_a]$, with $\sigma_a := \ell/(1+w_{\mathrm{lo}})$. On it put

$$f(\sigma) := \nu\, G(m,w) = f_1 f_2, \qquad f_1 := \nu\, \rho(w-1), \qquad f_2 := \Phi(m) e^{Q(m,w)}.$$

Here f_1 is increasing in σ (both ν and $\rho(w-1)$ increase as w decreases), and

$$|\partial_\sigma \log f_2| \le \eta_f(m,w) := \frac{\kappa_\Phi}{m^2} + |\partial_m Q| + \frac{(1+w)^2}{\ell}|\partial_w Q|,$$

since $dm/d\sigma = -1$ and $dw/d\sigma = -(1+w)^2/\ell$. Write $\sum_{p \le e^\sigma} 1/p = \log\sigma + M_a + E(\sigma)$, with the constant M_a chosen so that $E(\sigma_a) = 0$. Then $-E(\sigma) = \sum_{e^\sigma < p \le e^{\sigma_a}} 1/p - \log(\sigma_a/\sigma)$, so Lemma 5.1(iv) gives $|E(\sigma)| \le K'/\sigma$ on $[\sigma_J, \sigma_a]$. Partial summation gives $\sum_p f(\sigma_p)/p \le \int f\, d\sigma/\sigma + K'[f(\sigma_a)/\sigma_a + f(\sigma_J)/\sigma_J + \int |f'|\, d\sigma/\sigma]$. Using $|f'| \le f' + 2f\,|\partial_\sigma \log f_2|$ and integrating $\int f'\, d\sigma/\sigma$ by parts, the terms in $f(\sigma_J)$ cancel:

$$\sum_p \frac{f(\sigma_p)}{p} \le \int f \frac{d\sigma}{\sigma} + \frac{2K' f(\sigma_a)}{\sigma_a} + K' \int f \frac{d\sigma}{\sigma^2} + 2K' \int f\, |\partial_\sigma \log f_2| \frac{d\sigma}{\sigma}. \tag{35}$$

In w-variables $d\sigma/\sigma = dw/(1+w)$, so

$$\int f\, d\sigma/\sigma = \int G(m,w)\, dw/w \qquad \text{and} \qquad \int f\, d\sigma/\sigma^2 = \ell^{-1} \int G(m,w)(1+w)\, dw/w.$$

Now divide by $G(\ell,u)$. Put

$$D(w) := Q(m,w) - Q(\ell,u) - \frac{\kappa_\Phi}{w\ell}, \qquad \pi(dw) := \frac{\rho(w-1)\, dw}{w\, \rho(u-1)} \text{ on } [w_{\mathrm{lo}}, \infty),$$

$$g(w) := \frac{\rho(w-1)}{\rho(u-1)} e^{D(w)} \mathbf{1}[w \le U(m)].$$

The main term is $M := \int \pi\, e^D \mathbf{1}[w \le U(m)]$. The three error terms are

$$E_1 := \frac{2K'(1+w_{\mathrm{lo}})^2 g(w_{\mathrm{lo}})}{w_{\mathrm{lo}}\, \ell}, \qquad E_2 := \frac{K'}{\ell} \int g\, \frac{(1+w)\, dw}{w}, \qquad E_3 := 2K' \int g\, \eta_f\, \frac{dw}{w},$$

and T denotes the tail of Step 6, divided by $G(\ell,u)$. The induction closes if

$$\Sigma(\ell,u) := 1 - \big[M + E_1 + E_2 + E_3 + T\big] \ge 0 \qquad \text{for all } \ell > \ell_1,\ 0 \le u \le U(\ell). \tag{36}$$

Two ways of bounding M are used.

- $M = \int \pi\, \mathbf{1}[w \le U(m)] + J \le 1 + \mu_0 + J$, so $\Sigma \ge -[\mu_0 + J + E_1 + E_2 + E_3 + T]$, where $\mu_0 := \log(1/w_{\mathrm{lo}})$ for $u < 2$, and $\mu_0 := 0$ for $u \ge 2$, is the exact excess mass of π (by (R1)), and $J := \int \pi\, (e^D - 1)\, \mathbf{1}[w \le U(m)]$.
- M is at most the integral over $[w_{\mathrm{lo}}, \infty)$ of any positive majorant of $e^D \mathbf{1}[w \le U(m)]$.

The indicator cannot simply be dropped: D grows like $Bw^3 \log w/\ell$, so $\int \pi\, e^D$ diverges.

Step 6: the tail. The primes with $w > U(m)$ correspond exactly to the n that are e^{ℓ/u^*}-smooth, with $u^* := U(\ell) + \frac{1}{2}$. This uses $U(m) \ge U(\ell) - \frac{1}{2}$ for $m \ge \ell(1 - 1/U)$. Dropping the density condition and applying Lemma 5.3 gives a tail contribution to Ψ of at most $(\ell/\tau) C_0 \rho(u^*) e^{a_V(u^*)}$. Divide by $G(\ell,u)$, and use $\Phi_\infty \ge C_0$, $Q \ge a$ and $\rho(u^*)/\rho(u-1) \le e^{-r(u-1)(u^*-u+1)}$ (R3). With $y := \log(\ell/\tau) \ge \log 30$,

$$T \le \exp\big(\kappa_\Phi/\ell + a_V(u^*) + y - a(\ell,u) - r(u-1)(U(\ell) - u + \tfrac{3}{2})\big).$$

Maximising over the admissible y gives $T \le \exp(\kappa_{\max}/\ell + a_V(u^*) + E_T)$, where $\kappa_{\max} := 4\ell/30 + 1500 \ge \kappa_\Phi$ and, with $U_{30}(\ell) := u_0 + \ell^{1/4}\sqrt{2\log 30 + 4}$,

$$E_T := \begin{cases} \log 30 - a(\ell,u) - r(u-1)\big(U_{30} - u + \frac{3}{2}\big), & u \le U_{30}, \\ -2 - 1.5\, r(u-1), & u > U_{30}. \end{cases}$$

In the second case this uses $a(\ell, U(\ell)) = y + 2$.

Step 7: verification of (36). There are three regions. Throughout, $\tau \ge 10^8$ and $\ell > \ell_1 = 30\tau$.

(i) Region Z1: $u < 2$, every $\ell \ge 30\tau$. Here $\rho(u-1) = 1$, $w_{\rm lo} \in [w_1, 1)$ and $\pi(dw) = \rho(w-1)\,dw/w$. Since $u_0 = \ell^{1/4} \ge 234$ and $m \ge \ell w_1/(1+w_1) \ge 0.48\ell$, we have $m^{1/4} \ge 194$, so $D_a := a(m,w) - a(\ell,u) = 0$ for $w \le m^{1/4}$, in particular for $w \le 40$. Write $D = D' + D_\Phi$ with $D' := D_A + D_a$, $D_A := A(m,w) - A(\ell,u)$ and $D_\Phi := -\kappa_\Phi/(w\ell) \le 0$; then $e^D - 1 \le (e^{D'} - 1)_+ + (e^{D_\Phi} - 1)$.

The part $w > 40$ is negligible, uniformly in ℓ. There $\rho(w-1) \le \rho(39) e^{-r(39)(w-40)}$ by (R3), with $r(39) \ge 5.09$ and $\rho(39) \le \exp(-\sum_{k=1}^{38} r_{\rm low}(k)) \le 1.2 \cdot 10^{-65}$; and on $[40, U(m)]$ the slope of D in w is at most 0.25. The same holds for D', whose slope is at most that of D because D_Φ is increasing in w; as $D \le D'$ and $D'(40) \le 9.7 \cdot 10^{-4}$, both D and D' are at most $10^{-3} + 0.25(w-40)$. Every term below is multiplied by ℓ, so we check that the ℓ-scaled contributions of $w > 40$ are bounded independently of ℓ.

- For $40 < w \le 0.83\ell^{1/4} \le m^{1/4}$ we have $D' = D_A \le (B/\ell)h(w)$, with h as below, so $\ell(e^{D'} - 1)_+ \le B\, h(w)\, e^{10^{-3} + 0.25(w-40)}$; integrated against π this is at most $1.2 \cdot 10^{-65}\, B \int_{40}^\infty h(w) e^{10^{-3} - 4.84(w-40)}\, dw/w \le 2 \cdot 10^{-61}$.
- For $w > 0.83\ell^{1/4}$ (where D_a may be non-zero), the factor $\rho(w-1) \le 1.2 \cdot 10^{-65} e^{-5.09(w-40)}$ is at most $e^{-4(0.83\ell^{1/4} - 40)}$ times $e^{-1.09(w-40)}$, which beats the factor ℓ, the factor $e^{0.25(w-40)}$ and every polynomial in w, since $\ell^{1/4} \ge 234$.
- In ℓE_2 and ℓE_3 the weights are $K'(1+w)/w$, $K'(\kappa_\Phi/\ell)(1+w)^2/w^3$ and the A- and a-derivative terms of η_f, which are at most polynomial in w and in $\log \ell$ after the factor ℓ.

So the part $w > 40$ contributes at most 10^{-50} to each ℓ-scaled term below. Finally, $U(m) \ge m^{1/4} \ge 194$ and $1 - e^{D_\Phi} \le \kappa_\Phi/(w\ell)$, so $\ell \int \pi\, (1 - e^{D_\Phi}) \mathbf{1}[w > U(m)] \le \kappa_\Phi\, \pi(w > 194)/194 \le 10^{-50}\kappa_\Phi$.

- *The A-part.* With $h(w) := (1+w)^4 \log(2+w)/w$ one has $A(m,w) = (B/\ell)h(w)$ exactly, so $D' = D_A \le (B/\ell)h(w) \le 9.7 \cdot 10^{-4}$ for $w \le 40$ and $e^{D_A} - 1 \le 1.001 D_A$ there. On $[w_{\rm lo}, 2]$, $\rho(w-1) = 1$ and $h \le h(2) = 56.145$. On $[2, \infty)$, $\rho(w-1) \le e^{-(w-2)}$ by (R3) with $r(1) = 1$, and $h(w)/w \le (3+s)^4(\log 4 + s/4)/4$ with $s = w - 2$. So $\ell\, E_\pi[(e^{D'} - 1)_+ \mathbf{1}[w \le U(m)]] \le 1.001\, B\, (56.145 \log(2/w_{\rm lo}) + J_2) + 10^{-50}$, with $J_2 := \int_0^\infty e^{-s}(3+s)^4(\log 4 + s/4)\, ds/4 \le 200.6$ (by the Gamma moments).
- *The Φ-benefit.* Put $\Phi_{\rm ben} := E_\pi[1 - e^{-\kappa_\Phi/(w\ell)}]$, so that $\ell \int \pi\, (e^{D_\Phi} - 1)\mathbf{1}[w \le U(m)] \le -\ell\Phi_{\rm ben} + 10^{-50}\kappa_\Phi$. Using $1 - e^{-y} \ge y(1 - y/2)$, $\kappa_\Phi/\ell \le 0.13334$ and $w \ge w_1 \ge 29/31 - 10^{-8}$, $\Phi_{\rm ben} \ge (\kappa_\Phi/\ell)(1 - \kappa_\Phi/(2w_1\ell)) E_\pi[1/w] \ge 0.92873\, (\kappa_\Phi/\ell)\, c_2^-$, where

$$E_\pi[1/w] \ge c_2^- := \int_1^2 \frac{dw}{w^2} + \int_2^3 \frac{(1 - \log(w-1))\, dw}{w^2} = \frac{2}{3} + \frac{\log 2}{3} - \log\frac{4}{3} \ge 0.61003.$$

- *The excess mass.* With $x' := (\tau + \log 2)/\ell \le 1/30 + 2.4 \cdot 10^{-10}$, $\mu_0 = \log(1/w_{\rm lo}) \le \log(1/w_1) = \log\frac{1+x'}{1-x'} \le 2x' + x'^3$, so $\ell\mu_0 \le 2.0012\,\tau + 1.4$.
- *The remaining terms.* On $[w_{\rm lo}, 40]$ we have $e^D \le 1.001$, and $(1+w)^2/w$ is decreasing on $(0,1)$.
  - $\ell E_1 \le 2.002 K'(1+w_1)^2/w_1 \le 4.41 \cdot 4.0045 \le 17.7$.

– $\ell E_2 \le 1.001K'\big[\int_{w_1}^2 (1+w)\frac{dw}{w} + \int_2^\infty \rho(w-1)\,dw + \int_2^\infty \rho(w-1)\frac{dw}{w}\big]$. The first integral is $2+\log 2 - w_1 - \log w_1 \le 1.8244$. The second is $\int_1^\infty \rho \le \int_1^2 (1-\log v)\,dv + \rho(2)/r(2) \le 0.8021$, by (R3); the third is $\rho(2) = 1-\log 2$, by (R1). So $\ell E_2 \le 2.203 \cdot 2.9334 \le 6.5$.
– In η_f, the Φ-term is $\kappa_\Phi/m^2 = \kappa_\Phi(1+w)^2/(\ell w)^2$, and since $a(m,\cdot) = 0$ here, the Q-terms come from A alone: $|\partial_m A| = A/m$ and $(1+w)^2\ell^{-1}|\partial_w A|$ are together at most $4B(1+w)^6/\ell^2$ on $[w_1, 40]$. Since $\int \rho(w-1)(1+w)^6 dw/w \le 7820$, the latter contribute at most $4.41 \cdot 4B \cdot 7820/\ell \le 10^{-3}$ to ℓE_3. Hence

$$\ell E_3 \le 2.002K'\frac{\kappa_\Phi}{\ell}\Big[\int_{w_1}^2 \frac{(1+w)^2}{w^3}dw + \frac{9}{4}\int_2^\infty \rho(w-1)\frac{dw}{w}\Big] + 10^{-3}$$
$$\le 4.41 \cdot 0.13334 \cdot (2.3442 + 0.6905) + 10^{-3} \le 1.8.$$

– $\ell T \le e^{-200}$. Indeed $G(\ell,u) \ge \Phi(\ell) \ge C_0 e^{-\kappa_\Phi/\ell}$ here, and $\rho(u^*) \le e^{1-u^*}$ because $r \ge 1$ on $(1,\infty)$. So by Step 6, $\ell T \le \exp\big(\log\ell + \log(\ell/\tau) + \kappa_\Phi/\ell + a_V(u^*) + 1 - u^*\big)$, and $u^* \ge \ell^{1/4}(1 + \sqrt{2\log(\ell/\tau)+4}) \ge 234$, $a_V(u^*) \le 0.2$.

Altogether $\ell(E_1 + E_2 + E_3 + T) \le 26.1 \le 33$.

Altogether, since $\kappa_\Phi \ge 4\tau$ and $\log(2/w_{\rm lo}) \le \log(2/w_1) \le 0.76$,

$$\ell\Sigma \ge \tau\big(4(0.92873\,c_2^- - 10^{-50}) - 2.0012\big) - 1.001\,B\,(56.145\cdot 0.76 + J_2) - 33 - 1.4 \ge 0.265\,\tau - 2714 > 0,$$

where the 10^{-50} from the A-part is included in the 33.

(ii) Large ℓ: $\ell \ge 10^{18}$, every $u \ge 2$. This region is analytic, apart from a one-variable inequality checked in ball arithmetic. Put

$$s := w - u, \qquad r_0 := r(u-1), \qquad \Lambda(u) := (1+u)^2\log(2+u),$$
$$h_A(u,w) := h(w) - (1+u)^3\log(2+u), \qquad D_A = (B/\ell)\,h_A,$$
$$D_a := a(m,w) - a(\ell,u), \qquad D_\Phi := -\kappa_\Phi/(w\ell), \qquad D' := D_A + D_a.$$

Write E_π for integration against π, whose total mass is $1 + \mu_0 \le 1.7$. For $s \ge 0$ the density of π is at most $e^{-r_0 s}/u$, by (R3). The *first-order profile* is

$$c(u) := \ell\big(\Phi_{\rm ben} - \mu_0\big) - B\,j_A(u) - 2K'e_1(u) - K'e_2(u),$$

where

- $\Phi_{\rm ben} := E_\pi[1 - e^{-\kappa_\Phi/(w\ell)}]$ (exact, not linearised);
- $j_A := E_\pi h_A$;
- $e_1 := (1+w_{\rm lo})^2\rho(w_{\rm lo}-1)/(w_{\rm lo}\rho(u-1))$;
- $e_2 := \int (1+w)\rho(w-1)\,dw/(w\rho(u-1))$.

For $u \ge 2$, c depends on ℓ only through $\Phi_{\rm ben}$.

Lemma 5.7 (remainder, in ratio form)**.** *For $\tau \ge 10^8$, $\ell \ge \max(10^{18}, 30\tau)$ and $2 \le u \le U(\ell)$,*

$$\ell\Sigma \ge \big(1 - \varepsilon(\ell)\big)\big(c(u) + \ell s_a\big) - 60 \cdot \mathbf{1}[u \ge u_0 - 2] - 9 \cdot 10^{17}/\ell,$$

where $s_a := -E_\pi D_a$ is the slack of a, $a'(\ell,u) := \partial_u a = (u-u_0)_+/\sqrt{\ell}$, and

$$\varepsilon(\ell) := 2 \cdot 10^4\,U^2\log(2+U)/\ell + 30\,a'(\ell,U) + \ell^{-1/8} \le 0.03,$$

with $U = U(\ell)$; ε is decreasing in ℓ. Moreover $s_a \ge 0.67\,a'(\ell,u) - 0.54/\sqrt{\ell}$ for $u \ge u_0 - 2$, and $-0.01 \le \ell s_a \le 0$ for $u < u_0 - 2$.

The region $u < 2$ is region Z1, which (i) covers for every $\ell \ge 30\tau$; so the lemma is stated for $u \ge 2$, where $w_{\rm lo} = u - 1$, $\mu_0 = 0$ and π is a probability measure.

Proof. Notation and sizes. Put $\bar U := U(\ell)$, $\alpha' := a'(\ell,\bar U)$, $a' := a'(\ell,u) \le \alpha'$, $\lambda := \bar U^2 \log(2+\bar U)/\ell$, $s := w-u \in [-1,\infty)$ and $r_0 := r(u-1)$. Since $\tau \ge 10^8$ and $\ell \ge 10^{18}$, and all of the following decrease in ℓ (the worst case is $\tau = 10^8$),

$$\alpha' \le 2.24\cdot 10^{-4}, \qquad \lambda \le 8.2\cdot 10^{-7}, \qquad B\Lambda(u)/\ell \le 11.0001\lambda \le 9.1\cdot 10^{-6}, \qquad 1/\sqrt{\ell} \le 10^{-5}\alpha',$$

the last because $\alpha' = \sqrt{2\log(\ell/\tau)+4}\,\ell^{-1/4} \ge 3.28\,\ell^{-1/4}$. Iterating (R3) from $r(2)$ gives $r_0 \ge 1$, $r_0 \ge 4.59$ for $u \ge 27$, and $r_0 \ge r_{\mathrm{low}}(31619) \ge 12.82$ for $u \ge u_0 - 2$ (as $u_0 \ge 31622.7$). The same iteration gives $r(x) \ge \log x$ for every $x \ge 2$: it holds on $[2,3]$ since $r(2) \ge \log 3$, and if $r(x-1) \ge \log(x-1)$ then $r(x) \ge \log(x-1)\cdot \frac{x-1}{x-2} \ge \log x$.

Moments. Let $u \ge 2$. By (R3), the density of π at $s \ge 0$ is at most $e^{-r_0 s}/u$, and $\pi([u-1,u]) \le 1$, where $|s| \le 1$. Hence, for every $k \ge 0$,

$$M_k := E_\pi|s|^k \le 1 + \frac{k!}{u\,r_0^{k+1}}, \qquad \text{and} \qquad E_\pi\big[|s|^k e^{\gamma s_+}\big] \le 1 + \frac{k!}{u\,(r_0-\gamma)^{k+1}} \quad (0 \le \gamma \le r_0/2).$$

Also $\pi(w \le u+1) \ge 1 - e^{-r_0}/(ur_0) \ge 0.8$, and $E_\pi(u-w)_+ = 1 - 1/r(u)$ by (R1).

Decomposition. Write $D = D_A + D_a + D_\Phi$ and $D' := D_A + D_a$. Since $U(m) \le \bar U$, the main term is at most $\int \pi\, e^D \mathbf{1}[w \le \bar U]$. Write $e^D = e^{D_\Phi} + e^{D_\Phi}(e^{D'}-1)$ and $e^{D'} - 1 \le D' + \frac12 D'^2 e^{D'_+}$. Then $\int \pi\, e^{D_\Phi}\mathbf{1}[w \le \bar U] \le 1 - \Phi_{\mathrm{ben}}$, and $\int \pi\, e^{D_\Phi} D'\mathbf{1}[w \le \bar U] \le E_\pi D' + \rho_1$, because $D' \ge 0$ for $w > \bar U$ (there $h(w) > h(\bar U)$ and $a(m,w) \ge a(\ell,\bar U) \ge a(\ell,u)$). With $E_\pi D' = (B/\ell) j_A - s_a$, this gives

$$\ell\Sigma \ge c(u) + \ell s_a - \ell\big(\rho_1 + \rho_2 + \rho_3 + \rho_4 + E_3 + T\big),$$

where $\rho_1 := E_\pi[(1-e^{D_\Phi})|D'|]$, $\rho_2 := \frac12 E_\pi[D'^2 e^{D'_+}\mathbf{1}[w \le \bar U]]$, $\rho_3 := E_1 - E_1^{\mathrm{lin}}$ and $\rho_4 := E_2 - E_2^{\mathrm{lin}}$. Here E_i^{lin} is E_i with $e^D\mathbf{1}[w \le U(m)]$ replaced by 1 in g, so that $\ell E_1^{\mathrm{lin}} = 2K' e_1 = 2K' u^2 r_0$ and $\ell E_2^{\mathrm{lin}} = K' e_2$.

Pointwise bounds. (P1) and (P2) hold for all $w \ge u-1$, and (P3) for $u-1 \le w \le \bar U$. Put $\varphi_A(s) := 8$ for $s \le 0$ and $\varphi_A(s) := 1.5 + 8s(1+s)^3$ for $s \ge 0$.

(P1) $|D_A| \le (B\Lambda(u)/\ell)\varphi_A(s)$, $D_A \le 1.5B\Lambda(u)/\ell$ for $s \le 0$, and $D_A(u-1) < 0$. This uses $h_A(u) = (1+u)^3\log(2+u)/u \le 1.5\Lambda(u)$, that h is increasing on $[1,\infty)$ with $h'(w) \le 8\Lambda(w)$, that $\Lambda(w) \le \Lambda(u)(1+s)^3$ for $s \ge 0$, and $u^4\log(1+u)/(u-1) < (1+u)^3\log(2+u)$.

(P2) $D_a = \Delta\phi + \sigma_{\mathrm{sh}}$, where $\phi(x) := (x-u_0)_+^2/(2\sqrt{\ell})$, $\Delta\phi := \phi(w) - \phi(u)$ and $\sigma_{\mathrm{sh}} := a(m,w) - \phi(w) \ge 0$ (the shift from $m < \ell$). By convexity of ϕ,

$$-|s|a' \le \Delta\phi \le -|s|a' + \frac{s^2}{2\sqrt{\ell}} \quad (s \le 0), \qquad 0 \le \Delta\phi \le s\,a' + \frac{s^2}{2\sqrt{\ell}} \quad (s \ge 0).$$

The shift. Recall $m = \ell w/(1+w)$, so $m^{1/4} = u_0(1-x)^{1/4}$ and $\sqrt{m} = \sqrt{\ell}\,(1+1/w)^{-1/2}$ with $x := 1/(1+w) \le \frac12$. Put $A := (w-u_0)_+$ and $c_w := u_0 - m^{1/4}$.

- For $w \le u_0 - 3$: by concavity, $(1-x)^{1/4} \ge 1 - x/3$ on $[0,\frac12]$, so $c_w \le u_0/(3(1+w))$; and $w + u_0/(3(1+w)) < u_0$ on $[1, u_0-3]$ (a convex function of w, checked at both ends). So $w < m^{1/4}$ and $\sigma_{\mathrm{sh}} = 0$.
- For $w \ge u_0 - 3 \ge 31619$: $x \le 3.2\cdot 10^{-5}$, so $(1-x)^{1/4} \ge 1 - x/4 - 0.1x^2 \ge 1 - 0.2501x$ and $c_w \le 0.2501 u_0/(1+w)$. In particular $c_w \le 0.25012$, so $\sigma_{\mathrm{sh}} = 0$ again for $w \le u_0 - 0.26$. Also $1/\sqrt{m} \le (1+\frac{1}{2w})/\sqrt{\ell}$ and $(w - m^{1/4})_+ \le A + c_w$. Hence

$$\sigma_{\mathrm{sh}} \le \frac{(A+c_w)^2(1+\frac{1}{2w}) - A^2}{2\sqrt{\ell}} = \frac{A}{\sqrt{\ell}}\Big(c_w + \frac{A}{4w} + \frac{c_w}{2w}\Big) + \frac{c_w^2}{2\sqrt{\ell}}\Big(1 + \frac{1}{2w}\Big).$$

If $A > 0$ then $w > u_0$, and the bracket is at most $0.2501\frac{u_0}{1+w} + \frac14 - \frac{u_0}{4w} + 10^{-5} \le \frac14 + 0.0001\frac{u_0}{w} + 10^{-5} \le 0.2502$. For $w \ge u_0 - 0.26$, $c_w \le 0.2501$, so the last term is at most $0.0314/\sqrt{\ell}$.

Since $(w-u_0)_+ \le (u-u_0)_+ + s_+$, in all cases

$$0 \le \sigma_{\rm sh} \le \Big(0.2503\,\frac{(w-u_0)_+}{\sqrt{\ell}} + \frac{0.0314}{\sqrt{\ell}}\Big)\mathbf{1}[w \ge u_0 - 0.26] \le 0.2503\Big(a' + \frac{s_+}{\sqrt{\ell}}\Big) + \frac{0.0314}{\sqrt{\ell}}.$$

Hence, using $-|s|a' \le \Delta\phi \le 0$ for $s \le 0$ (as ϕ is non-decreasing) and the displayed bound for $s \ge 0$, $|D_a| \le (|s|+0.2503)a' + q(s)/\sqrt{\ell}$ and $(D_a)_+ \le (s_+ + 0.2503)a' + (q(s)+0.5)/\sqrt{\ell}$, with $q(s) := s_+^2/2 + 0.2503 s_+ + 0.0314$.

(P3) $D' \le 0.7\cdot 10^{-4} + 3\cdot 10^{-4} s_+$. For $s \le 0$ this follows from (P1), $\Delta\phi \le 0$ and $\sigma_{\rm sh} \le 0.2503\alpha' + 10^{-10}$, which give $D' \le 1.37\cdot 10^{-5} + 5.61\cdot 10^{-5}$. For $s \ge 0$, the slope of D_A in w is $(B/\ell)h' \le 8B\Lambda(\bar U)/\ell \le 7.3\cdot 10^{-5}$, and that of $a(m,w)$ is at most $(w - m^{1/4})_+/\sqrt{m} \le 1.0003\alpha'$ (the m-dependence only lowers it).

So $e^{D'_+} \le 1.0001\, e^{\gamma s_+}$ with $\gamma := 3\cdot 10^{-4}$, and the moment bounds apply. In particular, for $u = 2$ (the worst case of $r_0 \ge 1$) and for $u \ge 27$ ($r_0 \ge 4.59$) respectively,

$$E_\pi[\varphi_A e^{\gamma s_+}] \le 205.1,\ 8.07; \qquad E_\pi[\varphi_A^2 e^{\gamma s_+}] \le 2.71\cdot 10^6,\ 66.5; \qquad E_\pi[(1+s_+)^5 e^{\gamma s_+}] \le 165,\ 1.03.$$

The slack s_a. Let $u \ge u_0 - 2$. Then $r_0 \ge 12.82$, and $r(u) \ge r_{\rm low}(31620) \ge 12.8225$, so $1 - 1/r(u) \ge 0.92201$. By (P2) and the moments, $-E_\pi\Delta\phi \ge a'\big(E_\pi(u-w)_+ - E_\pi s_+\big) - E_\pi s^2/(2\sqrt{\ell}) \ge 0.9219a' - 0.5001/\sqrt{\ell}$ and $E_\pi\sigma_{\rm sh} \le 0.2503a' + 0.0315/\sqrt{\ell}$. So $s_a \ge 0.6716a' - 0.532/\sqrt{\ell}$. Let $u < u_0 - 2$. Then $a' = 0$, $D_a \ge 0$, and $D_a \ne 0$ only for $s \ge d := u_0 - 0.26 - u \ge 1.74$, where $D_a \le (s^2/2 + 0.2503 s + 0.0314)/\sqrt{\ell}$. If $u \ge u_0/2$, then $r_0 \ge \log(u_0/2 - 1) \ge 9.6$ and $\ell|s_a| \le \sqrt{\ell}\cdot 5e^{-1.74 r_0}/(u r_0) \le 4u_0^{-0.74} \le 0.01$; if $u < u_0/2$, then $d \ge u_0/2 - 0.26$ and $\ell|s_a| \le e^{-u_0/3}$. (Here $\int_d^\infty p(s)e^{-r_0 s}ds \le e^{-r_0 d}(p + p' + p'')(d)/r_0$ for the quadratic p. For $r_0 \ge 1$ the function $e^{-r_0 d}(p+p'+p'')(d)$ is decreasing in d, and $(p+p'+p'')(1.74) \le 4.98$, so the integral is at most $5e^{-1.74 r_0}/r_0$.)

The remainders.

(a) $\rho_1 \le 6\,\Phi_{\rm ben}\, E_\pi|D'|$, because $1 - e^{D_\Phi} \le 1 - e^{-\kappa_\Phi/((u-1)\ell)}$, $(1-e^{-y})/y$ is decreasing, $(u+1)/(u-1) \le 3$ and $\pi(w \le u+1) \ge 0.8 \ge \frac12$. By (P1), (P2) and the moments, $E_\pi|D'| \le 205.1\, B\Lambda(u)/\ell + 1.2504a' + 0.7/\sqrt{\ell}$. So $\ell\rho_1 \le (13540\lambda + 7.6\alpha')\,\ell\Phi_{\rm ben}$, and $\ell\rho_1 \le (540\lambda + 7.6\alpha')\,\ell\Phi_{\rm ben}$ if $u \ge 27$.

(b) $\rho_2 \le 1.0001\, E_\pi[D_A^2 e^{\gamma s_+}] + 1.0001\, E_\pi[D_a^2 e^{\gamma s_+}]$, from $D'^2 \le 2D_A^2 + 2D_a^2$ and (P3).

- The A-part is at most $1.0001 B^2\Lambda(u)^2 E_\pi[\varphi_A^2 e^{\gamma s_+}]/\ell$. For $u < 27$ this is at most $1.0001\cdot 121\cdot\Lambda(27)^2\cdot 2.71\cdot 10^6/\ell \le 2.3\cdot 10^{15}/\ell$. For $u \ge 27$, using $(1+u)^4\log^2(2+u) \le 1.088\,u^2\log(1+u)\Lambda(u)$, it is at most $1.0001\cdot(B^2/K')\cdot 1.088\cdot 66.5\cdot 1.00001\lambda\cdot K'u^2\log(1+u) \le 4000\lambda\,K'u^2\log(1+u)$.
- The a-part. For $u \ge u_0 - 2$, by (P2), $\ell E_\pi[D_a^2 e^{\gamma s_+}] \le 2\ell a'^2\cdot 1.5634 + 0.002$. If $s_a \ge 0$, the bound on s_a gives $a' \le 1.5 s_a + 0.81/\sqrt{\ell}$; if $s_a < 0$, then $a' \le 0.81/\sqrt{\ell}$. In both cases $\ell a'^2 \le 1.5\alpha'\,\ell(s_a)_+ + 0.81\alpha'\sqrt{\ell} + 0.66$. So the a-part of $\ell\rho_2$ is at most $4.7\alpha'\,\ell(s_a)_+ + 2.54\alpha'\sqrt{\ell} + 2.1$. For $u < u_0 - 2$, the tail estimate used for s_a makes it at most $\lambda K'u^2\log(1+u)$ if $u \ge 27$, and $e^{-u_0/3}$ if $u < 27$.

(c) $\rho_3 \le E_1^{\rm lin}\cdot 1.0001\, D'(u-1)_+$. By (P1) and (P2), $D'(u-1) \le \sigma_{\rm sh}(u-1)$, which vanishes unless $u \ge u_0 + 0.74$ and is at most $0.2503a' + 0.0314/\sqrt{\ell}$. So $\rho_3 = 0$ for $u < u_0 + 0.74$, and otherwise $\ell\rho_3 \le 2.0002 K'u^2 r_0(0.2504\alpha') \le 0.72\alpha'\,K'u^2\log(1+u)$, since $r_0 \le r_{\rm up}(u-1) \le \log u + \log\log u + 2 \le 1.42\log(1+u)$ there (R4).

(d) $\rho_4 \le (K'/\ell)\cdot 1.0001\,E_\pi[(1+w)D'_+e^{\gamma s_+}]$, with $1+w \le (1+u)(1+s_+)$ and $D'_+ \le (B\Lambda(u)/\ell)\varphi_A(s_+) + (s_+ + 0.2503)a' + (q(s)+0.5)/\sqrt{\ell}$. For $u \ge 27$ this gives $\ell\rho_4 \le 1.0001K'(1+u)(1.6\cdot 11.0001\lambda + 0.2601\alpha') \le (0.21\lambda + 0.004\alpha')K'u^2\log(1+u)$, as $(1+u)/(u^2\log(1+u)) \le 0.0116$. For $u < 27$, $a' = 0$, D_a vanishes except at $s \ge u_0 - 27$, and $\ell\rho_4 \le 1.0001\cdot 2.2\cdot 28\cdot 11\Lambda(27)\cdot 1049/\ell \le 2\cdot 10^9/\ell$.

(e) $E_3 \le 2.0002K'E_\pi[e^{\gamma s_+}\eta_f \mathbf{1}[w \le \bar U]]$, and $\eta_f = \kappa_\Phi/m^2 + |\partial_m Q| + (1+w)^2\ell^{-1}|\partial_w Q|$ with $Q = A + a$.

- The Φ-part: $\kappa_\Phi/m^2 = \kappa_\Phi(1+w)^2/(\ell w)^2 \le 4\kappa_\Phi/\ell^2$. Since $\ell\Phi_{\text{ben}} \ge 0.8\,(1-0.023)\,\kappa_\Phi/(u+1)$, its contribution is at most $2.0002K'\cdot 4\,E_\pi[e^{\gamma s_+}]\,\kappa_\Phi/\ell \le 26.5\kappa_\Phi/\ell \le 34(u+1)\ell^{-1}\,\ell\Phi_{\text{ben}} \le \lambda\,\ell\Phi_{\text{ben}}$.
- The A-part: $|\partial_m A| = A/m$ and $(1+w)^2\ell^{-1}|\partial_w A|$ together are at most $10B\Lambda(w)(1+w)^2/\ell^2 \le 10B\Lambda(u)(1+u)^2(1+s_+)^5/\ell^2$. For $u \ge 27$ the contribution to ℓE_3 is at most $44.01\cdot 11\cdot 1.03\,\Lambda(u)(1+u)^2/\ell \le 260\lambda\,K'u^2\log(1+u)$; for $u < 27$ it is at most $44.01\cdot 11\cdot\Lambda(27)\cdot 28^2\cdot 165/\ell \le 2\cdot 10^{11}/\ell$.
- The a-part vanishes for $w < u_0 - 0.26$. There $m \ge 0.9999\ell$ and $(w - m^{1/4})_+ \le 1.0001\alpha'\sqrt{\ell}$, so $\ell|\partial_m a| \le 0.2501(\alpha' u_0 + \alpha'^2 u_0^2)$ and $\ell\cdot(1+w)^2\ell^{-1}|\partial_w a| \le 1.0002\alpha'(1+u)^2(1+s_+)^2$. For $u \ge u_0 - 2$ the contribution to ℓE_3 is at most $4.41\big(0.2501(\alpha' u_0 + \alpha'^2 u_0^2) + 1.0004\alpha'(1+u)^2\big) \le 0.2\alpha'\,K'u^2\log(1+u)$, since $u \ge u_0 - 2$ and $\log(1+u) \ge 10.36$. For $u < u_0 - 2$ the factor $\pi(w \ge u_0 - 0.26) \le e^{-r_0 d}/(ur_0)$ makes it at most $10^{-3}(\lambda+\alpha')K'u^2\log(1+u)$ for $u \ge 27$, and $e^{-u_0/3}$ for $u < 27$.

(f) T. By Step 6, $T \le e^{0.1334}\exp\big(y - a(\ell,u) - r_0(\bar U - u + \tfrac{3}{2})\big)$, where $y = \log(\ell/\tau) = a(\ell,\bar U) - 2$, $\kappa_\Phi/\ell \le 0.13334$ and $a_V(u^*) \le 13\lambda$.

- If $\bar U - u \le (\bar U - u_0)/2$, then $u - 1 \ge 2.6u_0$ (as $\bar U - u_0 \ge 3.28u_0$), so $r_0 \ge \log(2.6u_0)$; also $y - a(\ell,u) \le \alpha'(\bar U - u) - 2$ (convexity of $a(\ell,\cdot)$), and $a' \ge \alpha'/2$, so $s_a \ge 0.3358\alpha' - 0.532/\sqrt{\ell} \ge 0.3357\alpha' \ge 1.103/u_0$. Hence $T \le e^{-1.8666}(2.6u_0)^{-1.5} \le 0.037\,u_0^{-1.5} \le 0.035\,\ell^{-1/8}s_a$.
- Otherwise $\bar U - u \ge 1.64u_0$, $y \le 4\log u_0$ and $r_0 \ge 1$, so $\ell T \le \exp(0.14 + 8\log u_0 - 1.64u_0) \le 1/\ell$.

Assembly. Write $c(u) = \ell\Phi_{\text{ben}} + c_0(u)$, and recall $\ell(s_a)_+$ and $\ell|s_a|$ from the bounds on s_a.

- $u \ge 27$. By the positivity argument below, $c_0(u) \ge K'u^2\log(1+u) > 0$, so both parts of c are non-negative; and $c \ge c_0 \ge 22.7\sqrt{\ell}$ when $u \ge u_0 - 2$, so that $2.54\alpha'\sqrt{\ell} \le 0.12\alpha' c$. Adding (a)–(f),

$$\ell(\rho_1 + \cdots + T) \le \varepsilon_1 c + \varepsilon_2\,\ell(s_a)_+ + 2.2\cdot\mathbf{1}[u \ge u_0 - 2] + 1/\ell,$$
$$\varepsilon_1 := 4300\lambda + 8.8\alpha', \quad \varepsilon_2 := 4.7\alpha' + 0.035\,\ell^{-1/8}.$$

- $2 \le u < 27$. Here $a' = 0$, $\rho_3 = 0$ and $c \ge 0.944\,\ell\Phi_{\text{ben}}$, by the Φ-dominance bound below. So (a)–(f) give $\ell(\rho_1 + \cdots + T) \le \varepsilon_1 c + 2.5\cdot 10^{15}/\ell$ with $\varepsilon_1 := 14400\lambda + \alpha'$.

In both cases $\varepsilon_1 \le 0.72\,\varepsilon$ and $\varepsilon_2 \le \varepsilon$. If $s_a \ge 0$, the claim follows at once. If $s_a < 0$, it needs in addition $\varepsilon\,\ell|s_a| \le (\varepsilon - \varepsilon_1)c$, which holds because $\ell|s_a| \le 0.54\sqrt{\ell} \le 0.024c$ for $u \ge u_0 - 2$, $\ell|s_a| \le 0.01 \le 0.2c$ for $27 \le u < u_0 - 2$, and $\ell|s_a| \le e^{-u_0/3}$ for $u < 27$. Finally, at $\tau = 10^8$ (the worst case), $\varepsilon(10^{18}) \le 0.0286$ and $\varepsilon(10^{22}) \le 0.0029$. The constants are crude, and nothing is optimised. □

Positivity of $c(u)$, for $\tau \ge 10^8$.

- $u \ge 27$: $c_0(u) := c(u) - \ell\Phi_{\text{ben}} \ge K'u^2\log(1+u)$. Here $\mu_0 = 0$, so $c_0 = -Bj_A - 2K'e_1 - K'e_2$ does not depend on ℓ or τ.

– On $[u-1, u]$, convexity of h ($h'' > 0$ on $[1,\infty)$) gives the chord bound $h(w) \le h(u) - h'(u-1)(u-w)$, with $h'(u-1) \ge 3u^2 \log(1+u)$, and (R1) gives $E_\pi(u-w)_+ = 1 - 1/r(u)$.
– On $w \ge u$, $h(w) - h(u) \le h'(w)s$ and (R3) bound the remaining part by $144\Lambda(u)/(ur_0^2)$.
– Hence $E_\pi h_A \le (1+u)^3 \log(2+u)/u - 3u^2 \log(1+u)(1 - 1/r(u)) + 144\Lambda(u)/(ur_0^2)$. Also $e_1 = u^2 r(u-1)$ and $e_2 \le u+1$.

With $B = 5K'$ this gives

$$\frac{c_0(u)}{K'u^2} \ge 5\log(1+u)\Big[3\big(1 - 1/r(u)\big) - \frac{(1+u)^3\log(2+u)}{u^3\log(1+u)} - \frac{144(1+u)^2\log(2+u)}{u^3 r_0^2 \log(1+u)}\Big] - 2r(u-1) - \frac{u+1}{u^2}.$$

With r enclosed in $[r_{\text{low}}, r_{\text{up}}]$, the right side is $\ge \log(1+u)$ on $[27, 4\cdot 10^5]$ (computer-assisted; see Appendix B: 9,609 cells in ball arithmetic, no failures, minimum margin 0.189). Beyond $4\cdot 10^5$ it is analytic: $r_{\text{low}} \ge 15.3$ gives $5\cdot 1.79 \log u - 2(\log u + \log\log u + 2) > \log(1+u)$.

- $2 \le u < 27$*: Φ-dominance.* By Jensen, $E_\pi[1/w] \ge 1/E_\pi w \ge 0.94/u$, since $E_\pi w = u - (1 - 1/r(u)) + E_\pi s_+ \le u - 1 + 1/r(2) + 1/(ur_0^2) \le u + 0.12$. With $1 - e^{-y} \ge y(1 - y/2)$ and $\kappa_\Phi/\ell \le 0.1334$, $\ell\Phi_{\text{ben}} \ge 0.933 \cdot 0.94\,\kappa_\Phi/u \ge 0.87 \cdot 4 \cdot 10^8/27 \ge 9.5 \cdot 10^6$. Meanwhile $B\,j_A \le B(1.5 + 392/u)\Lambda(u) \le 4.7 \cdot 10^5$ (by (R3) with $r_0 \ge 1$), $2K'e_1 = 2K'u^2 r(u-1) \le 2\cdot 10^4$ and $K'e_2 \le K'(u + 1.12) \le 62$. So $c_0 \ge -5.3\cdot 10^5$ and $c(u) \ge 0.944\,\ell\Phi_{\text{ben}} \ge 8.9\cdot 10^6$.
- $u < 2$*:* region Z1. Case (i) gives $\ell\Sigma \ge 0.265\tau - 2714 > 0$ for every $\ell \ge 30\tau$, in particular for $\ell \ge 10^{18}$.

So $c(u) \ge 5000$ for every $u \ge 2$, and $c(u) \ge c_0(u) \ge 22.7\sqrt{\ell}$ for $u \ge u_0 - 2$. By the bounds on s_a in Lemma 5.7, $c(u) + \ell s_a \ge 0.97\,c(u)$, and the lemma gives $\ell\Sigma \ge 0.94\,c(u) - 60\cdot\mathbf{1}[u \ge u_0 - 2] - 0.9 > 0$ for every $\ell \ge 10^{18}$ and $2 \le u \le U(\ell)$; with (i), every $u \le U(\ell)$ is covered.

(iii) Region Z2: $30\tau \le \ell < 10^{18}$ and $u \ge 2$ (computer-assisted; see Appendix B). Here we use a closed-form lower bound $S(\ell, u) \le \Sigma(\ell, u)$, built only from (R1)–(R4), the constant K' of Lemma 5.1, and convexity. Since $u \ge 2$, $w_{\text{lo}} = u - 1$ and $\mu_0 = 0$. Put $v := u - u_0$, $c_u := u_0/(3u)$, $k_u := 1 + 1/(2(u-1))$ and $\varphi(w) := (w - u_0 + c_u)_+^2/(2\sqrt{\ell})$.

- *Two elementary majorants.* First, $1/(2\sqrt{m}) = \sqrt{(1+w)/w}/(2\sqrt{\ell}) \le k_u/(2\sqrt{\ell})$ for $w \ge u - 1$. Second, $(w - m^{1/4})_+ \le (w - u_0 + u_0/(3(1+w)))_+ \le (w - u_0 + c_u)_+$. Together, $a(m, w) \le k_u\varphi(w)$, and φ is convex. Also $A(m, w) = (B/\ell)h(w)$ with h convex.
- *Part I, $w \in [u-1, u]$.* By the chord bounds for h and φ, together with $h(u) - h(u-1) \ge h'(u-1)$, $\varphi(u) - \varphi(u-1) \ge \varphi'(u-1)$, and $D_\Phi \le -\kappa_{\min}/(u\ell)$, we get $D \le \beta_I - \alpha(u - w)$, where:
  – $\beta_I := A(\ell, u)/u + \beta_a - \kappa_{\min}/(u\ell)$;
  – $\alpha := (B/\ell)u^3(3u-4)\log(1+u)/(u-1)^2 + k_u(v - 1 + c_u)_+/\sqrt{\ell}$;
  – $\beta_a := (k_u - 1)(v + c_u)_+^2/(2\sqrt{\ell}) + c_u(2v_+ + c_u)/(2\sqrt{\ell})$.
  Then use $e^{-\alpha t} \le 1 - (1 - e^{-\alpha})t$ and $E_\pi(u-w)_+ = 1 - 1/r(u)$ (R1).
- *Part II, $w \in [u, \bar{U}]$, $s = w - u$.* Here $D \le \beta_{II} + \gamma(s)s$, with $\beta_{II} := A(\ell, u)/u + \beta_a$ and a two-piece slope: $\gamma(s) = \gamma_1$ for $s \le 60$ and $\gamma(s) = \gamma_{\max}$ beyond. These are the slopes of $(B/\ell)h + k_u\varphi$ bounded via $h'(w) \le 8(1+w)^2\log(2+w)$ at $w = u + 60$ and at $w = \bar{U}$, respectively. This is combined with $\rho(w-1)/\rho(u-1) \le e^{-r(u-1)s}$ (R3). The integrals converge because $r(u-1) > \gamma_1$ and $r(u-1) > \gamma_{\max}$, which is checked in every cell.
- *What the induction needs.* It needs an upper bound for $\int \pi\, e^D \mathbf{1}[w \le U(m)] - 1$. The Part I/II functions are positive majorants of $e^D\mathbf{1}[w \le U(m)]$ on $[u-1, \bar{U}]$, and $\mathbf{1}[w \le$

$U(m)] = 0$ beyond $\bar{U}$. So their integrals bound it. The same majorants (with a two-piece bound for η_f) bound E_1, E_2 and E_3; T is bounded as in Step 6. The difference $M-1$ is written as a sum of small terms, so there is no cancellation against 1.

- *Uniformity in $\tau \geq 10^8$.* The bound uses $\kappa_{\min} := 4 \cdot 10^8$ where κ_Φ helps, $\kappa_{\max} = 4\ell/30 + 1500$ where it hurts, and the slopes at $\bar{U}(\ell) := U(\ell; \tau = 10^8)$, which is the largest junction.
- *The bound S.* With $r := r(u)$, $r_0 := r(u-1)$, $S_1 := 60$, $q := e^{-(r_0-\gamma_{\max})S_1}/(r_0-\gamma_{\max})$ and $d_+ := (d+c_u)_+$, the certificate evaluates

$$M - 1 \leq (e^{\beta_I}-1)\Big(1-\frac{1}{ur}\Big) + \frac{e^{\beta_{II}}-1}{ur} - e^{\beta_I}(1-e^{-\alpha})\Big(1-\frac{1}{r}\Big) + \frac{e^{\beta_{II}}}{u}\Big(\frac{\gamma_1}{r_0(r_0-\gamma_1)} + q\Big),$$

$$\ell E_1 \leq 2K' u^2 r_0 e^{\beta_I - \alpha}, \qquad \ell E_2 \leq K' \frac{u}{u-1}\Big((u-1)e^{\beta_I} + e^{\beta_{II}}\Big(\frac{1}{r_0-\gamma_1} + q\Big)\Big),$$

$$E_3 \leq 2K'\Big(\bar{\eta}(u,v)e^{\beta_I} + \frac{e^{\beta_{II}}}{u}\Big(\frac{\bar{\eta}(u+S_1, v+S_1)}{r_0-\gamma_1} + \bar{\eta}(\bar{U}, \bar{U}-u_0)\, q\Big)\Big),$$

where

$$\bar{\eta}(w,d) := \frac{4\kappa_{\max}}{\ell^2} + \frac{4B(1+w)^3\log(2+w)}{\ell^2} + \frac{d_+}{4(\ell/2)^{5/4}} + \frac{k_u d_+^2}{2\ell^{3/2}} + \frac{(1+w)^2}{\ell}\Big(\frac{2B(1+w)^2(3\log(2+w)+1)}{\ell} + \frac{\sqrt{2}\,d_+}{\sqrt{\ell}}\Big),$$

and $T \leq \exp(\kappa_{\max}/\ell + a_V(\bar{U}+\frac{1}{2}) + E_T)$ by Step 6. Then $S := -\big[(M-1)+E_1+E_2+E_3+T\big]$, with these bounds, satisfies $S \leq \Sigma$. Here the masses $\pi([u-1,u]) = 1-1/(ur)$ and $\pi([u,\infty)) = 1/(ur)$, and $\int_{u-1}^{u} \rho(w-1)\,dw = (u-1)\rho(u-1)$, come from (R1); on $w \geq u$, $\pi(dw) \leq e^{-r_0(w-u)}dw/u$ by (R3); and $(1+w)/w \leq u/(u-1)$ for $w \geq u-1$. Also $\bar{\eta} \geq \eta_f$, because $\kappa_\Phi \leq \kappa_{\max}$, $m \geq \ell/2$, $(1+w)/w \leq 2$ and $((1+w)/w)^{3/2} \leq 2k_u$ for $w \geq u-1 \geq 1$, and $(w-m^{1/4})_+ \leq (w-u_0+c_u)_+$. Since $\bar{\eta}$ is increasing in both arguments, its values at (u,v), $(u+S_1, v+S_1)$ and $(\bar{U}, \bar{U}-u_0)$ bound η_f on $[u-1,u]$, $[u,u+S_1]$ and $[u+S_1,\bar{U}]$. The code is `bs_cert_arb.py` (Appendix B.1).

- *The cover.* The range $\ell \in [3\cdot 10^9, 10^{18}]$ is split into 403 bands of ratio 1.05. In each band, u-cells cover $[2, \max(40, u_0/2)]$, and cells in $v = u - u_0(\ell)$ cover the rest up to $\bar{U}(\ell)$. The endpoints of the v-cells are rounded outward, so the seams between the two kinds of cells, and the top edge, are covered exactly for every ℓ in the band. On each cell, S is evaluated in ball arithmetic, with r_{low} and r_{up} computed in the same arithmetic. The enclosure bounds S on the whole cell, so no Lipschitz or interpolation argument is needed.
- *Result.* 302,013 cells (including adaptive bisections), no failures, and minimum certified $\ell S = 2.491 \cdot 10^6$.

Step 8: conclusion. By Steps 4–7, (34) holds for all $\ell \geq \tau$ and $0 \leq u \leq U(\ell)$, and Step 3 gives the theorem. The threshold $\ell_1 = 30\tau$ is internal to the induction: the base region $[\tau, 30\tau]$ is covered by Lemma 5.3. So the conclusion holds for every $Y \geq t$, with the same C_{BS}. For the $4t$-dense version, replace τ by $\tau + \log 4$ throughout; nothing else changes. □

The induction claim (34) is itself used later, in the proof of the short-interval Saias bound (Lemma 5.32). We record it.

Lemma 5.8 (BS-profile)**.** *Let $\tau = \log t \ge 10^8$, $\ell \ge \tau$ and $0 \le u \le U(\ell)$. Then*

$$W_t(e^\ell; e^{\ell/u}) := \sum_{n \in \mathcal{N} \cap (e^\ell, 2e^\ell],\ t\text{-dense},\ P^+(n) < e^{\ell/u}} \frac{1}{n} \le \frac{\tau}{\ell}\, \Phi_\infty\, \rho(u-1)\, e^{Q(\ell,u)},$$

with U, Q and $\Phi_\infty \le 38.3$ as in Step 3 of the proof of Theorem 5.6. Moreover, Φ_∞ is decreasing in τ.

Proof. This is (34), divided by $\max(1, \ell/\tau) = \ell/\tau$. Steps 4–7 prove it on the whole range $0 \le u \le U(\ell)$, not only at $u = 0$.

Uniformity in $\tau \ge 10^8$. The computer-assisted region is uniform in τ by construction ($\kappa_{\min}$, $\kappa_{\max}$, $\bar{U}$). The analytic regions are monotone in τ in the safe direction: $\ell\Sigma \ge 0.265\tau - 2714$ for $u < 2$, and for $\ell \ge 10^{18}$ the term $\ell\Phi_{\text{ben}}$ increases with κ_Φ. The base region uses Lemma 5.3 with Φ_∞ chosen for the given τ. □

Corollary 5.9 (Saias weights at every start)**.** *For every start $y \ge X_0$ and every $X_0^-/2 \le v \le y$,*

$$W_y^{\text{sh}}(v) := \sum_{e \in \mathcal{N}:\ y/e \in [v, 2v)} \frac{|c_e(y)|}{e} \le C_S\, \frac{\log(2v)}{\log y}, \qquad C_S := \max(2.01\, C_{\text{BS}},\, 1.4) \le 77.$$

Proof. If $c_e(y) \ne 0$, then by Lemma 5.5 $y/e \ge X_0^-$ (the cutoff in K_{pt}), e is squarefree and $\chi_y^t(e) = 1$. As $y/e \in [v, 2v)$, the density statement after Lemma 5.5 shows that e is $4v$-dense, including the case $j = 0$, $p_1 \le 4v$: with the primes of e in decreasing decision order $q_1 \succ \cdots \succ q_r$, the condition of χ_y^t at q_i reads $t(q_i) < y/(q_1 \cdots q_i) < 2v\, q_{i+1} \cdots q_r$, and the choice of p^* there converts this into the value-ordered condition.

- For $v \le \sqrt{y}/4$, apply Theorem 5.6 with $t := 4v \ge 2X_0^-$ and $Y := y/(2v)$. Then $Y \ge t$ and $\log Y \ge \frac{1}{2}\log y$, and
$$W_y^{\text{sh}}(v) \le C_{\text{BS}} \frac{\log(4v)}{\log(y/2v)} \le 2C_{\text{BS}} \Big(1 + \frac{\log 2}{\log X_0^-}\Big) \frac{\log(2v)}{\log y} \le 2.01\, C_{\text{BS}} \frac{\log(2v)}{\log y}.$$
- For $v > \sqrt{y}/4$, put $A := y/(2v) \in [\frac{1}{2}, 2\sqrt{y})$. Then $W_y^{\text{sh}}(v) \le \sum_{e \in (A, 2A]} 1/e$. This sum is at most 1 always (on $\mathbb{N}$ below X_0^-), and at most $\log 2 + 2.7\eta_1 + 1/A \le 0.6959$ if $A \ge y^{1/4}$ (partial summation as in the proof of Lemma 5.3). In the first case $\log(2v)/\log y \ge \frac{3}{4}$, and in the second $\log(2v)/\log y \ge \frac{1}{2} - \log 2/\log y$. Either way $W_y^{\text{sh}}(v) \le 1.4 \log(2v)/\log y$. □

5.5. **The operator kernels.** The long operator $\mathcal{R} = (1 + K_{\text{pt}}B)^{-1}$ of (30) differs from the sharp resolvent $(1 + K_{\text{pt}})^{-1}$ in two ways. After every latest-prime step it averages over a window of length H at the landing scale, and it evaluates the filters at the actual (averaged) centre, not at the nominal landing point. This subsection shows that the first difference costs only a constant factor in the Saias weights of Corollary 5.9, and that the second is supported on near-threshold steps, which are rare. The main tool is the Kantorovich–Rubinstein distance: it contracts under dilation by the factor d/n at every step, so that, unlike total variation, it can be iterated along chains of arbitrary length.

5.5.1. *Kernels and steps.* Fix a start x. For a cofactor $w \in \mathcal{N}$, $w \ge 2$, and a position v, define the *step*

$$Q_w(v, \cdot) := \mathbf{1}[\tau_w < v/w]\, \mathbf{1}[v/w \ge X_0^-]\, \text{box}_{H(v/w)}(\cdot - v/w), \tag{37}$$

where box_h is the uniform probability measure on $[-h/2, h/2)$. Thus the filter is evaluated at the actual centre v/w, and the window average comes after the filter. By the definitions of K_{pt}

and B in §5.1, for every function g,

$$(K_{\mathrm{pt}}Bg)(v)=\sum_{w\ge 2}\frac{1}{w}\,\mathbf{1}[\tau_w<v/w]\,\mathbf{1}[v/w\ge X_0^-]\,(Bg)(v/w)=\sum_{w\ge 2}\frac{1}{w}\int g\,dQ_w(v,\cdot).$$

Definition 5.10 (Kernels)**.** The kernels of $\mathcal{R}$ from the start x are the signed measures $C_n=C_n^x$ ($n\in\mathcal{N}$) given by the last-step recursion

$$C_1^x=\delta_x,\qquad C_n^x=-\sum_{d|n,\ d<n}\int Q_{n/d}(v,\cdot)\,dC_d^x(v)\quad(n\ge 2). \tag{38}$$

Expanding $\mathcal{R}=\sum_{k\ge 0}(-K_{\mathrm{pt}}B)^k$ (the series terminates at every point, by the remark after (30)) and grouping the chains $w_1,\dots,w_k$ according to their product $n=w_1\cdots w_k$ gives

$$(\mathcal{R}g)(x)=\sum_{n\in\mathcal{N}}\frac{1}{n}\int g\,dC_n^x. \tag{39}$$

We use the local exponent of the jitter width,

$$\varepsilon_*:=\sup_{v\ge X_0^-/2}\frac{d\log H}{d\log v}\le\max\Big(\frac{16\log\log(X_0^-/2)}{\log(X_0^-/2)},\,4\cdot 10^{-6}\Big)\le 4\cdot 10^{-6},$$

which follows from (18) for the standard width ($d\log H_{\mathrm{std}}/d\log v=16\log\log v/\log v$, decreasing in v) and from Theorem 6.22(c) on the raised range; so $H(vw)/H(v)\le w^{\varepsilon_*}$ for $v\ge X_0^-/2$, $w\ge 1$. Each C_n^x is supported in an interval of radius $R_n\le 1.02\,H(x/n)$ around x/n. Indeed, a step $d\to n$ maps the support of C_d, of radius R_d around x/d, to radius R_d/w around x/n and then adds half a window, so by induction, $R_n\le H(x/n)/2+\max_d 1.02\,H(x/d)/w\le H(x/n)\,(\frac12+1.02\cdot 2^{\varepsilon_*-1})\le 1.02\,H(x/n)$.

Nominal steps. Write $H_n:=H(x/n)$. For a step $d\to n$ with cofactor $w=n/d$, the *nominal* step

$$Q^{\mathrm{nom}}_{d\to n}(v,\cdot):=\kappa_{d\to n}\,\mathrm{box}_{H(v/w)}(\cdot-v/w),\qquad \kappa_{d\to n}:=\mathbf{1}[\tau_w<x/n]\,\mathbf{1}[x/n\ge X_0^-],$$

freezes the scalar filter at the nominal landing point x/n; the window width stays $H(v/w)$. The nominal kernels $C_n^{x,\mathrm{nom}}$ are defined by (38) with Q^{nom} in place of Q. Their masses $c_n:=C_n^{x,\mathrm{nom}}(\mathbb{R})$ satisfy $c_1=1$ and $c_n=-\sum_{d|n,d<n}\kappa_{d\to n}c_d$, which is the recursion of the sharp resolvent. Hence, by Lemma 5.5,

$$c_n=\mu(n)\chi_x^t(n)\,\mathbf{1}[x/n\ge X_0^-]. \tag{40}$$

Definition 5.11 (Near-threshold steps and terminal levels)**.** A step $d\to n$ ($w=n/d$) is *τ-near* if $|\tau_w-x/n|\le 2H_n$, and *X_0-near* if $|x/n-X_0^-|\le R_d/w$. A level n is *terminal* if $x/n<X_0^-+0.6H(X_0)$.

Lemma 5.12 (Structure of the near-threshold steps)**.** *(i) On a step that is neither τ-near nor X_0-near, $Q_w(v,\cdot)=Q^{\mathrm{nom}}_{d\to n}(v,\cdot)$ for every v in the support of C_d.*

(ii) $R_d/w\le 0.52\,H(X_0)$ for every X_0-near step, and every X_0-near step lands at a terminal level.

(iii) A terminal level has no children in either family: $Q_{w'}(v,\cdot)=Q^{\mathrm{nom}}_{n\to nw'}(v,\cdot)=0$ for every $w'\ge 2$ and every v in the support of C_n.

Proof. (i) On the support of C_d, the centre is $v/w\in[x/n-R_d/w,\,x/n+R_d/w]$, and $R_d/w\le 1.02H(x/d)/w\le 1.02\cdot 2^{\varepsilon_*-1}H_n\le 2H_n$. Off τ-near steps the filter $\mathbf{1}[\tau_w<v/w]$ is therefore constant there and equal to $\mathbf{1}[\tau_w<x/n]$; off X_0-near steps the same holds for $\mathbf{1}[v/w\ge X_0^-]$.

(ii) As in (i), $R_d/w\le 1.02\cdot 2^{\varepsilon_*-1}H(x/n)\le 0.5101\,H(x/n)$. If the step is X_0-near, then $x/n\le X_0^-+0.5101H(x/n)$. Since $H(v)/v$ is decreasing and $H(X_0)/X_0<10^{-10}$, this gives

$x/n \le X_0 + 0.02H(X_0)$, and then $H(x/n) \le H(X_0)(1+10^{-15})$ by the local exponent. Hence $R_d/w \le 0.52H(X_0)$ and $x/n \le X_0^- + 0.52H(X_0)$: the level n is terminal.

(iii) The support of C_n lies in $[x/n - R_n, x/n + R_n]$ with $R_n \le 1.02H_n$, so every child centre is at most $(X_0^- + 0.6H(X_0) + 1.03H(X_0))/2 < X_0^-$; both filters $\mathbf{1}[v/w' \ge X_0^-]$ and $\mathbf{1}[x/(nw') \ge X_0^-]$ vanish. □

Width variation. For $u \ge X_0^-$ put $\vartheta_u := \sup_{v \ge u/2} H'(v)$. Since $v \ge X_0^-/2$ there, $H'(v) \le \varepsilon_* H(v)/v$, and $H(v)/v$ is decreasing, so $\vartheta_u \le 2\varepsilon_* H(u/2)/u \le 10^{-5} H(u)/u$. Along any chain the nominal landing scales u_i decrease at least geometrically (by a factor ≥ 2) and stay $\ge X_0^-$, and $H(u)/u$ is decreasing, so

$$\prod_i (1 + 2\vartheta_{u_i}) \le \exp\Big(2 \sum_{j \ge 0} \vartheta_{2^j X_0^-}\Big) \le \exp\big(10^{-4} H(X_0)/X_0\big) \le \lambda_0 := \exp\big(2.1 H(X_0)/X_0\big). \tag{41}$$

Every position v in the support of C_d satisfies $v/w \ge (x/n)/2$, so $|H(v/w) - H_n| \le \vartheta_{x/n} |v/w - x/n|$.

5.5.2. *The Kantorovich–Rubinstein norm.* For a finite signed measure ν on $\mathbb{R}$ with $\nu(\mathbb{R}) = 0$ and compact support, put $F_\nu(t) := \nu((-\infty, t])$ and

$$\|\nu\|_{\mathrm{KR}} := \int_{\mathbb{R}} |F_\nu(t)|\, dt = \sup\Big\{ \int f\, d\nu : \ \mathrm{Lip}\, f \le 1 \Big\}.$$

The two expressions agree: for Lipschitz f, $\int f\, d\nu = -\int f'(t) F_\nu(t)\, dt$ (integration by parts, $\nu(\mathbb{R}) = 0$), which is at most $\int |F_\nu|$, with equality for $f' = -\operatorname{sgn} F_\nu$. Let D_w denote the push-forward under $t \mapsto t/w$, and box_h the uniform probability on $[-h/2, h/2)$.

Lemma 5.13 (Kantorovich–Rubinstein facts)**.** *Let ν have mass 0 and let ρ be a finite signed measure. Then:*

- *(K1)* $\|\rho * \nu\|_{\mathrm{KR}} \le \mathrm{TV}(\rho)\, \|\nu\|_{\mathrm{KR}}$*;*
- *(K2)* $\|D_w \nu\|_{\mathrm{KR}} = \|\nu\|_{\mathrm{KR}}/w$*;*
- *(K3)* $\mathrm{TV}(\mathrm{box}_h * \nu) \le (2/h)\, \|\nu\|_{\mathrm{KR}}$*;*
- *(K4)* $\|\rho - \rho(\mathbb{R})\delta_a\|_{\mathrm{KR}} \le \int |t-a|\, d|\rho|(t)$*; in particular* $\|\mathrm{box}_h - \delta_0\|_{\mathrm{KR}} = h/4$*;*
- *(K5) multiplying a measure by a $[0,1]$-valued (or $[-1,1]$-valued) function does not increase its total variation.*

Proof. (K1) $F_{\rho*\nu}(t) = \int F_\nu(t-s)\, d\rho(s)$, so $\int |F_{\rho*\nu}| \le \iint |F_\nu(t-s)|\, dt\, d|\rho|(s) = \mathrm{TV}(\rho)\|\nu\|_{\mathrm{KR}}$. (K2) $F_{D_w\nu}(t) = F_\nu(wt)$, and $\int |F_\nu(wt)|\, dt = \|\nu\|_{\mathrm{KR}}/w$. (K3) $\mathrm{box}_h * \nu$ has density $h^{-1}\big(F_\nu(t + h/2) - F_\nu(t - h/2)\big)$, whose L^1 norm is at most $(2/h)\int |F_\nu|$. (K4) For $\mathrm{Lip}\, f \le 1$, $\int f\, d(\rho - \rho(\mathbb{R})\delta_a) = \int (f(t) - f(a))\, d\rho(t) \le \int |t-a|\, d|\rho|$. For $\rho = \mathrm{box}_h$ and $a = 0$, $F(t) = t/h + \frac12 - \mathbf{1}[t \ge 0]$ on $[-h/2, h/2]$, and $\int |F| = 2\int_0^{h/2} (\frac12 - t/h)\, dt = h/4$. (K5) is immediate. □

Ordered factorisations. Write $H_{\mathrm{ord}}(m)$ for the number of ordered factorisations of $m \in \mathcal{N}$ into factors ≥ 2 ($H_{\mathrm{ord}}(1) = 1$), so that $H_{\mathrm{ord}}(m) = \sum_{g \mid m,\, g < m} H_{\mathrm{ord}}(g)$ for $m \ge 2$, and put

$$A_0(\varepsilon) := \sum_{m \ge 2} H_{\mathrm{ord}}(m) m^{\varepsilon - 2}, \qquad A_1(\varepsilon) := \sum_{m \ge 2} H_{\mathrm{ord}}(m) m^{\varepsilon - 2} \log m.$$

For $\mathcal{N} = \mathbb{N}$, $\sum_{m \ge 1} H_{\mathrm{ord}}(m) m^{-s} = 1/(2 - \zeta(s))$, whose abscissa of convergence is $\rho_0 = 1.7286\ldots < 1.97$, the root of $\zeta(\rho_0) = 2$; numerically $A_0(0.03) = 2.0673$ and $A_1(0.03) = 9.409$ (Appendix B.4). For $\mathcal{N}$ the values differ from these by less than $X_0^{-0.1}$. Indeed, the free semigroup $\mathcal{N}$ also has $\sum_m H_{\mathrm{ord}}(m) m^{-s} = 1/(2 - \zeta_{\mathcal{N}}(s))$, and every factorisation of $m < X_0^-$ uses only factors $< X_0^-$, where $\mathcal{N} = \mathbb{N}$; so the two sums agree on $m < X_0^-$. For the tails,

$\zeta_{\mathcal{N}}(1.8) \le \zeta(1.8) + X_0^{-0.7} \le 1.883$ by (21), so $\sum_m H_{\mathrm{ord}}(m) m^{-1.8} \le 8.6$ for both systems, and $\sum_{m \ge X_0^-} H_{\mathrm{ord}}(m) m^{\varepsilon-2}(1+\log m) \le 8.6\,(X_0^-)^{-0.16}$ for each.

5.5.3. *The KR bound for nominal kernels.* We first prove the bound for nominal kernels whose window widths are also frozen, and then transfer it to the nominal kernels of §5.5.1, whose widths $H(v/w)$ vary with the position v. The *frozen-width* nominal kernels $C_n^{x,\mathrm{fr}}$ are defined by (38) with the steps $Q^{\mathrm{fr}}_{d\to n}(v,\cdot) := \kappa_{d\to n}\,\mathrm{box}_{H_n}(\cdot - v/w)$; their masses are again the c_n of (40).

Theorem 5.14 (KR bound, nominal filters)**.** *For every start x and every $n \ge 2$,*

$$\mathrm{TV}(C_n^{x,\mathrm{fr}}) \le |c_n| + \tfrac{1}{2} \sum_{e\mid n,\, 1<e<n} |c_e|\, \frac{H_e}{H_n}\, H_{\mathrm{ord}}(n/e)\, \frac{e}{n},$$

$$\|C_n^{x,\mathrm{fr}} - c_n \delta_{x/n}\|_{\mathrm{KR}} \le \tfrac{1}{4} \sum_{e\mid n,\, e>1} |c_e|\, H_e\, H_{\mathrm{ord}}(n/e)\, \frac{e}{n}.$$

Proof. Write $C_n = C_n^{x,\mathrm{fr}}$ and $D_n := C_n - c_n\delta_{x/n}$, a measure of mass 0, with $D_1 = 0$. Since the dilation $D_{n/d}$ maps $\delta_{x/d}$ to $\delta_{x/n}$, (38) gives

$$D_n = c_n\big(\mathrm{box}_{H_n} - \delta_0\big)(\cdot - x/n) - \sum_{d\mid n,\, d<n} \kappa_{d\to n}\, \mathrm{box}_{H_n} * D_{n/d} D_d, \tag{42}$$

where $c_n = -\sum_{d<n} \kappa_{d\to n} c_d$ was used.

The KR bound. By (K4), (K1), (K2) and $|\kappa| \le 1$,

$$\|D_n\|_{\mathrm{KR}} \le \tfrac{1}{4}|c_n| H_n + \sum_{d\mid n,\, d<n} \frac{d}{n} \|D_d\|_{\mathrm{KR}}.$$

We show by induction on n that $\|D_n\|_{\mathrm{KR}} \le \frac{1}{4}\sum_{e\mid n, e>1} |c_e| H_e H_{\mathrm{ord}}(n/e)\, e/n$. Inserting this for $d < n$ (only $d > 1$ contribute, since $D_1 = 0$) and writing $d = eg$,

$$\begin{aligned}\sum_{d\mid n,\, d<n} \frac{d}{n} \sum_{e\mid d,\, e>1} |c_e| H_e H_{\mathrm{ord}}(d/e) \frac{e}{d} &= \sum_{e\mid n,\, 1<e<n} |c_e| H_e \frac{e}{n} \sum_{g\mid n/e,\, g<n/e} H_{\mathrm{ord}}(g) \\ &= \sum_{e\mid n,\, 1<e<n} |c_e| H_e H_{\mathrm{ord}}(n/e) \frac{e}{n},\end{aligned}$$

by the recursion $H_{\mathrm{ord}}(m) = \sum_{g\mid m,\, g<m} H_{\mathrm{ord}}(g)$. Adding the term $\frac{1}{4}|c_n|H_n$, which is the term $e = n$ ($H_{\mathrm{ord}}(1) = 1$), proves the claim.

The TV bound. From (42) and $C_n = c_n \mathrm{box}_{H_n}(\cdot - x/n) + (D_n - c_n(\mathrm{box}_{H_n} - \delta_0)(\cdot - x/n))$, by (K3) and (K2),

$$\mathrm{TV}(C_n) \le |c_n| + \frac{2}{H_n} \sum_{d\mid n,\, d<n} \frac{d}{n} \|D_d\|_{\mathrm{KR}} \le |c_n| + \frac{1}{2} \sum_{e\mid n,\, 1<e<n} |c_e| \frac{H_e}{H_n} H_{\mathrm{ord}}(n/e) \frac{e}{n},$$

by the KR bound for $d < n$ and the same divisor-sum identity. □

The bound is attained at $n = 4$ (void filter, constant H), where $\mathrm{TV}(C_4) = \frac{1}{4}$ while $\mu(4) = 0$. The point of the proof is that it never iterates the triangle inequality on total variation. It iterates on the KR distance, which dilation contracts by d/n at every step, and the filters enter only through the exact masses c_e.

Theorem 5.15 (KR bound, nominal filters, varying widths)**.** *Let $\lambda_0 = \exp(2.1H(X_0)/X_0)$ be as in (41). For every start x and every $n \geq 2$, the nominal kernels $C_n^{x,\mathrm{nom}}$ of §5.5.1 satisfy*

$$\mathrm{TV}(C_n^{x,\mathrm{nom}}) \leq |c_n| + \tfrac{1}{2}\lambda_0 \sum_{e|n,\,1<e<n} |c_e|\,\frac{H_e}{H_n}\,H_{\mathrm{ord}}(n/e)\,\frac{e}{n},$$

$$\|C_n^{x,\mathrm{nom}} - c_n\delta_{x/n}\|_{\mathrm{KR}} \leq \tfrac{1}{4}\lambda_0 \sum_{e|n,\,e>1} |c_e|\,H_e\,H_{\mathrm{ord}}(n/e)\,\frac{e}{n}.$$

Proof. For a cofactor w let $\mathcal{Q}_w\nu := \int \mathrm{box}_{H(v/w)}(\cdot - v/w)\,d\nu(v)$ be the nominal step without its scalar filter. We first prove two properties of $\mathcal{Q}_w$, for a mass-0 measure ν supported in an interval I with $\inf I/w \geq u/2$ and $H_* := \inf_{v\in I} H(v/w)$:

(Q1) $\|\mathcal{Q}_w\nu\|_{\mathrm{KR}} \leq (1+\vartheta_u)\|\nu\|_{\mathrm{KR}}/w$;
(Q2) $\mathrm{TV}(\mathcal{Q}_w\nu) \leq (2/H_*)(1+\vartheta_u)\|\nu\|_{\mathrm{KR}}/w$.

Proof of (Q1). For $\mathrm{Lip}\,f \leq 1$ put $g(v) := \int f\,d\mathrm{box}_{H(v/w)}(\cdot - v/w) = \int_{-1/2}^{1/2} f(v/w + sH(v/w))\,ds$. Then $|g'(v)| \leq \int_{-1/2}^{1/2}(1/w)(1+|s|H'(v/w))\,ds \leq (1+\vartheta_u)/w$, so $\int f\,d\mathcal{Q}_w\nu = \int g\,d\nu \leq (1+\vartheta_u)\|\nu\|_{\mathrm{KR}}/w$.

Proof of (Q2). Write $k_v(t) := H(v/w)^{-1}\mathbf{1}[|t - v/w| \leq H(v/w)/2]$ for the density of the window. Integrating by parts in v (ν has mass 0 and F_ν is its distribution function), $\mathcal{Q}_w\nu = \int k_v\,d\nu(v) = -\int F_\nu(v)\,\partial_v k_v\,dv$. For each v, $\partial_v k_v$ consists of two point masses at the ends of the window, of sizes $(1 \pm H'(v/w)/2)/(wH(v/w))$, and the density $-H'(v/w)/(wH(v/w)^2)$ on the window; its total variation is $\leq (2 + 2H'(v/w))/(wH(v/w)) \leq (2/H_*)(1+\vartheta_u)/w$. Hence $\mathrm{TV}(\mathcal{Q}_w\nu) \leq (2/H_*)(1+\vartheta_u)\int |F_\nu|/w$, which gives (Q2).

The recursion. With $D_n := C_n^{x,\mathrm{nom}} - c_n\delta_{x/n}$, the step from the point x/d lands at x/n with width exactly H_n, so

$$D_n = c_n\big(\mathrm{box}_{H_n} - \delta_0\big)(\cdot - x/n) - \sum_{d|n,\,d<n} \kappa_{d\to n}\,\mathcal{Q}_{n/d}D_d.$$

The support of D_d lies within R_d of x/d, so its image centres satisfy $v/w \geq (x/n)/2$, and on it $H(v/w) \geq H_n(1-\vartheta_{x/n})$ by the width-variation estimate of §5.5.1. By (K4) and (Q1),

$$\|D_n\|_{\mathrm{KR}} \leq \tfrac{1}{4}|c_n|H_n + (1+\vartheta_{x/n}) \sum_{d|n,\,d<n} \frac{d}{n}\|D_d\|_{\mathrm{KR}},$$

and by (Q2) and $(1+\vartheta)/(1-\vartheta) \leq 1 + 3\vartheta$,

$$\mathrm{TV}(C_n^{x,\mathrm{nom}}) \leq |c_n| + (2/H_n)(1+3\vartheta_{x/n}) \sum_{d<n} (d/n)\|D_d\|_{\mathrm{KR}}.$$

Let Θ_n be the largest product $\prod_i(1+3\vartheta_{u_i})$ over the nominal landing scales u_i of chains ending at n. The induction of Theorem 5.14, with every term multiplied by these factors, gives $\|D_n\|_{\mathrm{KR}} \leq \frac{1}{4}\Theta_n \sum_{e|n,e>1} |c_e|H_e H_{\mathrm{ord}}(n/e)\,e/n$ and the TV bound with $\frac{1}{2}\Theta_n$. Finally $\Theta_n \leq \exp(3\sum_j \vartheta_{2^j X_0^-}) \leq \exp(10^{-4}H(X_0)/X_0) \leq \lambda_0$, as in (41). □

Remark 5.16 (Other step kernels). The proof of Theorem 5.15 uses only (Q1), the bound $\|Q(x/d,\cdot) - \delta_{x/n}\|_{\mathrm{KR}} \leq \sigma H_n$ (for windows $\sigma = \frac{1}{4}$), and (Q2) with a constant τ_0'/H_* in place of $2/H_*$ (for windows $\tau_0' = 2$); the constant in front of the sum is then $\sigma\tau_0'\lambda_0$. The step order "box, then filter" satisfies these with $\sigma\tau_0' \leq 1$. The compensator kernel $\mathrm{box}(H(x)/m) * \mathrm{jitter}(H(c))$, whose jitter width is taken at the actual centre c (centre-dependent widths), satisfies (Q1) and (Q2) with 2ϑ in place of ϑ, which still gives $\Theta_n \leq \lambda_0$ in that proof by (41), and $\sigma \leq \frac{1}{2}(1+\vartheta_{x/n})$

(the sum of the two widths over 4, as $H(x)/m \le H_n$ and $H(c) \le (1+1.5\vartheta_{x/n})H_n$). So these two kernels obey Theorem 5.15 with $\frac{1}{2}\lambda_0$ replaced by λ_0 and by $(1+\vartheta_{X_0^-})\lambda_0$, respectively.

5.5.4. *Dyadic weights of the kernels.*

Corollary 5.17. *For a start x put $W_x^{\mathrm{nom}}(u) := \sum_{x/n\in[u,2u)} \mathrm{TV}(C_n^{x,\mathrm{nom}})/n$ and $W_x^{\mathrm{sh}}(v) := \sum_{x/e\in[v,2v)} |c_e|/e$.*

(a) $W_x^{\mathrm{nom}}(u) \le W_x^{\mathrm{sh}}(u) + \frac{1}{2}\lambda_0 \sum_{m\ge 2} H_{\mathrm{ord}}(m) m^{\varepsilon-2} W_x^{\mathrm{sh}}(um)$.

(b) Hence, by Corollary 5.9, for every start $y \in [X_0, x]$ and every $u \ge X_0/2$,

$$W_y^{\mathrm{nom}}(u) \le C_S\big[(1+\tfrac{1}{2}\lambda_0 A_0)\log(2u) + \tfrac{1}{2}\lambda_0 A_1\big]/\log y \le C_N \log(2u)/\log y,$$
$$C_N := C_S\big(2.04 + 4.71/\log X_0\big) \le 158.$$

(c) $\sum_n \mathrm{TV}(C_n^{x,\mathrm{nom}})/n \le 2.04(\log x + B_1)$*; this needs no Saias bound.*

Proof. Multiply Theorem 5.15 by $\mathbf{1}[x/n \in [u,2u)]/n$, sum over n, and substitute $n = em$ ($m \ge 2$). By (18), $H_e/H_{em} \le m^{\varepsilon}$, and $x/n \in [u,2u)$ if and only if $x/e \in [um, 2um)$. This gives (a). For (b), insert $W_y^{\mathrm{sh}}(v) \le C_S \log(2v)/\log y$ and use $\log(2um) = \log(2u) + \log m$; then $1 + \frac{1}{2}\lambda_0 A_0(0.03) \le 1 + 1.03367(1 + 3H(X_0)/X_0) \le 2.034 \le 2.04$ and $\frac{1}{2}\lambda_0 A_1(0.03) \le 4.7045(1 + 3H(X_0)/X_0) \le 4.71$ (with the values for $\mathcal{N}$, which differ by $< X_0^{-0.1}$), and $\log(2u) \ge \log X_0$. So C_N is the same constant as with frozen widths. For (c), sum (a) over all dyadic u and use $\sum_e |c_e|/e \le \log x + B_1$ (Lemma 5.5). Theorem 5.15 and the chain-sum identity hold verbatim from every start $y \le x$, with $c_m(y) = \mu(m)\chi_y^t(m)\mathbf{1}[y/m \ge X_0^-]$. □

5.5.5. *The true kernels.* The true kernels $C_n = C_n^x$ use the filter at the actual centre. They differ from the nominal ones only through the near-threshold steps of Definition 5.11. Write $P_{d\to n}$ and $P_{d\to n}^{\mathrm{nom}}$ for the step operators $\nu \mapsto -\int Q_{n/d}(v,\cdot)\,d\nu(v)$ and its nominal version, so that $C_n = \sum_{d<n} P_{d\to n} C_d$.

Proposition 5.18 (True kernels)**.** *Put $W_x^{\mathrm{op}}(u) := \sum_{x/n\in[u,2u)} \mathrm{TV}(C_n^x)/n$, and let $W_x^{\mathrm{op,nt}}(u)$ be the same sum restricted to non-terminal levels n. If $\lambda_0 C_N E_\eta \le \frac{1}{2}$ (Lemma 5.20), then on* $\mathrm{Good}(< x)$

$$W_x^{\mathrm{op,nt}}(v) \le 2C_N \log(2v)/\log x \quad \text{for every } v,$$
$$W_x^{\mathrm{op}}(u) \le 2C_N \log(2u)/\log x \quad (u \ge 2X_0),$$
$$W_x^{\mathrm{op}}(u) \le 5.1\, C_N \log(2u)/\log x \quad (u \in \{X_0/2, X_0\}),$$
$$\sum_n \mathrm{TV}(C_n^x)/n \le \frac{2.04(\log x + B_1)}{1 - 2.04\lambda_0\theta_*} + 3.2 \le 2.1(\log x + B_1) + 3.2.$$

Proof. Duhamel form. Put $\tilde{E}_n := C_n - C_n^{\mathrm{nom}}$. Subtracting the two recursions,

$$\tilde{E}_n = S_n + \sum_{d<n} P_{d\to n}^{\mathrm{nom}} \tilde{E}_d, \qquad S_n := \sum_{d<n} (P_{d\to n} - P_{d\to n}^{\mathrm{nom}}) C_d.$$

By Lemma 5.12(i), only τ-near and X_0-near steps contribute to S_n. On a step $d \to n$ the true and nominal filters at a centre c are $\varphi_\tau(c)\varphi_{X_0}(c)$ and $\kappa_\tau\kappa_{X_0}$. On X_0-near steps we assign the whole difference $\varphi_\tau\varphi_{X_0} - \kappa_\tau\kappa_{X_0}$ to $S_n^{X_0}$. On the other steps $\varphi_{X_0} = \kappa_{X_0}$ on the support, and the difference $(\varphi_\tau - \kappa_\tau)\kappa_{X_0}$ goes to S_n^τ; it vanishes unless the step is τ-near and $x/n \ge X_0^-$, which is exactly the notion of τ-near used in Lemma 5.20. Each part multiplies the measure $D_w C_d$ of

centres by a function with values in $[-1,1]$ and then places a window (a probability), so by (K5)

$$\mathrm{TV}(S_n^\tau) \le \sum_{\substack{d\to n \ \tau\text{-near, not } X_0\text{-near} \\ x/n \ge X_0^-}} \mathrm{TV}(C_d), \qquad \mathrm{TV}(S_n^{X_0}) \le \sum_{d\to n \ X_0\text{-near}} \mathrm{TV}(C_d).$$

By induction, $\tilde E_n = \sum_{n'|n} \mathcal{T}^{\mathrm{nom}}_{n'\to n} S_{n'}$, where $\mathcal{T}^{\mathrm{nom}}_{n'\to n}$ is the signed sum over nominal chains from level n' to level n ($\mathcal{T}_{n\to n} = \mathrm{id}$).

(i) X_0-near sources are terminal. By Lemma 5.12(ii)–(iii), $S^{X_0}_{n'}$ sits at a terminal level, which has no children; so $\mathcal{T}^{\mathrm{nom}}_{n'\to n} S^{X_0}_{n'} = 0$ for $n \ne n'$, and the parent d of an X_0-near step is non-terminal.

(ii) τ-sources: the fixed point. Fix n' and a point v in the support of $S^\tau_{n'}$, so $|v - x/n'| \le R_{n'}$. The measure $\mathcal{T}^{\mathrm{nom}}_{n'\to n}\delta_v$ is built by nominal steps from the start v with the scalar filters κ of the original chains. These are frozen filters, so the proof of Theorem 5.15 applies verbatim to it, with masses given by the sharp recursion of those filters, i.e. by $c_{n/n'}(x/n')$, and with widths $H(v/g) = H((x/n')/g)(1+O(\vartheta))$ for the cofactors g of the chain. Hence $\mathrm{TV}(\mathcal{T}^{\mathrm{nom}}_{n'\to n}\delta_v)$ is at most λ_0 times the right side of Theorem 5.15 for the start x/n' and level n/n', and Corollary 5.17(b) at the start x/n' gives

$$\sum_{x/n\in[u,2u)} \frac{\mathrm{TV}(\mathcal{T}^{\mathrm{nom}}_{n'\to n} S^\tau_{n'})}{n} \le \lambda_0 \frac{\mathrm{TV}(S^\tau_{n'})}{n'} \frac{C_N \log(2u)}{\log(x/n')}.$$

Summing over n' and grouping by the parent d of each τ-near step $d \to n'$ ($n' = dw$),

$$\sum_{x/n\in[u,2u)} \frac{\mathrm{TV}(\tilde E^\tau_n)}{n} \le \lambda_0 C_N \log(2u) \sum_d \frac{\mathrm{TV}(C_d)}{d}\, \tilde\eta(x/d), \qquad \tilde\eta(v) = \sum_{w \ \tau\text{-near at } v} \frac{1}{w\log(v/w)},$$

with d non-terminal. Let M be the least constant with $W^{\mathrm{op}}_x(v) \le M\log(2v)/\log x$ for all v, counting non-terminal levels and the τ-parts at terminal levels; $M < \infty$ since $n \le x/X_0^-$. Then $\sum_d \mathrm{TV}(C_d)/d \cdot \tilde\eta(x/d) \le \sum_v W^{\mathrm{op}}_x(v) \sup_{[v,2v)} \tilde\eta \le ME_\eta/\log x$, and $W^{\mathrm{op}}_x(u) \le W^{\mathrm{nom}}_x(u) + \lambda_0 C_N M E_\eta \log(2u)/\log x$. Hence $M \le C_N(1+\lambda_0 M E_\eta)$, and $M \le 2C_N$ since $\lambda_0 C_N E_\eta \le \frac12$. This is the bound on $W^{\mathrm{op,nt}}_x$, and also the bound $2C_N\log(2u)/\log x$ for $u \ge 2X_0$, where there are no terminal levels.

(iii) Terminal levels. They gain

$$\sum_{n \text{ terminal}} \frac{\mathrm{TV}(S^{X_0}_n)}{n} \le \sum_d \frac{\mathrm{TV}(C_d)}{d}\, \eta_{X_0}(x/d), \qquad \eta_{X_0}(v) := \sum_{w\in\mathcal{N}:\ |v - wX_0^-| \le 1.02H(v)} \frac{1}{w}.$$

- For $v \le X_0^2/2$ the w-window has length $2.04H(v)/X_0^- < 1$ and lies below X_0^-, where $\mathcal{N} = \mathbb{N}$; so it holds at most one w, and $\eta_{X_0}(v) \le 1.01X_0/v$.
- For $v > X_0^2/2$, the window I_v has length $\le 2.1H(v)/X_0$ at height v/X_0. By the lag bound Corollary 3.8, $|N - T| \le H$ at that height, and under the target (5), $T(I_v) \le 2|I_v| + 2 + \#\mathcal{A}(I_v)$; by Lemma 6.25, $\#\mathcal{A}(I_v) \le 2(v/X_0)^{\theta/2}$. So I_v holds at most $4.2H(v)/X_0 + 2H(v/X_0) + 4 + 2(v/X_0)^{\theta/2}$ elements, and $\eta_{X_0}(v) \le 1.01\big(4.2H(v) + 2X_0 H(v/X_0) + 4X_0 + 2X_0(v/X_0)^{\theta/2}\big)/v$.
- Since $\eta_{X_0}(v) = 0$ for $v < 2X_0^- - 1.1H(X_0)$, these give $\sum_{v \text{ dyadic}} \log(2v) \sup_{[v,2v)} \eta_{X_0} \le 1.52\log(2X_0)$ and $\sum_v \sup_{[v,2v)} \eta_{X_0} \le 1.52$: the first bullet contributes $0.506\log(2X_0)$ from the block $[X_0, 2X_0)$ and $1.01\sum_{j\ge1} 2^{-j}(\log(2X_0) + j\log 2)$ from the others, and the second bullet less than e^{-10^7}.

Terminal levels lie in the blocks $u \in \{X_0/2, X_0\}$. With (ii) they add at most $2C_N \cdot 1.52 \log(2X_0)/\log x$ there. Since $1.52\log(2X_0) \le 1.53\log(2u)$ for these u, the total there is

$$\le 5.1\, C_N \log(2u)/\log x.$$

The ∞-norm. Run (ii) with Corollary 5.17(c) in place of (b): the transfer from n' adds $\le 2.04\lambda_0(\log(x/n') + B_1)\,\mathrm{TV}(S^\tau_{n'})/n'$, and grouping by parents gives the weight

$$\theta(v) = \sum_{w\ \tau\text{-near at } v} (\log(v/w) + B_1)/w;$$

this yields the first term. For (iii) use the block bound $\sup_v W^{\mathrm{op}}_x(v) \le 2.1$, which needs no Saias bound: $W^{\mathrm{sh}}_x(v) \le \sum_{e\in(x/2v,x/v]} 1/e \le 1$, so $W^{\mathrm{nom}}_x(v) \le 1 + \frac12\lambda_0 A_0 \le 2.04$ by Corollary 5.17(a), and the τ-part adds $\le 2.04\lambda_0 \cdot 2.1 \sum_v \sup_{[v,2v)} \eta \le 10^{-6}$ (each transfer costs $\le 2.04\lambda_0$ without Saias weights, the parents contribute $\le \sup_v W^{\mathrm{op}}_x(v) \le 2.1$ per block, and $\sum_v \sup \eta \le E^0_\eta/\log X_0$). That gives $\le 2.1 \cdot 1.52 \le 3.2$.

No circularity. The inputs (Lemma 5.5, Corollary 5.9, Lemma 5.20, Corollary 3.8) concern $\mathcal{N}$ on $[1, x/X_0^-]$ only. □

Remark 5.19. Treating the X_0-near steps like the τ-near ones in the fixed point would put $\sum_v \log(2v)\sup\eta_{X_0} \approx 1.5\log X_0$ into E_η, and $C_N E_\eta \le \frac12$ would fail. They are harmless only because they are terminal.

Lemma 5.20 (τ-near energy)**.** *For $v \ge X_0^-$ put*

$$\eta(v) := \sum_{w\ \tau\text{-near at } v} 1/w, \quad \tilde\eta(v) := \sum_{w\ \tau\text{-near at } v} 1/(w\log(v/w))$$
$$\text{and} \quad \theta(v) := \sum_{w\ \tau\text{-near at } v} (\log(v/w) + B_1)/w,$$

where w is τ-near at v if $|\tau_w - v/w| \le 2H(v/w)$ and $v/w \ge X_0^-$. Put

$$E_\eta := \sum_{v\ dyadic} \log(2v) \sup_{[v,2v)} \tilde\eta, \quad E^0_\eta := \sum_{v\ dyadic} \log(2v) \sup_{[v,2v)} \eta,$$
$$E_\theta := \sum_{v\ dyadic} \log(2v) \sup_{[v,2v)} \theta, \quad \theta_* := \sup_v \theta(v).$$

Then, on Good$(<x)$*, at every height $\le x$ where these quantities are evaluated,*

$$E_\eta,\ E^0_\eta,\ E_\theta,\ \theta_* \le 10^4 H(X_0)\frac{\log^2 X_0}{X_0} = 10^4\Lambda^2 e^{8(\log\Lambda)^2-\Lambda} \le e^{-0.99\Lambda},$$

$$E^{(2)}_\eta := \sum_{v\ dyadic} (\log 3v)^2 \sup_{[v,2v)} \eta \le 10^4 H(X_0)\frac{\log^3 X_0}{X_0},$$

and $\sup_y \log y \cdot \eta(y) \le e^{-0.99\Lambda}$. (At $\Lambda = 10^8$ the first bound is $e^{-(10^8-2762)}$.)

Proof. Localisation. Let w be τ-near at v', and write $w = pm$ with p the prime of w that is largest in decision order, so that $\tau_w = t(p)$ and $P^+(m) \le p + H(p)$. Since $|t(p) - p| \le H(p)$, the condition $|t(p) - v'/(pm)| \le 2H(v'/w)$ gives $|p - v'/(pm)| \le 3H(p)(1+o(1))$, i.e. $p^2 \in v'/m \pm 3.1\,pH(p)$. So p lies in a window of length $3.1H(p_0)$ around $p_0 := \sqrt{v'/m} \ge X_0^-$.

Counting. Let $[P, 2P)$ be the dyadic block of p_0. The window lies in $[P - 1.6H(P),\ 2P + 1.6H(P)] \subset [P', 4P']$ with $P' := 0.51P$ and has length $\le 3.2H(P) \le 6H(P')$. If $P' \ge X_0$, Corollary 6.3(ii) at P' shows that it holds at most $13.4 \cdot 3.2H(P) + K_{\mathrm{loc}}(P') + K_{\mathrm{loc}}(2P') + 4 \le 43H(P) + 2K_{\mathrm{loc}}(2P') + 4 \le 46H(P)$ primes (K_{loc} is non-decreasing and $K_{\mathrm{loc}} \le H/100$). If

$P' < X_0$, then $P < 1.97X_0$. The part J_2 of the window in $[X_0, \infty)$ lies in $[X_0, 4X_0]$ and has length $\le 6H(X_0)$, so Corollary 6.3(ii) at X_0 bounds its primes by $13.4|J_2| + 2K_{\rm loc}(2X_0) + 4$; the part J_1 below X_0 holds at most $|J_1| + 1$ integers and at most $2H(X_0)$ jittered primes (§5.1). Since $|J_1| + |J_2| \le 3.2H(P)$ and $H(X_0), H(2X_0) \le 1.01H(P)$ (local exponent), the window again holds at most $13.4 \cdot 3.2H(P) + 2.05H(P) + 5 \le 46H(P)$ primes. Each has $1/p \le 1.01/P$ (as $p \ge P - 1.6H(P)$). For $v' \in [v, 2v)$ this forces $m \in (v/4P^2, 2v/P^2]$, and a pair (P, m) occurs for at most three dyadic v. The m are $4P$-smooth, so by Lemma 5.1, $\sum_m \log(8mP^2)/m \le 5.4 \log^2 P(1 + 5/\log P)$ and $\sum_m 1/m \le 1.9 \log P$.

E_η. Since $\log(2v)/\log(v/w) \le 1.01 \log(8mP^2)/\log P$, each P-block contributes at most $16.6 \cdot 46H(P) \log P/P$.

E_η^0, $E_\eta^{(2)}$, E_θ, θ_*. E_η^0 has no division by $\log(v/w)$; $E_\eta^{(2)}$ has one more factor $\le 1.01 \log(8mP^2)$; E_θ has the extra factor $\log(v/w) + B_1 \le 1.01 \log 4P$. So each P-block contributes $\le 16.6 \cdot 46H(P) \cdot 1.02 \log^2 P/P$ to E_η^0 and E_θ (one more $\log P$ to $E_\eta^{(2)}$), and θ_* is bounded by the same expression without the weight $\log(2v)$.

Summation. The blocks decay geometrically (ratio $\le 2^{\varepsilon - 1} \le 0.52$, by (18)). Summing from $P = X_0/2$ gives

$$\le 2.1 \cdot 16.6 \cdot 46 \cdot 1.02 \cdot H(X_0) \cdot 4 \log^2 X_0/X_0 \le 10^4 H(X_0) \log^2 X_0/X_0.$$

$\log y \cdot \eta(y)$. Fix y. As in the first step, every τ-near w at y is $w = pm$ with p in a window of length $\le 3.2H(P)$ around $p_0 = \sqrt{y/m} \in [P, 2P)$, $P \ge X_0/2$; the window holds $\le 46H(P)$ primes, each with $1/p \le 1.01/P$. Every prime q of m has $t(q) \le t(p)$, so $q \le p + 2H(p) \le 4P$, and $m \in (y/4P^2, y/P^2]$. Hence

$$\eta(y) \le \sum_{P \text{ dyadic} \ge X_0/2} \frac{47H(P)}{P} V_P(y), \qquad V_P(y) := \sum_{m \in (y/4P^2,\, y/P^2],\ P^+(m) \le 4P} \frac{1}{m}.$$

Put $L := \log 4P \ge 100$ and $u := \log(y/4P^2)/L$, so that $\log y \le (u + 2)L$ and $V_P(y)$ is at most two dyadic blocks $v(\ell, 4P)$ with $\ell \le (u + 1)L$.

- If $u < 1$: each of the two dyadic blocks has $\sum 1/m \le 1$ (the harmonic sum over $(Y, 2Y]$, by (21) above X_0^-), so $V_P(y) \le 2$ and $\log y \cdot V_P(y) \le 2(u + 2)L \le 6L$.
- If $1 \le u \le L$: Lemma 5.3 with $z = 4P$ gives $V_P(y) \le 2C_0 \rho(u) e^{a_V}$ with $a_V \le 4 \cdot 3.2\, u^2 \log(u + 2)/(uL) \le 12.8 \log(L + 2)$. Since $(u + 2)\rho(u) \le 3$, $\log y \cdot V_P(y) \le 4.2L(L + 2)^{12.8}$.
- If $u > L$: Lemma 5.30 gives $V_P(y) \le 2e^{14} L e^{-u}$, and $(u + 2)L \cdot 2e^{14} L e^{-u} \le 2e^{14} L^2 (L + 2) e^{-L} \le 1$.

In every case $\log y \cdot V_P(y) \le 5(L + 2)^{14}$. Summing over P (the terms $H(P)(\log 4P + 2)^{14}/P$ decay geometrically, with ratio ≤ 0.53, and the first is $\le 2.01H(X_0)(\Lambda + 4)^{14}/X_0$),

$$\log y \cdot \eta(y) \le 2.2 \cdot 47 \cdot 5 \cdot 2.01\, H(X_0)(\Lambda + 4)^{14}/X_0 \le e^{-\Lambda + 8(\log \Lambda)^2 + 14 \log(\Lambda + 4) + 7} \le e^{-0.99\Lambda}.$$

Numerics. $H(X_0) = e^{8(\log \Lambda)^2}$, which is $\le e^{2715}$ at $\Lambda = 10^8$, and $K_{\rm loc}(X_0) \le H(X_0)$. For every $\Lambda \ge 10^8$, $9.3 + 2 \log \Lambda + 8(\log \Lambda)^2 \le 0.01\Lambda$ and $7 + 14 \log(\Lambda + 4) + 8(\log \Lambda)^2 \le 0.01\Lambda$, which give the bounds $e^{-0.99\Lambda}$. □

Consequently $\lambda_0 C_N E_\eta \le \frac{1}{2}$ and $2.04\lambda_0 \theta_* \le 0.02$, with room e^{-10^7}. The per-scale constant $K_{\rm loc}(P)$ is essential: a constant growing with the current scale would make E_η grow with x. The quantity E_η^0 is what makes the τ-near term NT summable with the weights $\log(2u)$ of Lemma 5.34; a pointwise bound on $\log y \cdot \eta(y)$ alone would give a sum growing like $\log \log x$.

Lemma 5.21 (Filtered cofactor weight)**.** *For a start y and a scale $v \ge X_0/2$ put $\omega_y(v) := \sum_{w \in \mathcal{N}:\ y/w \in [v, 2v),\ \tau_w < y/w} 1/w$. Then:*

(i) $\omega_y(v) \le 1.2$*;*
(ii) for $y \ge 4v^2$ *the filter forces* $P^+(w) \le 3v$*. Put* $u_y := \log(y/2v)/\log 3v$*. If* $u_y \le \log 3v$*, then* $\omega_y(v) \le 2C_0\rho(u_y)e^{a_V}$ *with* $a_V \le 13u_y \log(u_y+2)/\log 3v$*; so* $\omega_y(v) \le 2C_0\rho(u_y)e^{0.02}$ *for* $u_y \le 30$ *and* $\omega_y(v) \le 2C_0 e^{-0.9u_y \log u_y}$ *for* $30 < u_y \le \log 3v$*. If* $u_y > \log 3v$*, then* $\omega_y(v) \le 5(\log 3v)^{1.65}e^{-u_y/2}$*;*
(iii) $\sum_{Y=2^k \ge v} \log(2Y) \sup_{y\in[Y,2Y)} \omega_y(v) \le C_S'(\log 3v)^2$ *with* $C_S' := 10$*.*

Proof. (i) $\omega_y(v) \le \sum_{w\in(y/2v,y/v]} 1/w \le \log 2 + 2\eta_1 + 2v/y \le 1.2$ by (21) (and ≤ 1 below X_0^-).

(ii) If w passes the filter then $\tau_w < y/w \le 2v$, so every prime of w is decided before slot $2v$ and is $\le 2v + H(2v) \le 3v$. Since $w \in (y/2v, y/v]$, Lemma 5.3 with $z = 3v$ and $\ell = \log(y/2v)$ gives the first bound (its hypothesis $u \le L_z$ is $u_y \le \log 3v$); the numerical forms follow from $a_V \le 13u\log(u+2)/\log 3v$ and $\log 3v \ge 10^8$, and from $\rho(u) \le e^{-u\log u}$. For $u_y > \log 3v$, Rankin's bound with exponent $1 - 1/(2\log 3v)$ and Lemma 5.1 gives $\omega_y(v) \le 5(\log 3v)^{1.65}e^{-u_y/2}$.

(iii) The range $Y < 4v^2$ contributes at most $1.2 \cdot \frac{3}{2\log 2}(1 + o(1))(\log 3v)^2 \le 2.7(\log 3v)^2$. The range $Y \ge 4v^2$ contributes at most $\frac{2\cdot 0.70\cdot e^{0.02}}{\log 2}(\log 3v)^2 \int_1^\infty \frac{1+u}{\Gamma(u+1)}\,du \le 7.11(\log 3v)^2$, using $\rho(u) \le 1/\Gamma(u+1)$ (from $\rho(u) \le \rho(u-1)/u$) and $\int_1^\infty (1+u)/\Gamma(u+1)\,du = 3.4479$. The parts with $u_y > 30$ add less than $10^{-20}(\log 3v)^2$. The total is $\le 9.82(\log 3v)^2$. □

The filter forces a dense cofactor ($v < \sqrt{y}$) to be $3v$-smooth, hence Dickman-small. No Riemann-sum or short-interval smooth-number input is needed, and Lemma 5.3 avoids the Rankin loss $(\log 3v)^{2+e}$.

5.6. **The noise, per scale.**

Inputs from §6. Each is used frozen at the scale where it is read.

- The queue bound $Q \le B(u)$ (Theorem 6.6, (71)); so $|\Delta Q(W)| \le B(u)$ on windows at scale u.
- The per-prime increment bound (Proposition 6.10, Corollary 6.11). On $E_{\mathrm{cl}}^{(u)}$, for every slot window $I \subset [u, 2u]$ with $|I| \le H(u)$, every non-absorbed latest prime $q \ge y(u)$, every $a \ge 1$ and every jitter offset, $X_q^{(a)}(I) \le 2M_1(u)$. The event $E_{\mathrm{cl}}^{(u)}$ fails on Good with probability $\le L_u^{-2}$.
- The local count (Corollary 6.3). For dyadic P and windows $J \subset [P, 4P]$ of length $\le 6H(P)$ (part (i)), the non-absorbed count, and so the prime count, is $\le 13.4|J| + K_{\mathrm{loc}}(P) + 2$, with $K_{\mathrm{loc}}(P) = 2B(4P) + 7\lambda_P/3 + 23$ and $\lambda_P = 4\log P$. For real $P \ge X_0$ (part (ii)) the same holds with $K_{\mathrm{loc}}(P) + K_{\mathrm{loc}}(2P) + 4$ in place of $K_{\mathrm{loc}}(P) + 2$.
- Absorption (Lemmas 6.25, 6.26, Proposition 6.27). Non-absorbed latest primes satisfy $q \ge y(u) = H(u)L_0L_u^6$, so $|I/q| \le L_u^{-6}$ for $|I| \le H(u)$. The small-q case S1 does not arise in the present regime.

The composites with $q^2 \mid n$: the source s_2. Recall $C^{(\ge 2)} = \mathrm{Comp}^{(2)} + \mathrm{Mart}^{(2)}$ (§5.3). Fix u and a window $I \subset [u, 2u]$, and put $\sigma := \frac{1}{2}\sqrt{L_u \log L_u}$.

- The latest-prime filter forces $\tau_m < t(q)$ for $q^a m$, so $P^+(m) \le 2q$.
- *Interchange the sums.* Fix $a \ge 2$ and m. Then $q^a m \in I$ means $q \in I_{m,a} := (I/m)^{1/a}$, and the q-sum of the landing probabilities is $\int_{I_{m,a}} \tilde\rho \le 1.01\,|I_{m,a} \cap [X_0^-, \infty)|$, since $\tilde\rho \le 1.01$ surely (see the variance bound below) and $\tilde\rho = 0$ below X_0^-. With $y = q^a m \in I$ we have $dq = dy/(amq^{a-1})$, and on $I_{m,a} \cap [X_0^-, \infty)$, $q \ge Q_{m,a} := \max(X_0^-, (u/m)^{1/a})$. Hence

$$\frac{\mathrm{Comp}^{(2)}(I)}{|I|} \le 1.01 \sum_{a\ge 2} \frac{1}{a} \sum_m \frac{1}{m\, Q_{m,a}^{a-1}},$$

where m runs over $\mathcal{N}$ with $m \le 2u/(X_0^-)^a$ and $P^+(m) \le 2(2u/m)^{1/a}$.

- *Blocks.* Split the m-sum into blocks $m \in (Y, 2Y]$, $Y = 2^{k-1}$ ($k \ge 0$); for each a, at most $1.443L_u$ of them are nonempty. Define s by $e^{as} = u/(2Y)$ and put $s_Y := \max(\log X_0^-, s)$. In the block, $Q_{m,a} \ge e^{s_Y}$, $\log Y = L_u - as - \log 2$, and the m are z-smooth with $\log z := s + \log 2 + (2\log 2)/a \le s + 1.39$. A nonempty block has $Y < 2u/(X_0^-)^a$, so $s > \log X_0^- - 0.7$ and $\log z \ge \log X_0^- - 0.01$. The block contributes at most $\frac{1}{a} e^{-(a-1)s_Y} v(\log Y, z)$, where $v(\ell, z)$ is the $1/n$-weight of the z-smooth $n \in \mathcal{N} \cap (e^\ell, 2e^\ell]$, as in Lemma 5.3.
- *First bound.* Always $v \le 1$ (the small heights in the proof of Lemma 5.3; the block $Y = \frac12$ holds only $m = 1$), and $s_Y \ge \log X_0^-$. So $\mathrm{Comp}^{(2)}(I)/|I| \le 1.01 \cdot 1.443 L_u (X_0^-)^{-1}\big(\frac12 + \sum_{a\ge3} (X_0^-)^{2-a}/a\big) \le 0.73 L_u/X_0^-$.
- *Second bound, for $\sigma > \log X_0^-$.* Then $L_u \ge 10^{15}$, since $\log X_0^- \ge 10^8 - 1$. We show that every block term is at most $\frac1a e^{-\sigma}(X_0^-)^{-(a-2)/2}$, that is, $(a-1)s_Y - \log v \ge \sigma + \frac12 (a-2)\log X_0^-$.
  - If $s_Y \ge \sigma$, or if $(a-2)s_Y \ge 2\sigma$, this holds with $v \le 1$.
  - Otherwise $s \le s_Y < \sigma$ and $as < 4\sigma$, so $\ell := \log Y \ge L_u - 4\sigma - 0.7 \ge (1 - 4\cdot 10^{-7})L_u$ and $\log z < \sigma + 1.39$. Since $(a-1)s_Y \ge \frac12 (a-2)\log X_0^-$, it suffices that $-\log v \ge \sigma$. Put $u' := \ell/\log z$ and $\omega := 2\sqrt{L_u/\log L_u}$, so that $\sigma = \frac14 \omega \log L_u$ and $u' \ge (L_u - 4\sigma - 0.7)/(\sigma + 1.39) \ge (1 - 4\cdot 10^{-7})\,\omega$. For every $w \ge 1$, iterating $\rho(w) \le \rho(w-1)/w$ (from (R1)) down to $w - \lfloor w \rfloor \in [0,1)$, where $\rho = 1$, gives $\rho(w) \le \Gamma(w - \lfloor w \rfloor + 1)/\Gamma(w+1) \le 1/\Gamma(w+1) \le e^{-w\log w + w}$ (Stirling).
  - If $u' \le \log z$: Lemma 5.3 gives $v \le C_0 \rho(u') e^{a_V}$ with $a_V = 12.8\, u' \log(u'+2)/\log z \le 1.3 \cdot 10^{-7} u' \log(u'+2)$. So

$$-\log v \ge u'(\log u' - 1) - 1.3\cdot 10^{-7} u' \log(u'+2) \ge (1 - 10^{-6})\,\omega\big(\tfrac12 \log L_u - \tfrac12 \log\log L_u - 0.31\big),$$

    which is $\ge 1.75\sigma$ for $\log L_u \ge 34.5$: with $x := \log L_u$ the ratio to σ is $(1-10^{-6})(2 - (2\log x + 1.24)/x)$, which increases in x, and $\sigma \ge \log X_0^-$ forces $L_u \ge 10^{15}$.
  - If $u' > \log z$: then $\sqrt{\ell} > \log z$, and $v(\ell, z) \le v(\ell, z')$ with $\log z' := \sqrt{\ell}$. Lemma 5.3 at the Dickman variable $\sqrt{\ell}$, where $a_V = 12.8 \log(\sqrt{\ell} + 2)$, gives $-\log v \ge \sqrt{\ell}(\log\sqrt{\ell} - 1) - 12.8\log(\sqrt{\ell}+2) \ge 5.5\sigma$; its ratio to σ also increases in $\log L_u$, and is ≥ 5.53 at $L_u = 10^{15}$.

  Summing, $\mathrm{Comp}^{(2)}(I)/|I| \le 1.01 \cdot 1.443 L_u\, e^{-\sigma}\big(\frac12 + \sum_{a\ge3} (X_0^-)^{-(a-2)/2}/a\big) \le 0.73 L_u\, e^{-\sigma}$.
- Together, for every window $I \subset [u, 2u]$,

$$\frac{\mathrm{Comp}^{(2)}(I)}{|I|} \le s_2(u) := 9L_u \exp\Big(-\max\big(\log X_0^-, \tfrac12\sqrt{L_u \log L_u}\big)\Big). \tag{43}$$

  In particular $s_2(u) \le 9L_u/X_0^- \le 9(1 + 10^{-9})L_u/X_0$ always, since $X_0/X_0^- \le 1 + H(X_0)/X_0$.
- $\sum_{u \text{ dyadic}} \log(2u)\, s_2(u) \le e^{-9\cdot 10^7}$, and the same holds with the weight $(\log 3u)^2$. For $L_u \le \Lambda^2$, use $s_2(u) \le 9L_u/X_0^-$: these at most $1.443\Lambda^2$ scales contribute $\le 14\Lambda^8/X_0^-$. For $L_u > \Lambda^2$, $\frac12\sqrt{L_u \log L_u} \ge 3.03\sqrt{L_u}$, and $10L^3 e^{-3.03\sqrt{L}}$ is decreasing, so these scales contribute $\le 10\Lambda^6 e^{-3.03\Lambda} + \frac{10}{\log 2}\int_{\Lambda^2}^\infty L^3 e^{-3.03\sqrt{L}}\, dL \le 10\Lambda^7 e^{-3.03\Lambda}$. The total is $\le e^{-0.99\Lambda}$.
- Put $A_u := \#\{a \ge 2 : (X_0^-)^a \le 2u\} \le 1.01 L_u/\Lambda$. By Corollary 6.11 (which covers every $a \ge 1$), the increment of a prime q to $\mathrm{Mart}^{(2)}(I)$ is at most $b_2 := 2A_u M_1(u)$.

The per-scale Freedman events. Put $\lambda_u' := 3\log u + 2\log\log u + 2 + \log H(u)$. For dyadic $u \ge X_0$, let $\mathcal{W}_u'$ be the set of windows $[a,b] \subset [u, 2u]$ with integer endpoints and $b - a \le 1.03H(u) + 1$; there are at most $(u+1)(1.03H(u)+2)$ of them. Let $\mathcal{W}_u \subset \mathcal{W}_u'$ consist of those with $b - a \ge H(u) - 1$.

- G_u^{NV} (the event E3 of Definition 6.8): every $W \in \mathcal{W}_u$ satisfies (44) and (45).
- G_u^{sw} (the event E3′): every $W \in \mathcal{W}'_u$ satisfies (44) and (45).

$$|\mathrm{Mart}(W)| \le \sqrt{2\bar{V}_W \lambda'_u} + \tfrac{4}{3} M_1(u) \lambda'_u, \qquad \bar{V}_W := 2M_1(u) \cdot 50 \log(2u) \cdot |W|; \tag{44}$$

$$|\mathrm{Mart}^{(2)}(W)| \le \sqrt{2\bar{V}_W^2 \lambda'_u} + \tfrac{2}{3} b_2 \lambda'_u, \qquad \bar{V}_W^2 := b_2 \cdot s_2(u) \cdot |W|. \tag{45}$$

Truncation. As in Theorem 6.6(i), Freedman's inequality is applied to the truncated increments $\tilde{X}_q(W) := \min(X_q(W), 2M_1(u))$ and $\tilde{X}_q^{(2)}(W) := \min(X_q^{(2)}(W), b_2)$, each compensated by its own conditional mean. These are martingales in decision order, with increments $\le 2M_1$ and $\le b_2$ surely. Freedman's inequality is used in the form

$$\mathbb{P}(\exists n : \ S_n \ge a, \ V_n \le v) \le \exp\Big(- \frac{a^2}{2(v + ba/3)} \Big),$$

for increments bounded by b [8, (4.1)] (with Bennett's bound for the exponent), with the deterministic a and $v = \bar{V}_W, \bar{V}_W^2$ of (44), (45).

- *Truncation is inactive on* $\mathrm{Good}(< t(W))$. For every latest prime q contributing to W, the set of cofactors (elements whose primes are all decided before q) is $\mathcal{F}_{q-}$-measurable. So "$X_q(W) \le 2M_1$ for every jitter offset of q" is an $\mathcal{F}_{q-}$-measurable event. It may not yet be certified at time s_q (for $q < \sqrt{2u}$). But on $\mathrm{Good}(< t(W))$ the event $E_{\mathrm{cl}}^{(u)}$ holds (its time $8u/L^{11} + H$ is $< t(W)$), and Corollary 6.11 then gives it for every q and every offset. So $\tilde{X}_q = X_q$ and $\mathbb{E}[\tilde{X}_q \mid \mathcal{F}_{q-}] = \mathbb{E}[X_q \mid \mathcal{F}_{q-}]$. The truncated and the true martingales therefore agree on $\mathrm{Good}(< t(W))$, and likewise for $\mathrm{Mart}^{(2)}$.
- *The variance bound holds on* $\mathrm{Good}(< t(W))$. We have $\mathrm{Var}(\tilde{X}_q \mid \mathcal{F}_{q-}) \le 2M_1 \mathbb{E}[X_q \mid \mathcal{F}_{q-}]$, so $V_W \le 2M_1 \mathrm{Comp}(W)$; likewise $V_W^2 \le b_2 \mathrm{Comp}^{(2)}(W) \le b_2 s_2 |W|$. Next, $\mathrm{Comp}(W) = \int_W (K_{\mathrm{pt}} \tilde{\rho} - K_{\mathcal{A}} \tilde{\rho} + \mathrm{NT}_{\mathrm{pt}}) \le 1.01 \sum_v \omega_y(v) \cdot |W| + \int_W |\mathrm{NT}_{\mathrm{pt}}| \le 50 \log(2u)\, |W|$. This uses $\sum_v \omega_y(v) \le 1.2 \log_2(4u/X_0)$ (Lemma 5.21(i), that is, $(\mathrm{H_{BS}})$ at heights $\le u$), and the sure bound $\tilde{\rho} \le 1.01$. The latter holds because each continuous slot decides at most one prime, and the starts of consecutive continuous slots are at least $1 - 10^{-8}$ apart (a continuous slot needs $\int T'_{\mathrm{cont}} = 1$, with $T'_{\mathrm{cont}} \le 1 + 10^{-8}$). No upper bound on the length of a slot is used in this section.
- So on $\mathrm{Good}(< t(W))$ the event $\{|\mathrm{Mart}(W)| >$ the bound of (44)$\}$ lies in Freedman's event for the truncated martingale; likewise for (45). No positivity of the margin is used.

Probability. With $a = \sqrt{2v\lambda'_u} + \frac{2}{3} b \lambda'_u$, Freedman's inequality gives, per window and per martingale, a two-sided failure probability $\le 2e^{-\lambda'_u} = 2e^{-2} u^{-3} L_u^{-2} H(u)^{-1}$. With four bounds per window,

$$\mathbb{P}\big((G_u^{\mathrm{NV}} \cap G_u^{\mathrm{sw}})^c \cap \mathrm{Good}(< t)\big) \le 8e^{-2}(u+1)(1.03H(u)+2) u^{-3} L_u^{-2} H(u)^{-1} \le 1.2\, u^{-2} L_u^{-2}.$$

Each window's bound is an event whose time is the moment the window closes ($\le 2u$). These are the events E3 and E3′ of Definition 6.8. Together with E_{cl} they enter $\mathrm{Good}(< t)$ and the budget $\mathbb{P}(\tau < \infty) \le 1.5/\Lambda$ of Lemma 6.9. The budget is dominated by E_{cl}: $\sum_u L_u^{-2} \le 1.45/\Lambda$, while the Freedman part is $\le 2X_0^{-2}$.

Lemma 5.22 (per-scale noise)**.** *For $y \in [u, 2u]$, work on* Good$(< y + H(y))$*. This event contains the E3/E3′ windows of the blocks $u' \in \{u/2, u, 2u\}$ that close by $y + H(y)/2$, and $E_{\mathrm{cl}}^{(u')}$. Put*

$$\nu_0(u') := \sqrt{\frac{206 M_1(u')\lambda'_{u'} \log(2u')}{H(u')}}\Big(1 + \frac{1}{H(u')}\Big) + \sqrt{\frac{4.2 A_{u'} M_1(u') s_2(u')\lambda'_{u'}}{H(u')}} + \frac{\frac{4}{3}(1 + A_{u'})M_1(u')\lambda'_{u'} + 3B(u') + 100\log(2u') + 6}{H(u')} + 1.1 s_2(u').$$

Then

$$|\nu(y)| \le \bar\nu(u) := 2.05 \max_{u' \in \{u/2, u, 2u\}} \nu_0(u').$$

For every slot window $I \subset [u, 2u]$ with $|I| \le H(u)$, put $E(I) := \sigma(I) - \int_I m$, where $m = T' - K_{\mathrm{pt}}\tilde\rho - \mathrm{NT}_{\mathrm{pt}}$ is the pointwise margin density (§5.3). Then

$$|E(I)| \le \sqrt{200 M_1(u)\lambda'_u \log(2u)(|I| + 2)} + \sqrt{4 A_u M_1(u) s_2(u)\lambda'_u(|I| + 2)} + \tfrac{4}{3}(1 + A_u)M_1(u)\lambda'_u + B(u) + 100\log(2u) + s_2(u)(|I| + 2) + 2.$$

Write $\hat\nu(u) := \sup_{|I| \le H(u)} |E(I)|/H(u)$. Both bounds depend on u only: nothing at the current scale x enters.

Proof. The bound on ν. By (28), $H(y)\nu(y) = \mathrm{Mart}(W_y) + \mathrm{Mart}^{(2)}(W_y) - \Delta Q(W_y) + \mathrm{Comp}^{(2)}(W_y) + E_{\mathrm{edge}}$.

- E_{edge} counts the two partial slot cells at the ends of W_y. Each holds at most one slot and at most $A_j \le B(u) + 1$ composites, so $|E_{\mathrm{edge}}| \le 2B(u) + 4$.
- $|\Delta Q| \le B(u)$ by Theorem 6.6, and $\mathrm{Comp}^{(2)}(W_y) \le s_2|W_y|$ by (43).
- Choose $W^- \subset W_y \subset W^+$ in $\mathcal{W}_u$ with $|W^+ \setminus W^-| \le 2$. Counts and compensators are monotone in the window, so $\mathrm{Mart}(W^-) - \mathrm{Comp}(W^+ \setminus W^-) \le \mathrm{Mart}(W_y) \le \mathrm{Mart}(W^+) + \mathrm{Comp}(W^+ \setminus W^-)$, with $\mathrm{Comp}(W^+ \setminus W^-) \le 100\log(2u)$. The same holds for $\mathrm{Mart}^{(2)}$, with $2s_2$ in place of $100\log(2u)$.
- Apply (44) and (45) with $|W^\pm| \le 1.03H(u) + 1$, and divide by $H(y) \ge H(u)$.
- If W_y crosses u or $2u$ (that is, y is within $H(y)/2$ of it), split W_y there. Each piece lies in one dyadic block u' and has length $\le 1.03H(u')$, so the above applies to it in block u'. The two bounds add, and $H(y)/H(u') \le 1.02$ by (18); the factor 2.05 covers this.

The bound on $E(I)$. The bookkeeping (26) on the slot window I has no partial cells. By the computation of (28) without the B-average, $\sigma(I) = \int_I m - \mathrm{Comp}^{(2)}(I) - \mathrm{Mart}(I) - \mathrm{Mart}^{(2)}(I) + \Delta Q(I) \pm 2$. Now sandwich I between windows of $\mathcal{W}'_u$ as above, and apply (44), (45) and (43). □

The (SW) sums. Under the tolerance $M_1 \le H/L^{14}$ of (65), with $B \le 28 M_1 L^2$, $A_u \le 1.01 L_u/\Lambda$ and $\lambda'_u \le 18.2 L_u$ (since $\log H \le 15.1\Lambda$ on the raised range),

$$\bar\nu(u) \le 128 L_u^{-6} + 2.3 \max_{u'} s_2(u'), \qquad \hat\nu(u) \le 62 L_u^{-6} + 1.1 s_2(u),$$

and

(46)
$$\sum_{u \text{ dyadic} \ge X_0} \log(2u)\,\bar\nu(u/2) \le 10^{-30}, \qquad \sum_v (\log 3v)^2\,\bar\nu(v/2) \le 10^{-22}, \qquad \sum_v \log(3v)\,\hat\nu(v) \le 10^{-30}.$$

- The leading term is $\sqrt{206 \cdot 18.2}\, L\,(M_1/H)^{1/2} \le 61.3 L^{-6}$. The linear terms are $\le 140 L^{-12}$: $\frac{4}{3}(1 + A_u)M_1\lambda' \le 50 L^2 M_1$ and $3B \le 84 L^2 M_1$. Then $\sum_j 128(\Lambda + j\log 2)^{-5} \le 46.2/\Lambda^4$ and $\sum_j 128(\Lambda + j\log 2)^{-4} \le 61.6/\Lambda^3$.

- On the standard range, where $H = H_{\mathrm{std}}$ and $M_1 \le \frac{1}{2}H^{1/2}$, the same sums are $\le e^{-600}$; the first term is then $\approx 18L\,e^{-2(\log L)^2}$.
- The sums converge whenever $H(v)/M_1(v) \ge (\log v)^{8+\delta}$. This is what fixes the tolerance $M_1 \le H/L^{14}$.

5.7. **A local prime number theorem.**

Lemma 5.23 (threshold mass)**.** *For squarefree n put $\tau(n) := \max_{p\mid n} t(p)\prod_{q\mid n,\,t(q)\ge t(p)} q$, so that $\chi_y^t(n) = 1$ if and only if $\tau(n) < y$. Assume $|N - T| \le D := B + \frac{1}{2}H(1 + \kappa_+/\log x) + 4$ at the relevant heights (Corollary 3.8). For $x \ge X_0^3$ and $6H(x)x^{-1/3} \le \delta \le \frac{1}{4}$,*

$$\sum_{n:\ \tau(n)\in[x(1-\delta),x)} \frac{1}{n} \le 55\,\delta\log x\,\big(\log H(x) + \log(1/\delta) + 2\big) + 14\big(K_{\mathrm{loc}}(\sqrt{x}) + 2\big)\log x\,\sqrt{\frac{H(x)}{\delta x}}.$$

It is used only in Proposition 5.35(B), where $x \ge X_{\mathrm{sw}}$ with $\log X_{\mathrm{sw}} = X_0^{1/100} > 3\Lambda$.

Proof. Let p be the prime attaining $\tau(n) = t(p)\,m$, with $m := \prod_{t(q)\ge t(p)} q$, and write $n = k\cdot m$, where all primes of k are t-below p. The map $n \mapsto (k, m)$ is injective. A prime q with $t(q) < t(p)$ has $q < P := p + 3H(p)$, and $q < p$ if $p < X_0^-$ (then q and p are rational primes, with $t(q) = q$). Hence $\sum_k 1/k \le \prod_{q<P}(1 - 1/q)^{-1}$, which we bound as follows.

- If $p < X_0^-$: for $p < 400$ a direct computation gives $\prod_{q<p}(1 - 1/q)^{-1} \le 1.8\log p + 0.25$ (Appendix B, `finite-n`); for $p \ge 400$, Rosser and Schoenfeld [16, Thm. 8, (3.29)] give $\prod_{q\le p}(1 - 1/q)^{-1} \le e^{\gamma}\log p\,(1 + 1/(2\log^2 p)) \le 1.7811\log p + 0.16$.
- If $p \ge X_0^-$: split at $y_0 := X_0^- - 1$, below which all primes are rational. The rational part is at most $e^{\gamma}\log y_0\,(1 + 1/(2\log^2 y_0))$ by the same bound. For the rest, $-\log(1 - 1/q) \le 1/q + 1/q^2$, and Lemma 5.1(iv) with $y = y_0$, $x = P$ gives $\sum_{y_0<q<P} 1/q \le \log(\log P/\log y_0) + 2.2/\log y_0$, while $\sum_{q>y_0} 1/q^2 \le 3/y_0$. So the product is at most $1.7811\log P\,(1 + 3\cdot 10^{-8}) \le 1.782\log p$.

In all cases $\sum_k 1/k \le 1.8\log p + 1 \le 2\log x$, as $p \le x$ and $\log x \ge 10^8$. Also $1/m \le 2p/x$.

Write $m = p\,m'$, where m' is the product of the primes $q \ne p$ of n with $t(q) \ge t(p)$. Then $p^2m' = \tau(n)\,p/t(p) \in I_p := [x(1-\delta), x)\cdot p/t(p)$. The jitter is centred: $|t(p) - p| \le H(t(p))/2$ (and $t(p) = p$ for a rational prime p).

(i) $p \le \sqrt{\delta x/(2H(x))}$. The interval I_p/p^2 has length $\ge 2D$, so it holds at most $4\delta x/p^2$ elements m' (by $|N - T| \le D$). These pairs contribute at most $4\delta x\sum_{p\le\sqrt{x}} 1/p$ to $\sum p$.

(ii) $p > \sqrt{\delta x/(2H(x))}$; then $p \ge \sqrt{3}\,x^{1/3}$, since $\delta \ge 6H(x)x^{-1/3}$.

- *The jitter.* $H(p)/p \le H(x)/(\sqrt{3}x^{1/3}) \le \delta/(6\sqrt{3})$. Also $t(p) \le p + H(t(p))/2 \le p + t(p)/2$, so $t(p) \le 2p$, and $H(t(p)) \le 2^{4\cdot 10^{-6}}H(p) \le 1.000003\,H(p)$ (by (18) if $t(p) > p$, by monotonicity otherwise). So $|t(p) - p| \le 0.0482\,\delta p \le \delta p/20$, and $p/t(p) \in [1/(1 + \delta/20),\ 1/(1 - \delta/20)]$.
- $m' = 1$. A prime q of m' has $t(q) \ge t(p)$, so $q \ge t(q)/2 \ge t(p)/2 \ge p/3$ (as $H(z) \le z$ here, and $p \le t(p) + H(t(p))/2 \le 1.5\,t(p)$). Then $p^2m' \ge p^3/3 \ge \sqrt{3}\,x$, while $p^2m' < x/(1 - \delta/20) < 1.02\,x$ by I_p. So $m' = 1$.
- *The interval.* Now $p^2 \in [x(1 - \delta)/(1 + \delta/20),\ x/(1 - \delta/20))$. This p-interval has length $\ell := f(\delta)\sqrt{x}$, where $f(\delta) := (1 - \delta/20)^{-1/2} - ((1 - \delta)/(1 + \delta/20))^{1/2}$. On $[0, \frac{1}{4}]$, f is convex with $f(0) = 0$ (the first term is convex; the second is $\sqrt{h}$ with $h := (1 - \delta)/(1 + \delta/20)$, and $2hh'' - h'^2 < 0$), so $f(\delta)/\delta$ is non-decreasing and $f(\delta) \le 4f(\frac{1}{4})\,\delta = 0.58259\,\delta$. Hence $\ell \le 0.6\,\delta\sqrt{x}$.

- *The local count.* The p-interval has minimum $\geq ((1-\delta)/(1+\delta/20))^{1/2}\sqrt{x} \geq 0.86\sqrt{x} \geq X_0$ (as $x \geq X_0^3$), maximum $\leq (1-\delta/20)^{-1/2}\sqrt{x} \leq 1.0064\sqrt{x}$, and endpoint ratio $\big((1+\delta/20)/((1-\delta)(1-\delta/20))\big)^{1/2} \leq 1.17$. Let $P_1 \leq \sqrt{x}$ be the dyadic scale with $P_1 \leq \min < 2P_1$; then $P_1 \geq X_0$ and the interval lies in $[P_1, 2.34P_1) \subset [P_1, 4P_1)$. Cut it into $\lceil \ell/(6H(P_1))\rceil$ windows of length $\leq 6H(P_1)$ and apply the local count at the dyadic scale P_1 (Corollary 6.3(i)) to each. This gives at most $13.4\ell + \lceil \ell/(6H(P_1))\rceil(K_{\mathrm{loc}}(P_1)+2) \leq 13.5\ell + K_{\mathrm{loc}}(\sqrt{x}) + 2$ primes, since $K_{\mathrm{loc}} \leq H/100$ and K_{loc} is non-decreasing.

Each pair (p, m') contributes at most $2\log x \cdot 2p/x$. Collecting, with $p \leq 1.0064\sqrt{x}$ and $\ell \leq 0.6\,\delta\sqrt{x}$ in case (ii), the sum is at most

$$\frac{4\log x}{x}\Big[4\delta x(\log\log x + 1) + 1.0064\sqrt{x}\,\big(8.1\,\delta\sqrt{x} + K_{\mathrm{loc}}(\sqrt{x}) + 2\big)\Big]$$
$$\leq \delta\log x\,(16\log\log x + 48.7) + 4.05\big(K_{\mathrm{loc}}(\sqrt{x})+2\big)\frac{\log x}{\sqrt{x}}.$$

Since $\log H \geq 8(\log\log x)^2 \geq 16\log\log x$ and $H(x)/\delta \geq 1$, this is at most the stated bound. □

Notation for the local prime number theorem.

- Π is the realised prime measure (prime powers weighted $1/k$), so that $N = \exp^*(\Pi)$; and $\Pi_0 := \log^*(\delta_1 + du)$, whose density is $(1-1/u)/\log u$.
- $T_0 := T - (\#\mathcal{A} - \int \bar\rho_{\mathcal{A}})$ is the $\mathcal{A}$-free target. Its Mellin transform is $\hat T_0(s) = s/(s-1) + \hat R(s) + \hat E(s)$ (Lemma 4.3), where $\hat R = \sum_k \hat R_k$ is the transform of the designed oscillation (16).
- $\theta' := \max(\theta + 0.01,\ 8(\log\log X_0)^2/\log X_0 + 0.005) \leq 0.51$ is a fixed exponent (only $\theta' < 1$ is needed), and $\Lambda = \log X_0$.

Theorem 5.24 (local prime number theorem)**.** *Let $y \geq X_0$, and assume:*

(LP-a) $N = T_0$ *below* X_0^-, *and* $|N(u) - T_0(u)| \leq D_0(u) := B(u) + 3H(u) + 4 + 3.1u^{\theta/2}(\log u)^{-4} \leq u^{\theta'}$ *for* $X_0^- \leq u \leq y$. *(On* $\mathrm{Good}(<y)$ *this holds by Theorem 6.6, Corollary 3.8 and (7).)*

(LP-b) $|\hat E(s)| \leq C_8(1+\kappa|s|)e^{\gamma}\Lambda$ *for* $\sigma \geq 1 - 8/\Lambda$, *with* $\log C_8 < 440$ *(Lemma 4.3 with* $a = 8$*).*

(LP-c) $|\hat R(s)| \leq |s|$ *for* $\sigma \geq 1 - 8/\Lambda$. *(By (16): there* $\mathrm{Re}\,w = \sigma - \theta \geq 0.4$, *so* $|\hat R_k(s)| \leq C\alpha_k|s|x_k^{\theta-\sigma} \leq 2C|s|x_k^{-0.4}$, *which is summable over k since* $x_1 \geq \exp(X_0^2)$; *see also (17).)*

Then there are $c_{\mathrm{zf}} \geq 0.088/\Lambda$, $c_{\mathrm{LP}} := c_{\mathrm{zf}}/5 \geq 0.0176/\Lambda$, *an effective K with* $\log K = O(\Lambda)$, *and a threshold* X_{LP} *with* $\log X_{\mathrm{LP}} = O(\Lambda^2\log^2\Lambda)$, *such that for* $X_{\mathrm{LP}} \leq v \leq y$:

(i) $\Big|\displaystyle\int_{[1,v]} \frac{d(\Pi - \Pi_0)(u)}{u}\Big| \leq K\,e^{-c_{\mathrm{LP}}\sqrt{\log v}}$;

(ii) $|\Pi(v) - \Pi_0(v)| \leq K\,v\,e^{-c_{\mathrm{LP}}\sqrt{\log v}}$;

(iii) (short windows) *let* $\Lambda_p := \log v \geq 76\Lambda^3$, $\eta \in [e^{-2\Lambda}, \tfrac12]$ *with* $v(1+2\eta) \leq y$, *and* $w := \eta\Lambda_p^{-2}$. *Put*

$$k := \big\lceil \log(1/\eta) + 20\log\Lambda_p \big\rceil, \qquad L_1 := 2\log\Lambda_p + \log(1/\eta) + \log k + 3.08,$$

and let $\varphi \geq \mathbf{1}_{[v,v(1+\eta)]}$ *be the window of Step 10 (built with this k):* $0 \leq \varphi \leq 1$, *supported in* $[v(1-w), v(1+\eta+w)]$. *Then* $\int\varphi\,d\Pi \leq (1+\mathrm{Rel})\int\varphi\,d\Pi_0$, *with*

$$\mathrm{Rel}(\Lambda_p,\eta) \leq 18\,G(L_1)\,\Lambda_p^3\,\eta^{-1}\exp\Big(-\frac{0.9\,c_{\mathrm{zf}}\Lambda_p}{L_1 + 442 + 2\log\Lambda + \log L_1}\Big) + \Lambda_p^{-2}, \tag{47}$$

where $G(L) := 400(L + 450 + 2\log\Lambda)$; *so* $\log G(L) \leq \Lambda$ *for* $L \leq e^{\Lambda/2}$.

Proof of Theorem 5.24. This is Landau's argument [14, Lemmas 6.2–6.3 and the proof of Thm. 6.6], with four changes. It is run on an extension of N beyond y, which is not a Beurling system. Positivity is therefore imported from a finite Euler-type product over the primes $\le y$ (Step 2). Landau's discs have radius $\asymp 1/\Lambda$. And the region near the pole is handled with the fine form of $\hat{E}$. Throughout, $s = \sigma + it$.

Step 1: the extension. Put $D := N - T_0$, so $D = 0$ below X_0^- and $|D(u)| \le u^{\theta'}$ on $[X_0^-, y]$ by (LP-a). Define $N^\sharp := N$ on $[1, y]$ and $N^\sharp(u) := N(y) + T_0(u) - T_0(y)$ for $u > y$. Then $D^\sharp := N^\sharp - T_0$ equals $D(\min(u, y))$, and

$$\zeta^\sharp(s) := \int_{[1,\infty)} u^{-s}\, dN^\sharp = \hat{T}_0(s) + \hat{D}^\sharp(s), \qquad \hat{D}^\sharp(s) = s\int_{X_0^-}^{y} D(u)u^{-s-1}du + D(y)y^{-s}.$$

So $\hat{D}^\sharp$ is entire, with $|\hat{D}^\sharp(s)| \le |s|(X_0^-)^{\theta'-\sigma}/(\sigma-\theta') + y^{\theta'-\sigma}$ for $\sigma > \theta'$. Let $\Pi^\sharp := \log^*(dN^\sharp)$, a signed measure (the series converges in the weighted norm $\int u^{-\sigma}|\cdot|$ for large σ). Since $\log^*$ is causal, $\Pi^\sharp = \Pi$ on $[1, y]$.

Step 2: upper bounds on $[1, y]$. Lemma 5.1 holds on $[1, y]$: there $T_0 - \lfloor\cdot\rfloor = E_{\mathrm{rig}} + R + \{\cdot\}$ on $[X_0, y]$, so (LP-a), (8) and $|R| \le 2u^\theta \log^* u$ (Lemma 3.4) give ($\mathrm{H_{BS}}$), as in §5.2. Write $d\psi = \log u\, d\Pi$ and $S(z) = \int_{[1,z]} d\psi/u \le \log z + 0.43$. By Lemma 5.1(iv) and the values for $\mathbb{N}$, $\int_{[1,y]} d\Pi/u \le \log\log y + 4.5$; the same argument gives $\int_{[1,z]} d\Pi/u \le \log\log z + 4.5$ for $e \le z \le y$. Let $N_y := \exp^*(\Pi|_{[1,y]})$, the convolution exponential of the restricted logarithmic measure. It is a positive Euler proxy, not the counting measure of the y-smooth elements: for a prime $\sqrt{y} < p \le y$ it gives p^2 the weight $\frac{1}{2}$. Only the following properties are used.

(a) $N_y = N$ on $[1, y]$, since $N = \exp^*(\Pi)$ and $\exp^*$ is causal (all measures live on $[1, \infty)$).
(b) $dN_y \ge 0$, since $\exp^*(\mu) = \sum_{k\ge 0} \mu^{*k}/k!$ and $\Pi \ge 0$.
(c) $P_y(s) := \int u^{-s} dN_y = \exp(\int_{[1,y]} u^{-s} d\Pi)$, since the Mellin transform turns $*$ into products.
(d) $|(dN_y)^{*-1}| = |\exp^*(-\Pi|_{[1,y]})| \le \exp^*(\Pi|_{[1,y]}) = dN_y$, term by term.

By (c), $P_y(1) \le e^{4.5}\log y$.

Step 3: positivity transfer. Put $\Delta := \zeta^\sharp - P_y = \int_{(y,\infty)} u^{-s}(dT_0 - dN_y)$, and let $\sigma = 1 + \eta$ with $0 < \eta \le 1/8$.

- *On* $[1, y]$, $N_y = N$, and ($\mathrm{H_{BS}}$) gives

$$\int_{[1,y]} u^{-\sigma} dN \le \sum_{n\le y} n^{-\sigma} + \varepsilon_1(y)y^{-\eta} + \sigma\int_{X_0^-}^{y} \varepsilon_1(u)u^{-1-\eta}du \le 1/\eta + 1 + 2\eta_1 \le 1/\eta + 2.01.$$

- *The* T_0 *part on* (y, ∞). There $dT_0 = d\Psi(\cdot, X_0) + (1 - \rho_{\mathrm{det}} + R')\, du \ge 0$ ((5), (6)), and $T_0(u) - u = E_{\mathrm{rig}}(u) + R(u)$ with $|E_{\mathrm{rig}}(u)| \le \kappa X_0 + 2 + \kappa u$ (8) and $|R(u)| \le 2u^\theta \log^* u$ (Lemma 3.4). Integrate by parts against $u^{-\sigma}$: the term u of T_0 gives exactly $y^{-\eta}/\eta$, and the rest, including the boundary term at y, is at most $\big(\kappa(1 + 2\eta) + 2(\kappa X_0 + 2)\eta/y + O(y^{\theta-1}\log y)\big)y^{-\eta}/\eta \le 3\kappa\, y^{-\eta}/\eta$, since $y \ge X_0$ and $y^{\theta-1}\log y \le X_0^{-0.49} \ll \kappa$. The same computation against $u^{-\sigma}\log u$ (decreasing for $u \ge e$), whose main term is $\int_y^\infty u^{-\sigma}\log u\, du = y^{-\eta}(\log y/\eta + 1/\eta^2)$, gives

$$\int_{(y,\infty)} u^{-\sigma} dT_0 \le (1 + 3\kappa)\frac{y^{-\eta}}{\eta}, \qquad \int_{(y,\infty)} u^{-\sigma}\log u\, dT_0 \le (1 + 3\kappa)\, y^{-\eta}\Big(\frac{\log y}{\eta} + \frac{1}{\eta^2}\Big).$$

- *The* N_y *part on* (y, ∞) uses only $dN_y \ge 0$ and $u^{-\eta} \le y^{-\eta}$ for $u > y$: $\int_{(y,\infty)} u^{-\sigma} dN_y \le y^{-\eta}P_y(1)$, and $\int_{(y,\infty)} u^{-\sigma}\log u\, dN_y \le y^{-\eta}(-P_y'(1)) = y^{-\eta}P_y(1)S(y)$, since $-P_y'/P_y(1) = \int_{[1,y]} d\psi/u = S(y)$.

Hence, as $1 + 3\kappa \le 2$,

$$P_y(\sigma) \le 1/\eta + 2.01 + y^{-\eta}P_y(1), \qquad |\Delta(s)| \le 2y^{-\eta}/\eta + y^{-\eta}P_y(1),$$
$$|\Delta'(s)| \le 2y^{-\eta}(\log y/\eta + 1/\eta^2) + y^{-\eta}P_y(1)\,S(y), \qquad |P_y(s)| \ge 1/P_y(\sigma), \qquad |P_y'/P_y(s)| \le S(y).$$

The last two hold because $\Pi \ge 0$. Set $\eta_y := 17\log\log y/\log y$, so that $y^{-\eta} \le (\log y)^{-17}$ for $\eta \ge \eta_y$. Then, for $1 + \eta_y \le \sigma \le 1 + 1/8$,

$$P_y(\sigma) \le 2/\eta, \qquad |\Delta/P_y| \le 10^{-2}, \qquad \Big|\frac{d}{ds}\log(1 + \Delta/P_y)\Big| \le (\eta\log y)^{-1}. \tag{48}$$

Let $F := (dT_0 - dN_y)|_{(y,\infty)} * (dN_y)^{*-1}$. Since $|(dN_y)^{*-1}| \le dN_y$, the same bounds give $\int u^{-\sigma}|dF| \le 10^{-2}$ for $1 + \eta_y \le \sigma \le 1 + 1/8$, and so for every $\sigma \ge 1 + \eta_y$, since F lives on (y,∞) and $\int u^{-\sigma}|dF|$ is non-increasing in σ. Hence $\Pi^\sharp = \Pi|_{[1,y]} + \log^*(\delta_1 + F)$ satisfies $\int u^{-\sigma}|d\Pi^\sharp| < \infty$, $\zeta^\sharp$ has no zeros in $\sigma \ge 1 + \eta_y$, and $\log\zeta^\sharp = \log P_y + \log(1 + \Delta/P_y)$ is the Mellin transform of $\Pi^\sharp$ there. The coefficients of $-P_y'/P_y$ form the positive measure $d\psi|_{[1,y]}$. So for the nonnegative polynomial

$$(1 + \cos\phi)^4 = \tfrac{1}{8}\big(35 + 56\cos\phi + 28\cos 2\phi + 8\cos 3\phi + \cos 4\phi\big),$$

with $(a_0, \dots, a_4) := (35, 56, 28, 8, 1)$, we get

$$\sum_{k=0}^{4} a_k \operatorname{Re}\Big(-\frac{\zeta^{\sharp\prime}}{\zeta^\sharp}\Big)(\sigma + ikt) \ge -\frac{128\,\varepsilon_f}{\eta}, \qquad \varepsilon_f := 5\cdot 10^{-4} \ \ (\ge 1/\log y), \qquad \sigma = 1 + \eta \ge 1 + \eta_y. \tag{49}$$

Step 4: the pole term. By Step 3 and Lemma 5.1(ii), with partial summation, $-P_y'/P_y(1+\eta) = \int_{[1,y]} u^{-\eta}dS(u) \le \eta\int_1^\infty (\log u + 0.43)u^{-1-\eta}du = 1/\eta + 0.43$. Hence, by (48),

$$-\frac{\zeta^{\sharp\prime}}{\zeta^\sharp}(1 + \eta) \le \frac{1 + 10^{-8}}{\eta} + 0.43 \qquad (\eta \ge \eta_y). \tag{50}$$

This replaces the usual use of the pole of $\zeta^\sharp$; the crude bound $|\hat{E}| \le C_8 e^\gamma\Lambda$ is useless near $s = 1$.

Step 5: Landau's lemma. Let f be analytic on $|z - z_0| \le r$, with $|f| \le M|f(z_0)|$ there, and suppose every zero of f in the disc has real part $\le \operatorname{Re} z_0$. Then for any zero ρ_1 in the disc,

$$-\operatorname{Re}\frac{f'}{f}(z_0) \le \frac{2}{r}\log M - \operatorname{Re}(z_0 - \rho_1)\Big(\frac{1}{|z_0 - \rho_1|^2} - \frac{1}{r^2}\Big).$$

Proof. Divide f by the Blaschke factors of its zeros, normalised on the circle of radius r; the quotient g has no zeros and $|g/g(z_0)| \le M$ on the circle. Borel–Carathéodory for $h = \log(g/g(z_0))$ on radii $R \uparrow r$ gives $|h'(z_0)| \le 2\log M/r$. Each zero contributes $\operatorname{Re}(z_0 - \rho)(1/|z_0 - \rho|^2 - 1/r^2) \ge 0$ to $-\operatorname{Re} f'/f(z_0)$; drop all but ρ_1.

Step 6: the region near the pole. For $|t| \le X_0$ and $\sigma \ge 1 - 8/\Lambda$, the proof of Lemma 4.3 gives the fine form $\hat{E}(s) = \zeta(s) - 1/(s-1) - 1 + O(X_0^{-\sigma}) + O(\kappa(1 + |s|)e^{\mathrm{Ein}(8)}e^\gamma\Lambda)$. Since $s/(s-1) = 1 + 1/(s-1)$, (17) (that is, $|\hat{R}(s)| \le |s|e^{-0.39X_0^2}$) and Step 1 give

$$\zeta^\sharp(s) = \zeta(s) + E(s), \qquad |E(s)| \le 3X_0^{-0.3}(1 + |t|) \qquad (|t| \le X_0,\ \sigma \ge 1 - 8/\Lambda),$$

using $\kappa \le X_0^{-1/3}$ and $e^{\mathrm{Ein}(8)} \le e^{438}$. On $|s-1| \le 1/8$ the Laurent series gives $|\zeta(s) - 1/(s-1)| \le \gamma + 0.02 \le 0.6$ (as $|\gamma_1| \le 0.073$), so $|(s-1)\zeta^\sharp(s) - 1| \le 0.6/8 + 10^{-9} < 1$, and $\zeta^\sharp$ has no zero there. This covers every point with $|t| < 1/16$ and $1 - 8/\Lambda \le \sigma \le 1.1$; for $\sigma > 1.1$, $\zeta^\sharp \ne 0$ by Step 3.

Step 7: the zero-free region for $|t| \geq 1/16$. Take Landau discs of radius $r := a/\Lambda$ with $a := 6.8 \leq 8$, so that they stay in $\sigma \geq 1 - 8/\Lambda$, where (LP-b) and (LP-c) apply. For $|t| \geq 1/16$ they do not meet the pole. Put

$$L_t := \log(|t|+3), \qquad X(t) := L_t + 442 + 2\log\Lambda, \qquad Y(t) := X(t) + \log X(t) + 0.7.$$

- *Growth.* Let $|t| \geq 1/16$, $\eta_y \leq \eta \leq 10^{-3}/\Lambda$, $k \in \{1,2,3,4\}$ and $z_0 = 1 + \eta + ikt$. Step 3 gives $|\zeta^\sharp(z_0)| \geq (1-10^{-2})/P_y(1+\eta) \geq 0.495\,\eta$. On the disc $|z - z_0| \leq r$:
  - $|z/(z-1)| \leq 1 + 1/(k|t| - r) \leq 17.1$;
  - $|\hat{R}(z)| \leq |z| \leq k|t| + 1.01$, by (LP-c);
  - $|\hat{E}(z)| \leq e^\gamma C_8 \Lambda + 10^{-10^7}|z|$, by (LP-b), since $\kappa \leq X_0^{-1/3}$ and $\log C_8 < 440$;
  - $|\hat{D}^\sharp(z)| \leq 2.1|z|X_0^{-0.48} + 1$, by Step 1, since $\sigma - \theta' \geq 0.48$.

  So $|\zeta^\sharp| \leq 2(1 + 3X_0^{-0.48})\max(k|t| + 19.2,\ e^\gamma C_8\Lambda)$ there, for every $|t| \geq 1/16$. Now $\log(4|t| + 19.2) \leq L_t + 1.857$ and $\log(e^\gamma C_8\Lambda) < 440.578 + \log\Lambda$; since $\max(a,b) \leq a + b - 2.975$ when $a, b \geq 2.975$, the ratio $M_k := \max_{|z-z_0|\leq r} |\zeta^\sharp(z)|/|\zeta^\sharp(z_0)|$ satisfies (the constant is $440.855 + \log(1 + 3X_0^{-0.48}) \leq 440.86$)

$$\log M_k \leq L_t + 440.86 + \log\Lambda + \log(1/\eta). \tag{51}$$

- *The inequality.* Suppose $\zeta^\sharp(\beta + it) = 0$ with $1 - \beta \leq \eta/\kappa_L$ for a parameter $\kappa_L > 0$, and $\eta \geq \eta_y$. Apply Step 5 at $z_0 = 1 + \eta + it$ with this zero, and at $z_0 = 1 + \eta + ikt$ ($k = 2,3,4$) with no zero term. This is legitimate because all zeros have real part $< 1 + \eta_y \leq 1 + \eta$ (Step 3), so the dropped terms are ≥ 0. Insert these and (50) into (49). Using $1/(\eta + 1 - \beta) \geq \kappa_L/((\kappa_L + 1)\eta)$, we get

$$\frac{N_0(\kappa_L)}{\eta} \leq 15.05 + \frac{2}{r}\sum_{k=1}^{4} a_k \log M_k + \frac{56\,\eta(1 + 1/\kappa_L)}{r^2}, \qquad N_0(\kappa_L) := \frac{56\kappa_L}{\kappa_L + 1} - 35(1 + 10^{-8}) - 128\varepsilon_f.$$

- *The constant.* Since $\sum_{k=1}^4 a_k = 93$, the inequality gives, as $|t| \to \infty$, the zero-free width $\text{merit} \cdot r/(2\log|t|)$, with

$$\text{merit} := \max_{\kappa_L} \frac{N_0(\kappa_L)}{93\,\kappa_L} = 0.0262291 \qquad (\kappa_L \approx 3.79).$$

  So $\text{merit} \cdot r/2 = 0.0891790/\Lambda$, and $\text{merit} \cdot a/2 \geq 0.088$ needs $a \geq 6.71$. We fix $\kappa_L := 3.79$, for which $N_0 \geq 9.2449$, and put

$$c_{\text{zf}} := 0.088/\Lambda.$$

  The slack between 0.08917 and 0.088 absorbs the lower-order terms of (51) at finite height, as follows. The constant merit does not depend on Λ.
- *Explicit width.* Let $|t| \geq 1/16$, and suppose that $\eta := 0.33769/(\Lambda Y(t))$ satisfies $\eta \geq \eta_y$. Then $\zeta^\sharp(\sigma + it) \neq 0$ for

$$\sigma \geq 1 - \delta_t, \qquad \delta_t := \eta/\kappa_L = \frac{0.0891}{\Lambda\, Y(t)}. \tag{52}$$

  Indeed, otherwise the inequality holds with this η. Here $\log(1/\eta) = \log\Lambda + \log Y + 1.0857$, and $\log Y - \log X \leq (\log X + 0.7)/X \leq 0.0144$ (as $X \geq 479.96$), so the right side of (51) is at most $X - 0.054 + \log Y \leq X + \log X - 0.039 \leq Y$. Multiply the inequality by η, and use $\eta\Lambda = 0.33769/Y \leq 7.1\cdot 10^{-4}$: the right side becomes at most $(2\cdot 93/6.8)\cdot 0.33769 + 10^{-6} \leq 9.2369 < 9.2449 \leq N_0$, a contradiction.
- *Two consequences.* Write $\delta_{\text{zf}}(t) := \delta_t$.
  - Since $0.088(X + \log X + 0.7) \leq 0.0891(X + 18)$ for $X \geq 479$, we have $\delta_{\text{zf}}(t) \geq c_{\text{zf}}/(L_t + 460 + 2\log\Lambda)$.

– *Finite height.* Let $L \ge 5$ and $|t| \le e^L$. Then $L_t \le L + 3e^{-L}$, and $0.088(X + \log X + 0.7) \le 0.0891(L + 442 + 2\log\Lambda + \log L)$: the difference of the two sides increases with L and with Λ, and is positive at $L = 5$, $\Lambda = 10^8$. So at height e^L the zero-free width is $\delta_{\rm zf} \ge c_{\rm zf}/(L + 442 + 2\log\Lambda + \log L)$. Every use below has $L \ge 40$.

- *The height constraint.* Step 3 requires $\eta = \kappa_L \delta_{\rm zf}(t) \ge \eta_y = 17\log\log y/\log y$, and under the present scaling this binds. Assume

$$\log y \ge 10^4\Lambda^2 \log\log y, \qquad \text{so that} \quad Q_y := \frac{\log y}{\Lambda \log\log y} \ge 10^4\Lambda, \tag{53}$$

and put $\log T_y := Q_y/51 - 500$. For $|t| \le T_y$ we have $X(t) \le Q_y$ and $Y(t) \le Q_y/51 - 57.3 + 2\log\Lambda + \log Q_y \le Q_y/50.35$, because $(1/50.35 - 1/51)Q_y \ge 2.53\cdot 10^{-4} Q_y \ge 2\log\Lambda + \log Q_y$. Since $0.33769 \cdot 50.35 \ge 17.002$, this gives $\eta = 0.33769/(\Lambda Y) \ge 17/(\Lambda Q_y) = \eta_y$. So the zero-free region is

$$\sigma \ge 1 - \frac{c_{\rm zf}}{\log(|t|+3) + 460 + 2\log\Lambda} \qquad \text{for } |t| \le T_y, \qquad \log T_y := \frac{\log y}{51\Lambda\log\log y} - 500. \tag{54}$$

Together with Step 6 this covers all $|t| \le T_y$. Every application below satisfies (53).

- By (LP-c), $\hat R$ enters only through the growth bound, like $\hat D^\sharp$. The region (54) is consistent with the designed zeros, which have $1 - \beta_k \asymp \log\log\gamma_k/\log\gamma_k$ (§4).

Step 8: $\log\zeta^\sharp$ in the region. Assume (53), and let $\mathfrak{G}_1(s) := \log(\zeta^\sharp(s)(s-1)/s)$, the branch that is real on $(1,\infty)$; it is analytic in the zero-free region, including near $s = 1$. Fix $f := 0.9$, and for $1/16 \le |t| \le T_y$ let $s_1 := 1 + \eta_t + it$ with $\eta_t := \kappa_L\delta_t$. Consider the discs centred at s_1 of radii $R_1 = \eta_t + \frac{1+f}{2}\delta_t$ and $r_1 = \eta_t + f\delta_t$. They avoid the pole, and they lie in the zero-free region, since $\delta_{\rm zf}$ changes by a relative 10^{-8} over them. The function $h := \log\zeta^\sharp - \log\zeta^\sharp(s_1)$ vanishes at s_1, and $\operatorname{Re} h \le \log M_1 \le Y(t)$ on the larger disc, by (51) with $\eta = \eta_t$ (as in Step 7). Borel–Carathéodory gives $|h| \le \frac{2r_1}{R_1 - r_1} Y(t) = \frac{4(\kappa_L + f)}{1-f} Y(t) \le 187.6\, Y(t)$ on the smaller disc. At the anchor, Step 3 gives $|\log\zeta^\sharp(s_1)| \le \log P_y(1+\eta_t) + 0.02 \le \log(2/\eta_t) + 0.02 \le 0.5\, Y(t)$. Finally $|\log((s-1)/s)| \le \log 17 + \pi \le 6$ for $|t| \ge 1/16$ and $\sigma \le 1.1$. Hence

$$|\mathfrak{G}_1(s)| \le 190\, Y(t) \qquad \text{for } |s - s_1| \le r_1.$$

A point $s = \sigma + it$ of the contour $\sigma = 1 - f\delta_{\rm zf}(T_1)$, with $1/16 \le |t| \le T_1 \le T_y$, satisfies $|s - s_1| = \eta_t + f\delta_{\rm zf}(T_1) \le r_1$, because $\delta_{\rm zf}$ decreases in $|t|$. For $|t| < 1/16$ the contour lies in $|s - 1| \le 1/16 + 10^{-8}$. There Step 6 gives $|\log((s-1)\zeta^\sharp(s))| \le -\log(1 - 0.075) \le 0.078$, and $|\log s| \le -\log(1 - |s-1|) \le 0.0646$, so $|\mathfrak{G}_1| \le 0.1426$. Since $Y \le 2X$, it follows that for $5 \le L \le \log T_y$, on the contour $\sigma = 1 - f\delta_{\rm zf}(e^L)$, $|t| \le e^L$,

$$|\mathfrak{G}_1| \le 190\, Y(e^L) \le G(L) := 400\,(L + 450 + 2\log\Lambda);$$

in particular $\log G(L) \le \Lambda$ for $L \le e^{\Lambda/2}$.

Step 9: Perron's formula; proof of (i) and (ii). Let $\mathfrak{G}(s) := \log\big(\zeta^\sharp(s+1)\, s/(s+1)\big)$. It is analytic in the shifted zero-free region, $\mathfrak{G}(0) = 0$ (the residue of $\zeta^\sharp$ at 1 is 1, and $\zeta^\sharp > 0$ on $(1,\infty)$), and $\mathfrak{G}$ is the Mellin transform of $(\Pi^\sharp - \Pi_0)/u$ on $\operatorname{Re} s \ge \eta_y$. Let $\Phi^\sharp(w) := \int_{[1,w]} d(\Pi^\sharp - \Pi_0)/u$; it equals the left side of (i) for $w \le y$.

- Take $h_0 \in C_c^\infty(-2,-1)$ with $h_0 \ge 0$, $\int h_0 = 1$, and put $h := Th_0(T\cdot)$. Smoothing to the left keeps everything inside $[1, y]$:

$$\Phi^h(v) := \int \Phi^\sharp(ve^\tau) h(\tau)\, d\tau = \frac{1}{2\pi i}\int_{(\eta_y)} \mathfrak{G}(s)\, \hat h_0(s/T)\, v^s\, \frac{ds}{s}, \qquad |\hat h_0(w)| \le C_m e^{2|\operatorname{Re} w|}(1+|w|)^{-m}.$$

- Shift the contour to $\operatorname{Re} s = -f\delta_{\rm zf}(T_1)$ for $|t| \le T_1 \le T_y$. There is no residue at $s = 0$, because $\mathfrak{G}(0) = 0$. Take $T := e^{\sqrt{\log v}}$, $T_1 := e^{2\sqrt{\log v}}(\log y)^{2/m}$ and $m := 19$.

- The contributions are as follows. The shifted line gives at most $K_8\, v^{-f\delta_{\mathrm{zf}}(T_1)}\log^2(T_1+3)$. The horizontal segments, and the line $\operatorname{Re} s=\eta_y$ beyond T_1, give at most
$$K_9 v^{\eta_y}(\log\log y)(T/T_1)^m \le e^{-\sqrt{\log v}},$$
using $v^{\eta_y}\le(\log y)^{17}$. The smoothing error is
$$|\Phi^\sharp(v)-\Phi^h(v)| \le \int_{(ve^{-2/T},v]} \frac{d\Pi+d\Pi_0}{u} \le \frac{6}{T}+3v^{\theta'-1},$$
using $d\Pi\le dN$.
- The heights used lie below T_v once $\log v\ge 1.1\cdot 10^4\Lambda^2(\log\log v)^2$; here we apply the argument with $y':=v$ in place of y, which is allowed since (LP-a) holds on $[1,v]$; and $y'=v$ satisfies (53). For $\log v\ge(1000+4\log\Lambda)^2$ the offset in (54) at most doubles the denominator. Together this gives (i) with $c_{\mathrm{LP}}=c_{\mathrm{zf}}/5$. The threshold X_{LP} is chosen above both heights, so $\log X_{\mathrm{LP}}=O(\Lambda^2\log^2\Lambda)$, and $\log K=O(\Lambda)$.
- *Proof of (ii), from (i).* Put $\Phi(w):=\int_{[1,w]} d(\Pi-\Pi_0)/u$; then $\Phi=\Phi^\sharp$ on $[1,y]$, Φ is right-continuous, and $\Phi(1^-)=0$. Integrating by parts, for $X_{\mathrm{LP}}\le v\le y$,
$$\Pi(v)-\Pi_0(v)=\int_{[1,v]} u\,d\Phi(u)=v\,\Phi(v)-\int_1^v \Phi(u)\,du.$$
  - On $[1,X_{\mathrm{LP}}]$: both $\int_{[1,u]} d\Pi/w$ and $\int_{[1,u]} d\Pi_0/w$ are non-decreasing in u. By Step 2 at height X_{LP}, the first is $\le \log\log X_{\mathrm{LP}}+4.5$. The density of Π_0/w is $(1-1/w)/(w\log w)\le 1/w$ on $[1,e]$ and $\le 1/(w\log w)$ beyond, so the second is $\le \log\log X_{\mathrm{LP}}+1$. Hence $|\Phi(u)|\le 2\log\log X_{\mathrm{LP}}+5.5$ there.
  - On $[X_{\mathrm{LP}},v]$: $|\Phi(u)|\le Ke^{-c_{\mathrm{LP}}\sqrt{\log u}}$ by (i). With $u=e^x$, the function $e^{x-c_{\mathrm{LP}}\sqrt{x}}$ has derivative $\ge 0.99\,e^{x-c_{\mathrm{LP}}\sqrt{x}}$, so $\int_{X_{\mathrm{LP}}}^v e^{-c_{\mathrm{LP}}\sqrt{\log u}}du\le 1.02\,v\,e^{-c_{\mathrm{LP}}\sqrt{\log v}}$.
  - $v\,e^{-c_{\mathrm{LP}}\sqrt{\log v}}$ is increasing, so $X_{\mathrm{LP}}\le e^{c_{\mathrm{LP}}\sqrt{\log X_{\mathrm{LP}}}}\,v\,e^{-c_{\mathrm{LP}}\sqrt{\log v}}$.

  Hence
$$|\Pi(v)-\Pi_0(v)|\le K'v\,e^{-c_{\mathrm{LP}}\sqrt{\log v}},\qquad K':=2.02K+e^{c_{\mathrm{LP}}\sqrt{\log X_{\mathrm{LP}}}}\big(2\log\log X_{\mathrm{LP}}+5.5\big).$$
  Since $\log X_{\mathrm{LP}}=O(\Lambda^2\log^2\Lambda)$ and $c_{\mathrm{LP}}=c_{\mathrm{zf}}/5=0.0176/\Lambda$ (Step 7), both $c_{\mathrm{LP}}\sqrt{\log X_{\mathrm{LP}}}$ and $\log\log X_{\mathrm{LP}}$ are $O(\log\Lambda)$, so $\log K'=O(\Lambda)$. As $K'\ge K$, (i) and (ii) hold together with K' in place of K, and with the same c_{LP}. No second contour integral is needed.

Step 10: short windows; proof of (iii).

- *The window.* Put $a:=\log v$, $b:=\log(v(1+\eta))$, $\omega:=w/4=\eta\Lambda_p^{-2}/4$, and let $\kappa:=\beta_\omega^{*k}$ be the k-fold convolution of the box $\beta_\omega:=(k/2\omega)\mathbf{1}_{[-\omega/k,\,\omega/k]}$; so $\kappa\ge 0$, $\int\kappa=1$ and κ is supported in $[-\omega,\omega]$. Put $\psi:=\mathbf{1}_{[a-\omega,\,b+\omega]}*\kappa$ and $\varphi(u):=\psi(\log u)$. Then $0\le\psi\le 1$, $\psi=1$ on $[a,b]$ and $\psi=0$ outside $[a-2\omega,b+2\omega]$; so $\varphi\ge\mathbf{1}_{[v,v(1+\eta)]}$, $0\le\varphi\le 1$, and φ is supported in $[ve^{-2\omega},v(1+\eta)e^{2\omega}]\subset[v(1-w),v(1+\eta+w)]\subset[1,y]$.
- *Its Mellin transform.* For $s=\sigma+it$ with $0<\sigma\le 2$, $\tilde\varphi(s):=\int_0^\infty\varphi(u)u^{s-1}du=\int\psi(x)e^{sx}dx=\hat I(s)\,\hat\kappa(s)$, where $\hat I(s)=\int_{a-\omega}^{b+\omega}e^{sx}dx$ and $\hat\kappa(s)=\hat\beta_\omega(s)^k$ with $\hat\beta_\omega(s)=\sinh(s\omega/k)/(s\omega/k)$. Hence
$$|\hat I(s)|\le\min\Big(1.0001\,\eta,\ \frac{2}{|t|}\Big)e^{\sigma(b+\omega)},\qquad |\hat\kappa(s)|\le e^{\sigma\omega}\min\Big(1,\frac{k}{|t|\omega}\Big)^k,$$
using $b-a+2\omega\le 1.0001\eta$, $|e^{s\beta'}-e^{s\alpha'}|\le 2e^{\sigma\beta'}$, $|\hat\beta_\omega(s)|\le e^{\sigma\omega/k}$ and $|\hat\beta_\omega(s)|\le e^{\sigma\omega/k}k/(|t|\omega)$. Also $e^{\sigma(b+2\omega)}\le 1.51\,v^\sigma$ for $\sigma\le 1.0001$, as $\eta\le\frac12$. Since $T_1\omega=e^{L_1}\eta\Lambda_p^{-2}/4=e^{3.08}k/4\ge 2ek$, we have $(k/(|t|\omega))^k\le(2e)^{-k}(T_1/|t|)^k$ for $|t|\ge T_1$.

- *The contour.* Put $c := 1 + 17\log\Lambda_p/\Lambda_p$. As $y \ge v$, $c - 1 \ge 17\log\log y/\log y = \eta_y$, and $c - 1 \le 1/8$; so by Step 3, $\Pi^\sharp$ has $\int u^{-c}|d\Pi^\sharp| < \infty$, its Mellin transform there is $\log\zeta^\sharp$, and that of Π_0 is $\log(s/(s-1))$. Since φ is supported in $[1, y]$, where $\Pi^\sharp = \Pi$, Mellin inversion gives

$$\int \varphi\, d(\Pi - \Pi_0) = \frac{1}{2\pi i}\int_{(c)} \tilde\varphi(s)\,\mathfrak{G}_1(s)\,ds, \qquad \mathfrak{G}_1(s) = \log\frac{\zeta^\sharp(s)(s-1)}{s}.$$

  Put $T_1 := e^{L_1}$ and $c' := 1 - f\delta_{\mathrm{zf}}(T_1)$, with $f = 0.9$ and $\delta_{\mathrm{zf}}(T_1)$ the width of Step 7. The function $\mathfrak{G}_1$ is analytic on the rectangle $[c', c] \times [-T_1, T_1]$: it lies in $|s-1| \le 1/8$ for $|t| < 1/16$ (Step 6), and in the zero-free region of Step 7 for $1/16 \le |t| \le T_1 \le T_y$ (by the Range bullet below). By Cauchy's theorem the line (c) may be replaced by: (A) the segment $\sigma = c'$, $|t| \le T_1$; (B) the two horizontal segments $c' \le \sigma \le c$, $t = \pm T_1$; and (C) the two half-lines $\sigma = c$, $|t| \ge T_1$.
- *Bounds for $\mathfrak{G}_1$.* On (A), $|\mathfrak{G}_1| \le G(L_1)$ by Step 8. On (B), the part with $\sigma \le 1 + 2\eta_{T_1} + f\delta_{T_1}$ lies in the disc $|s - s_1| \le r_1$ of Step 8 (with $t = \pm T_1$), where $|\mathfrak{G}_1| \le 190Y(T_1) \le G(L_1)$; on the rest, $\sigma - 1 \ge 2\eta_{T_1} \ge \eta_y$, and Step 3 gives $|\mathfrak{G}_1(s)| \le \log P_y(\sigma) + 0.02 + \log(|s|/|s-1|) + \pi \le 2\log(1/(\sigma-1)) + 4 \le G(L_1)$ (as $\sigma - 1 \le 1/8$). On (C), $|\mathfrak{G}_1(c+it)| \le 2\log(1/(c-1)) + 4 \le 2\log\Lambda_p$, in the same way.
- *The main term.* $\int \varphi\, d\Pi_0 \ge \int_v^{v(1+\eta)}(1 - 1/u)\,du/\log u \ge \eta v/(1.0001\Lambda_p)$.
- *(A).* By the bounds above, $\int_{|t|\le T_1} |\tilde\varphi(c'+it)|\,dt \le 1.51 v^{c'}\big[2.0002 + 4\log(\eta T_1)\big]$ (use 1.0001η for $|t| \le 1/\eta$ and $2/|t|$ beyond), and $\log(\eta T_1) = 2\log\Lambda_p + \log k + 3.08$. With $v^{c'-1} = \exp(-f\delta_{\mathrm{zf}}(T_1)\Lambda_p)$ and the finite-height width of Step 7 at $L = L_1 \ge 5$, the contribution of (A), divided by the main term, is at most

$$0.2404\, G(L_1)\,\Lambda_p\big(14.33 + 8\log\Lambda_p + 4\log k\big)\,\eta^{-1}\exp\Big(-\frac{0.9\, c_{\mathrm{zf}}\Lambda_p}{L_1 + 442 + 2\log\Lambda + \log L_1}\Big),$$

  which is at most the first term of (47), since $0.2404(14.33 + 8\log\Lambda_p + 4\log k) \le 18\Lambda_p^2$.
- *(C).* $\int_{|t|\ge T_1} |\tilde\varphi(c+it)|\,dt \le 1.51 v^c \int_{|t|\ge T_1} \frac{2}{|t|}(2e)^{-k}\Big(\frac{T_1}{|t|}\Big)^k dt = 6.04\, v^c (2e)^{-k}/k$, with $v^c = v\Lambda_p^{17}$. So (C), divided by the main term, is at most

$$\frac{1}{2\pi}\cdot 2\log\Lambda_p \cdot 6.04 \cdot 1.0001\,\frac{\Lambda_p^{18}(2e)^{-k}}{k\eta} \le 1.93\,\frac{\Lambda_p^{18}\log\Lambda_p\, e^{-k}}{k\eta} \le 0.1\Lambda_p^{-2},$$

  because $e^{-k} \le \eta\Lambda_p^{-20}$ and $\log\Lambda_p \le k/20$.
- *(B).* On the segments, $|\tilde\varphi| \le 1.51 v^c (2/T_1)(2e)^{-k}$ and the length is $c - c' \le 1$; so (B), divided by the main term, is at most $\frac{2}{2\pi}\cdot G(L_1)\cdot 3.02 \cdot 1.0001\,\Lambda_p^{18}(2e)^{-k}/(T_1\eta) \le \Lambda_p^{-2} G(L_1)/T_1 \le 10^{-3}\Lambda_p^{-2}$.

Adding (A)–(C) gives (47).

- *Range.* Step 7 gives the width at height e^{L_1} if (53) holds and $L_1 \le \log T_y$. Both hold for $\Lambda_p \ge 76\Lambda^3$. First, $\log y \ge \Lambda_p \ge 76\Lambda^3$, and $\log y/\log\log y$ is increasing, so (53) holds. Second, as $k \le 2\Lambda + 20\log\Lambda_p + 1$, $L_1 + 500 \le 2\Lambda + 2\log\Lambda_p + \log(2\Lambda + 20\log\Lambda_p + 1) + 503.1$, while $\log T_y + 500 \ge \Lambda_p/(51\Lambda\log\Lambda_p)$; the latter exceeds the former for every $\Lambda_p \ge 76\Lambda^3$, since both sides increase, the right side faster, and at $\Lambda_p = 76\Lambda^3$ the right side is $\ge \Lambda^2/(3\log\Lambda + 5) \gg 2\Lambda$. Also $L_1 \ge 2\log\Lambda_p \ge 40$. So $5 \le L_1 \le \log T_y$, and Step 8 applies at $L = L_1$.
- *Size.* Let $\eta \ge e^{-2\Lambda}$ and $L_1 \le e^{\Lambda/2}$, so that $\log G(L_1) \le \Lambda$. If $0.9 c_{\mathrm{zf}}\Lambda_p/(L_1 + 442 + 2\log\Lambda + \log L_1) \ge 3\Lambda + 3\log\Lambda_p + 6$, then the first term of (47) is $\le 18e^{-6} \le 0.045$, and

Rel ≤ 0.05. This inequality holds for $\Lambda_p \ge 76\Lambda^3$: at $\Lambda_p = 76\Lambda^3$ (with the largest L_1, at $\eta = e^{-2\Lambda}$) the left side is at least 1.003 times the right side for every $\Lambda \ge 10^8$, and the left side grows linearly in Λ_p while the right side and L_1 grow logarithmically. If $L_1 > e^{\Lambda/2}$, then $\log \Lambda_p \ge e^{\Lambda/2}/3$ and $G(L_1) \le \Lambda_p$, and the same comparison holds with room to spare. The threshold $76\Lambda^3$ is below $\Lambda_1 - 1 = 100\Lambda^3 - 1$ of Lemma 5.32. □

Remark 5.25. Positivity cannot be dropped: $N = \lfloor u \rfloor + \frac{2}{3}\cos(3\log u + \pi/2)$ has $|N - u| \le 2$, but its $\zeta^\sharp$ has a zero near $1 + 3i$. The theorem uses only the realised N on $[1, y]$, so it holds pathwise on Good($< y$), and the adaptedness of T plays no role. Parts (i) and (ii) are used only in Proposition 5.35(B), whose range starts at $X_{\rm sw} \gg X_{\rm LP}$; part (iii) is used in Lemma 5.32.

5.8. **The bridge, the jump at X_0, and positivity of the sharp reference.**

Proposition 5.26 (the bridge: operator versus sharp)**.** *Let o_u be the supremum of $\mathrm{osc}(T_{\rm c}')$ over windows of length $2.04H(u)$ meeting $[u, 2u)$: the H-window oscillation of the jump-free source. Only the kernels C_n with $n \ge 2$ use it, so it is read at heights $\le x/2 + 1.1H(x)$. For $x \ge 2X_0$ split*

$$(\mathcal{R}T')(x) - \rho^{\rm sh}(x) = \mathrm{BR}_c(x) + J_{X_0}(x), \qquad J_{X_0}(x) := j_0\Big[(\mathcal{R}\mathbf{1}_{[X_0,\infty)})(x) - \sum_n c_n(x)\mathbf{1}[x/n \ge X_0]/n\Big],$$

where $\mathrm{BR}_c := \mathcal{R}T_{\rm c}' - \sum_n c_n T_{\rm c}'(\cdot/n)/n$. *Then*

$$\log x \cdot |\mathrm{BR}_c(x)| \le (5.1C_N + C_S) \sum_{u\ dyadic} \log(2u)\, o_u + 8.4\, C_N E_\theta =: E_{\rm BR}.$$

Proof. Write $\mathrm{BR}_c(x) = \sum_n \frac{1}{n}\big[\int T_{\rm c}'\, dC_n^{\rm nom} - c_n T_{\rm c}'(x/n)\big] + \sum_n \frac{1}{n}\int T_{\rm c}'\, d\tilde{E}_n$, in the notation of §5.5.

- *The Kantorovich–Rubinstein part.* $\int T_{\rm c}'\, dC_n^{\rm nom} - c_n T_{\rm c}'(x/n) = \int (T_{\rm c}' - T_{\rm c}'(x/n))\, dC_n^{\rm nom}$, an integral over $[x/n \pm R_n]$ with $R_n \le 1.02H(x/n)$. Let u be the dyadic scale with $x/n \in [u, 2u)$; then $u \ge X_0/2$, since $x/n \ge X_0^-$. Each half $[x/n - R_n, x/n]$, $[x/n, x/n + R_n]$ has length $\le 1.02 \cdot 2^{\varepsilon} H(u) \le 2.04H(u)$ and meets $[u, 2u)$, so the integral is at most $2o_u\, \mathrm{TV}(C_n^{\rm nom})$. Summing over n with Corollary 5.17(b) (start x), this part is at most $\sum_u 2o_u W_x^{\rm nom}(u) \le 2C_N \sum_u \log(2u)\, o_u/\log x$, which is inside the first term.
- *The near-threshold part.* For τ-near steps, by the Duhamel form in Proposition 5.18 and $|T_{\rm c}'| \le 2$, it is at most $4.2\sum_d (\mathrm{TV}(C_d)/d)\, \theta(x/d) \le 8.4 C_N E_\theta/\log x$ (Lemma 5.20). For X_0-near (terminal) steps, $T_{\rm c}'$ is within o_{X_0} of $T_{\rm c}'(X_0) = 0$ on the support, so their part is at most $o_{X_0} \cdot 2C_N \cdot 1.52 \log(2X_0)/\log x$, which is inside the first term.

Only the oscillation of the smooth source enters, never the H-scale roughness of the noise. The jump $j_0\mathbf{1}_{[X_0,\infty)}$ has no small oscillation, and its whole effect is J_{X_0}, which is treated next. □

5.8.1. *The jump of the target at X_0.* The source T' vanishes below X_0 and equals $j_0 = 1 - \rho_{\rm det}(X_0)$ just above it (§5.1). A jump cannot be handled by the oscillation bounds of Proposition 5.26: the kernels C_n^x with x/n within about $H(X_0)$ of X_0 straddle it, and their weight in a dyadic block is of order $\log X_0/\log x$, which is not small compared with the margin: $\rho^{\rm sh}(x)\log x \ge 0.97 - o(1)$ for $x \ge X_0^{1.01}$, but only $\ge 0.82 - o(1)$ on $[X_0 + H(X_0), X_0^{1.01}]$ (Proposition 5.35 and Lemma 7.3(d)). We therefore treat the jump separately and exactly. Recall $J = j_0\mathbf{1}_{[X_0,\infty)}$, $T_{\rm c}' = T' - J$, and put

$$\rho_1^{\rm sh} := (1 + K_{\rm pt})^{-1}\mathbf{1}_{[X_0,\infty)}, \qquad \text{so that} \qquad \rho^{\rm sh} = \rho_{\rm c}^{\rm sh} + j_0\rho_1^{\rm sh}, \qquad J_{X_0} = j_0\big(\mathcal{R}\mathbf{1}_{[X_0,\infty)} - \rho_1^{\rm sh}\big),$$

with $\rho_{\rm c}^{\rm sh} := \sum_n c_n T_{\rm c}'(\cdot/n)/n$ and J_{X_0} as in Proposition 5.26. Throughout this subsection

$$\Lambda := \log X_0 \ge 10^8, \qquad \delta := H(X_0)/X_0 = e^{8(\log\Lambda)^2 - \Lambda}, \qquad \tau_0 := \log(4.01X_0), \qquad N := y/X_0.$$

By (18), $\varepsilon \le 0.03$. Near X_0 the jitter width is the standard one, and its local exponent $\varepsilon_* := \sup_{X_0 \le v \le X_0^2} d\log H/d\log v \le 16\log\Lambda/\Lambda \le 2.95\cdot 10^{-6}$ ($16\log\log v/\log v$ is decreasing), so that $H(vw)/H(v) \le w^{\varepsilon_*}$ for $X_0 \le v \le vw \le X_0^2$. By Huxley's theorem (80), $j_0 \le 1.0001/\Lambda$ (§5.1).

Lemma 5.27 (The X_0 jump in exact chain form)**.** *Let $y \ge 2X_0$. For a level n write $H_n := H(y/n)$. For a pair (d,w) with $d,w \in \mathcal{N}$, $d,w \ge 2$, $n = dw$, let $D_d := C_d^{y,\mathrm{nom}} - c_d\,\delta_{y/d}$ (a measure of mass 0; $C_d^{y,\mathrm{nom}}$ as in §5.5.1), and define*

$$\mathbb{B}_0 := \{n : \ |y/n - X_0| \le H_n/2\},$$
$$\mathbb{B}_1 := \{(d,w) : \ \text{the step } d \to n \text{ is } X_0\text{-near, i.e. } |y/(dw) - X_0^-| \le R_d/w\},$$
$$\mathbb{B}_2 := \{(d,w) \notin \mathbb{B}_1 : \ |y/n - X_0| \le (1+\vartheta_{X_0})H_n/2 + R_d/w\}.$$

Then, on Good$(<y)$,

$$\big|\mathcal{R}\mathbf{1}_{[X_0,\infty)}(y) - \rho_1^{\mathrm{sh}}(y)\big| \le \mathfrak{B}(y) + \frac{4.3\,C_N E_\theta}{\log y}, \tag{55}$$

$$\mathfrak{B}(y) := \sum_{n\in\mathbb{B}_0}\frac{|c_n|}{2n} + \sum_{(d,w)\in\mathbb{B}_1}\frac{\mathrm{TV}(D_d)}{2dw} + \lambda_0\sum_{(d,w)\in\mathbb{B}_2}\frac{\|D_d\|_{\mathrm{KR}}}{dw\cdot wH_n}. \tag{56}$$

By Lemma 5.20, $E_\theta \le 10^4\Lambda^2 e^{8(\log\Lambda)^2-\Lambda}$, so with $C_N \le 158$

$$4.3\,C_N E_\theta \le 4.3\cdot 158\cdot 10^4\,\Lambda^2 e^{8(\log\Lambda)^2-\Lambda} \le e^{-0.99\Lambda} \qquad (\Lambda \ge 10^8), \tag{57}$$

since $15.8 + 2\log\Lambda + 8(\log\Lambda)^2 \le 0.01\Lambda$. If no step from the start y is τ-near, the last term of (55) may be omitted.

Proof. All kernels below start at y; write $C_n = C_n^y$ for the true and C_n^{nom} for the nominal kernels, so that $\mathcal{R}\mathbf{1}_{[X_0,\infty)}(y) = \sum_n n^{-1}C_n([X_0,\infty))$ by (39), and $\rho_1^{\mathrm{sh}}(y) = \sum_n n^{-1}c_n\mathbf{1}[y/n \ge X_0]$.

Step 1: the decomposition of Proposition 5.18. Write $\tilde E_n := C_n - C_n^{\mathrm{nom}}$. Then $\tilde E_n = S_n + \sum_{d<n} P_{d\to n}^{\mathrm{nom}}\tilde E_d$ with sources $S_n = S_n^\tau + S_n^{X_0}$ on τ-near and X_0-near steps, where $P_{d\to n}$, $P_{d\to n}^{\mathrm{nom}}$ are the step operators of §5.5.5. By Lemma 5.12, the X_0-near sources sit at terminal levels and are never transferred, so

$$C_n = C_n^{\mathrm{nom}} + S_n^{X_0} + \tilde E_n^\tau, \qquad \tilde E_n^\tau := \sum_{n'|n}\mathcal{T}_{n'\to n}^{\mathrm{nom}} S_{n'}^\tau,$$

where $\mathcal{T}_{n'\to n}^{\mathrm{nom}}$ is the nominal transfer (the signed sum over nominal chains from level n' to n). On a step $d \to n$ the true and nominal filters at the centre c are $\varphi_\tau(c)\varphi(c)$ and $\kappa_\tau\kappa_{X_0}$, where $\varphi(c) := \mathbf{1}[c \ge X_0^-]$, $\kappa_{X_0} := \mathbf{1}[y/n \ge X_0^-]$, and $\varphi_\tau, \kappa_\tau$ are the τ-filters. As in the proof of Proposition 5.18, on X_0-near steps the whole difference goes to $S_n^{X_0}$, and on the other steps the difference $(\varphi_\tau - \kappa_\tau)\kappa_{X_0}$ goes to S_n^τ; so $S_n^{X_0} = -\sum_{d\to n\ X_0\text{-near}} \mathcal{Q}\big((\varphi_\tau\varphi - \kappa_\tau\kappa_{X_0})\,D_w C_d\big)$, where $\mathcal{Q}$ places the window $\mathrm{box}_{H(c)}$ at each centre c. Split $C_d = C_d^{\mathrm{nom}} + (C_d - C_d^{\mathrm{nom}})$ and write $S_n^{X_0} = S_n^{X_0,\mathrm{nom}} + S_n^{X_0,\tau}$ accordingly.

Step 2: the nominal part, level by level. For a centre c let U_c be uniform on $[-H(c)/2, H(c)/2)$ and $G(c) := \mathbb{P}(c + U_c \ge X_0)$, so that $\int \mathbf{1}_{[X_0,\infty)}\,d(\mathcal{Q}\mu) = \int G\,d\mu$ for every measure μ of centres. Near X_0, $G(c) = \min\big(1, \max\big(0, \frac12 + (c - X_0)/H(c)\big)\big)$, which is Lipschitz with constant $\le (1 + 2\vartheta_{X_0})/H_n \le \lambda_0/H_n$ on the support of D_wD_d (width variation, §5.5.1), and $G(y/n) = \mathbb{P}(y/n + U \ge X_0)$ with U uniform on $[-H_n/2, H_n/2)$. By (38) for the nominal kernels and $D_wC_d^{\mathrm{nom}} = c_d\delta_{y/n} + D_wD_d$,

$$C_n^{\mathrm{nom}}([X_0,\infty)) = c_n\,G(y/n) - \sum_{d|n,\,d<n}\kappa_{d\to n}\int G\,d(D_wD_d).$$

Adding

$$S_n^{X_0,\mathrm{nom}}([X_0,\infty)) = -\sum_{X_0\text{-near}} \int G\,(\varphi_\tau\varphi - \kappa_\tau\kappa_{X_0})\,d(D_w C_d^{\mathrm{nom}}),$$

and using $(\varphi_\tau\varphi-\kappa_\tau\kappa_{X_0})\,c_d\delta_{y/n}=0$ (the filters at the nominal point equal their nominal values),

$$\begin{aligned} &C_n^{\mathrm{nom}}([X_0,\infty)) + S_n^{X_0,\mathrm{nom}}([X_0,\infty)) - c_n\mathbf{1}[y/n\ge X_0] \\ &\qquad = c_n\big(G(y/n)-\mathbf{1}[y/n\ge X_0]\big) - \sum_{d\mid n,\,d<n}\int G\psi_{d\to n}\,d(D_wD_d), \end{aligned} \tag{58}$$

where $\psi_{d\to n}:=\varphi_\tau\varphi$ on X_0-near steps and $\psi_{d\to n}:=\kappa_\tau\kappa_{X_0}$ (a constant) on the others. Since $D_1=0$, only $d\ge 2$ occurs.

- $|G(y/n)-\mathbf{1}[y/n\ge X_0]|\le\frac12$, and it vanishes unless $|y/n-X_0|\le H_n/2$, i.e. unless $n\in\mathbb{B}_0$.
- D_wD_d has mass 0, total variation $\mathrm{TV}(D_d)$, and support of radius R_d/w around y/n. The function $G\psi_{d\to n}$ takes values in $[0,1]$, so $|\int G\psi_{d\to n}\,d(D_wD_d)|\le\frac12\,\mathrm{TV}(D_d)$ (subtract the constant $\frac12$).
- If the step is not X_0-near, then $G\psi_{d\to n}=\kappa_\tau\kappa_{X_0}G$, and G is Lipschitz with constant $\le\lambda_0/H_n$; by Lemma 5.13(K2), $|\int G\,d(D_wD_d)|\le\lambda_0\|D_wD_d\|_{\mathrm{KR}}/H_n=\lambda_0\|D_d\|_{\mathrm{KR}}/(wH_n)$. If moreover $|y/n-X_0|>(1+\vartheta_{X_0})H_n/2+R_d/w$, the support lies where G is constant (there $H(c)\le(1+\vartheta_{X_0})H_n$) and the term vanishes.

Multiplying (58) by $1/n=1/(dw)$ and summing over n gives at most $\mathfrak{B}(y)$.

Step 3: the τ-part. What remains is

$$\sum_n n^{-1}\big(\tilde E_n^\tau + S_n^{X_0,\tau}\big)([X_0,\infty)),$$

which is at most $\sum_n\mathrm{TV}(\tilde E_n^\tau)/n+\sum_n\mathrm{TV}(S_n^{X_0,\tau})/n$. As in the proof of Proposition 5.18 (the ∞-norm step), the first sum is $\le 2.04\lambda_0\sum_d(\mathrm{TV}(C_d)/d)\,\theta(y/d)\le 4.1\,C_NE_\theta/\log y$, by Corollary 5.17(c) at every start, the bound on $W_y^{\mathrm{op,nt}}$ in Proposition 5.18 (the parents of τ-near steps are non-terminal), and Lemma 5.20. The second is $\le\sum_d(\mathrm{TV}(\tilde E_d)/d)\,\eta_{X_0}(y/d)$ with η_{X_0} as in the proof of Proposition 5.18; with the block bound $\sum_{y/d\in[v,2v)}\mathrm{TV}(\tilde E_d)/d\le 2\lambda_0C_N^2E_\eta\log(2v)/\log y$ from step (ii) of that proof and $\sum_v\log(2v)\sup\eta_{X_0}\le 1.52\log(2X_0)$, it is

$$\le 3.1\,C_N^2\tau_0E_\eta/\log y\le 0.1\,C_NE_\theta/\log y,$$

because $\tilde\eta\le\theta/\log^2 X_0^-$ pointwise. Together, the τ-part is at most $4.3\,C_NE_\theta/\log y$; if no step is τ-near, then $S^\tau=0$, $\tilde E^\tau=0$ and $C_d=C_d^{\mathrm{nom}}$ for every parent of an X_0-near step, so the τ-part vanishes. □

Remark 5.28. A cruder bound by the level sum $\sum_n(\mathrm{TV}(C_n)+|c_n|)/n$ over the band does not suffice: at $n_0=6$ it is about 0.74, while $\mathfrak{B}(6X_0)\le 5/12$ (Proposition 5.31). The gain comes from three facts used above: the direct chain ($d=1$) has no centre smear; the defect of a mass-0 measure against a $[0,1]$-valued function is at most half its total variation; and the window average contributes only half a jump.

Two counting tools. The ranges below use two elementary bounds. The first is a local count on the realised semigroup.

Lemma 5.29 (Reciprocal sums over short intervals)**.** *On* Good$(<y)$*, for every* $1\le M\le y/2$ *and* $0<a\le 0.01$*,*

$$\sum_{e\in\mathcal{N}\cap M[1-a,1+a]}\frac{1}{e}\le 28\,a+7M^{-3/4}.$$

Proof. Let $I := M[1-a, 1+a] \subset [M/2, 2M]$; it suffices to show $\#(\mathcal{N} \cap I) \le 13.5|I| + K^*(M)$ with $K^*(M)/M \le 6M^{-3/4}$, since then $\sum_{e \in I} 1/e \le (27aM + K^*(M))/(M(1-a)) \le 28a + 7M^{-3/4}$.

- If $M < X_0/4$, then $I \subset [1, X_0^-)$, where $\mathcal{N} = \mathbb{N}$, and $\#(\mathcal{N} \cap I) \le |I| + 1$; take $K^*(M) := 1$.
- If $M \ge 2X_0$, cut $I \subset [P, 4P]$ ($P = M/2 \ge X_0$) into $\lceil |I|/6H(P) \rceil$ pieces J of length $\le 6H(P)$. By Corollary 6.3(ii), $N(J) \le T(J) + 11.4|J| + K_{\mathrm{loc}}(P) + K_{\mathrm{loc}}(2P)$, and under the target (5), $T(J) \le 2|J| + 2 + \#\mathcal{A}(J)$. Since $K_{\mathrm{loc}} \le H/100$, summing over the pieces gives $\#(\mathcal{N} \cap I) \le 13.42|I| + K_{\mathrm{loc}}(P) + K_{\mathrm{loc}}(2P) + 2 + \#\mathcal{A}(I)$. By Lemma 6.25, each window $[u, u + h(u)]$, $h(u) = u^{\theta/2}$, holds at most $h(u)L_u^{-4}$ absorbed elements; covering I by $\lceil |I|/h(M/2) \rceil$ such windows, $\#\mathcal{A}(I) \le 0.01|I| + (2M)^{\theta/2}$. As $K_{\mathrm{loc}}(P) + K_{\mathrm{loc}}(2P) + 2 \le M^{1/8}$ and $\theta < \frac{1}{2}$, take $K^*(M) := 2M^{1/4}$.
- If $X_0/4 \le M < 2X_0$, apply the first case to $I \cap [1, X_0^-)$ and the second, with $P := X_0$, to $I \cap [X_0, \infty) \subset [X_0, 4X_0]$; in between, $\mathcal{N} \cap [X_0^-, X_0)$ consists of integers and of at most $2H(X_0)$ jittered primes (§5.1). This gives $K^*(M) := 4X_0^{1/4} \le 4(4M)^{1/4}$.

In all cases $K^*(M)/M \le 6M^{-3/4}$. □

The second is a Rankin bound for smooth elements, valid at every height.

Lemma 5.30 (Rankin bound)**.** *Assume* (21) *on* $[1, z]$ *(for instance, work on* Good$(< z)$*; cf. Lemma 5.1). Let* $z \ge e^{10}$, $\ell \ge 0$, *and* $v(\ell, z) := \sum_{n \in \mathcal{N} \cap (e^\ell, 2e^\ell],\, P^+(n) \le z} 1/n$. *Then* $v(\ell, z) \le e^{14} (\log z)\, e^{-\ell/\log z}$.

Proof. Put $\beta := 1/\log z$. Since $\mathcal{N}$ is the free semigroup on $\mathcal{P}$, $v(\ell, z) \le e^{-\beta\ell} \sum_{P^+(n) \le z} n^{\beta - 1} = e^{-\beta\ell} \prod_{p \le z} (1 - p^{\beta-1})^{-1}$. For $p \le z$, $p^\beta \le e$, and $e^{\beta \log p} - 1 \le (e-1)\beta \log p$. So

$$\log \prod_{p \le z} (1 - p^{\beta-1})^{-1} \le \sum_{p \le z} \frac{1}{p} + (e-1)\beta \sum_{p \le z} \frac{\log p}{p} + \sum_p \frac{p^{2\beta-2}}{1 - p^{\beta-1}}.$$

By Lemma 5.1(iv) and (iii), the first two sums are $\le \log\log z + 4.1$ and $\le (e-1)(1 + 0.43/\log z) \le 1.8$; the last is ≤ 1.3: the primes $p \le 7$ contribute ≤ 0.95 (using $p^\beta \le p^{1/10}$), and those with $p \ge 11$ contribute $\le 1.33\, e^2 \sum_{p \ge 11} p^{-2} \le 0.31$. Hence the product is $\le e^{7.2} \log z$, and the claim follows with room. □

The three ranges. Put

$$j_{X_0}(x) := \sup_{y \in W_x} \log y \cdot j_0\Big(\mathfrak{B}(y) + \frac{4.3\, C_N E_\theta}{\log y}\Big), \tag{59}$$

so that $\log y \cdot |J_{X_0}(y)| \le j_{X_0}(x)$ for every $y \in W_x$, by Lemma 5.27.

Proposition 5.31 (The X_0 band in three ranges)**.** *Let* $\Lambda \ge 10^8$, $x \ge 2X_0$ *and* $N := x/X_0$. *On* Good$(< x)$*:*

(i) (small) if $N \le e^{0.01\Lambda}$, *then* $j_{X_0}(x) \le 0.4209$ *(computer-assisted; see Appendix B.3);*
(ii) (middle) if $e^{0.01\Lambda} \le N \le e^{600\Lambda^4}$, *then* $\mathfrak{B}(y) \le 200\delta + 400N^{-1/10}$ *for* $y \in W_x$, *and* $j_{X_0}(x) \le e^{-0.0009\Lambda}$;
(iii) (huge) if $N \ge e^{600\Lambda^4}$, *then* $\mathfrak{B}(y) \le (400\delta + e^{-60})\, \tau_0/\log N$ *for* $y \in W_x$, *and* $j_{X_0}(x) \le e^{-59}$.

The three ranges cover $x \ge 2X_0$ *for every* $\Lambda \ge 10^8$.

Throughout the proof, $y \in W_x$, and we write N also for y/X_0, which differs from x/X_0 by a factor $1 + O(H(x)/x)$. We use the following consequences of Theorem 5.15 for the nominal

kernels (with $H_e/H_d \le (d/e)^\varepsilon$ from (18), and $|c_e| \le \mu^2(e)$):

$$\begin{aligned} \tfrac{1}{2}\,\mathrm{TV}(D_d) &\le |c_d| + \tfrac{\lambda_0}{4} \sum_{m|d,\,1<m<d} H_{\mathrm{ord}}(m) m^{\varepsilon-1} |c_{d/m}|, \\ \frac{\lambda_0 \|D_d\|_{\mathrm{KR}}}{w H_n} &\le \frac{\lambda_0^2 w^{\varepsilon-1}}{4} \sum_{m|d,\,m<d} H_{\mathrm{ord}}(m) m^{\varepsilon-1} |c_{d/m}| \quad (n = dw). \end{aligned} \tag{60}$$

For the first, $\mathrm{TV}(D_d) \le |c_d| + \mathrm{TV}(C_d^{\mathrm{nom}})$; for the second, $H_e/H_n \le (n/e)^\varepsilon = (wm)^\varepsilon$ with $m = d/e$. Since $\lambda_0^2 \le 1 + 7\delta$, the factors λ_0, λ_0^2 change the numerical constants below by a relative amount $\le 7\delta \le e^{-9\cdot 10^7}$, and we absorb them. Consequently, for every set I of levels,

$$\begin{aligned} \sum_{d\in I} \frac{\mathrm{TV}(D_d)}{2d} &\le \sum_{d\in I} \frac{|c_d|}{d} + \frac{\lambda_0}{4} \sum_{m\ge 2} H_{\mathrm{ord}}(m) m^{\varepsilon-2} \sum_{e\in I/m} \frac{|c_e|}{e}, \\ \sum_{d\in I} \frac{\lambda_0 \|D_d\|_{\mathrm{KR}}}{d\, w H_{dw}} &\le \frac{\lambda_0^2 w^{\varepsilon-1}}{4} \sum_{m\ge 1} H_{\mathrm{ord}}(m) m^{\varepsilon-2} \sum_{e\in I/m} \frac{|c_e|}{e}. \end{aligned} \tag{61}$$

Proof of (i): the small range. A single level. Let $N \le e^{0.01\Lambda}$. If $n \in \mathbb{B}_0$ then $|n - N| \le nH_n/(2X_0) \le 0.51\,(1.01N)\delta$. If $(d,w) \in \mathbb{B}_1 \cup \mathbb{B}_2$ then $|y/n - X_0| \le H_n/2 + R_d/w + (X_0 - X_0^-) \le 1.6H(X_0)$, so $|n - N| \le 2N\delta$. Since $N\delta \le e^{0.01\Lambda + 8(\log\Lambda)^2 - \Lambda} < 10^{-10}$, every level occurring in $\mathfrak{B}(y)$ is the integer n_0 nearest to N, and $n_0 < X_0$. Below X_0^- the semigroup is $\mathbb{N}$, so all divisors d of n_0 are rational integers, all cofactors are rigid, and no step below level n_0 is τ-near or X_0-near; in particular the τ-term of (55) vanishes.

The bound $B(n_0)$. For $2 \le n$, put

$$B(n) := \frac{1}{n}\Big[\tfrac{1}{2}\mu^2(n) + \tfrac{1}{2} \sum_{d|n,\,1<d<n} \Big(2\mu^2(d) + \tfrac{1}{2}\lambda \frac{X(d)}{d}\Big)\Big],$$

$$X(d) := \sum_{e|d,\,1<e<d} \mu^2(e)\, e\, H_{\mathrm{ord}}(d/e), \qquad \lambda := 1.0001.$$

Each pair $(d, n_0/d)$ lies in at most one of $\mathbb{B}_1, \mathbb{B}_2$. By (60) with ε_* in place of ε and $\lambda_0 (d/e)^{\varepsilon_*} \le \lambda$ for $d \le 10^6$, the $\mathbb{B}_1$-weight of d is $\le \mu^2(d) + \frac{1}{4}\lambda X(d)/d$; the $\mathbb{B}_2$-weight is $\le \frac{1}{8}\lambda(\mu^2(d) + X(d)/d)$, which is smaller. Hence $\mathfrak{B}(y) \le B(n_0)$ for $n_0 \le 10^6$.

Computation, $n \le 10^6$. The certificate of Appendix B.3 computes μ^2, H_{ord}, X and the divisor sums exactly in integers, and evaluates $B(n)$ as a sum of at most $\tau(n)$ nonnegative terms, inflated by $1 + 10^{-9}$. The maximum over $2 \le n \le 10^6$ is $B(6) = (\frac{1}{2} + \frac{1}{2}(2+2))/6 = 5/12$; the next values are $B(12) = 0.2778$ and $B(2) = B(4) = B(10) = \frac{1}{4}$, and $\sup_{n\ge 100} B(n) = 0.0881$.

Analytic tail, $10^6 < n \le X_0^{0.01}$. Here we use $H_e/H_d \le (d/e)^{\varepsilon_*}$ directly instead of λ (all heights are $\le X_0^{1.01}$). Write $d = em$ with $e > 1$ squarefree, $m \ge 2$. Then $\sum_{d|n} \frac{1}{n} \sum_{e|d} \mu^2(e) H_{\mathrm{ord}}(m) m^{\varepsilon_* - 1}$ is a sum over factorisations $n = mk$ ($k \ge 2$) of $H_{\mathrm{ord}}(m) m^{\varepsilon_* - 2} \cdot 2^{\omega(k)} k^{-1}$, which for $\sigma := 0.2$ is at most

$$n^{-\sigma} \sum_{m\ge 2} H_{\mathrm{ord}}(m) m^{\sigma + \varepsilon_* - 2} \cdot \sup_{k\ge 2} 2^{\omega(k)} k^{\sigma - 1} = n^{-0.2}\, A(1.8 - \varepsilon_*) \cdot 2^{0.2},$$

where $A(s) := 1/(2 - \zeta(s)) - 1$ and the supremum is attained at $k = 2$ (since $2p^{-0.8} < 1$ for $p \ge 3$). In ball arithmetic, $A(1.8 - 3\cdot 10^{-6}) \le 7.4915$ (the value is 7.49142; Appendix B.4). With $2^{\omega(n)} \le 1.64\sqrt{n}$ (the maximum of $\prod_{p|n} 2p^{-1/2}$ is at $n = 6$),

$$B(n) \le \frac{\frac{1}{2} + 1.64\sqrt{n}}{n} + \frac{1}{4} \cdot 7.4915 \cdot 2^{0.2}\, n^{-0.2} \le \frac{\frac{1}{2} + 1.64\sqrt{n}}{n} + 2.152\, n^{-0.2} \le 0.14 \qquad (n > 10^6).$$

Conclusion. $\log y \le \Lambda + \log N + 10^{-10}$, so $\log y \cdot j_0 \le 1.0001\,(1.01 + 10^{-18}) \le 1.010102$, and $4.3 C_N E_\theta \le e^{-0.99\Lambda}$ by (57). So $j_{X_0}(x) \le 1.010102 \cdot \frac{5}{12} + e^{-0.99\Lambda} \le 0.42088 \le 0.4209$. □

Proof of (ii): the middle range. Let $e^{0.01\Lambda} \le N \le e^{600\Lambda^4}$. Every set below is an interval of relative half-width ≤ 0.01, so Lemma 5.29 applies to it, and we use $|c_e| \le 1$.

(α) The levels $\mathbb{B}_0$. Since $H_n \le H(1.01X_0) \le 1.01 H(X_0)$ there, $\mathbb{B}_0 \subset N[1 - 0.51\delta, 1 + 0.51\delta]$, so this part of $\mathfrak{B}(y)$ is $\le \frac12(28 \cdot 0.51\delta + 7N^{-3/4}) \le 7.2\delta + 3.5N^{-3/4}$.

(β_1) X_0-near pairs with $w \le \sqrt{N}$. For fixed w the condition $|y/d - wX_0^-| \le R_d \le 1.02H(y/d)$ and (18) give $d \in I_w := M_w[1 - 1.01\delta_w, 1 + 1.01\delta_w]$, with $M_w := y/(wX_0^-)$ and $\delta_w := 1.03 w^{\varepsilon-1}\delta$. By (61) and Lemma 5.29,

$$\sum_{d \in I_w} \frac{\mathrm{TV}(D_d)}{2d} \le 28.3\delta_w\Big(1 + \frac{A_0}{4}\Big) + 7M_w^{-3/4} + \frac74 M_w^{-3/4} \sum_{m \le 2M_w} H_{\mathrm{ord}}(m) m^{\varepsilon - 1.25} \le 43\,\delta_w + 41\,M_w^{-0.2},$$

using $A_0 = A_0(\varepsilon) \le 2.07$ and $\sum_{m \le X} H_{\mathrm{ord}}(m) m^{\varepsilon-1.25} \le X^{0.55} A(1.8 - \varepsilon) \le 13.1 X^{0.55}$, where $A(1.77) = 13.08$ (the abscissa of $\sum H_{\mathrm{ord}}(m) m^{-s}$ is $1.7286 < 1.77$). Weighting by $1/w$ and summing over $2 \le w \le \sqrt{N}$, with $M_w \ge N/w$ and $\sum_{w \ge 2} w^{\varepsilon-2} \le \zeta(1.97) - 1 = 0.674$, gives $\le 43 \cdot 1.03 \cdot 0.674\,\delta + 41 N^{-0.2} \cdot 5 N^{0.1} \le 30\,\delta + 205\,N^{-0.1}$.

(β_2) Pairs in $\mathbb{B}_2$ with $w \le \sqrt{N}$. Here $|y/d - wX_0| \le wH_n/2 + R_d \le 1.04\, wH(X_0)$, so d ranges over an interval of relative half-width $\le 1.05\delta$ around $y/(wX_0)$. By (61) and Lemma 5.29, for fixed w this part is $\le \frac14 w^{\varepsilon-1}\big(29.4\delta(1 + A_0) + 142(w/N)^{0.2}\big)$, and summing over w gives $\le \frac14 \cdot 29.4 \cdot 3.07 \cdot 0.674\,\delta + 36 N^{-0.2} \sum_{w \ge 2} w^{-1.77} \le 15.3\,\delta + 34\,N^{-0.2}$.

(γ) Pairs with $w > \sqrt{N}$. Then $d < 1.01\sqrt{N}$; fix d and count w. For X_0-near pairs the w-window $\{w : |y/d - wX_0^-| \le R_d\}$ has relative half-width $\le 1.02\delta(d/N)^{1-\varepsilon}$ at height $M' := y/(dX_0^-) \ge 0.98\sqrt{N}$, so $\sum_w 1/w \le 29\delta(d/N)^{1-\varepsilon} + 7M'^{-3/4}$. For pairs in $\mathbb{B}_2$ the window has relative half-width $\le 1.05\delta$, and by (60) the weight carries $\frac14 w^{\varepsilon-1} \le \frac14 N^{-0.485}$. By Theorem 5.15, $\sum_{d \le D} \mathrm{TV}(D_d)/(2d) \le 1.52(\log D + 1)$ and $\sum_{d \le D} d^{-1} \sum_{m \mid d} H_{\mathrm{ord}}(m) m^{\varepsilon-1} |c_{d/m}| \le 3.07(\log D + 1)$. Hence this part is $\le (\log N + 1)\big(1.52(29\delta N^{-0.485} + 7.2N^{-3/8}) + 0.77 N^{-0.485}(30\delta + 7.2N^{-3/8})\big) \le N^{-1/4}$.

Total. $\mathfrak{B}(y) \le 53\,\delta + 3.5N^{-3/4} + 205N^{-0.1} + 34N^{-0.2} + N^{-1/4} \le 200\,\delta + 400\,N^{-1/10}$. Finally $\log y \le 601\Lambda^4$, so $\log y \cdot j_0 \le 1.0001 \cdot 601\Lambda^3$, and with $N \ge e^{0.01\Lambda}$ and $4.3 C_N E_\theta \le e^{-0.99\Lambda}$ by (57),

$$j_{X_0}(x) \le 1.0001 \cdot 601\Lambda^3\big(200\, e^{8(\log\Lambda)^2 - \Lambda} + 400\, e^{-0.001\Lambda}\big) + e^{-0.99\Lambda} \le e^{-0.0009\Lambda},$$

because $0.0001\Lambda \ge 12.4 + 3\log\Lambda$ for $\Lambda \ge 10^8$. □

Proof of (iii): the huge range. Let $J := \log N \ge 600\Lambda^4$ and $W_1 := e^{100}$. The main pieces use Lemma 5.32 below, with $J_1 = 300\Lambda^4$. Recall from Lemma 5.5 that $c_e(y) \ne 0$ forces e to be squarefree and $4v$-dense whenever $y/e \in [v, 2v)$, including the condition $p_1 \le 4v$ on its smallest prime. Applying this with v just above $y/(2e)$, so that $4v \le 2.03\, sX_0$, an element e with $y/e \le 1.01\, sX_0$ is t-dense with $t = 4.01\, sX_0$.

(α) The levels $\mathbb{B}_0$. $\mathbb{B}_0 \subset [N(1 - 0.51\delta), N] \cup [N, N(1 + 0.51\delta)]$. Lemma 5.32 with $t = 4.01X_0$ and $\eta = 0.52\delta \in [e^{-2\Lambda}, \frac12]$ gives $\frac12 \cdot 2 \cdot 61.1(1 + 2.1\tau_0/J) \cdot 0.52\delta\,\tau_0/(J - 1) \le 32\,\delta\tau_0/J$.

(β_1) X_0-near pairs with $w \le W_1$. As in (ii), $d \in I_w = M_w[1 - 1.01\delta_w, 1 + 1.01\delta_w]$ with $\delta_w = 1.03 w^{\varepsilon-1}\delta \ge \delta e^{-100} \ge e^{-2\Lambda}$ and $\log M_w \ge J - 101$. In (61), the element d is t_w-dense with $t_w := 4.01 wX_0$, and the element $e \in I_w/m$ is t_{wm}-dense with $t_{wm} := 4.01 wmX_0$.

- For $m \le N^{1/100}/w$: the height of I_w/m is $\ge N^{0.99}/1.01$, so its logarithm J' is $\ge 0.99J - 1 \ge J_1$; and $t_{wm} \le 4.01 N^{1/100} X_0 \le (N^{0.99})^{1/10}$ because $X_0 \le N^{1/(600\Lambda^3)}$. So Lemma 5.32

applies, with $\tau_{wm}/J' \le 0.0103$ and factor $1 + 2.1\tau_{wm}/J' \le 1.022$: $\sum_{e\in I_w/m} |c_e|/e \le 2\cdot 61.1\cdot 1.022\cdot 1.02\,\delta_w\,\tau_{wm}/J'$, with $\tau_{wm} = \tau_0 + \log w + \log m$.

- For $m > N^{1/100}/w$: Lemma 5.29 gives $\sum_{e\in I_w/m} 1/e \le 29\delta_w + 7$, and

$$\sum_{m>N^{1/100}/w} H_{\mathrm{ord}}(m) m^{\varepsilon-2} \le (N^{1/100}/W_1)^{-0.2} A(1.77) \le e^{22.6} N^{-0.002}.$$

With $A_0 \le 2.07$ and $A_1 = A_1(\varepsilon) \le 9.42$, and $J' \ge 0.99J - 1$,

$$\sum_{d\in I_w} \frac{\mathrm{TV}(D_d)}{2d} \le \frac{127.4\,\delta_w}{0.99J-1}\Big(\tau_w\big(1+\tfrac{A_0}{4}\big) + \tfrac{A_1}{4}\Big) + e^{26}N^{-0.002}, \qquad \tau_w := \tau_0 + \log w.$$

Weighting by $1/w$ and summing over $2 \le w \le W_1$, with

$$\sum_{w\ge 2} w^{-1.97} = 0.674 \quad \text{and} \quad \sum_{w\ge 2} w^{-1.97}\log w = -\zeta'(1.97) \le 1.0001$$

(ball arithmetic; Appendix B.4), gives

$$\le \frac{132.5\,\delta}{0.99J-1}\big(0.674(1.518\tau_0 + 2.36) + 1.518\cdot 1.0001\big) + e^{31}N^{-0.002} \le 138\,\delta\tau_0/J.$$

(β_2) Pairs in $\mathbb{B}_2$ with $w \le W_1$. Now d ranges over an interval of relative half-width $\le 1.05\delta$ around $y/(wX_0)$, and by (61) with the factor $\frac14 w^{\varepsilon-1}$ and the same use of Lemma 5.32 (now including $m = 1$), this part is $\le \frac14 \sum_{2\le w\le W_1} w^{\varepsilon-2}\cdot \frac{2\cdot 61.1\cdot 1.022\cdot 1.06\,\delta}{0.99J-1}\big(\tau_w(1+A_0)+A_1\big) + e^{31}N^{-0.002} \le 72\,\delta\tau_0/J$.

(γ') Pairs with $w > W_1$. Fix d; then $y/d > W_1X_0/2$. Group d by the dyadic block $y/d \in [v, 2v)$, $v \ge W_1X_0/2$. For X_0-near pairs, the w-window has relative half-width $\le 1.02H(v)/v \le 1.02\delta(X_0/v)^{1-\varepsilon}$ at height $\ge v/X_0$, so $\sum_w 1/w \le 29\delta(X_0/v)^{0.97} + 7(X_0/v)^{3/4}$ by Lemma 5.29. For pairs in $\mathbb{B}_2$ the window has relative half-width $\le 1.05\delta$, and the weight carries $\frac14 w^{\varepsilon-1} \le \frac14 (X_0/v)^{0.97}$. By Proposition 5.18 and Corollary 5.9 (and Corollary 5.17 for the m-sums), the blocks satisfy

$$\sum_{y/d\in[v,2v)} \mathrm{TV}(D_d)/(2d) \le (C_N + C_S)\log(2v)/\log y$$

$$\text{and} \quad \sum_{y/d\in[v,2v)} d^{-1}\sum_{m\mid d} H_{\mathrm{ord}}(m) m^{\varepsilon-1}|c_{d/m}| \le C_S\big((1+A_0)\log(2v) + A_1\big)/\log y.$$

Summing the geometric series over $v = 2^j W_1X_0/2$,

$$(\gamma') \le \frac{2.5\,(C_N+C_S)(\tau_0+102)}{\log y}\Big(29\delta\big(\tfrac{2}{W_1}\big)^{0.97} + 7\big(\tfrac{2}{W_1}\big)^{3/4}\Big) + \frac{2.5\cdot 0.77\,C_S(\tau_0+102)}{\log y}\big(\tfrac{2}{W_1}\big)^{0.97}(30\delta+7) \le e^{-60}\,\frac{\tau_0}{J}.$$

Total. $\mathfrak{B}(y) \le (32 + 138 + 72)\,\delta\tau_0/J + e^{-60}\tau_0/J \le (400\delta + e^{-60})\,\tau_0/\log N$. Since $\log y \le J + \Lambda + 1$ and $\tau_0 \le 1.0001\Lambda$, $\log y \cdot j_0 \cdot \tau_0/J \le 1.0003$, so with $4.3C_NE_\theta \le e^{-0.99\Lambda}$ by (57), $j_{X_0}(x) \le 1.0003(400\delta + e^{-60}) + e^{-0.99\Lambda} \le e^{-59}$. □

Contiguity. The three ranges meet at $N = e^{0.01\Lambda}$ and $N = e^{600\Lambda^4}$ and so cover $x \ge 2X_0$ for every $\Lambda \ge 10^8$. No step uses a short-interval count at relative width below $e^{-2\Lambda}$, and no sum of the form $\sum_w 1/(w\log(N/w))$ arises: the pairs with $w > W_1$ are counted by fixing d.

A short-interval Saias bound. The huge range rests on the following lemma. Saias' asymptotic for integers with t-dense divisors has relative error $O(1/\log t)$, which is far larger than the relative widths $\eta \approx \delta$ needed here, so it cannot be used directly. Instead we combine the joint "t-dense and smooth" profile of Lemma 5.8 with the short-window prime counts of Theorem 5.24(47).

Lemma 5.32 (Short-interval Saias bound)**.** *Put* $\Lambda_1 := 100\Lambda^3$ *and* $J_1 := 300\Lambda^4$. *Let* $J := \log N \ge J_1$, $t \in [X_0, N^{1/10}]$, $\tau := \log t$ *and* $\eta \in [e^{-2\Lambda}, \frac12]$. *Then, on* Good$(< y)$ *for every* $y \ge N$,

$$S_t\big([N, N(1+\eta)]\big) := \sum_{\substack{n \in \mathcal{N} \cap [N, N(1+\eta)] \\ n \text{ squarefree, } t\text{-dense}}} \frac{1}{n} \le 61.1\Big(1 + \frac{2.1\,\tau}{J}\Big)\frac{\eta\,\tau}{J}. \tag{62}$$

The trivial bound is about 27η; (62) is the Saias density τ/J inside an interval of relative width η, with no loss in η.

Proof. Let $I := [N, N(1+\eta)]$ and let $n \in I$ be squarefree and t-dense. Since $n > t$, n is not prime. Put $p := P^+(n)$ and $m := n/p$. Then m is squarefree and t-dense with $P^+(m) < p \le tm$ (Theorem 5.6, Step 1 of its proof); in particular $p^2 \le t\,mp = tn$, so $m = n/p \ge \sqrt{n/t} \ge \sqrt{N/t}$ (this uses the lower end $n \ge N$ of I). Put $P_1 := e^{\Lambda_1}$ and split according to $p < P_1$ or $p \ge P_1$.

(B) $p < P_1$. Then n is P_1-smooth, and $I \subset (N/2, N] \cup (N, 2N]$. By Lemma 5.30, this part is $\le 2e^{14}\Lambda_1 e^{-(J-1)/\Lambda_1}$. This is $\le 10^{-6}\eta\tau/J$, because $\eta \ge e^{-2\Lambda}$, $\tau \ge \Lambda$, and $(J-1)/\Lambda_1 \ge 3\Lambda - 10^{-20} \ge 2\Lambda + \log J + 3\log\Lambda + 34$ at $J = J_1$, while the left side grows linearly and the right logarithmically in J.

(A) $p \ge P_1$: the prime sum. For fixed m, p runs over primes in $I/m = [X, X(1+\eta)]$ with $X := N/m$ and $\Lambda_p := \log X \ge \Lambda_1 - 1$; in particular $X \ge X_{\mathrm{LP}}$, since $\log X_{\mathrm{LP}} = O(\Lambda^2 \log^2\Lambda)$. Since $m \ge 2$ and $\eta \le \frac12$, also $X(1+2\eta) \le N \le y$, which is the hypothesis $v(1+2\eta) \le y$ of Theorem 5.24(iii) with $v = X$. We apply Theorem 5.24(iii), which is stated for $\Lambda_p \ge 76\Lambda^3$; here $\Lambda_p \ge \Lambda_1 - 1 \ge 76\Lambda^3$. There Rel $\le 0.05 + \Lambda_p^{-2} \le 0.1$ as soon as the exponent in (47) is at least $\log(360\,G(L_1)\Lambda_p^3\eta^{-1})$. Put $\ell_p := \log\Lambda_p$, so that $k \le \log(1/\eta) + 20\ell_p + 1 \le 2\Lambda + 20\ell_p + 1$ and $L_1 = 2\ell_p + \log(1/\eta) + \log k + 3.08 \le 2\Lambda + 2\ell_p + \log(2\Lambda + 20\ell_p + 1) + 3.08$, and write $D := L_1 + 442 + 2\log\Lambda + \log L_1$ for the denominator of the exponent. Since $c_{\mathrm{zf}} \ge 0.088/\Lambda$, the exponent is at least $0.0792\,\Lambda_p/(\Lambda D)$.

- *Case $\ell_p \le e^{\Lambda/2}/3$.* Then $L_1 \le e^{\Lambda/2}$, so $\log G(L_1) \le \Lambda$ by Theorem 5.24(iii). Write $\Lambda_p = 76\Lambda^3 r$ with $r \ge 1$, and $\ell_0 := \log(76\Lambda^3) \le 0.001\Lambda$. Then $\ell_p = \ell_0 + \log r$; since k and L_1 are at most $3\Lambda(1 + \log r)$ and $1 + \log r \le r$, both $\log k$ and $\log L_1$ are at most $\log(3\Lambda) + \log r$. So

  $$D \le 2.003\Lambda + 4\log r, \qquad \log\big(360\,G(L_1)\Lambda_p^3\eta^{-1}\big) \le 3\Lambda + 3\ell_p + 5.9 \le 3.002\Lambda + 3\log r,$$

  since $2\ell_0 + 446 + 2\log\Lambda + 2\log(3\Lambda) \le 0.003\Lambda$ and $3\ell_0 + 5.9 \le 0.002\Lambda$. Using $\log r \le r$ and $\log^2 r \le r$,

  $$\begin{aligned}\Lambda D\,(3.002\Lambda + 3\log r) &\le \Lambda\big(6.0131\Lambda^2 + 18.02\Lambda\log r + 12\log^2 r\big) \le 6.0132\,\Lambda^3 r \\ &< 0.0792 \cdot 76\,\Lambda^3 r = 0.0792\,\Lambda_p,\end{aligned}$$

  which is the required inequality.
- *Case $\ell_p > e^{\Lambda/2}/3$.* Then $\ell_p \ge \Lambda^2$, so $k \le 21\ell_p$ and $L_1 \le 2\ell_p + 2\Lambda + \log(21\ell_p) + 3.08$. Since $2\Lambda + 457 + 2\log\Lambda + 2\log(21\ell_p) \le \ell_p$ (with huge room, as $\ell_p \ge \Lambda^2 \ge 10^{16}$), both $L_1 + 450 + 2\log\Lambda \le 3\ell_p$ and $D = L_1 + 442 + 2\log\Lambda + \log L_1 \le 3\ell_p$. So $G(L_1) \le 400 \cdot 3\ell_p \le \Lambda_p$, hence $\log(360\,G\Lambda_p^3\eta^{-1}) \le 4\ell_p + 2\Lambda + 5.9 \le 5\ell_p$, and the required inequality $0.0792\,\Lambda_p \ge \Lambda \cdot 3\ell_p \cdot 5\ell_p$ holds because $190\Lambda\ell_p^2 \le \ell_p^3 \le e^{\ell_p} = \Lambda_p$.

So Rel ≤ 0.1 for every $\Lambda_p \ge \Lambda_1 - 1$. The window φ of Theorem 5.24(iii) majorises $\mathbf{1}_{[X,X(1+\eta)]}$, takes values in $[0,1]$ and is supported in $[X(1-w), X(1+\eta+w)]$ with $w = \eta\Lambda_p^{-2}$. As $d\Pi_0 \le du/\log u$, this gives $\Pi([X,X(1+\eta)]) \le \int \varphi\, d\Pi \le 1.1 \int \varphi\, d\Pi_0 \le 1.1\,(1+3\Lambda_p^{-2})\,\eta X/\log X$: the support has length $\eta(1+2\Lambda_p^{-2})X$, and $\log u \ge \log X - 2w$ on it. As $\Lambda_p \ge 10^{25}$, $1.1(1+3\Lambda_p^{-2}) \le 1.1001$. Hence

$$\sum_{p\in I/m} \frac{1}{p} \le \frac{1}{X}\,\Pi\big([X,X(1+\eta)]\big) \le 1.1001\,\frac{\eta}{\log X} = \frac{1.1001\,\eta}{\log(N/m)}.$$

(A) $p \ge P_1$: the m-sum. Group m into dyadic blocks $(M_k, 2M_k]$, $M_k := 2^k M_0$ with $M_0 := \frac12\sqrt{N/t}$, up to $M_k \le N(1+\eta)/P_1$. In a block, $P^+(m) < p \le N(1+\eta)/m < 2N/M_k$ and $\log(N/m) \ge \sigma_k := \log(N/2M_k)$, so

$$S_A \le 1.1001\,\eta \sum_k \frac{W_t(M_k;\,2N/M_k)}{\sigma_k}, \qquad W_t(M;z) := \sum_{\substack{m\in\mathcal{N}\cap(M,2M]\\ m \text{ sqf, } t\text{-dense, } P^+(m)<z}} \frac{1}{m}.$$

Put $\ell_k := \log M_k$ and $u_k := \ell_k/\log(2N/M_k)$. Since $\tau \le J/10$, $\ell_k \ge (J-\tau)/2 - 2\log 2 \ge \tau$. Recall from Lemma 5.8 the profile $U(\ell) = \ell^{1/4}\big(1+\sqrt{2\log(\ell/\tau)+4}\big)$, $Q(\ell,u) = A + a$ with $A = 11(1+u)^3\log(2+u)/\ell$ and $a = (u-\ell^{1/4})_+^2/(2\sqrt{\ell})$, and $\Phi_\infty \le 38.3$.

Region 1: $u_k \le U(\ell_k)$. By Lemma 5.8, $W_t(M_k;2N/M_k) \le (\tau/\ell_k)\,\Phi_\infty\,\rho(u_k-1)\,e^{Q(\ell_k,u_k)}$, with $\rho(v) := 1$ for $v \le 1$.

- For $u \le \ell^{1/4}$: $a = 0$ and $A \le 11\cdot 1.01\,\ell^{-1/4}(\frac14\log\ell + 0.01) \le 7\cdot 10^{-7}$, because $\ell \ge 0.449J \ge 10^{34}$ (as $J \ge 300\Lambda^4 \ge 3\cdot 10^{34}$) and $\ell^{-1/4}\log\ell$ is decreasing. For $\ell^{1/4} < u \le U(\ell)$: $e^Q \le e^2\ell/\tau$ and $\rho(u-1) \le 1/\Gamma(\ell^{1/4})$, so these k contribute less than $10^{-6}\tau/J$ in total.
- With $\tilde J := J + \log 2$ we have $u_k = \ell_k/(\tilde J - \ell_k)$ and $\sigma_k = \tilde J - \ell_k - 2\log 2$. The function $f(\ell) := \rho(u(\ell)-1)/(\ell(\tilde J - \ell))$ satisfies
$$\int f(\ell)\,d\ell = \frac{1}{\tilde J}\int \rho(u-1)\frac{du}{u} \qquad (\ell = \tilde J u/(1+u)).$$
The sum over $\ell_k \in \ell_0 + (\log 2)\mathbb{Z}$ differs from $(\log 2)^{-1}\int f$ by at most the total variation of f on Region 1, which is $\le 4\sup f \le 10^{-6}/J$, since there $\tilde J - \ell \ge \ell/U(\ell)$.
- The lower limit is $u_{\min} \ge (J-\tau-6)/(J+\tau+8)$, and by $\int_a^\infty \rho(w-1)\,dw/w = \rho(a)$ $(a \ge 1)$,
$$\int_{u_{\min}}^\infty \rho(u-1)\frac{du}{u} = \log\frac{1}{u_{\min}} + \rho(1) \le 1 + \log\frac{J+\tau+8}{J-\tau-6} \le (1+10^{-6})\Big(1+\frac{2.1\tau}{J}\Big),$$
using $\log\frac{1+x}{1-x} = 2x(1+\frac{x^2}{3}+\frac{x^4}{5}+\cdots) \le 2.1x$ for $0 \le x \le 0.1$.

Hence Region 1 contributes at most $1.1001\cdot\frac{38.3}{\log 2}\,(1+3\cdot10^{-6})\big(1+\frac{2.1\tau}{J}\big)\frac{\eta\tau}{J} \le 60.81\big(1+\frac{2.1\tau}{J}\big)\frac{\eta\tau}{J}$.

Region 2: $u_k > U(\ell_k)$. Drop the density condition: by Lemma 5.30, $W_t(M_k;2N/M_k) \le e^{14}\log(2N/M_k)\,e^{-u_k}$, and $\log(2N/M_k) \le 1.01\sigma_k$. Since $U(\ell) \ge 3\ell^{1/4} \ge 3(0.449J)^{1/4}$ and there are at most $J/\log 2 + 1$ blocks, Region 2 contributes $\le 1.1001\,\eta\,e^{14}\cdot 1.01\,(J/\log 2 + 1)\,e^{-3(0.449J)^{1/4}} \le 10^{-6}\eta\tau/J$.

Adding (B), Region 1 and Region 2 gives (62), since $60.81 + 2\cdot 10^{-6} \le 61.1$. □

Remark 5.33. No cancellation of the Möbius function and no short-interval asymptotic for smooth numbers is used. The smooth part needs no density saving: its Rankin factor e^{-J/Λ_1} beats η/J outright. The thresholds Λ_1 and J_1 depend only polynomially on Λ; if the zero-free constant of Theorem 5.24 were only $c_{\mathrm{zf}} \ge \Lambda^{-A}$, it would suffice to take $\Lambda_1 = 100\Lambda^{2+A}$ and $J_1 = 300\Lambda^{3+A}$, and the middle range of Proposition 5.31 would still close.

Oscillation of the target. All bounds are for T_c', and none uses a local count at the height where the oscillation is read.

(63)
$$\sum_u \log(2u)\, o_u \le e^{-83}, \qquad \sum_v (\log 3v)^2\, o_v \le e^{-41}, \qquad \sup_v \log v \cdot o_v \le e^{-120}, \qquad E_{\mathrm{BR}} \le e^{-76}.$$

Proof.
- Near $u = X_0$, T_c' is continuous: it is 0 below X_0 and $1 - \rho_{\det} - j_0$ above, which vanishes at X_0.
- $o_{\det} \le (2.04H(u) + 2)X_0^{5/12}/u$, from the linear window of length κu in (3) and the unit spacing of the rigid set.
- On Good$(< u)$, the $\mathcal{A}$-part of T_c' is $\sigma_{\mathcal{A}} = K_{\mathcal{A}}\tilde\rho - \rho_{\mathcal{A}}$ (as $\bar\rho_{\mathcal{A}} = \rho_{\mathcal{A}}$, Remark 3.1). So $o_{\mathcal{A}} \le 2\sup|\sigma_{\mathcal{A}}| \le 4L_u^{-4}$, by Lemma 6.26(c) (from parts (a), (b) and Lemma 6.25). Lemma 6.25 holds on Good$(< t)$ at heights up to $t + 1.01H(t)$, using only E_{cl} at lower heights and Chebyshev below $2y$. So reading $o_{\mathcal{A}}$ at the current height t costs no event later than t. Moreover $o_{\mathcal{A}}$ vanishes unless $\log\log u > 2\log\Lambda + 5.97$ ($= 42.8$ at $\Lambda = 10^8$), since $\mathcal{A}$ is empty below that height.
- $o_R \le \sum_{k:\, x_k \le u} \gamma_k\kappa_k H(u)u^{-1}(x_k/u)^{1-\theta} \le H(u)u^{-\theta}$, since $\gamma_k\kappa_k \le x_k^{1-\theta}$ and $u \ge x_1 \ge \exp(X_0^2)$.
- Hence $\sum_u \log(2u)\, o_u \le 4.2H(X_0)X_0^{-7/12}\log X_0 + 5e^{-2\cdot 42.8} \le e^{-83}$, and $E_{\mathrm{BR}} \le 883e^{-83} + 8.4C_N E_\theta \le e^{-76}$. Also $\sum_v (\log 3v)^2 o_v \le 5.8e^{-42.8} \le e^{-41}$ and $\sup_v \log v \cdot o_v \le 4e^{-3\cdot 42.8} \le e^{-120}$. All of these improve with Λ.

No triangular windows and no Lipschitz bound are needed. □

Lemma 5.34 (the noise terms, in predictable form)**.** *With* (31)*,*

$$M(x) - (B\mathcal{R}T')(x) = (\vartheta/H)(x) - (BK_{\mathrm{pt}}B\mathcal{R}F_r)(x) - (B\mathcal{R}G)(x), \tag{64}$$

and, on Good$(< x)$*,*

$$\log x \cdot |M - B\mathcal{R}T'|(x) \le \varepsilon_N := 5.1C_N \sum_u \log(2u)\bar F_r(u) + 15.3C_N C_S' \sum_v (\log 3v)^2 F_r^+(v) + \sup_x \frac{6\log x}{H(x)} \le 10^{-13},$$

where $\bar F_r(v) := o_v + 6/H(v) + \bar\nu(v) + 54\eta(v)$ *bounds* $|F_r|$ *on* $[v, 2v)$*, and* $F_r^+ := 2\bar F_r$ *bounds* $|(1-B)F_c|$*. Every sum is over dyadic scales* $\ge X_0/2$*; there is no dependence on* x*, and no event at the current scale is used.*

Proof. The identity. By (30) for F_c and $K_{\mathrm{pt}}B\mathcal{R} = 1 - \mathcal{R}$, $K_{\mathrm{pt}}\tilde\rho_c = F_c - \mathcal{R}F_c + \mathcal{R}G$, where $\tilde\rho_c := \tilde\rho - (1 + BK_{\mathrm{pt}})^{-1}BJ$. So $M = B\mathcal{R}J + BT_c' + \vartheta/H - BF_c + B\mathcal{R}F_c - B\mathcal{R}G$. Insert $F_c = T_c' + F_r$ and $\mathcal{R}T_c' + \mathcal{R}J = \mathcal{R}T'$. In particular, the current-scale terms $-B(\nu + \mathrm{NT})(x)$ of BF_c cancel exactly against those of $B\mathcal{R}F_c$ (its $n = 1$ term). Every term on the right of (64) involves ν, NT and $\tilde\rho$ only at scales $\le x/2 + 2H$.

The bounds.
- $\log x \cdot |(BK_{\mathrm{pt}}B\mathcal{R}F_r)(x)| \le 5.1C_N \sum_{u \le x/2+2H} \log(2u)\bar F_r(u)$. The kernels of $K_{\mathrm{pt}}B\mathcal{R} = 1 - \mathcal{R}$ are the C_n with $n \ge 2$, so Proposition 5.18 applies ($5.1C_N$ bounds both kinds of blocks).
- $\log x \cdot |(B\mathcal{R}G)(x)| \le 15.3C_N C_S' \sum_v (\log 3v)^2 F_r^+(v)$, since $|G(y)| \le 3\sum_v \omega_y(v)\sup_{[v,3v]} F_r^+$ (Lemma 5.21(iii)).
- $|\vartheta/H| \le 6/H(x)$, and $\sup\tilde\rho \le 1.01$ surely (§5.6), so $|\mathrm{NT}| \le 2.02\eta \le 54\eta$ by (27).

By (46) (note that $\sum_u \log(2u)\bar\nu(u)$ and $\sum_v (\log 3v)^2\bar\nu(v)$ are dominated by the sums there), Lemma 5.20 (E^0_η and $E^{(2)}_\eta$) and (63), $\varepsilon_N \le 10^{-13}$. The largest part is $30.6 C_N C'_S e^{-41}$, from the oscillation of $\sigma_{\mathcal{A}}$. □

Design facts used. (§3).

- $R = \sum_k R_k$, with $R_k \equiv 0$ on $[1, x_k + 2H(x_k)]$, $x_1 \ge \exp(X_0^2)$, $x_{k+1} \ge \exp(x_k)$, and $\gamma_k \in [\Gamma_k, 2\Gamma_k]$ with $\Gamma_k = \kappa_0 x_k^{1-\theta}/(2\log^3 x_k)$; so $\log\gamma_k \asymp (1-\theta)\log x_k$.
- (19): $|R'_k(v)| \le \kappa_k (v/x_k)^{\theta-1}$ with $\kappa_k = 2\kappa'_k/\log x_k$, $\kappa'_k = \kappa_0/\log^2 x_k$, $\kappa_0 \le \frac14$.

A squarefree average. $F_\theta := \sup_y y^{\theta-1}\sum_{n\in\mathcal{N},\, n\le y} \mu^2(n) n^{-\theta} \le 1.33$ for $\theta \le \frac12$ (in particular for $\theta \in [\frac14, \theta_1]$):

- $F_\theta \le F_{1/2}$, since $y^{\theta-1}n^{-\theta} = y^{-1}(y/n)^\theta$ increases with θ;
- a direct computation of $F_{1/2}$ on $\mathbb{N}$ for $y \le 10^6$ gives the maximum 1.32964, at $y = 7$ (see Appendix B, `finite-n`);
- for $y \ge 10^6$, the elementary bound $|Q(u) - 6u/\pi^2| \le 3\sqrt{u}$ for the squarefree count Q gives $y^{\theta-1}\sum_{n\le y}\mu^2(n)n^{-\theta} \le (6/\pi^2)/(1-\theta) + 3y^{-1/2} + 1.5y^{-1/2}\log y \le 1.25$; and ($\mathrm{H_{BS}}$) changes the count for $\mathcal{N}$ by a relative $O(\eta_1)$.

Proposition 5.35 (positivity of the sharp reference)**.** *On* Good$(< x)$,

$$\rho^{\mathrm{sh}}(x)\cdot\log x \ge \begin{cases} 0.82 - o(1), & X_0 + H(X_0) \le x \le X_0^{1.01}, \\ 0.97 - o(1), & x \ge X_0^{1.01}/2. \end{cases}$$

The cutoff X_0^- in K_{pt} does not affect ρ^{sh}: $c_n(y) = \mu(n)\chi_y^t(n)\mathbf{1}[y/n \ge X_0^-]$ (Lemma 5.5), and $T' = 0$ on $[X_0^-, X_0)$. The proof has two ranges. Put $\log X_{\mathrm{sw}} := X_0^{1/100}$. For every $\Lambda \ge 10^8$, $X_0^{1.01} \le X_{\mathrm{sw}}$, $\log X_{\mathrm{sw}} \le (0.0048/\kappa)^{1/5}$ (the limit for μ_0 below, which is $\ge X_0^{0.079}$), and $\log X_{\mathrm{sw}} \gg \log X_{\mathrm{LP}} = O(\Lambda^2\log^2\Lambda)$.

Proof for range (A)*: $X_0 + H \le x \le X_{\mathrm{sw}}$.* No designed oscillation is active, since $x_1 \ge \exp(X_0^2) > X_{\mathrm{sw}}$. So $T' = 1 - \rho_{\mathrm{det}} + \sigma_{\mathcal{A}}$ there. Lemma 7.3 ((81), (82)) gives $(1+K_{\mathrm{pt}})^{-1}(1-\rho_{\mathrm{det}}) = \mathrm{MA}[\mathbb{P}] + \rho_{\mathrm{edge}} + (1+K_{\mathrm{pt}})^{-1}(\mu_0 + \mu_1)$, where:

(a) $\mathrm{MA}[\mathbb{P}](q) := \mathrm{MA}_{\kappa q}[\text{rational primes} \ge X_0](q)$ is the density of the new primes (the rational primes $\ge X_0$) in the linear window at q (§7); it vanishes for $(1+\kappa)q \le X_0$. By (80) and Lemma 7.3(a) (Huxley and Heath-Brown, $X_0 \ge X_{\mathrm{Hux}}$), $\mathrm{MA}[\mathbb{P}](x)\log x \ge 0.99$ for every $x \ge X_0$;

(b) $|\mu_0(u)| \le 2\kappa(\log u)^3$ for $u \ge X_0$ with $\log\log u \le \Lambda/8$, which contains range (A) (Lemma 7.3(b)). This smooth part contains the drift $+(\kappa/2)\Psi$, the partial windows, the order mismatch $H(q)/(\kappa q)$, and the $O(1/(\kappa x))$ term of (81);

(c) $\sum_{u \text{ dyadic} \ge X_0} \log(2u)\sup_{[u,2u]}|\mu_1| \le 7\cdot 10^{-6}$ (83);

(d) $\rho_{\mathrm{edge}} := (1+K_{\mathrm{pt}})^{-1}\mu_{\mathrm{edge}}$ is the edge at X_0, where μ_{edge} is the part of $K_{\mathbb{Q}}\mathrm{MA}[\mathbb{P}]$ that lies below X_0 (82). By Lemma 7.3(d), $\rho_{\mathrm{edge}}(x)\log x \ge -0.167$ for $X_0 \le x \le X_0^{1.01}$, and $|\rho_{\mathrm{edge}}(x)|\log x \le 3X_0^{-0.01} + X_0^{-0.34}$ for $X_0^{1.01}/2 \le x \le X_{\mathrm{sw}}$.

Now bound the responses.

- μ_0: by Lemma 5.5 ($|c_n| \le 1$ and $\sum|c_n|/n \le \log x + B_1$, with $B_1 = 1$),
$$\log x\cdot|(1+K_{\mathrm{pt}})^{-1}\mu_0(x)| \le \log x\,(\log x + 1)\sup_{u\le x}|\mu_0(u)| \le 2.1\,\kappa(\log x)^5 \le X_0^{-0.34}$$
for $\log x \le X_0^{1/100}$, since $\kappa = X_0^{-5/12+2\varepsilon_\kappa} \le X_0^{-0.3967}$.
- μ_1: by Corollary 5.9, $\log x\cdot|(1+K_{\mathrm{pt}})^{-1}\mu_1(x)| \le C_S\cdot 7\cdot 10^{-6} \le 6\cdot 10^{-4}$.

- $\sigma_{\mathcal{A}}$: likewise, at most $C_S \sum_v \log(2v) \cdot 2L_v^{-4}\mathbf{1}[\log\log v > 2\log\Lambda + 5.97] \le e^{-80}$.

So, pointwise, for $X_0 + H \le x \le X_0^{1.01}$,

$$\rho^{\mathrm{sh}}(x)\log x \ge 0.99 - 0.167 - 6\cdot 10^{-4} - X_0^{-0.34} - e^{-80} \ge 0.822,$$

and for $X_0^{1.01}/2 \le x \le X_{\mathrm{sw}}$,

$$\rho^{\mathrm{sh}}(x)\log x \ge 0.99 - 6\cdot 10^{-4} - 2X_0^{-0.34} - 3X_0^{-0.01} - e^{-80} \ge 0.989 \ge 0.97,$$

for every $X_0 \ge \max(e^{10^8}, X_{\mathrm{Hux}})$; no constant here depends on Λ. By Lemma 7.3(d)(i), the loss 0.167 occurs only on the windows $N\cdot(X_0/(1+\kappa), X_0)$ with $\mu(N) = 1$, each of relative length κ. □

Proof for range (B)*:* $x \ge X_{\mathrm{sw}}$. Call γ_j *resolvable at* x if $\gamma_j \le \Gamma(x) := e^{c_{\mathrm{LP}}\sqrt{\log x}/4}$, and let γ_* be the largest resolvable γ_j ($\gamma_* := 1$ if there is none). Put

$$\delta := \min\big(\gamma_*^{-2}\log^{-4} x,\ X_0^{-1/2}\big), \qquad J := [x(1-2\delta),\, x(1-\delta)].$$

The window J ends $\delta x \gg H$ before x, so the proof uses only the realised data before $x - H$, on Good($< x$). All old oscillations $j < k$ are resolvable: $\log\gamma_j \asymp (1-\theta)\log x_j$, while $c_{\mathrm{LP}}\sqrt{\log x} \ge c_{\mathrm{LP}}\sqrt{x_j} \gg \log x_j$.

Step 1: the window average of $\tilde\rho$.

- $\int_J \tilde\rho = \#\{q : s_q \in J\} \pm (H(x)+2)$, the error coming from the partial boxes at the two ends of J; and slots lie within $H/2$ of values. By the local count, and since the prime powers $p^k \in J$ with $k \ge 2$ number at most $2\sqrt{x}$, $\int_J \tilde\rho = \Pi(J) \pm (27H(x) + 2K_{\mathrm{loc}} + 6 + 2\sqrt{x})$.
- By Theorem 5.24(ii), $\Pi(J) = \delta x\,\pi_0(x)(1 + O(\delta)) + O(Kxe^{-c_{\mathrm{LP}}\sqrt{\log x}})$, where $\pi_0(x) := (1 - 1/x)/\log x$.
- Since $1/\delta \le \Gamma^2\log^4 x + X_0^{1/2}$, and $c_{\mathrm{LP}}\sqrt{\log x} \ge (0.0176/\Lambda)X_0^{0.005}$ exceeds $\log K + 2\Lambda + 5\log\log x$ (recall $\log K = O(\Lambda)$), this gives $\mathrm{avg}_J\tilde\rho = \pi_0(x) + o(1/\log x)$.

Step 2: ρ^{sh} *against* $\tilde\rho$ *on average.*

- $\tilde\rho - \rho^{\mathrm{sh}} = [\tilde\rho + \nu + \mathrm{NT} - B\mathcal{R}T'] - \nu - \mathrm{NT} + B[\mathcal{R}T' - \rho^{\mathrm{sh}}] + (B\rho^{\mathrm{sh}} - \rho^{\mathrm{sh}})$.
- Lemma 5.34 and Proposition 5.26 bound the first and fourth terms by $(\varepsilon_N + E_{\mathrm{BR}} + j_{X_0}(x))/\log x$. In range (B), $\log(x/X_0) \ge X_0^{1/100} - \Lambda \ge 600\Lambda^4$, so x lies in the huge range of Proposition 5.31, where $j_{X_0} \le e^{-59}$. Also $|\nu| \le \bar\nu$ and $|\mathrm{NT}| \le 54\eta$.
- Averaged over J, $\int_J(Bf - f)$ is a boundary term $\le 2H(x)\sup|f|$, and $\sup|\rho^{\mathrm{sh}}| \le 2(\log x + B_1)$.
- Hence $\mathrm{avg}_J\rho^{\mathrm{sh}} \ge \pi_0(x) - (\varepsilon_N + E_{\mathrm{BR}} + e^{-59} + o(1))/\log x$.

Step 3: pointwise against average, for all components except the current oscillation. For $y \in J$, $\rho^{\mathrm{sh}}(x) - \rho^{\mathrm{sh}}(y) = \sum_n [c_n(x)T'(x/n) - c_n(y)T'(y/n)]/n$.

- *Threshold changes:* at most 2·(Lemma 5.23 at 2δ) $= o(1/\log x)$, since $|T'| \le 2$, $\delta \le \log^{-4} x$ and $\log H(x) = 8(\log\log x)^2$. The lemma applies at 2δ: $2\delta \le 2X_0^{-1/2} \le \frac14$; and $1/\delta \le \Gamma(x)^2\log^4 x + X_0^{1/2} \le x^{0.02}$ (as $\gamma_* \le \Gamma(x)$ and $\log x \ge X_0^{1/100}$), while $6H(x)x^{-1/3} \le x^{-0.3}$ by Theorem 6.22(c), so $2\delta \ge 6H(x)x^{-1/3}$.
- *The jump of* T' *at* X_0*:* only $n \in [y/X_0, x/X_0]$ see it. Put $N_* := x/X_0$, $I := [N_*(1-2\delta), N_*]$ and $L_* := \log(N_*/2)$. By the lag bound, $|N - T| \le H(N_*)$ at both ends of I, and under the target (5), $T(I) \le 2|I| + 2 + \#\mathcal{A}(I)$. Cover I by $\lceil |I|/h(N_*(1-2\delta))\rceil$ consecutive windows $[u, u+h(u)]$, $h(u) = u^{\theta/2}$; by Lemma 6.25 each holds at most $h(u)L_u^{-4}$ absorbed elements, so $\#\mathcal{A}(I) \le (2.01\,\delta N_* + N_*^{\theta/2})L_*^{-4}$. Hence $\#(\mathcal{N}\cap I) \le 4.01\,\delta N_* + 2H(N_*) + 2 + N_*^{\theta/2}$, and this part is at most $j_0\sum_{n\in\mathcal{N}\cap I} 1/n \le j_0\big(4.1\,\delta + 3H(N_*)/N_* + 2N_*^{\theta/2-1}\big) = o(1/\log x)$.

- ρ_{det}, *smooth change:* for $|v-v'|\le \delta v$, $|\rho_{\mathrm{det}}(v)-\rho_{\mathrm{det}}(v')|\le \min\big(2(\delta(1+\kappa)v+1)/(\kappa v),\ 3\Psi(2v,X_0)/(\kappa v)\big)$. Put $u:=\log v/\Lambda$ and $U_0:=\Lambda$. Use the first bound for $u<U_0$ and the second beyond; for $u\ge U_0$, Rankin's bound (exponent $1-1/(2\Lambda)$) gives $\Psi(2v,X_0)\le 10v\Lambda^{1.65}e^{-u/2}$. With the weights of Corollary 5.9 ($\sum_{v<X_0^{U_0}}\log(2v)\le 1.5(U_0\Lambda)^2$) and $1/\kappa\le X_0^{5/12}$, the weighted sum is at most

$$C_S\Big[1.5\Lambda^4(2\delta+2/X_0)/\kappa+\sum_{u\ge\Lambda}\log(2v)\cdot 30\Lambda^{1.65}e^{-u/2}/\kappa\Big]\le C_S\big[5\Lambda^4X_0^{-1/12}+90\Lambda^{4.7}e^{-\Lambda/12}\big]\le e^{-10^6},$$

  using $\delta\le X_0^{-1/2}$. Both terms decrease with Λ, so this holds for every $\Lambda\ge 10^8$.
- *Resolvable R_j', smooth change:* at most $\delta\gamma_j\kappa_j/(1-\theta)\le \kappa_j/((1-\theta)\gamma_j\log^4 x)$, which is summable over j.
- $\sigma_{\mathcal{A}}$: $|\sigma_{\mathcal{A}}|\le 2L_v^{-4}$, and it vanishes unless $\log\log v>2\log\Lambda+5.97$. So the weighted sum is at most $4C_S\sum_v\log(2v)L_v^{-4}/\log x\le C_Se^{-80}/\log x$.

Step 4: the current oscillation, if it is not resolvable. Let $k:=\max\{j: x_j\le x\}$; R_j vanishes on $[1,x]$ for $j>k$. Put $\rho_{\mathrm{osc}}^{(k)}(y):=\sum_n c_n(y)R_k'(y/n)/n$.

- *Separation.* If γ_k is not resolvable, then $\gamma_k>\Gamma(x)$. Every $j<k$ is resolvable, and $\gamma_*\le\gamma_{k-1}\le x_{k-1}\le\log x$ (as $x_k\ge\exp(x_{k-1})$). So $\gamma_k\delta\ge\Gamma(x)\min(\log^{-6}x, X_0^{-1/2})\ge e^{c_{\mathrm{LP}}\sqrt{\log x}/5}$, since $c_{\mathrm{LP}}\sqrt{\log x}\gg\log X_0+\log\log x$ in range (B).
- *Average.* Where $c_n(y)$ is constant on J, $\int_J R_k'(y/n)\,dy/n=R_k(x(1-\delta)/n)-R_k(x(1-2\delta)/n)$, and $|R_k(v)|\le 2\kappa_k v(v/x_k)^{\theta-1}/((1-\theta)\gamma_k)$ (integrate the oscillation at frequency γ_k). So, with Lemma 5.5, the J-average is at most $4\kappa_k(\log x+B_1)/((1-\theta)\gamma_k\delta)$, plus the threshold part $\le\kappa_k\cdot R(2\delta)$, where $R(\cdot)$ is the bound of Lemma 5.23. Both are $o(1/\log x)$.
- *Pointwise.* $|c_n|\le\mu^2(n)$, and $R_k'(x/n)\ne 0$ forces $n\le x/x_k$. So

$$|\rho_{\mathrm{osc}}^{(k)}(x)|\le\kappa_k(x/x_k)^{\theta-1}\sum_{n\le x/x_k}\mu^2(n)n^{-\theta}\le\kappa_kF_\theta.$$

  - For $x\le x_k^2$: $\kappa_kF_\theta\log x\le 4\kappa_k'F_\theta=4\kappa_0F_\theta/\log^2 x_k=o(1)$.
  - For $x_k^2<x<\exp(C'\log^2 x_k)$, with $C':=(4(1-\theta)/c_{\mathrm{LP}})^2$:

$$\kappa_kF_\theta\log x\le 2\kappa_k'F_\theta C'\log x_k=2\kappa_0F_\theta C'/\log x_k=o(1),$$

    since $\log x_k\ge X_0^2$ and $C'\asymp\Lambda^2$.
- For $x\ge\exp(C'\log^2 x_k)$, $\gamma_k\le x_k^{1-\theta}$ is resolvable, and Step 3 covers it.

Step 5. Combining Steps 1–4, $\rho^{\mathrm{sh}}(x)\log x\ge\pi_0(x)\log x-\varepsilon_N-E_{\mathrm{BR}}-e^{-59}-o(1)\ge 1-10^{-12}-o(1)\ge 0.97$. □

Remark 5.36. No twisted Möbius estimate is used: old oscillations are *resolved* by the window of the local prime number theorem, not cancelled. Range (A) uses no local prime number theorem, and range (B) uses no Huxley-type input beyond $j_0\le 1.0001/\Lambda$. The ranges fit: $\log X_{\mathrm{LP}}=O(\Lambda^2\log^2\Lambda)\ll\log X_{\mathrm{sw}}=X_0^{1/100}\ll(0.0048/\kappa)^{1/5}$.

5.9. **The margin: window average, pointwise, and R1′.**

Theorem 5.37 (the margin on H-windows, uniformly in x)**.** *On* Good$(<x)$*, for every $x\ge 2X_0$, the compensator margin* (29) *satisfies*

$$\log x\cdot M(x)=\log x\cdot(\tilde T'-BK_{\mathrm{pt}}\tilde\rho)(x)\ge c_{\mathrm{sh}}(x)-o(1)-\varepsilon_N-E_{\mathrm{BR}}-j_{X_0}(x),\qquad \varepsilon_N+E_{\mathrm{BR}}\le 10^{-13},$$

where $c_{\rm sh}(x) := 0.82$ *for* $2X_0 \le x < X_0^{1.01}$ *and* $c_{\rm sh}(x) := 0.97$ *for* $x \ge X_0^{1.01}$, *and* j_{X_0} *is defined in* (59). *By Proposition 5.31,* $j_{X_0} \le 0.4209$ *for* $x \le X_0 e^{0.01\Lambda}$, $j_{X_0} \le e^{-0.0009\Lambda}$ *for* $X_0 e^{0.01\Lambda} \le x \le X_0 e^{600\Lambda^4}$, *and* $j_{X_0} \le e^{-59}$ *for* $x \ge X_0 e^{600\Lambda^4}$.

Proof.
- By (64) and Lemma 5.34, $\log x \cdot M(x) \ge \log x \cdot (B\mathcal{R}T')(x) - \varepsilon_N$.
- By Proposition 5.26, Lemma 5.27 and (59) at each point y of W_x, $\mathcal{R}T'(y) \ge \rho^{\rm sh}(y) - (E_{\rm BR} + j_{X_0}(x))/\log y$. (For $y < 2X_0^-$, $\mathcal{R}T' = \rho^{\rm sh} = T'$.)
- By Proposition 5.35 at each point of W_x, $B\rho^{\rm sh}(x) \cdot \log x \ge c_{\rm sh}(x) - o(1)$. For $x \ge X_0^{1.01}$ we have $W_x \subset [X_0^{1.01}/2, \infty)$, where its bound is $0.97 - o(1)$; for $2X_0 \le x < X_0^{1.01}$ both of its ranges give at least $0.82 - o(1)$ on $W_x \subset [X_0 + H(X_0), \infty)$.
- Every noise input lives at scales $\le x/2 + 2H$. Proposition 5.35 at points of W_x uses only data up to $x - H/2$ (its windows J end before x). □

This is the place where the divergence of the sup-norm noise response is avoided. A bound of the form $|Ke| \le (1+\log x)^{2K+1} \sup|\nu|$ diverges, because ν is of size $\asymp H(X_0)^{-1/2}$ at $u \approx X_0$. Here the noise enters only through (46), whose weights $\log(2u)$ and $(\log 3v)^2$ come from Corollary 5.9 and Lemma 5.21. The exponent of $1/\log x$ in Theorem 5.6 is exactly 1, which is what makes these sums independent of x.

Proposition 5.38 (the pointwise margin)**.** *Let* $m(t) = T'(t) - (K_{\rm pt}\tilde\rho)(t) - {\rm NT}_{\rm pt}(t)$ *be the pointwise margin density (§5.3), so that* $M = Bm + B\,{\rm NT}_{\rm pt} + \vartheta/H$. *Put* $v(t) := \sup_{|h| \le 1.01H(t)} |m(t+h) - m(t)|$, $N := t/X_0$, *and*

$$E_S := \sup_v \log v \cdot o_v + 2 \sup_y \log y \cdot \eta(y) + 27 \sum_{v\ dyadic} \log(3v)\,\hat\nu(v) + E_\delta \qquad (\le 10^{-28}),$$

$$E_\delta := 4.3 \sum_{v\ dyadic,\, v \ge X_0/2} \log(3v) \sup_{[v,2v)} \delta, \qquad \delta(z) := 6.6\,\varepsilon_H \frac{H(z)}{z} \log z, \qquad \varepsilon_H := 4 \cdot 10^{-6},$$

$$V_{X_0}(t) := 1.02\,(1 + \log N/\Lambda) \sum_{M \in I_t'} H_{\rm ord}(M)\, M^{\varepsilon' - 2}, \qquad I_t' := \{M \in \mathcal{N} : |t/M - X_0| \le 5.5H(X_0)\},$$

with $\varepsilon' := \varepsilon + 0.015 \le 0.045$ *and* $H_{\rm ord}(M)$ *the number of ordered factorisations of* M *in* $\mathcal{N}$. *On* Good$(< t)$*:*

(a) *for* $t \ge X_0^{1.01}$: $v(t)\log t \le 6.5E_S + V_{X_0}(t)$ *with* $V_{X_0}(t) \le e^{-0.0015\Lambda}$, *and*

$$m(t)\log t \ge 0.97 - o(1) - \varepsilon_N - E_{\rm BR} - 54 \sup_y \log y \cdot \eta(y) - 6.5E_S - j_{X_0}(t) - V_{X_0}(t) \ge 0.96 - o(1);$$

(b) *for* $2X_0 \le t \le X_0^{1.01}$ *the jump is paid once:* $m(t)\log t \ge \rho^{\rm sh}(t)\log t - j_{X_0}(t) - 10^{-12} \ge 0.82 - 0.4209 - 10^{-12} \ge 0.399$*;*

(c) *for* $X_0 \le t < 2X_0$: $m(t)\log t \ge 0.73$, *surely (no event is used). Below* X_0 *there are no slots.*

Proof. Step 1: short windows. For a slot window $I \subset [u, 2u]$ with $|I| \le H(u)$, Lemma 5.22 gives $\sigma(I) = \int_I m + E(I)$, with $|E(I)| \le \hat\nu(u)H(u)$ on $G_u^{\rm sw}$. Write $\varepsilon_s(z) := \sup|E(I)|$ over windows with $|I| \le s$ at z.

Step 2: the sub-H increments of $\tilde\rho$ *are strip differences of* m, *up to* δ. Fix z and $0 \le s \le 1.01H(z)$, write $H := H(z)$, and put $I_\pm := [z \pm H/2,\, z \pm H/2 + s)$ and $\delta_z := \varepsilon_H H(z)/z$. By (9) and (18) (whose exponent is ε_H), H is continuous and non-decreasing, $H(y)/H(x) \le (y/x)^{\varepsilon_H}$ for $X_0^-/2 \le x \le y$, and $H(x)/x$ is decreasing; so $0 \le H(y) - H(x) \le 1.01\,\varepsilon_H H(x)(y-x)/x$ for $x \le y \le 2x$. (Within $2H$ of z the points stay above $X_0^-/2$, even when z is near X_0.)

- *Exact form.* The boxes are half-open, $[s_q - H(s_q)/2,\, s_q + H(s_q)/2)$ (§5.1); put $\varphi_\pm(x) := x \pm H(x)/2$. By the regularity of H, $\varphi_\pm$ is continuous and strictly increasing, and its difference quotients lie in $[1 - 0.52\delta_z, 1 + 0.52\delta_z]$ within $2H$ of z. The box of q contains y if and only if $\varphi_-(s_q) \le y < \varphi_+(s_q)$, so $\mathbf{1}[z + s \in \text{box}] - \mathbf{1}[z \in \text{box}] = \mathbf{1}[\varphi_-(s_q) \in (z, z+s]] - \mathbf{1}[\varphi_+(s_q) \in (z, z+s]]$. Hence, exactly,

$$\tilde\rho(z+s) - \tilde\rho(z) = \int_{A_+} \frac{d\sigma(x)}{H(x)} - \int_{A_-} \frac{d\sigma(x)}{H(x)}, \qquad A_+ := \varphi_-^{-1}\big((z, z+s]\big), \quad A_- := \varphi_+^{-1}\big((z, z+s]\big).$$

- *Position and length.* A_+ is an interval starting at $a_+ := \varphi_-^{-1}(z) = z + H(a_+)/2$, and A_- one starting at $a_- := \varphi_+^{-1}(z) = z - H(a_-)/2$. So $|a_\pm - (z \pm H/2)| \le 0.26\,\delta_z H$ and $|A_\pm| = s(1+\theta_\pm)$ with $|\theta_\pm| \le 0.52\,\delta_z$; in particular $|A_\pm \triangle I_\pm| \le 1.05\,\delta_z H$ and $|A_\pm| \le 1.0001 s$.
- *Weights.* $A_+ \subset [z, z + 1.53H]$ and $A_- \subset [z - 0.51H, z + 0.52H]$, so $|H(z)/H(x) - 1| \le 1.6\,\delta_z$ on $A_\pm$. Each continuous slot decides at most one prime, and continuous-slot starts are at least $1 - 10^{-8}$ apart (§5.6), so $\sigma(A_\pm) \le 1.01|A_\pm| + 1 \le 1.03H$. Hence $H(z)[\tilde\rho(z+s) - \tilde\rho(z)] = \sigma(A_+) - \sigma(A_-) + R_1$ with $|R_1| \le 1.6\,\delta_z(\sigma(A_+) + \sigma(A_-)) \le 3.3\,\delta_z H$.
- *Strips.* By Step 1, $\sigma(A_\pm) = \int_{A_\pm} m + E(A_\pm)$. Replacing $A_\pm$ by $I_\pm$ changes $\int m$ by at most $1.05\,\delta_z H\, m^*(z)$ each, where $m^*(z) := \sup_{|y-z|\le 2H} |m(y)|$. Surely $|T'| \le 2$, $0 \le K_{\mathrm{pt}}\tilde\rho(y) \le 1.01 \sum_{2 \le w \le y/X_0^-} 1/w \le 1.01 \log y$ and $|\mathrm{NT}_{\mathrm{pt}}(y)| \le 2.02\,\eta(y) \le 2.02(\log y + 1)$ (as $\tilde\rho \le 1.01$), so $|m(y)| \le 3.1 \log y + 4.1$ and $m^*(z) \le 3.1 \log z + 4.2$.

Altogether,

$$\tilde\rho(z+s) - \tilde\rho(z) = \frac{1}{H(z)} \int_0^s \big[m(z + H/2 + r) - m(z - H/2 + r)\big]\, dr + \frac{E(A_+) - E(A_-)}{H(z)} + R(z),$$

with $|R(z)| \le (3.3 + 2.1\, m^*(z))\,\delta_z \le 6.52\,\delta_z \log z \le \delta(z)$, since $\log z \ge 10^8$. This holds also near X_0, where $\sigma = 0$ and $m = T' = 0$ below X_0. For $0 \le r \le s$ the shift between the two strips is $H(z) \le 1.01 H(z - H/2 + r)$, so the integrand is at most $v(z - H/2 + r)$. Hence

$$|\Delta_s \tilde\rho(z)| \le \frac{s}{H(z)} \sup v + \frac{2\varepsilon_{1.0001s}(z)}{H(z)} + \delta(z),$$

with the supremum over $[z - H, z + 0.6H]$.

Step 3: the recursion. Write $\kappa_w := \mathbf{1}[\tau_w < t/w]$. Then

$$\Delta_h m(t) = \Delta_h T'(t) - \sum_w (\kappa_w/w) \Delta_{h/w} \tilde\rho(t/w) + (\text{filter switching}) + \Delta_h \mathrm{NT}_{\mathrm{pt}}.$$

- Filter switching and NT occur only for w τ-near at t, and cost at most $2\eta(t)$. The cutoff X_0^- switches nothing on, since $\tilde\rho(X_0^-) = 0$.
- $h/w \le 1.01 H(t/w) w^{\varepsilon - 1}$ by (18). So

$$v(t) \le S(t) + \sum_{w \ge 2} 1.01\,\kappa_w\, w^{\varepsilon - 2} \sup\{v(t') : |t' - t/w| \le 2.1 H(t/w)\},$$

$$S(t) := o_t + 2\eta(t) + 2 \sum_w \frac{\kappa_w}{w} \cdot \frac{\varepsilon_{1.02H(t)/w}(t/w)}{H(t/w)} + 1.0001 \sum_w \frac{\kappa_w}{w}\, \delta(t/w).$$

- For $h < 0$, $\Delta_{h/w}\tilde\rho(t/w) = -\Delta_{|h|/w}\tilde\rho(z')$ with $z' := t/w - |h|/w \ge t/w - 0.51 H(t/w)$, and Step 2 is applied at z', which may lie in the dyadic block below that of t/w. There $H(t/w)/H(z') \le 1 + 10^{-9}$, which Step 5 absorbs (as $1.01(1 + 10^{-9}) \le 2^{0.015}$), and

$\delta(z') \le 1.0001\,\delta(t/w)$: as H is non-decreasing and $z' \ge t/w - 0.51H(t/w)$,

$$\frac{H(z')}{z'} \le \frac{H(t/w)}{z'} \le \Big(1 + \frac{0.52H(t/w)}{z'}\Big)\frac{H(t/w)}{t/w},$$

and $\log z' \le \log(t/w)$. This is the factor 1.0001 in $S(t)$.

- Base case $t < 2X_0^-$: there $K_{\rm pt}\tilde\rho = \mathrm{NT}_{\rm pt} = 0$, so $m = T' = T'_{\rm c} + J$ and $v(t) \le o_t + j_0\mathbf{1}[|t - X_0| \le 1.01H(t)]$.
- $v(t)$ depends on v and on the noise only at points $\le t/2 + 2.1H(t/2) < t$. So the recursion is a well-founded induction on t, and each short-window event is used only at strictly earlier times. This is the pointwise form of the event ordering in Definition 6.8 (event E3′).

Step 4: $\log t \cdot S(t) \le E_S$. Since $H(t) \le w^{\varepsilon_H} H(t/w)$ by (18), $1.02H(t)/w \le 0.52H(t/w)$ for $w \ge 2$, and Lemma 5.22 gives $\varepsilon_{1.02H(t)/w}(t/w)/H(t/w) \le 2\hat\nu(t/w)$, so the third term of S is at most $4\sum_{v'} \omega_t(v')\hat\nu(v')$, and the last term is at most $1.0001\sum_{v' \ge X_0/2} \omega_t(v') \sup_{[v',2v')} \delta$ (the points $t/w \ge X_0^-$ lie in blocks $v' \ge X_0/2$).

- For $v' \ge \sqrt{t}/2$ use $\omega \le 1.2$ and $\log t \le 2\log(2v')$.
- For $v' < \sqrt{t}/2$ use Lemma 5.21(ii): $\log t \cdot \omega_t(v') \le 2.1 \max_u (1+u)\rho(u) \cdot \log 3v' \le 4.3 \log 3v'$.
- In both cases $1.0001 \log t \cdot \omega_t(v') \le 4.3 \log 3v'$ (the constants above are 2.4 and $2.1 \cdot 2 = 4.2$), so the last term contributes at most E_δ to $\log t \cdot S(t)$. Since $H(v)/v$ is decreasing, $H(vw) \le w^{\varepsilon_H} H(v)$ and $\log H(X_0) \le 8(\log\Lambda)^2 + 10^{-6}\Lambda$ (Theorem 6.22(c)), we get $E_\delta \le 28.4\,\varepsilon_H \cdot 10\Lambda^2 H(X_0)/X_0 \le e^{-0.9\Lambda}$; so $E_S \le 10^{-28}$ still holds.
- With $\log t \cdot o_t \le \sup_v \log v \cdot o_v$ and $\log t \cdot 2\eta(t) \le 2\sup_y \log y \cdot \eta(y)$, this gives $\log t \cdot S(t) \le E_S$.

Step 5: solving the recursion. Iterating Step 3, a chain with product M has weight at most $H_{\rm ord}(M) M^{\varepsilon'-2}$, and it ends at a base point within $\sum_i 2.1H(t_i)(t_k/t_i) \le 4.4H(X_0)$ of t/M. So $v(t) \le \sum_M H_{\rm ord}(M) M^{\varepsilon'-2} E_S/\log(t/M) + v_{X_0}(t)$, where v_{X_0} collects the base terms $j_0\mathbf{1}[\cdots]$.

- Split at $M = \sqrt{t}$. The part $M \le \sqrt{t}$ is at most $2(1 + A_0(0.045))E_S/\log t = 6.44E_S/\log t$, since $A_0(\varepsilon') := \sum_{M \ge 2} H_{\rm ord}(M) M^{\varepsilon'-2}$ has $A_0(0.045) = 2.218$. The tail is at most $27t^{-0.1}(\log t/\Lambda)E_S/\log t \le X_0^{-0.09} E_S/\log t$, by Rankin's trick at $s = 1.75$ (as $1/(2 - \zeta(1.75)) = 26.5$).
- *The jump.* The base term $j_0\mathbf{1}[\cdots]$ is reached only by chains with $M \in I'_t$. So $v_{X_0}(t) \le j_0 \sum_{M \in I'_t} H_{\rm ord}(M) M^{\varepsilon'-2}$, and $j_0 \log t \le 1.01(1 + \log N/\Lambda)$. For $N \ge e^{0.01\Lambda}$, every $M \in I'_t$ has $M \ge N/2$, so $\sum_{M \in I'_t} H_{\rm ord}(M) M^{\varepsilon'-2} \le (N/2)^{-0.155} \sum_M H_{\rm ord}(M) M^{-1.8} \le 8.5(N/2)^{-0.155}$, using $\sum_M H_{\rm ord}(M) M^{-1.8} = 1/(2 - \zeta_{\mathcal N}(1.8)) \le 8.4912$, since $\zeta_{\mathcal N}(1.8) \le \zeta(1.8) + X_0^{-0.7}$ (§5.5) and $\zeta(1.8) = 1.8822296\ldots$. This gives $V_{X_0}(t) \le e^{-0.0015\Lambda}$.

Step 6(a). $m \ge Bm - v$ pointwise, and $Bm = M - B\,\mathrm{NT}_{\rm pt} - \vartheta/H \ge M - 54\eta - 6/H$. Insert Theorem 5.37.

Step 6(b): the small range; the jump is paid once.

- For $y \in W_t$, every n with $c_n(y) \ne 0$ or $C_n^y \ne 0$ has $n \le y/X_0^- < X_0$, so n is a rigid integer. Hence $\tau_w = P^+(w) < y/w$ always: the τ-filter of $K_{\rm pt}$ is void, $\mathrm{NT} = \mathrm{NT}_{\rm pt} = 0$, and there are no τ-near steps.
- *Split.* $m = m_c + m_j$ and $\tilde\rho = \tilde\rho_c + B\mathcal{R}J$, with $m_j := \mathcal{R}J$ exactly (since the J-part of $\tilde\rho$ is $B\mathcal{R}J$, and $1 - K_{\rm pt}B\mathcal{R} = \mathcal{R}$).
- *The jump part.* By Lemma 5.27, pointwise, $m_j(t) \ge j_0\rho_1^{\rm sh}(t) - j_0\mathfrak{B}(t)$; no step is τ-near here, so the τ-term of (55) is absent. Here $\rho_1^{\rm sh} = (1 + K_{\rm pt})^{-1}\mathbf{1}_{[X_0,\infty)}$ and $\mathfrak{B}$ is defined there.

- *The smooth part.* Steps 1–5 apply to $(m_c, \tilde\rho_c)$. Indeed $\tilde\rho_c = \tilde\rho - Bm_j$, and for a bounded f,

$$\Delta_s(Bf)(z) = \frac{1}{H(z)}\int_0^s \big[f(z+H/2+r) - f(z-H/2+r)\big]\,dr + R_B, \qquad |R_B| \le 2.02\,\varepsilon_H \frac{H(z)}{z}\sup|f|,$$

  for $0 \le s \le 1.01H(z)$. Indeed $0 \le H(z+s)/H(z) - 1 \le (1+s/z)^{\varepsilon_H} - 1 \le \varepsilon_H s/z$, so the two window edges move by $(H(z+s)-H(z))/2 \le \varepsilon_H H(z)s/(2z)$ each and the weight $1/H$ changes by a relative $\le \varepsilon_H s/z$; this gives $|R_B| \le 2\varepsilon_H(s/z)\sup|f|$. With $f = m_j = \mathcal{R}J$ and $|\mathcal{R}J| \le j_0 \sum_n \mathrm{TV}(C_n)/n \le 2.2$ in this range (Proposition 5.18), Step 2 holds for $(m_c, \tilde\rho_c)$ with remainder $R(z) - R_B$, and $|R(z) - R_B| \le (6.52\log z + 4.45)\,\delta_z \le \delta(z)$. The base case is $m_c = T_c'$, which has no jump, so $v_c(t)\log t \le 6.5E_S$. By (64) and Proposition 5.26, $M_c := M - B\mathcal{R}J$ satisfies $\log t \cdot M_c \ge \log t \cdot B\rho_c^{\mathrm{sh}} - \varepsilon_N - E_{\mathrm{BR}}$, where $\rho_c^{\mathrm{sh}} := \sum_n c_n T_c'(\cdot/n)/n$; and $Bm_c = M_c - \vartheta/H$.
- ρ_c^{sh} *is flat on* W_t. Here $c_n(y) = \mu(n)\mathbf{1}[y/n \ge X_0^-]$, and $|T_c'| \le o_{X_0}$ on $[X_0^-, X_0 + H]$. So $|\rho_c^{\mathrm{sh}}(y) - \rho_c^{\mathrm{sh}}(t)| \le \sum_u o_u + o_{X_0} \le 11H(X_0)X_0^{-7/12}$, which is at most $e^{-10^7}/\log t$. (Here $o_R = 0$, and $o_{\mathcal{A}} = 0$ because $\mathcal{A}$ is empty while $\log\log n \le 2\log\Lambda + 5.97$.)
- *Sum.* $m(t) \ge \rho_c^{\mathrm{sh}}(t) + j_0\rho_1^{\mathrm{sh}}(t) - j_0\mathfrak{B}(t) - (\varepsilon_N + E_{\mathrm{BR}} + 6.5E_S + e^{-10^7})/\log t - 6/H$, and $\rho_c^{\mathrm{sh}} + j_0\rho_1^{\mathrm{sh}} = \rho^{\mathrm{sh}}$. Then use $\rho^{\mathrm{sh}}(t)\log t \ge 0.82$ (Proposition 5.35(A); the range $2X_0 \le t \le X_0^{1.01}$ lies in range (A) for every Λ, and the proof for range (A) gives 0.822 there with no $o(1)$ term). The edge term of Lemma 7.3(d) is part of ρ^{sh}, and the two lower bounds are simply added; so it does not matter whether its dips and the worst case of j_{X_0} occur at the same t.

Step 6(c): below $2X_0$. Here $\mathcal{A} = \emptyset$, $R = 0$, and $K_{\mathrm{pt}}\tilde\rho(t)$ can be nonzero only through w with $t/w \ge X_0^-$, i.e. $w = 2$.

- For $X_0 \le t < 2X_0^-$: $K_{\mathrm{pt}}\tilde\rho(t) = \mathrm{NT}_{\mathrm{pt}}(t) = 0$, so $m(t) = T'(t) = 1 - \rho_{\mathrm{det}}(t)$. The integers in $[t, (1+\kappa)t]$ that are not X_0-smooth include every prime there, so $1 - \rho_{\mathrm{det}}(t) \ge \mathrm{MA}[\mathbb{P}](t) - 1/(\kappa t) \ge (0.99 - 10^{-9})/\log t$, by (80) and Lemma 7.3(a).
- For $2X_0^- \le t < 2X_0$: $m(t) = T'(t) - \frac12\tilde\rho(t/2)$, with $t/2 \in [X_0^-, X_0)$ (the filter $\tau_2 = 2 < t/2$ is void, and $\mathrm{NT}_{\mathrm{pt}} = 0$). Since slots start at X_0 and boxes are centred, $\tilde\rho(t/2)$ counts the primes decided at slots in $[X_0, X_0 + H/2]$, at most one per continuous slot. So $\tilde\rho(t/2) \le (\int_{X_0}^{X_0+H/2} T'_{\mathrm{cont}} + 3)/H(X_0) \le \frac12(j_0 + o_{X_0}) + 3/H \le 0.5001/\Lambda$, since $T'_{\mathrm{cont}} \le j_0 + o_{X_0}$ there and $j_0 \le 1.0001/\Lambda$. With $T'(t) \ge (0.99 - 10^{-9})/\log t$ as above and $\log t \le \Lambda + 0.7$,

$$m(t)\log t \ge 0.99 - 0.2501(1 + 0.7/\Lambda) - 10^{-9} \ge 0.739.$$

- These bounds use no event at all, only (80) and the slot rule. □

Remark 5.39. The contraction in Step 5 comes from compressing a shift h at scale t to a shift h/w at scale t/w: a relative shift $w^{\varepsilon-1}$, times the weight $1/w$. In the small range the jump is compared pointwise with its sharp value (Lemma 5.27) and never enters a variation; bounding it as an additive variation term as well would pay for it twice.

Corollary 5.40 (R1′ on every window)**.** *Let I be any window: a slot window, or a physical interval $[a, b)$ with arbitrary (for instance integer, deterministic) endpoints, of any length, anywhere in $[1, \infty)$. It may start below X_0 and straddle $2X_0$ or $X_0^{1.01}$. Let $\ell_{\mathrm{jit}}(I) := \#\{$continuous slots in $I\} \le 1.01|I \cap [X_0, \infty)| + 1$, and let $L \ge \log\max(I \cup \{X_0\})$. On* $\mathrm{Good}(< \max I)$,

$$\ell_{\mathrm{jit}}(I) - \mathrm{Comp}(I) = \int_I m \pm 2 \ge 0.399\,\frac{|I \cap [X_0, \infty)|}{L} - 2 \ge \frac{c'}{L}\,\ell_{\mathrm{jit}}(I) - K_d, \qquad \mathrm{Comp}^{(2)}(I) \le s_2\,|I|,$$

with $c' = 0.15$, $K_d = 3$, and s_2 the supremum of $s_2(u)$ (43) over the dyadic scales meeting I.

Proof.
- $\int_I m = \ell_{\mathrm{jit}}(I) - \mathrm{Comp}(I) \pm 2$ is the bookkeeping of §5.3 without the B-average: $\#\{\text{continuous slots in } I\} = \int_I T'_{\mathrm{cont}} \pm 2$ and $\mathrm{Comp}(I) = \int_I (K_{\mathrm{pt}}\tilde\rho - K_{\mathcal{A}}\tilde\rho + \mathrm{NT}_{\mathrm{pt}})$ exactly, for any interval I. Below X_0 there are no slots, and below $2X_0^-$ no jittered composites, so $m = 0$ on $[1, X_0)$ and that part contributes nothing.
- Comp is the compensator $\mathrm{Comp}_{\mathrm{jit}}$ of §6 (latest prime dividing once). The q^2-part is the separate source $\mathrm{Comp}^{(2)}$, bounded by (43).
- $m \log t \geq 0.399$ pointwise on all of $[X_0, \infty)$, by Proposition 5.38(a)–(c): it is ≥ 0.73 on $[X_0, 2X_0)$, ≥ 0.399 up to $X_0^{1.01}$, and ≥ 0.959 beyond. Since $0.399 \geq 1.01c' = 0.1515$, the $+1$ in the slot count only enters K_d.

In the form used in Theorem 6.6: $\mathrm{Comp}(I) \leq \ell_{\mathrm{jit}}(I)(1 - c'/L) + K_d$ for every such I with $L \geq \log \max I$. Above $X_0^{1.01}$ the hole rate is at least $0.959/\log x$, independently of the block length L_0 and of the tolerance $M_1 \leq H/L^{14}$. □

6. The queue

This section proves that, on an event of probability greater than $\frac{1}{2}$ (Lemma 6.9), the Lindley queue of Section 3 satisfies $Q_j \leq B(s_j)$ at every slot (Theorem 6.6). Here $B(x)$ is quasi-polylogarithmic in x, and is bounded by $H(x)^{O(1)}$ on the raised range of Section 6.7. Consequently (L-a) holds: $|D(x)| \leq B(x) + \frac{H(x)}{2}(1 + \kappa_+/\log x) + 4 \leq x^\eta$. The ingredients are:

- the landing count and the local count (Lemma 6.1 and Corollary 6.3);
- the good event and the first-failure budget (Definition 6.8 and Lemma 6.9);
- the cluster event E_{cl} and its per-prime increment bound (Proposition 6.10 and Corollary 6.11);
- the chain of lemmas proving E_{cl} at every scale (Sections 6.5, 6.6 and 6.7, together with Appendix A);
- the analysis of the absorbed set $\mathcal{A}$: its pointwise sparsity (Lemma 6.25) and the consequences (Section 6.8).

6.1. **Conventions, parameters and standing facts.**

Scales. A *dyadic scale* is $x = 2^m X_0$ with $m \geq 0$. We write $L := \log x$ and $\ell := \log L$. All parameters are *frozen per scale*: they are evaluated at x and used for heights in $[x, 2x)$. The one exception is the jitter width $H(\cdot)$, together with the functions $c(\cdot)$, $\bar r_0(\cdot)$ and $\bar M_1(\cdot)$ that enter it: these are functions of the height itself, so that H is continuous. Where a statement at scale x writes H without an argument, it means $H(x)$. We write $\Lambda := \log X_0 \geq 10^8$ (Section 2.2).

Parameters at scale x. These are the choices of Section 2.2 and Section 3.

- $\delta_1 := 2L^{-6}$. The *grid windows* at height $z \in [x, 2x]$ are the intervals $[k\delta_1, (k+2)\delta_1)$, $k \in \mathbb{Z}$, of length $2\delta_1 = 4L^{-6}$.
- $r_0 := \lceil L/(3\ell) \rceil + 3$ is the rank threshold (Theorem A.12).
- $c := c(x)$ is the capacity threshold. Here $c(\cdot)$ is the continuous, non-decreasing function of the height defined in Theorem 6.22. We put $Q := c^{1/2}$ and $\varphi(u) := c + uQ + u(u+1)/2$.
- $M_1 := r_0 + \varphi(r_0) + 1$ is the cluster bound. Then $M_1 \leq 2c$, and $M_1 \geq L$ for $L \geq L_*$. The integer r_0 and the cluster bound M_1 are used for all counting (Theorem A.12, Lemma 6.13, Theorem 6.22(a), the cluster events).
- For the jitter width we use smooth majorants. For a height $v \geq X_0$ put

$$\bar r_0(v) := \frac{\log v}{3 \log\log v} + 4, \qquad \bar M_1(v) := \bar r_0 + \varphi_v(\bar r_0) + 1,$$

where φ_v is φ with $c = c(v)$. Then $\bar{r}_0$ is continuous and increasing, $r_0 \le \bar{r}_0(x) \le r_0 + 1$, and $\bar{M}_1$ is continuous and non-decreasing. Moreover $M_1 \le \bar{M}_1(x) \le M_1 + c^{1/2} + \bar{r}_0 + 2 \le M_1(1 + 2c^{-1/2})$ and $\bar{M}_1 \le 2c(v)$, since $\bar{r}_0 \le L \ll c^{1/2}$.

- $H(v) := \max(H_{\mathrm{std}}(v), \bar{M}_1(v)^2)$, where $H_{\mathrm{std}}(v) := \exp(8(\log\log v)^2)$. This is the width of the box jitter of Section 3. It is continuous and non-decreasing in v, and $H(x) \ge \bar{M}_1(x)^2 \ge M_1^2$.
  - Where $c(v) = \frac{1}{4}H_{\mathrm{std}}(v)^{1/2}$, that is, off the raised range, we have $\bar{M}_1 \le 2c = \frac{1}{2}H_{\mathrm{std}}^{1/2}$, so $H = H_{\mathrm{std}}$. This holds in particular for $v \in [X_0, X_0^2]$: there $s \le \log 2 - \log\Lambda$, so $e^{s/2} \le \sqrt{2}\,\Lambda^{-1/2}$, $\Lambda_s \ge e^{-s} \ge \Lambda/2$ and $A(\Lambda_s, s) \le 2\ell$, whence $\gamma(s)\Lambda \le 10.4\ell + 5.2 < 4\ell^2 - \log 4$. For heights $v < X_0$ we put $H(v) := H_{\mathrm{std}}(v)$; thus H is continuous at X_0, and $H = H_{\mathrm{std}}$ on $[X_0^-/2, X_0^2]$.
  - The *raised range* is the set of heights where $c(v) > \frac{1}{4}H_{\mathrm{std}}(v)^{1/2}$, that is, where $\gamma(s)\Lambda > 4\ell^2 - \log 4$ (Theorem 6.22). There $H = \max(H_{\mathrm{std}}, \bar{M}_1^2) \ge H_{\mathrm{std}}$.
  - In all cases the tolerance $M_1 \le H/L^{14}$ of Section 2.2 holds.
- $L_0 := \max(L^5, 48M_1L^3/c'^2)$ is the block length. We put $q_{\min} := L_0L^6$ and $y := H(x) \cdot L_0 \cdot L^6$. The latter is the absorption threshold of Section 3, including its ramp. Like M_1, both are frozen per scale and built from the integer r_0. They are used only through per-scale upper and lower bounds, through monotonicity across scales, and (in Lemma 6.25) through $\sup_{[u,2u]} y$, so their jumps cause no difficulty.
- $B := B(x)$ is the queue bound of Theorem 6.6. The constants $c' = 0.15$ and $K_d = 3$ are the drift and edge constants of the drift bound R1′ (Corollary 5.40).
- $\lambda := 4L$ in Theorem 6.6. For the budget (Lemma 6.9) we use, per scale u, $\lambda_u := 3\log u + 2\log\log u$ and $\lambda'_u := \lambda_u + 2 + \log H(u)$.

In summary,

(65)
$$r_0 = \left\lceil \frac{L}{3\ell} \right\rceil + 3, \qquad M_1 = r_0 + \varphi(r_0) + 1, \qquad L_0 = \max\left(L^5, \frac{48M_1L^3}{c'^2}\right), \qquad y = H(x)L_0L^6,$$
$$\bar{r}_0(v) = \frac{\log v}{3\log\log v} + 4, \qquad \bar{M}_1(v) = \bar{r}_0 + \varphi_v(\bar{r}_0) + 1, \qquad H(v) = \max(H_{\mathrm{std}}(v), \bar{M}_1(v)^2).$$

Filtration. Slots $s_1 < s_2 < \dots$ (Section 3) are processed in increasing order. $\mathcal{F}_t$ is the σ-field of all decisions and jitter offsets at slots $\le t$, and $\mathcal{F}_{s-}$ is the same for slots $< s$. For a decision slot s, $\mathcal{N}_{\prec s}$ denotes the set of elements of $\mathcal{N}$ all of whose primes are decided at slots $< s$. It is $\mathcal{F}_{s-}$-measurable.

The threshold X_0^-. With centred placement (Section 3), jittered primes have values at least $X_0^- := X_0 - H(X_0)/2$. Since $H(X_0)/X_0 = e^{8\log^2\Lambda - \Lambda} \le e^{-0.98\Lambda}$, we have $\log X_0^- \ge \Lambda - e^{-0.98\Lambda}$. Every lower bound for a jittered prime below is stated with X_0^-.

The Lindley recursion. By Lemma 3.7, the queue is $Q_0 = 0$ and

$$Q_j = \max(0,\, Q_{j-1} + A_j - 1), \tag{66}$$

where A_j is the number of arrivals assigned to slot s_j. The *excess* of a slot window W is $\mathrm{exc}(W) := C(W) - \#\{\text{slots of } W\}$. By the excess identity (Lemma 3.9), $\mathrm{exc}(W) = C_{\mathrm{jit}}(W) - \ell_{\mathrm{jit}}(W) \pm 2$. Here C_{jit} counts the non-absorbed composites whose latest-decided prime is jittered, and $\ell_{\mathrm{jit}}(W)$ counts the jittered (continuous) slots of W. Lindley's formula gives

$$Q_j = \max\left(0,\, \max_{k<j} \mathrm{exc}((k,j])\right). \tag{67}$$

Standing facts at height z. All of the following are sure (pathwise) statements, except (F2), which is distributional.

(F1) At most $H(t)(1+\kappa_+/\log t)+3$ decided primes have jitter windows containing a given point t, since the slot density satisfies $T' \le 1+\kappa_+/\log t$ (Section 3). Consequently, the label sums over slots in Lemma A.5 carry the factor $\theta := 1 + 1/\Lambda$, which bounds $1 + \kappa_+/\log q + 3/H$ for jittered primes $q \ge X_0^-$ (with $\kappa_+ = 0.8$).

(F2) *(Conditional law of positions.)* Whether a prime is decided at slot s, and the slot s itself, are $\mathcal{F}_{s-}$-measurable (Lemmas 3.2 and 3.5). Given $\mathcal{F}_{s-}$, the position of the prime decided at s is uniform on $[s - H(s)/2,\, s + H(s)/2)$ (centred box jitter), so its conditional density is at most $1/H(s)$. The slots and positions of distinct primes are *not* independent: which slots carry primes depends on the history of the greedy process. Joint statements follow by iterated conditioning along the decision order. Let $j_1 < \cdots < j_k$ be slot indices, let D be the event that primes are decided at all of them, with positions $q_1, \dots, q_k$, and let $h_1, \dots, h_k$ be $\mathcal{F}_{s_{j_1}-}$-measurable with $H(s_{j_i}) \ge h_i$ surely (for instance $h_i := H(s_{j_1})$, as H is non-decreasing). Then, for every Borel $g \ge 0$ on $\mathbb{R}^k$,

$$\mathbb{E}\big[\mathbf{1}_D\, g(q_1, \dots, q_k) \,\big|\, \mathcal{F}_{s_{j_1}-}\big] \le \prod_{i=1}^{k} h_i^{-1} \int_{\mathbb{R}^k} g(t)\, dt.$$

Indeed, given $\mathcal{F}_{s_{j_k}-}$ the event that $j_1, \dots, j_{k-1}$ are decision slots, the positions $q_1, \dots, q_{k-1}$, the event that j_k is a decision slot, and the box at s_{j_k} are all determined, while q_k is uniform on a box of length at least h_k. So the conditional expectation given $\mathcal{F}_{s_{j_k}-}$ is at most h_k^{-1} times $\mathbf{1}_{D'} \int g(q_1, \dots, q_{k-1}, t)\, dt$, where D' is the event for $j_1, \dots, j_{k-1}$; now induct on k.

(F3′) *(Harmonic sums, $\mathcal{F}_{s-}$-form.)* On Good$(< s)$ (Definition 6.8), for every $Y \ge 1$,

$$\sum_{c \in \mathcal{N}_{\prec s},\ c \le Y} \frac{1}{c} \;\le\; A_s(Y) := \begin{cases} \log Y + 2, & \text{if } Y + H(Y) \le s, \\ e^\gamma \log s \cdot \exp\big(2.21/\Lambda + 5H(s)/s\big), & \text{otherwise.} \end{cases}$$

Moreover $A_s(Y) \le A' := 1.8(\log z + 1)$ for $s \le 2z$ and $Y \le 2z$.

Proof. If $Y + H(Y) \le s$, then all of $\mathcal{N} \cap [1, Y]$ is decided before s, and the bound is the sure harmonic bound of Lemma 5.1(iv), which holds on Good$(< s)$. Otherwise we bound the sum by $\prod_{p \text{ decided before } s}(1 - 1/p)^{-1}$. Put $y_0 := X_0^- - 1$. Every realised prime $p \le y_0$ is rational, since jittered primes are at least X_0^-. We split the primes decided before s into three groups.

- The rational primes $p \le y_0$ give at most $e^\gamma \log y_0\, (1 + 1/(2\log^2 y_0)) \le e^\gamma \Lambda(1 + 1/(2\Lambda^2))$ [16, Thm. 8, (3.29)], valid for $y_0 \ge 286$.
- The realised primes in $(y_0, s - H(s)]$ are the rational primes in (y_0, X_0), the jittered primes that land in $[X_0^-, X_0)$, and all realised primes in $[X_0, s - H(s)]$. By the sure local Mertens bound of Lemma 5.1(iv) with $y = y_0$, which sums over all realised primes, they give at most $(\log s/\log y_0)\, e^{2.20/\log y_0 + 1/y_0}$. That bound reads $\mathcal{N}$ only on $[1, s - H]$ and holds on Good$(< s)$. Since $\log y_0 \ge \Lambda - H(X_0)/X_0$ and $H(X_0)/X_0 = e^{8\log^2\Lambda - \Lambda} \le e^{-0.98\Lambda}$, this is at most $(\log s/\Lambda)\, e^{2.20/\Lambda + 1/X_0 + e^{-0.97\Lambda}}$.
- The remaining primes decided before s have values in $(s - H, s + H/2]$, hence values greater than $s/2$. There are at most $2H + 1$ of them, so they contribute a factor at most $(1 - 2/s)^{-(2H+1)} \le e^{5H(s)/s}$. Since $\log H \le 8\ell^2 + 10^{-6}\log s$ (Theorem 6.22(c)), this factor is at most $e^{10^{-9}}$.

These groups cover every prime decided before s: such a prime has value at most $s + H(s)/2$, and a value below X_0^- is a rational prime. Multiplying the three bounds gives $e^\gamma \log s \cdot \exp\big(2.20/\Lambda + 1/(2\Lambda^2) + 1/X_0 + e^{-0.97\Lambda} + 5H(s)/s\big)$, and $1/(2\Lambda^2) + 1/X_0 + e^{-0.97\Lambda} \le 0.01/\Lambda$. □

(F4) *(Sure Mertens and Chebyshev bounds on* Good$(<s)$*.)* These are the realised-prime bounds of Lemma 5.1(iv), used inside (F3′) and in Appendix A.

Absorbed and non-absorbed elements. $\mathcal{A}$ (see (4)) is the extended absorption set: the elements n with $\log n \geq L_* = 10^7$ whose latest jittered prime r satisfies $r < y(n)$. In Lemma 6.25, $y(n)$ is replaced by $\sup_{[u,2u]} y$ on the dyadic block of n. Absorbed elements are counted in the target T and are not arrivals. All statements about arrivals concern $\mathcal{N} \setminus \mathcal{A}$.

- *Stopped process.* Every statement of this section is proved on Good$(<t)$ for the relevant t, that is, for the process stopped at the first failure time τ. There Lemma 6.25 gives $\rho_{\mathcal{A}} \leq L^{-4} < 0.1/\log u$, so the capped density $\bar{\rho}_{\mathcal{A}} = \min(\rho_{\mathcal{A}}, 0.1/\log u)$ of (5) equals $\rho_{\mathcal{A}}$. No argument of this section uses $\rho_{\mathcal{A}}$ off the good event: Lemma 6.25, Proposition 6.27, the compensator of $\sigma_{\mathcal{A}}$ (Lemma 6.26(b)) and (7) all run on the good event.
- *Slot lengths.* This section uses ρ_{det} only through $T' \leq 1 + 1/(\kappa u) \leq 1.01$, in the linear normalisation of Section 3 (see (3) and Lemma 7.1(1)): $\rho_{\mathrm{det}} \leq$ (rigid fraction) $+1/(\kappa u)$, and the positivity margin is (6), with no $\kappa/2$ loss. This enters as slot spacing at least 0.99 in (F1) and Appendix A, and in the candidate counts of Lemma 6.1. No $\kappa/2$ form is used.

6.2. **Landing count and local count.** Fix a dyadic P and put $\lambda_P := 4\log P$, $H_- := H(P/2)$, $H_+ := H(8P)$ and $\rho_H := H_+/H_-$. Then $\rho_H \leq 1.35$ for $\log P \geq 1100$, and $\rho_H \leq 1 + 1.2\cdot 10^{-5}$ for $\log P \geq 10^8$, the raise of Theorem 6.22 included. Indeed, on $[X_0/2, \infty)$ both one-sided derivatives of $\log H$ in $\log z$ are at most $16\ell/L + 10^{-7} \leq 3.1\cdot 10^{-6}$: above X_0 by Theorem 6.22(c), and below X_0 because $H = H_{\mathrm{std}}$ there (Section 6.1). So $\rho_H \leq 16^{3.1\cdot 10^{-6}} \leq 1 + 1.2\cdot 10^{-5}$. For a window $J \subset [P, 4P+1]$ with $|J| \leq 6H(P)$ we define:

- the *candidates* $\mathcal{C}(J)$: the decision slots s with $|s - J| \leq H(s)/2$. (Selection is by decision slot, which is predictable; selection by the value q would not be.)
- $\Lambda(J)$: the number of primes decided at slots in $\mathcal{C}(J)$ that land in J;
- the compensator $\mu(J) := \sum_{s\in\mathcal{C}(J)} \mathbb{P}(\text{the prime at } s \text{ lands in } J \mid \mathcal{F}_{s-})$;
- $\bar{\mu}(J) := (|J| + H_+ + 2)\min(1, |J|/H_-)$. By (F1)–(F2) this is a *sure* bound: $\mu(J) \leq \bar{\mu}(J) \leq (1+\rho_H)|J| + 2$. Here the slot-density excess $\kappa_+/\log P + 1/(\kappa P)$ in (F1) is absorbed by the gap between H_+ and the widths $H(s) \leq H(4P + 1 + H)$ of the candidates. Above x_1, where $R' \neq 0$ (Section 3.2), $H = H_{\mathrm{std}}$ and this gap is at least $11\ell H/L$, more than the excess $7\cdot 0.8H/L$ over the at most $7H$ candidate slots. Below x_1, $R' = 0$ and the excess is at most $7H/(\kappa P) \ll 1$.

Lemma 6.1 (Landing count)**.** *Let* Land(J) *be the event*

$$|\Lambda(J) - \mu(J)| \leq \sqrt{2\bar{\mu}(J)\lambda_P} + \tfrac{2}{3}\lambda_P, \qquad \textit{hence} \qquad \Lambda(J) \leq \min\big(\#\mathcal{C}(J),\, 2\bar{\mu}(J) + \tfrac{7}{6}\lambda_P\big). \tag{68}$$

Then $\mathbb{P}(\mathrm{Land}(J)^c) \leq 2e^{-\lambda_P} = 2P^{-4}$*, with no conditioning. Summed over the landing windows* $J \in \mathcal{J}_P$ *defined below, the total is at most* $146H(P)P^{-3}$*. The event* Land(J) *is* $\mathcal{F}_t$*-measurable with* $t(\mathrm{Land}(J)) := \max J + H(\max J)$*.*

Proof. Take the indicators $\mathbf{1}(\text{the prime at } s \text{ lands in } J)$, $s \in \mathcal{C}(J)$, in slot order. Their conditional means sum to $\mu(J) \leq \bar{\mu}(J)$ surely, and the increments are at most 1. Apply Freedman's inequality [8, (4.1)] to both tails. We use it in the form

$$\mathbb{P}(\exists n:\ S_n \geq a,\ V_n \leq v) \leq \exp\Big(-\frac{a^2}{2v + 2ab/3}\Big), \tag{69}$$

for a martingale S_n with increments at most b and predictable quadratic variation V_n, which follows from [8, (4.1)] by scaling and Bennett's bound $(1+u)\log(1+u) - u \geq u^2/(2(1+u/3))$. Take $v = \bar{\mu}$ and $b = 1$. The second form in (68) follows from $\sqrt{2\bar{\mu}\lambda} \leq \bar{\mu} + \lambda/2$. □

Remark 6.2. A bound of the form $\Lambda(J) \le |J|(1+|J|/H)+\lambda$ with failure probability $e^{-\lambda}$ would need $\lambda \ge 3.92\mu$, which is false for $\lambda \lesssim |J|$; it would also condition on later decisions. It is not used.

Landing windows. Let $\mathcal{J}_P$ be the set of windows $[P + i/2,\, P + j/2)$ with integers $0 \le i < j$, $j - i \le 12H(P)$ and $P + j/2 \le 4P + 1$. There are at most $(6P + 3) \cdot 12H(P) \le 73P\,H(P)$ of them. Every $J \subset [P, 4P]$ with $|J| \le 6H(P) - 1$ is contained in a smallest $J^+ \in \mathcal{J}_P$, and $|J^+| \le |J| + 1$.

Corollary 6.3 (Local count)**.** (i) *Let P be dyadic, and let $J \subset [P, 4P]$ with $|J| \le 6H(P)$ (including $|J| < 1$). On the queue bound at the slots $s_j \le \max J + H(\max J)$, and, if $|J| \le 6H(P) - 1$, on* Land(J^+),

$$|N(J)-T(J)| \le 2B(4P)+2\Lambda(J)+2|J|+5 \le (4\rho_H+6)|J|+K_{\mathrm{loc}}(P) \le 11.4|J|+K_{\mathrm{loc}}(P), \tag{70}$$

where $K_{\mathrm{loc}}(P) := 2B(4P) + \frac{7}{3}\lambda_P + 23$. Under the target (5) of Section 3, $T(J) \le 2|J| + 2 + \#\mathcal{A}(J)$, so the non-absorbed count satisfies $N(J) - \#\mathcal{A}(J) \le 13.4|J| + K_{\mathrm{loc}}(P) + 2$.

(ii) *Let $P \ge X_0$ be real, let P_1 be the dyadic scale with $P_1 \le P < 2P_1$, and let $J \subset [P, 4P]$ with $|J| \le 6H(P)$. Split J at $4P_1$ into $J_1 := J \cap [P_1, 4P_1)$ and $J_2 := J \cap [4P_1, 8P_1)$. On the queue bound at the slots $s_j \le \max J + H(\max J)$, and on the events* Land(J_i^+) *of (i) at the dyadic scales P_1 and $2P_1$ (for the pieces with $|J_i| \le 6H(2^{i-1}P_1) - 1$),*

$$|N(J) - T(J)| \le 11.4|J| + K_{\mathrm{loc}}(P) + K_{\mathrm{loc}}(2P),$$
$$N(J) - \#\mathcal{A}(J) \le 13.4|J| + K_{\mathrm{loc}}(P) + K_{\mathrm{loc}}(2P) + 4.$$

The events used are the queue constraints of Definition 6.8 at those slots and the events E4 at the dyadic scales P (in (i)), respectively P_1 and $2P_1$ (in (ii)).

Proof. (i) The count identity (Lemma 3.6), applied to the slots meeting J, gives $|\Delta Q| \le B$, $|\Delta C| \le B + 1$, $|\Delta(j - T)| \le 2$ and $|\Delta L_\pm(J)| \le \Lambda(J) + \#\{\text{slots in } J\} \le \Lambda(J) + |J| + 1$. This gives the first inequality. Now let $|J| \le 6H(P) - 1$. Since $J \subset J^+$, we have $\mathcal{C}(J) \subset \mathcal{C}(J^+)$ and $\Lambda(J) \le \Lambda(J^+)$. On Land$(J^+)$, by (68),

$$2\Lambda(J) \le 4\bar{\mu}(J^+) + \tfrac{7}{3}\lambda_P \le 4(1+\rho_H)(|J|+1) + 8 + \tfrac{7}{3}\lambda_P \le 4(1+\rho_H)|J| + \tfrac{7}{3}\lambda_P + 17.4,$$

using $\rho_H \le 1.35$; this gives the second inequality. If $6H(P) - 1 < |J| \le 6H(P)$, then $|J| \ge H$ and Remark 6.4 gives the bound directly.

(ii) We have $J_1 \subset [P_1, 4P_1]$ and $J_2 \subset [2P_1, 8P_1]$, so (i) applies to J_1 at the dyadic scale P_1 and to J_2 at the dyadic scale $2P_1$, provided $|J_i| \le 6H(2^{i-1}P_1)$. A piece with $6H(2^{i-1}P_1) < |J_i| \le 6H(P)$ has $|J_i| \ge H$, and Remark 6.4 together with $T(J_i) \le 2|J_i|+2+\#\mathcal{A}(J_i)$ gives the same two bounds for it. Add the bounds for the two pieces, and use $K_{\mathrm{loc}}(P_1) \le K_{\mathrm{loc}}(P)$ and $K_{\mathrm{loc}}(2P_1) \le K_{\mathrm{loc}}(2P)$, since K_{loc} is non-decreasing. □

Remark 6.4. Windows longer than $6H(P)$ need no landing event. The sure lag bound (11) for $x \ge X_0 + 2H$, and below that the start-up bound $|D| \le B(X_0) + \frac{H(X_0)}{2}(1 + \kappa_+/\log X_0) + 5$ from the proof of Proposition 4.1, give $|N(J) - T(J)| \le 2\sup|D| \le 2B + H(1 + \kappa_+/L) + 10 \le 1.01|J| + 2B + 6$ for $|J| \ge H$, since $H(0.01 - \kappa_+/L) \ge 4$. Alternatively, splitting J into $\lceil |J|/H \rceil$ pieces gives $N(J) - \#\mathcal{A}(J) \le 13.5|J| + K_{\mathrm{loc}}(P) + 2$.

Remark 6.5 (Uses of Corollary 6.3, and non-circularity). Corollary 6.3 at height z reads the queue bound at slots up to $z + H(z)$ and the event Land at height z. It is therefore used only by steps that run after $z + 7H(z)$, and always at a strictly smaller height. Its uses are:

- Lemma 5.23, step (ii), through part (i) at a dyadic scale $P_1 \le \sqrt{x}$;
- Lemma 5.20, through part (ii) at $P' = 0.51P$ (or at X_0), at heights $p_0 \le \sqrt{x}$;
- Lemma 5.29, through part (ii) at $P = M/2$, at heights $M \le y/2$;

- Proposition 5.35, step 1, for a window that ends $\delta x \gg H$ before x;
- optionally, step (iv) of Lemma 6.25, through part (ii), at heights $u/d \le u^{1-\theta/2}$.

It is *not* used for the per-prime increment bound in Theorem 6.6. A local count there would force $B(x) \ge (3/c')L^2 B(x/q)$, which is incompatible with (71). The increment bound comes instead from E_{cl} (Corollary 6.11).

6.3. **The queue bound.** Fix a dyadic scale x and put $\lambda := 4L$. Recall that $\ell_{\mathrm{jit}}(\cdot)$ counts continuous (jittered-prime) slots.
Windows.

- *Blocks are measured in continuous slots.* $\mathcal{W}_x$ is the set of slot windows $W = (k, j]$ with right end $s_j \in [x, 2x)$ and $\ell_{\mathrm{jit}}(W) \le L_0$, of any total slot count. By (6), continuous slots have density at least $0.09/L$, so such a W has physical length at most $(L_0+1)L/0.09 \le 12L\,L_0$.
- *Freedman's inequality is applied to deterministic windows.* $\mathcal{I}_x$ is the set of physical windows $I = [a, b)$ with $a, b \in \mathbb{Z}$, $b \in [x,\, 2x + 12L + 2]$ and $b - a \le 12L(L_0+1) + 2$.

For $I \in \mathcal{I}_x$ and a latest prime q, let $X_q^{(1)}(I)$ count the non-absorbed composites $n \in I$ with $q \,\|\, n$ and q latest; let $X_q^{(2)}(I)$ count those with $q^a \,\|\, n$, $a \ge 2$; and put $\tilde{X}_q := \min(X_q^{(1)}(I), 2M_1)$. These are $\mathcal{F}_q$-measurable: $n = q^a m$ with m decided before q, and absorption is decided by $q < y(n)$. They depend neither on T nor on the slot positions. Composites have $q \le b/2$, so every $X_q(I)$ is decided by $b/2 + H(b/2)$. The event $\mathrm{Fr}(I)$ below also reads $\ell_{\mathrm{jit}}(I)$ and uses R1′ on I, which holds on $\mathrm{Good}(< \max I)$ (Corollary 5.40); so $\mathrm{Fr}(I)$ carries the time $t(I) := b$.

For $I \in \mathcal{I}_x$, let $\mathrm{Fr}(I)$ be the event that, with $\ell := \ell_{\mathrm{jit}}(I)$ (an integer, possibly random),

(a) $\sum_q \big(\tilde{X}_q - \mathbb{E}[\tilde{X}_q \mid \mathcal{F}_{q-}]\big) \le \sqrt{2v_\ell\lambda} + \frac{4}{3}M_1\lambda$, where $v_\ell := 2M_1(\ell + K_d)$;

(b) $\#\{q :\ X_q^{(2)}(I) \ge 1\} \le \lambda$.

The sums run over latest primes, selected by decision slot.

Theorem 6.6 (Queue bound)**.** *Let x be a dyadic scale. The drift bound R1′ is used in the form of Corollary 5.40, on every window (slot or physical) in $[1, \infty)$, with $L := \log(2x + 12\log x + 2)$; this exceeds $\log x$ by less than 0.7, which R1′'s slack absorbs.*

(i) *For each $I \in \mathcal{I}_x$, $\mathbb{P}(\mathrm{Fr}(I)^c \cap \mathrm{Good}(< t(I))) \le 2(L_0+6)e^{-\lambda}$, with $t(I) = b$. Hence $\sum_{I\in\mathcal{I}_x} \mathbb{P}(\mathrm{Fr}(I)^c \cap \mathrm{Good}(< t(I))) \le 30L(L_0+1)(L_0+6)x^{-3} \le x^{-2}$.*

(ii) *(Deterministic part.) Put $t_j := \lceil s_{j+1} \rceil + 1$, the time of the queue constraint at slot j (Definition 6.8). On $\mathrm{Good}(< t_j) \cap \bigcap_{I\in\mathcal{I}_x,\, t(I)\le t_j} \mathrm{Fr}(I)$, with $s_j \in [x, 2x)$,*

$$Q_j \le B(x) := \frac{4M_1L^2}{c'} + \frac{16M_1L}{3} + 4\sqrt{M_1L(K_d+4)} + 8M_1L\Big(\frac{L}{\Lambda} + 1\Big) + K_d + 9. \tag{71}$$

In particular $B(x) \le 5M_1L^2/c' \le 34M_1L^2$. So $B \le L^{O(1)}H_{\mathrm{std}}^{1/2}$ off the raised range, and $B \le X_0^{7.6}L^2$ on it (Theorem 6.22). Moreover $B \le H$ at every scale: $\log M_1 \le \frac{1}{2}\log H$ since $H \ge M_1^2$, and $34L^2 \le H_{\mathrm{std}}^{1/2} = e^{4\ell^2}$, so $B \le (M_1^2 H_{\mathrm{std}})^{1/2} \le H$.

Proof of (i). Peel over the value of $\ell_{\mathrm{jit}}(I)$. For each deterministic $\ell \in \{0, \dots, L_0 + 5\}$, apply Freedman's inequality in the form (69), with deterministic $a(\ell) := \sqrt{2v_\ell\lambda} + \frac{4}{3}M_1\lambda$, $v := v_\ell$ and $b := 2M_1$ (by truncation).

- On $\{\ell_{\mathrm{jit}}(I) = \ell\} \cap \mathrm{Good}(< t(I))$, the predictable variance satisfies $V \le 2M_1\,\mathrm{Comp}_{\mathrm{jit}}(I) \le 2M_1(\ell + K_d) = v_\ell$, by R1′ (Corollary 5.40) applied to I. So (a) fails there only on the event $\{S \ge a(\ell),\, V \le v_\ell\}$, which has probability at most $e^{-\lambda}$.
- Windows with $\ell_{\mathrm{jit}}(I) > L_0 + 5$ are never used in (ii).

- For (b): take $b = 1$. On the good event, $\sum_q \mathbb{P}(X_q^{(2)}(I) \geq 1 \mid \mathcal{F}_{q-}) \leq \mathrm{Comp}^{(2)}(I) \leq s_2(x)|I| \ll 1$. This is the source s_2 of (43), restated in Corollary 5.40, with
$$s_2 \leq 9L \exp\big(-\max(\log X_0^-, \tfrac{1}{2}\sqrt{L \log L})\big).$$
Since $\log(12L\,L_0) \leq \log M_1 + 4\ell + 10.2$ and $\log M_1 + 5\ell + 13 \ll \frac{1}{2}\sqrt{L \log L}$ at every scale (Theorem 6.22), $s_2 \cdot 12L\,L_0 \ll 1$. Freedman's inequality then gives $\#\{q : X_q^{(2)} \geq 1\} \leq 1 + \sqrt{2\lambda} + \frac{2}{3}\lambda \leq \lambda$, except with probability $e^{-\lambda}$.
- Count: $\#\mathcal{I}_x \leq (x + 12L + 2)(12L(L_0 + 1) + 3)$, and $e^{-\lambda} = x^{-4}$. □

Proof of (ii). Sandwich. Let $W \in \mathcal{W}_x$ have physical interval $[s_{k+1}, s_{j+1})$, and let $I \in \mathcal{I}_x$ be the smallest window with integer endpoints containing it. Then:

- $C_{\mathrm{jit}}(W) \leq \sum_q \big(X_q^{(1)}(I) + X_q^{(2)}(I)\big)$, since C_{jit} is monotone in the window;
- $\ell_{\mathrm{jit}}(I) \leq \ell_{\mathrm{jit}}(W) + 4$, since each of the two unit edges holds at most 2 slots (slot spacing at least 0.99);
- $t(I) = b = \lceil s_{j+1} \rceil \leq t_j$, and $b \leq s_j + 11.3L + 1 \leq 2x + 12L + 2$. Indeed a single slot has length at most $11.3L$ by (6); slot lengths up to 2 occur at primes in $(X_0, 2X_0)$, and longer ones where $T'_{\mathrm{cont}} < \frac{1}{2}$. So $I \in \mathcal{I}_x$.

Hence $\mathrm{exc}(W) \leq C_{\mathrm{jit}}(I) - \ell_{\mathrm{jit}}(I) + 6$, by the excess identity (Lemma 3.9).

Truncation is inactive. On $E_{\mathrm{cl}}^{(x)}$ (whose time $8x/L^{11} + H$ is less than s_j), Corollary 6.11 applies to I, since $|I|/q \leq (12L(L_0 + 1) + 2)/y \leq 13/(HL^5) \ll \delta_1$. It gives $X_q^{(1)}(I)(U) \leq 2M_1$ for every value U of the jitter offset of q. So $\tilde{X}_q \equiv X_q^{(1)}(I)$, and the compensators agree.

Short windows. Let $\ell := \ell_{\mathrm{jit}}(I) \leq \ell_{\mathrm{jit}}(W) + 4 \leq L_0 + 4$. By R1′, $\mathrm{Comp}_{\mathrm{jit}}(I) \leq \ell(1 - c'/L) + K_d$. Together with (a),
$$\mathrm{exc}(W) \leq -\frac{c'\ell}{L} + \sqrt{16M_1L(\ell + K_d)} + \frac{4}{3}M_1\lambda + (\text{the } a \geq 2 \text{ part}) + K_d + 6.$$
The $a \geq 2$ part is at most $\lambda \cdot 2a_{\max}M_1 \leq 8M_1L(L/\Lambda + 1)$, by (b), Corollary 6.11 and $a_{\max} \leq L/\Lambda + 1$. Maximising over $\ell \geq 0$ gives $\mathrm{exc}(W) \leq B(x)$.

Full blocks. If $\ell_{\mathrm{jit}}(W) = L_0$, then $\ell \geq L_0$ and
$$\frac{c'L_0}{L} \geq \sqrt{16M_1L(L_0 + 4 + K_d)} + \frac{16M_1L}{3} + 8M_1L\Big(\frac{L}{\Lambda} + 1\Big) + L^3 + K_d + 10. \tag{72}$$
Indeed, since $M_1 \geq c \geq \frac{1}{4}H_{\mathrm{std}}^{1/2} \gg L^2$, we have $L_0 = 48M_1L^3/c'^2$ at every scale. Then $c'L_0/L = 48M_1L^2/c'$ and $\sqrt{16M_1LL_0} = \sqrt{768}\,M_1L^2/c' \leq 27.72M_1L^2/c'$, so the left side minus the first term on the right is at least $20.28M_1L^2/c' \geq 135M_1L^2$, which exceeds the rest. So $\mathrm{exc}(W) \leq 0$ on full blocks.

Lindley. Let $k' := \max\{k < j : \ \ell_{\mathrm{jit}}((k, j]) = L_0\}$, if it exists. By (67),
$$\begin{aligned} Q_j &\leq \max\Big(Q_{k'} + \mathrm{exc}((k', j]),\ \max\{\mathrm{exc}(W) : \ W \in \mathcal{W}_x,\ W = (k, j],\ \ell_{\mathrm{jit}}(W) < L_0\}\Big) \\ &\leq \max(B(s_{k'}), B(x)) = B(x). \end{aligned}$$
This uses: additivity of exc over slots; $Q_{k'} \leq B(s_{k'})$ on $\mathrm{Good}(< t_j)$, which holds because $t_{k'} \leq s_{k'+1} + 2 \leq s_j$ (a full block has physical length at least $0.99L_0 \gg 12L$); and that B is non-decreasing across scales, because M_1 and H are non-decreasing in x (Theorem 6.22(c)). A window $(k, j]$ with $\ell_{\mathrm{jit}} > L_0$ splits at k'. □

Remark 6.7 (Base scales). R1′ is used on windows $I \subset [x - 12L(L_0 + 1) - 2,\ 2x + 12L + 2]$ from $x = X_0$ on. Below $2X_0^-$ there are no jittered composites, so $C_{\mathrm{jit}} = \mathrm{Comp}_{\mathrm{jit}} = 0$ and R1′ is trivial there. On $[2X_0^-, 2X_0)$ it is part of Corollary 5.40.

6.4. **The probability skeleton.**

Definition 6.8 (Good event; first failure time)**.** Every event E below carries a time $t(E)$, which is a slot, and is $\mathcal{F}_{t(E)}$-measurable. Put

$$\mathrm{Good}(<t) := \{Q_j \le B(s_j) \text{ for all } j \text{ with } t_j < t\} \cap \bigcap_{E:\ t(E)<t} E,$$

$$t_j := \lceil s_{j+1} \rceil + 1, \qquad \tau := \min\{t :\ \mathrm{Good}(\le t) \text{ fails}\},$$

with $B(s) := B(x)$ for $s \in [x, 2x)$. Then τ is an $(\mathcal{F}_t)$-stopping time. The events are listed per dyadic scale $u = 2^m X_0$, with parameters frozen at u, in Table 1.

event	windows at scale u	$\mathbb{P}(E^c \cap \mathrm{Good}(<t(E)))$	$t(E)$
E1: $E_{\mathrm{cl}}^{(u)}$ (Proposition 6.10)	grid windows, left end in $[X_0/2,\ 4u/L^{11}]$	$\le L_u^{-2}$	$8u/L^{11} + H$
E2: $\mathrm{Fr}(I)$ (Theorem 6.6)	grid windows $I \in \mathcal{I}_u$ (integer endpoints, right end $\le 2u + 12L + 2$, length $\le 12L(L_0+1)+2$), peeled over $\ell_{\mathrm{jit}}(I)$	$\le u^{-2}$	$b = \max I$
E3: G_u^{NV} (Lemma 5.22), including the bound (45)	windows with integer endpoints in $[u, 2u]$, length in $[H-1,\ 1.03H+1]$, two-sided, $\lambda'_u := 3\log u + 2\log\log u + 2 + \log H(u)$	$\le 1.2u^{-2}L_u^{-2}$ for E3 and E3′ together (on $E_{\mathrm{cl}}(u)$; sure variance bound $V_I \le 2M_1\,\mathrm{Comp}(I)$)	each window when it closes, $\le 2u$
E3′: G_u^{sw} (Lemma 5.22), including (45)	windows with integer endpoints in $[u, 2u]$, length $\le 1.03H + 1$, λ'_u	(with E3)	each window when it closes, $\le 2u$
E4: $\mathrm{Land}(J)$ (Lemma 6.1)	landing windows $J \in \mathcal{J}_u$ (half-integer grid in $[u, 4u+1]$, length $\le 6H(u)$)	$\le 146H(u)u^{-3}$	$\max J + H(\max J)$

TABLE 1. The events of Definition 6.8 at the dyadic scale $u = 2^m X_0$.

There is no separate event for small latest primes: since $\Lambda \ge 10^8 > L_*$, these are absorbed (Section 6.8), and the small-q case of S1 needs $L < L_*$ (Proposition 6.27).

Lemma 6.9 (First failure; probability budget)**.**

$$\mathbb{P}(\tau < \infty) \le \sum_{m \ge 0} \sum_{E \text{ at scale } 2^m X_0} \mathbb{P}(E^c \cap \mathrm{Good}(<t(E))) \le \frac{1.5}{\Lambda} < \frac{1}{2}.$$

On $\{\tau = \infty\}$, $Q_j \le B(s_j)$ for all j, every event of Table 1 holds, and (12) holds pathwise: $|D(x)| \le B(x) + \frac{H(x)}{2}(1 + \kappa_+/\log x) + 4 \le x^\eta$ for all $x \ge X_0 + 2H$.

Proof. (*) Times of the events. Each bound in Table 1 holds on $\mathrm{Good}(<t(E))$. No conditioning is used: Freedman's inequality is applied to deterministic windows, with deterministic (a, v) after peeling over ℓ_{jit} (E2; likewise E3 and E3′ in Lemma 5.22). Every consumer of an event E runs at a time at least $t(E)$:

- E1 is used by Theorem 6.6, by the noise bounds of Lemma 5.22 and by Lemma 6.25, at scales $\ge u$;
- E2 is used by the queue constraint at slot j, whose time t_j is at least $t(I)$;
- E3 is used by ν on $[u, 2u]$, read at times $\ge 2u$;
- E3′ is used by Proposition 5.38 (the pointwise margin);

- E4 is used by the local counts, read at times at least 4 times their height.

(D) The queue constraint at slot j carries the time $t_j := \lceil s_{j+1} \rceil + 1$.

- t_j is predictable, because T is adapted (Lemma 3.2). The outer window $I \supset W$ of the proof of Theorem 6.6(ii) ends at $b = \lceil s_{j+1} \rceil < t_j$, and R1′ on I holds on $\mathrm{Good}(< \max I)$.
- $\mathrm{Good}(< t_j) \cap \bigcap_{t(E)=t_j} E \subset \{Q_j \le B(s_j)\}$ by Theorem 6.6(ii), and the inputs of that theorem have times at most t_j.
- $t_j \le s_j + 11.3L + 2$. Every consumer of a queue constraint reads it with margin at least $H \gg 12L$: Corollary 6.3 at height z reads slots $\le z + H$ and runs after $z + 7H$; the Lindley step reads $Q_{k'}$ with $t_{k'} \le s_j$; and (12), Section 4 and Section 5 read queue bounds at heights $\le x/2$ or with margin at least $H/2$.
- No event with time at most t_j reads Q_j.

First failure. On $\{\tau = t\}$, $\mathrm{Good}(< t)$ holds and $\mathrm{Good}(\le t)$ fails. If some E with $t(E) = t$ fails, then $E^c \cap \mathrm{Good}(< t(E))$ occurred. Otherwise the queue constraint with time t fails, which (D) excludes.

Summation. Summing Table 1 over m, E1 contributes $\sum_m L_u^{-2} \le \sum_{m \ge 0} (\Lambda + m \log 2)^{-2} \le 1.45/\Lambda$. Everything else contributes $\le \sum_u O(u^{-2}) \ll 1/X_0$.

The target T is adapted (Section 3), and every bound above holds for any adapted T. The union runs over dyadic scales, with finitely many windows per scale, never over all x; note that $\sum_x 1/x$ diverges. □

6.5. **The cluster event and its reduction.** Fix a dyadic height $z \ge X_0/2$; this covers the band $[X_0^-, X_0)$ of centred placement. Put $L := \log z$, and let J be a grid window at height z, so $|J| = 2\delta_1 = 4L^{-6} < 1$. Since $|J| < 1$, J contains at most one element with no jittered prime. Write $S(J)$ for the set of elements of $\mathcal{N}$ in J that have a jittered prime. For $n = M \cdot d$, with M the jittered part and d the rigid (X_0-smooth) part, the *exponent vector* $f_n \in \mathbb{Z}_{\ge 0}^{\mathcal{P}_{\mathrm{jit}}}$ of n is that of M. *Rank*, *span* and *degree* $\deg f := \|f\|_1$ refer to these vectors. Every degree is at most $D := \log 2z / \log X_0^-$, which equals $\log 2z/\Lambda$ up to a relative error at most $e^{-0.98\Lambda}$.

- *(V1) One element per vector.* Two elements of J with the same exponent vector f have the same jittered part $M_f \ge X_0^-$. They differ by $M_f|d - d'| \ge X_0^- > |J|$ unless $d = d'$. So distinct elements of $S(J)$ have distinct exponent vectors.
- *Capacity.* For $S' \subset S(J)$ of rank r, put $\mathrm{Cap}(S') := |S'| - r$.

Proposition 6.10 (The cluster event)**.** *Let $E_{\mathrm{cl}}(z)$ be the event that every grid window J at height z satisfies $\#(\mathcal{N} \cap J) \le M_1(z)$. Then, with $t_z := 2z + H(2z)$,*

$$\mathbb{P}(E_{\mathrm{cl}}(z)^c \cap \mathrm{Good}(< t_z)) \le L^{-2}.$$

The event $E_{\mathrm{cl}}^{(u)} := \bigcap_{X_0/2 \le z \le 4u/L_u^{11}} E_{\mathrm{cl}}(z)$ (over dyadic z) is the event E1 of Definition 6.8. Its new heights at scale u form one dyadic block, so its failure probability on the good event is at most $L_u^{-2}(1 + o(1))$.

Proof. Suppose $\#(\mathcal{N} \cap J) \ge M_1 + 1$, so $|S(J)| \ge M_1 = r_0 + \varphi(r_0) + 1$. Let $r := \mathrm{rank}\, S(J)$.

- *Low heights,* $2z < (X_0^-)^2$, in particular every $z \in [X_0/2, X_0)$. Every element of $S(J)$ is $q \cdot d$ with exactly one jittered prime q, because two jittered primes give a product at least $(X_0^-)^2 > 2z$. (Heights $z \ge X_0$ are in any case covered by the general argument below.) So the exponent vectors are unit vectors, distinct by (V1). Hence $\mathrm{rank}\, S(J) = |S(J)|$ and $\mathrm{Cap} = 0$, so the case $r < r_0$ below is vacuous. The case $r \ge r_0$ is the direct (F1)–(F2) count $E \le (1.01|J|)^{r_0}/r_0!$, to which Theorem A.12 applies; Theorem A.12 is stated for $z \ge X_0/2$.

- *Case $r \geq r_0$.* $S(J)$ contains a rank-full r_0-subset. By Theorem A.12 with $k = r_0$ and $\delta = |J| = 4L^{-6}$, the expected number of rank-full r_0-subsets of $\mathcal{N} \cap J$ on $\mathrm{Good}(< t_z)$ is at most $(e\,\theta A_R \delta)^{r_0} e^{(3H(X_0)+2)r_0/X_0}(1 + r_0/X_0)^{1.02r_0} e^{2.2r_0/\Lambda}$; the factor $e^{(3H(X_0)+2)r_0/X_0}$ is absorbed in the per-element factor 1.01. There are at most $2z/\delta$ windows. The total is at most L^{-4} for $L \geq 465$, by Corollary A.14.
- *Case $r < r_0$.* Then $|S(J)| > r_0 + \varphi(r_0) \geq r + \varphi(r)$. So J is a window as in Theorem 6.22(a): with V a heavy subspace of $S(J)$ of least dimension, Lemma 6.13 gives $\mathrm{Cap}(S(J) \cap V) \geq c$. By Theorem 6.22(a), the probability that some window at height z is of this kind is at most L^{-3}.

Summing the two cases gives the claim. □

Corollary 6.11 (Per-prime increment bound)**.** *Let $u \geq X_0$ be a height. On $E_{\mathrm{cl}}^{(u)}$ the following holds for every window $I \subset [u, 2u + 12L_u + 2]$ (slot window or physical window) with $|I| \leq \max(H(u), 12L_u(L_0(u) + 1) + 2)$, every non-absorbed latest prime $q \geq y(u)$, every $a \geq 1$ and* every *value of the jitter offset of q:*

$$X_q^{(a)}(I) := \#\{n \in I \setminus \mathcal{A} :\ q^a \,\|\, n,\ q \text{ latest}\} \leq 2M_1(u).$$

Proof. The cofactors $m = n/q^a$ lie in I/q^a, which has length at most

$$|I|/y \leq \max\left(1/(L_0 L^6),\, 13/(HL^5)\right) \leq \delta_1(u/q),$$

since $y = H \cdot L_0 \cdot L^6$. They are q-smooth, so they are decided no later than q.

- If I/q^a meets $[X_0^-, \infty)$, it lies at height at least $X_0/2$. It is covered by two grid windows of E_{cl} at that height, each holding at most $M_1(u/q^a) \leq M_1(u)$ elements, since M_1 is non-decreasing.
- If I/q^a lies below X_0^-, the cofactors have no jittered prime, so they are rigid integers. Since I/q^a has length less than 1, it holds at most one of them. □

Remark 6.12. Corollary 6.11 is all that Section 5 (Lemma 5.22, Proposition 5.38) and Theorem 6.6 take from E_{cl}. Shorter windows only make it easier. It covers every window of length at most L^{-6} at any height $\geq X_0$, including the cofactor windows at height u/q^2 of the q^2-composites in Section 5.

Lemma 6.13 (Heavy subspace)**.** *Let $S \subset \mathbb{Z}_{\geq 0}^m \setminus \{0\}$ be finite, of rank less than r_0, with $|S| \geq r_0 + \varphi(r_0)$. Call a subspace W spanned by elements of S* heavy *if $\#(S \cap W) \geq \varphi(\dim W)$. Let V be a heavy subspace of least dimension r. Then:*

(i) $\mathrm{Cap}(S \cap V) \geq \varphi(r) - r \geq c$;
(ii) *every proper subspace $W' \subsetneq V$ spanned by elements of S satisfies $\#(S \cap V \setminus W') \geq Q + r + 1$;*
(iii) *if the elements of $S \cap V$ of degree at most k number at most Q, then the elements of $S \cap V$ of degree at least $k + 1$ span V.*

Proof. $\operatorname{span} S$ is heavy, since φ is increasing and $|S| \geq \varphi(r_0) \geq \varphi(\operatorname{rank} S)$. So V exists, and (i) follows from $\varphi(r) \geq c + r$.

(ii) W' is not heavy, by minimality of V, so $\#(S \cap W') \leq \varphi(\dim W') - 1 \leq \varphi(r-1) - 1$; and $\varphi(r) - \varphi(r-1) = Q + r$.

(iii) Otherwise the elements of high degree lie in a proper element-spanned subspace W', so $S \cap V \setminus W'$ consists of elements of low degree only, of which there are at most $Q < Q + r + 1$. This contradicts (ii). □

6.6. **Capacity forces thin covers.** Throughout this subsection $S \subset \mathbb{Z}^m_{\geq 0} \setminus \{0\}$ is finite, of rank r, and every vector in S has ℓ^1-norm at most D. Put $P(S) := \bigcup_{v \in S} \operatorname{supp} v$. A *cover* is a subset $C \subset S$ with $\bigcup_{y \in C} \operatorname{supp} y = P(S)$, and $\kappa(S)$ is the least size of a cover. For a basis $B \subset S$ and $y \in S \setminus B$ (an *extra*), write $e_y = \sum_{i \in B} \alpha_i e_i$, $I_+ := \{i : \alpha_i > 0\}$, $I_- := \{i : \alpha_i < 0\}$ and $\operatorname{supp}_B(y) := I_+ \cup I_-$. No affine-hyperplane assumption is used in this subsection, except in the last statement of Theorem 6.17.

Lemma 6.14 (Covering lemma; minimal covers)**.** (a) *Every basis $B \subset S$ is a cover.*
(b) *If S has an extra y that is not proportional to a basis vector, then $\kappa(S) \leq r - 1$.*
(c) *A minimal cover C (one with no proper subset that covers) is linearly independent.*

Proof. (a) Every e_y is a combination of the e_i, $i \in B$, and nonnegativity gives $\operatorname{supp} e_y \subset \bigcup_{\alpha_i \neq 0} \operatorname{supp} e_i$.

(b) Rearrange as $e_y + \sum_{l \in I_-} |\alpha_l| e_l = \sum_{i \in I_+} \alpha_i e_i$. All coefficients are positive and all vectors are nonnegative, so

$$\operatorname{supp}(y) \cup \bigcup_{l \in I_-} \operatorname{supp}(e_l) = \bigcup_{i \in I_+} \operatorname{supp}(e_i). \tag{73}$$

If $I_- \neq \emptyset$, then $B \setminus I_-$ covers. If $I_- = \emptyset$, then $\{y\} \cup (B \setminus I_+)$ covers, and it has size at most $r - 1$ when $|I_+| \geq 2$. The remaining case, $|I_+| = 1$ with $I_- = \emptyset$, is the proportional case.

(c) Each $c \in C$ has a prime p_c that lies in no other element of C, for otherwise $C \setminus \{c\}$ covers. The $C \times \{p_c\}$ submatrix is diagonal with positive diagonal. □

Lemma 6.15 (Coefficient support)**.** *If $\kappa(S) \geq r - j$, then for every basis B and every extra y we have $|I_-| \leq j$ and $|I_+| \leq j + 1$. Hence $|\operatorname{supp}_B(y)| \leq 2j + 1$.*

Proof. If $|I_-| \geq 1$, then $B \setminus I_-$ covers by (73), so $r - |I_-| \geq \kappa \geq r - j$. In all cases $\{y\} \cup \{e_l : l \in I_-\} \cup (B \setminus (I_+ \cup I_-))$ covers by (73). It has size $r + 1 - |I_+|$, so $|I_+| \leq j + 1$. □

Theorem 6.16 (Capacity forces thin covers)**.** *For every $j \geq 0$,*

$$\kappa(S) \geq r - j \implies \operatorname{Cap}(S) := |S| - r \leq N(r, D, j) := \sum_{s=1}^{2j+1} \binom{r}{s} \binom{D+s}{s}.$$

Equivalently, $\operatorname{Cap}(S) > N(r, D, j)$ *implies* $\kappa(S) \leq r - 1 - j$. *Also* $N(r, D, j) \leq e\,(r(D+1))^{2j+1}$.

Proof. Fix a basis B. By Lemma 6.15, each extra lies in $\operatorname{span}\{e_i : i \in I\}$ for some $I \subset B$ with $|I| = s \leq 2j + 1$. Choose s coordinates on which this span projects injectively. The extras with support I then map injectively into $\{v \in \mathbb{Z}^s_{\geq 0} : \|v\|_1 \leq D\}$, which has exactly $\binom{D+s}{s}$ points. Sum over I. □

Theorem 6.17 (Thin-Cover Lemma)**.** *If $\kappa(S) \geq r - 1$, then* $\operatorname{Cap}(S) \leq 4r(D+1)^3$. *If S lies on an affine hyperplane $w^{\mathsf{T}} e = 1$ not through 0, the bound improves to* $\operatorname{Cap}(S) \leq 3r(8D + 4)$.

This is the case $j = 1$ of Theorem 6.16, with $O(r)$ supports instead of $\binom{r}{3}$. It is recorded for completeness; Section 6.7 uses Theorem 6.16.

Proof. Supports have size at most 3. This is Lemma 6.15 with $j = 1$. A support of size 1 means that y is proportional to a basis vector; there are at most r such supports, each holding at most $D + 1$ points.

Exchange rule. Write $I(y) := \operatorname{supp}_B(y)$. Let $I(y) \cap I(y') = \{h\}$ for extras y, y'. Put $B' := B - e_h + y$ and substitute for e_h in y'. Nothing cancels, so $|\operatorname{supp}_{B'}(y')| = |I(y)| + |I(y')| - 1 \leq 3$, and hence $|I(y)| = |I(y')| = 2$. In particular a path of three 2-supports $\{1, 2\}, \{2, 3\}, \{3, 4\}$ is

impossible: after exchanging e_2 for y_{12} it produces a 3-support meeting a 2-support in exactly one point.

Components. In the hypergraph of supports of sizes 2 and 3, each component on v_c vertices of B is one of: a star or a triangle of pairs; a book $\{i, j, l_t\}$, possibly together with the pair $\{i, j\}$; or a family inside a 4-set. Each has at most $3v_c$ supports, so there are at most $3r$ in all.

Points per support. A support of size s contributes at most $(D+1)^s$ points, as in Theorem 6.16. So $\mathrm{Cap} \le r(D+1) + 3r(D+1)^3 \le 4r(D+1)^3$.

Hyperplane case. By Lemma 6.15 with $j = 1$, S intersected with a plane is weakly convex: no point of S lies strictly inside a triangle of points of S. Projecting to $[0, D]^2$ gives at most $4(2D+1)$ points per plane. □

6.7. **High-capacity configurations are rare at every scale.**
Setting. Fix a dyadic height $z \ge X_0$, and put $L := \log z$, $\ell := \log L$ and $D := \log 2z / \log X_0^-$, as in Section 6.5; every degree is at most D, and $L/\Lambda \le D \le 2L/\Lambda$. Let J range over the $2z/\delta$ grid windows at height z, where $\delta := 2\delta_1 = |J|$. A *configuration* is a rank-full set $B' \subset S(J)$ of rank $r < r_0$, together with a minimum cover $C \subset B'$ of $P(B')$, so $|C| = \kappa$. Write $R := B' \setminus C$ for the *rows outside the cover*. For $y \in B'$ write $n_y = M_y d_y$, with M_y jittered and d_y rigid. Every prime of B' lies in $P(C)$, and every row has degree at most D, so $|P(C)| \le \pi := \kappa D$.
Skeletons. The greedy pivots of B' (Appendix A) are r primes of $P(C)$. Since E_P is invertible, some term of its Leibniz expansion is nonzero; it matches the rows bijectively with the pivots, each row with a pivot occurring in it. Choose the matching by a fixed rule (say, the lexicographically first permutation with a nonzero Leibniz term), and index the rows by the decision rank of their pivots, the i-th pivot being the i-th earliest. Index the primes of $P(C)$ by decision rank, earliest first; these are the *labels*. The *skeleton* $\mathfrak{s}$ of a configuration records:

(s1) r, and the indices of the rows of C;
(s2) the labels of the r pivots;
(s3) the exponents of the pivots in the rows of C;
(s4) for each row $y \in R$, its monomial M_y in the labels, and its tail d_y.

The skeleton determines the pivot pattern E_P in decision order, the run structure (two consecutive pivots lie in one run if and only if their labels are adjacent) and the number of free primes in each block. It does not record the free primes of the rows of C, their exponents in the rows of C, or the tails of the rows of C; these are summed by Lemma A.5. Skeletons are decision-ordered, and the decision order of the pivots is recorded by (s2), whose count is part of Lemma 6.20(ii).

Proposition 6.18 (Cover Bound)**.** *For every skeleton $\mathfrak{s}$, let $X_{\mathfrak{s}}$ be the number of configurations with skeleton $\mathfrak{s}$ in J, and let $\bar{S}_{\mathfrak{s}}$ be the envelope of Lemma A.5 (with $t_J := t_z$). Then*

$$\mathbb{E}\big[X_{\mathfrak{s}};\ \mathrm{Good}(<t_z)\big] \le \theta^r \bar{S}_{\mathfrak{s}} \le (\theta\delta)^r \bar{\Sigma}_{\mathfrak{s}}, \qquad \bar{\Sigma}_{\mathfrak{s}} := (1+\kappa\lambda)^{s_R} \frac{\kappa^\kappa}{\kappa!} \mathrm{Ch}^\kappa \cdot \prod_{i=1}^{r} \tau_i^{-e_i} \cdot z^{-|R|}, \tag{74}$$

where:

- *the constant* $\mathrm{Ch} := e^\gamma \log(2z + H(2z)) \exp(2.21/\Lambda + 10^{-9})$ *is at least the constant* A_{top} *of Lemma A.6 (S3), as* $5H(X_0)/X_0 \le 10^{-9}$*;*
- $\lambda := \log(A_{\mathrm{top}}/\mathrm{Eu}_{\mathrm{rat}})$*, and* $s_R \le |R|$ *is the number of runs at which no row of C joins (in the staging of Lemma A.8);*
- $e_i \ge 0$ *is the total exponent of the i-th pivot in the rows of C, minus* 1*;*
- $\tau_i := \max(X_0^-, \sigma_i - 2H(\sigma_i))$ *with* $\sigma_i := X_0 + 0.99(i-1)$*, as in (86); the i-th pivot has value at least τ_i. More generally, the label of rank j has value at least τ_j, because the primes of $P(C)$ are distinct jittered primes in distinct slots (Section A.1).*

Proof. From Lemma A.5. The first inequality in (74) is Lemma A.5, applied with $t_J := t_z$ (every element of J has all its primes decided before $2z + H(2z)$) and with the data (s1)–(s4) fixed in advance. Its recursion processes the runs latest first. For each run P_t it (i) records the label sum at u_t, with $\chi_t := \mathbf{1}(u_t \le s_{\max})\mathbf{1}(G_{<u_t})$; (ii) integrates the pivots of the run; and (iii) bounds the sum over the contents of the late block below the run by Lemma A.6 (S3)(a). The chain rows are the rows of C, each starting at the stage where it joins; the covering hypothesis of (S3) is Lemma A.8 (3). The block sums run over the contents compatible with (s1)–(s4), and are bounded by the sums over all contents. The entries of the rows of R in the block columns are fixed by (s4) and carry no weight. The end step treats the early block and the tails of the rows of C by (S3)(c). So $\bar{S}_{\mathfrak{s}}$ is deterministic.

The chain recursion. Let K_t be the number of rows of C in $\mathcal{R}_{\le t}$, and j_t the number of rows of C joining at stage t; $K_T = \kappa$. By Lemma A.6 (S3), the free primes of the rows of C, in all blocks, and their tails, contribute at most $\mathrm{Fin} := G_T(\mathrm{Eu}_{\mathrm{rat}}) + \mathrm{Eu}_{\mathrm{rat}}^{j_T} G_{T-1}(\mathrm{Eu}_{\mathrm{rat}})$, with the recursion (88). At a stage with $j_t = 0$, since G_{t-1} is non-increasing and $E \ge \mathrm{Eu}_{\mathrm{rat}}$,

$$G_t(E) = \int_E^{A_{\mathrm{top}}} \frac{K_t}{x} G_{t-1}(x)\, dx \le K_t \lambda\, G_{t-1}(E),$$

and at the end, if $j_T = 0$, $\mathrm{Fin} \le (1+\kappa\lambda) G_{T-1}(\mathrm{Eu}_{\mathrm{rat}})$. Every step of the recursion is positive and monotone, so these factors can be pulled out. What remains is the recursion of Theorem A.12 for the composition of κ formed by the non-zero j_t, with the same K_t. By the proof of Theorem A.12, its output is at most $F_\kappa(\rho) A_{\mathrm{top}}^\kappa \le (\kappa^\kappa/\kappa!) A_{\mathrm{top}}^\kappa$. Hence $\mathrm{Fin} \le (1+\kappa\lambda)^{s_R} (\kappa^\kappa/\kappa!)\, \mathrm{Ch}^\kappa$. (Every stage has $K_t \ge 1$, by Lemma A.8 (3) applied to the pivot columns of P_1; so no stage has $K_t = 0$.)

Exact identity. By the Volume Lemma (Lemma A.2), the pivot-integral of $\prod_{y \in B'} \mathbf{1}(n_y \in J)$ equals $\int_{J^r} \prod_{\mathrm{piv}} q / \big(|\det E_P| \prod_y n_y\big)\, dn$.

- Order the pivots latest first, and stage the rows by Lemma A.8 (nested lexicographically-first bases with the rows of C listed first).
- The Staged Volume Lemma (Theorem A.10) then integrates the pivots stage by stage. Every late block is handled by the block bound (S3)(a), against the weight $\prod_g g^{-a_g}$ of the rows of C of its stage. This is the condition that Lemma A.8 guarantees.
- $\prod_{\mathrm{piv}} q / \prod_y n_y = \prod_{y \in C} c_y^{-1} \cdot \prod_{y \in R} n_y^{-1} \cdot \prod_{\mathrm{piv}} q^{1 - s_p^C}$, where $c_y := n_y/(\text{pivot part of } n_y)$ and s_p^C is the column sum of p over the rows of C.

Allocation. Every prime has an occurrence in some row of C, since C covers $P(B')$. So:

- each pivot p cancels the factor q_p^{+1} of the Jacobian against one occurrence in a row of C, and its remaining $e_i = s_p^C - 1$ occurrences there contribute at most $\tau_i^{-e_i}$, by the caps of Lemma A.5;
- the free primes of the rows of C keep their weight g^{-a_g} in the block bounds, and the tails of the rows of C their weight $1/d_y$; together they give Fin;
- a row of R gets $1/n_y \le 1/z$; its monomial and tail are fixed by (s4).

Bookkeeping. The factor $\theta = 1 + 1/\Lambda$ per pivot comes from (F1)–(F2) and the slot sums, each row's n_y is integrated over J once, which gives δ^r, and $|\det E_P| \ge 1$. □

Remark 6.19. Every row of the basis B' is a Jacobian row. Only extras outside B' would be mere indicators, and none occur (Lemma 6.20(i)). Being a Jacobian row is what supplies the factor $1/n_y \le 1/z$ of a row of R, and it does not cost the rigid count. The per-row weights below never credit the excess weights $\prod_i \tau_i^{-e_i}$ beyond the pattern sum over the rows of C in Lemma 6.20(ii).

Lemma 6.20 (Reduction to configurations; type union)**.** (i) *(Extras drop out.) Let* $S' := S(J) \cap V$ *with* V *as in Lemma 6.13, of rank* r *and with* $\mathrm{Cap}(S') \ge c$. *Then* $\kappa(S') \le$

$r-1-j(c)$, *where* $j(c):=\max\{j:\ e(r(D+1))^{2j+1}<c\}$ *(Theorem 6.16). A minimum cover* C *of* S' *is independent (Lemma 6.14(c)), so it extends to a basis* $B'\subset S'$ *with* $P(B')=P(S')$ *(Lemma 6.14(a)). So* $|R|=r-\kappa\geq 1+j(c)$*. If moreover the elements of* S' *of degree at most* k *number at most* Q*, the completion can be chosen with every row of* R *of degree at least* $k+1$ *(Lemma 6.13(iii) and Steinitz exchange).*

(ii) *(Type union.) Sum (74) over skeletons and multiply by the* $2z/\delta$ *windows. For* $L\geq 60$*, the result is, per height, at most*

$$13L^{5.1}\cdot z\cdot\prod_{y\in R}w_y,\qquad w_y:=z^{-1}\sum_{M_y}\Psi(2z/M_y,X_0), \tag{75}$$

where M_y *runs over the monomials of the degree class of the row in the* π *labels, with the* j*-th label evaluated at* τ_j*, as in (74).*

(iii) *(Consequence.) The probability that some window at height* z *contains a configuration with* $|R|\geq 1+j$ *and all* $w_y\leq w$ *is at most* $r_0\cdot 13L^{5.1}z\cdot w^{1+j}$*. It is at most* L^{-3} *as soon as*

$$(1+j)\log(1/w)\geq L+10\ell+10. \tag{76}$$

Proof. Proof of (ii). Fix r. The quantity $\bar\Sigma_{\mathfrak s}$ of (74) is deterministic. We sum it over the data (s1)–(s4) of the skeletons:

- (s1): at most 2^r choices;
- (s2): the labels of the pivots form an r-subset of $\{1,\dots,\pi\}$, so there are at most $\binom{\pi}{r}\leq(e\pi/r)^r\leq(eD)^r$ choices, as $\kappa\leq r$;
- (s3): if the i-th pivot is matched to a row of C, its exponent there is at least 1, and the sum of $\tau_i^{-e_i}$ over its exponents in the κ rows of C is $(1-1/\tau_i)^{-\kappa}$. If it is matched to a row of R, it still occurs in a row of C, and the sum is $\tau_i\big((1-1/\tau_i)^{-\kappa}-1\big)\leq\kappa(1-1/\tau_i)^{-\kappa-1}$. By the estimate in the proof of Lemma A.4 (with r in place of k), $\prod_{i\leq r}(1-1/\tau_i)^{-1}\leq B:=e^{(3H(X_0)+2)/X_0}(1+r/X_0)^{1.02}\leq 2.1L^{1.02}$, as $r\leq L$. So (s3) contributes at most $\kappa^{|R|}B^{\kappa+1}\leq\kappa^{|R|}B^{2\kappa}$;
- (s4): the label of rank j has value at least τ_j, so a row of R with $n_y\leq 2z$ has $M_y(\tau)d_y\leq 2z$, where $M_y(\tau)$ is M_y with the j-th label evaluated at τ_j. With the factor z^{-1} of $\bar\Sigma_{\mathfrak s}$, the sum over (M_y,d_y) is at most w_y.

In $\bar\Sigma_{\mathfrak s}$ we have $\kappa^\kappa/\kappa!\leq e^\kappa$, $\mathrm{Ch}\leq e^\gamma(L+0.7)(1+10^{-7})\leq 2L$, and $1+\kappa\lambda\leq L$ for $L\geq 60$, since $\mathrm{Eu}_{\mathrm{rat}}\geq 1$ gives $\lambda\leq\log(2L)$, and $\kappa\leq r-1\leq r_0-2\leq L/(3\ell)+2$. Also $\kappa\leq r\leq L$ and $D\leq 2L/\Lambda\leq L$. Distributing the factors over the rows, a row of C costs at most $\theta\delta C'$, with

$$C':=2\cdot eD\cdot e\cdot\mathrm{Ch}\cdot B^2\leq 2e^2\cdot L\cdot 2L\cdot 4.41L^{2.04}\leq 131L^{4.04}\leq 6L^{5.1}\qquad(L\geq 60),$$

and a row of R costs at most $\theta\delta\cdot 2\cdot eD\cdot\kappa(1+\kappa\lambda)\cdot w_y\leq 8e\theta L^{-3}w_y\leq w_y$. Hence the sum is at most

$$\frac{2z}{\delta}(\theta\delta C')^\kappa\prod_{y\in R}w_y\leq 2z\cdot\theta C'\cdot(\theta\delta C')^{\kappa-1}\prod_{y\in R}w_y\leq 13L^{5.1}z\prod_{y\in R}w_y,$$

since $\kappa\geq 1$ and $\theta\delta C'=24\theta L^{-0.9}\leq 1$ for $L\geq 60$. The rows are indexed by their pivots, so the decision order costs nothing beyond (s2).

Proof of (iii). Sum (ii) over the fewer than r_0 possible ranks. Since $w\leq 1$, we have $\prod_{y\in R}w_y\leq w^{|R|}\leq w^{1+j}$. As $z=e^L$, $r_0\leq L=e^\ell$ and $\log 13\leq 2.6$, the bound $r_0\cdot 13L^{5.1}z\,w^{1+j}$ is at most L^{-3} once $(1+j)\log(1/w)\geq L+9.1\ell+2.6$, which (76) implies. □

Lemma 6.21 (Row weights)**.** *Let* $y\in R$ *have degree* $k\geq 1$*, so that* $M_y\geq(X_0^-)^k$ *and* $d_y\leq 2z/(X_0^-)^k$*.*

(a) $w_y \le \sum_{k' \ge k} \binom{\pi+k'-1}{k'} \Psi(2z/(X_0^-)^{k'}, X_0)/z$.

(b) *(High degree.) If* $e(\pi+k)/(kX_0^-) \le 1/3$, *then* $w_y \le 3\,(e(\pi+k)/k)^k (X_0^-)^{-k}$; *that is,* $\log(1/w_y) \ge k(\log X_0^- - 1 - \log((\pi+k)/k)) - \log 3$.

(c) *(Smooth tails.) Put* $u := \log Y/\Lambda$. *For* $\max(10^4, \log\Lambda) \le u \le X_0^{0.3}$ *we have* $\Psi(Y, X_0) \le Y \exp(-0.9u \log u)$. *For* $u > X_0^{0.3}$ *we have*

$$\Psi(Y, X_0) \le \exp\big(1.26 X_0 \log(2\log Y)/\Lambda\big).$$

(d) *(Low degree, smooth tails.) Put* $u_1 := \max(10^4, \log\Lambda)$. *If* $2u_1 \le D-1$, $D \le X_0^{0.3}$ *and* $\pi(X_0^-)^{-1}(eD)^{0.9} \le 1/3$, *then* $\log(1/w_y) \ge g := \log X_0^- - \log(3\pi) + 0.9(D-1)\log(D-1) - e^{-0.4\Lambda}$.

(e) *(Joint Rankin bound at exponent* $\frac{3}{4}$*; any* L*.)*

$$\log(1/w_y) \ge \frac{L}{4} - \log 2 - \log\Lambda - 2 - \frac{5.2 X_0^{1/4}}{\Lambda} - 5.5(X_0+\pi)^{1/4}.$$

Proof. (a) Count the pairs (M_y, d_y).

(b) Use $\Psi(Y, X_0) \le Y$. The terms $\binom{\pi+k'-1}{k'} \cdot 2(X_0^-)^{-k'}$ have consecutive ratio $(\pi+k')/((k'+1)X_0^-) \le 1/(3e)$, by the hypothesis. So the sum over $k' \ge k$ is at most $2.3\binom{\pi+k-1}{k}(X_0^-)^{-k} \le 3(e(\pi+k)/k)^k (X_0^-)^{-k}$.

(c) We use Rankin's bound $\Psi(Y, y) \le Y^\sigma \prod_{p \le y}(1-p^{-\sigma})^{-1}$ with $y = X_0$ and $\sigma = 1-\eta$, where $\eta := \log(u\log u)/\Lambda$. Since $u \le X_0^{0.3}$, we have $\eta \le 0.3 + \log(0.3\Lambda)/\Lambda \le 0.31$, so $\sigma \ge 0.69$.

- $Y^\sigma = Y \exp(-u\log u - u\log\log u)$.
- Write $\log \prod_{p \le X_0}(1-p^{-\sigma})^{-1} = \sum_{p \le X_0} p^{-\sigma} + R(\sigma)$ with $R(\sigma) := \sum_p \big(-\log(1-p^{-\sigma}) - p^{-\sigma}\big)$. This is decreasing in σ, and $R(0.69) \le 0.77$ and $R(0.75) \le 0.63$ (computer-assisted: the sum over $p \le 10^7$ in interval arithmetic, plus the tail $\sum_{n > 10^7} n^{-2\sigma}/(2(1-n^{-\sigma}))$; see Appendix B.5).
- $\sum_{p \le X_0} p^{-\sigma} = \sum_{p \le X_0} 1/p + \sum_{p \le X_0} g(p)$ with $g(t) := (t^\eta - 1)/t$. The first sum is at most $\log\Lambda + 0.27$ [16, Thm. 5, (3.18)]. The function g decreases for $t \ge t_0 := (1-\eta)^{-1/\eta}$, and $t_0 \in [e, e^{1.45}]$. Partial summation with $\pi(t) \le 1.26t/\log t$ [16, (3.6)], dropping the negative part of $-\int \pi g'$ on $[2, t_0]$, makes the boundary terms at X_0 cancel and gives

$$\sum_{p \le X_0} g(p) \le 1.26\Big(\frac{t_0^\eta - 1}{\log t_0} + \int_{t_0}^{X_0} \frac{t^\eta - 1}{t\log t}\,dt\Big) \le 1.26\Big(\frac{\eta}{1-\eta} + \mathrm{Ein}(\eta\Lambda)\Big),$$

$$\mathrm{Ein}(x) := \int_0^x \frac{e^w - 1}{w}\,dw.$$

  Here $\eta\Lambda = \log(u\log u) \ge 11.4$ and $e^{\eta\Lambda} = u\log u$. For $x \ge 11$, $\mathrm{Ein}(x) \le \mathrm{Ei}(x) \le (e^x/x)(1 + 2/x)$: integrating by parts twice on $[11, x]$ gives $\mathrm{Ei}(x) \le (e^x/x)(1 + 1/x + 2.75/x^2) + 134$, and $2.75/x + 134x^2 e^{-x} \le 1$ there.
- Altogether, with $1.26\eta/(1-\eta) \le 0.6$,

$$\log \prod_{p \le X_0}(1-p^{-\sigma})^{-1} \le \log\Lambda + 0.27 + 0.6 + 1.26\,\frac{u\log u}{\log(u\log u)}\Big(1 + \frac{2}{\log(u\log u)}\Big) + 0.77$$
$$\le \log\Lambda + 1.3u + 2,$$

  since $\log u\,(1 + 2/\log(u\log u)) \le \log(u\log u)$ when $\log\log u \ge 2$.
- So, for $u \ge \max(10^4, \log\Lambda)$,

$$\log(\Psi/Y) \le -u\log u - u\log\log u + 1.3u + \log\Lambda + 2 \le -0.9u\log u.$$

This needs $0.1u\log u + u\log\log u - 1.3u \geq \log\Lambda + 2$, which holds there. (At $u = 10^4$ alone it would need $\log\Lambda \leq 1.84\cdot 10^4$; this is why the lower limit is $\max(10^4, \log\Lambda)$.)

- For $u > X_0^{0.3}$: an X_0-smooth $d \leq Y$ has at most $1 + \log Y/\log 2$ choices of exponent for each of the at most $1.26X_0/\Lambda$ primes $p < X_0$.

(d) The k'-th term of (a) has $Y = 2z/(X_0^-)^{k'}$, so $u = \log Y/\Lambda = (D-k')\log X_0^-/\Lambda$. For $k' = 1$ this gives $(D-1)(1-\varepsilon) \leq u \leq D-1$ with $\varepsilon := e^{-0.98\Lambda}/\Lambda$, so $u_1 \leq u \leq X_0^{0.3}$ and (c) applies. Since $D-1 \leq e^{0.3\Lambda}$, we have $u\log u \geq (D-1)\log(D-1) - (D-1)\varepsilon(\log(D-1)+2) \geq (D-1)\log(D-1) - e^{-0.48\Lambda}$. So the $k' = 1$ term is at most $2\pi\exp(-0.9(D-1)\log(D-1) + e^{-0.48\Lambda})/X_0^-$.

- Consecutive values of u differ by $\log X_0^-/\Lambda \leq 1$. So, while the next term still has $u \geq u_1$, the ratio of consecutive terms is at most $\pi(X_0^-)^{-1}(eD)^{0.9}$, which is at most $1/3$ by hypothesis.
- Once $u < u_1$, bound the term by $\Psi \leq Y$ instead. The first such term is at most $\pi(X_0^-)^{-1}e^{0.9(u_1+1)\log(u_1+1)}$ times the previous one, which is far less than $\frac{1}{3}$, and the following ratios are at most $(\pi + k')/((k'+1)X_0^-) \leq \frac{1}{3}$.
- So $w_y \leq \frac{3}{2}(1+e^{-\Lambda})\cdot 2\pi\exp(-0.9(D-1)\log(D-1) + e^{-0.48\Lambda})/X_0^-$, which gives g. Where (d) is used (Steps 3–4 of the proof of Theorem 6.22), $D - 1 \geq \Lambda e^{s_H+1} - 1 \geq 2u_1$ and $D \leq 2L/\Lambda \leq X_0^{0.3}$, as $\ell \leq 0.3\Lambda$ there.

(e) Rankin's bound $\mathbf{1}(Md \leq 2z) \leq (2z/(Md))^{3/4}$ gives

$$w_y \leq z^{-1}\sum_{(M,d)}\Big(\frac{2z}{Md}\Big)^{3/4} \leq 2z^{-1/4}\prod_{j\leq\pi}(1-\tau_j^{-3/4})^{-1}\prod_{p<X_0}(1-p^{-3/4})^{-1}.$$

- The rigid product: its logarithm is $\sum_{p<X_0}p^{-3/4} + R(\frac{3}{4})$, with $R(\frac{3}{4}) \leq 0.63$ as in (c). Partial summation with $\pi(t) \leq 1.26t/\log t$ gives $\sum_{p<X_0}p^{-3/4} \leq 1.26\big(X_0^{1/4}/\Lambda + \frac{3}{4}\int_2^{X_0} t^{-3/4}dt/\log t\big)$, and $\int_2^{X_0} t^{-3/4}dt/\log t = \mathrm{Ei}(\Lambda/4) - \mathrm{Ei}(\frac{1}{4}\log 2) \leq 4X_0^{1/4}(1+8/\Lambda)/\Lambda + 1$, by the bound on Ei in (c) and $\mathrm{Ei}(\frac{1}{4}\log 2) \geq -1$. So the logarithm is at most $5.05X_0^{1/4}/\Lambda + 0.95 + 0.63 \leq \log\Lambda + 2 + 5.2X_0^{1/4}/\Lambda$.
- The labels are evaluated at $\tau_i = \max(X_0^-, \sigma_i - 2H(\sigma_i)) \geq \sigma_i - 2H(\sigma_i) \geq \frac{2}{3}\sigma_i = \frac{2}{3}(X_0 + 0.99(i-1))$, since $H(v) \leq v/6$. With $(3/2)^{3/4} \leq 1.3555$ and $\tau_i^{-3/4} \leq X_0^{-0.7}$,

$$\begin{aligned}\log\prod_{j\leq\pi}(1-\tau_j^{-3/4})^{-1} &\leq (1+2X_0^{-0.7})\,1.3555\sum_{i=1}^{\pi}(X_0+0.99(i-1))^{-3/4}\\ &\leq (1+2X_0^{-0.7})\,1.3555\Big(X_0^{-3/4} + \frac{4\big((X_0+\pi)^{1/4} - X_0^{1/4}\big)}{0.99}\Big),\end{aligned}$$

  which is at most $5.5(X_0+\pi)^{1/4}$.
- Since $\log(2z^{-1/4}) = \log 2 - L/4$, the three bounds give the claim. □

Theorem 6.22 (High-capacity configurations at every scale; the capacity parameter)**.** *Let $\Lambda = \log X_0 \geq 10^8$ be arbitrary (Section 2.2), and let $\theta \in [\frac{1}{4}, \theta_1]$ with $\theta_1 = \frac{1}{2} - 10^{-4}$, as in Section 2, so that $\theta_b = \theta/2 \geq \frac{1}{8}$. For a height $z \geq X_0$ (not necessarily dyadic) put $\ell := \log\log z$ and $s := \ell - 2\log\Lambda$, so that $L = \Lambda^2 e^s$. Define*

$$A(\Lambda,s) := \log\big(2\Lambda^3 e^{2s}/(2\log\Lambda + s)\big) = \log\big(2L^2/(\ell\Lambda)\big), \qquad G(\Lambda,s) := \frac{2\sqrt{A(A+1)}\,e^{s/2}}{\sqrt{\Lambda - A - 1}},$$

$$\Lambda_s := \max(10^8, e^{-s}), \qquad \gamma_*(s) := 1.3\,G(\Lambda_s, s).$$

Let $s_H = 5.9745\ldots$ be the root of $\gamma_*(s_H) = \frac{1}{3} - 10^{-5}$. Put $\gamma(s) := \gamma_*(s)$ for $s \le s_H$, $\gamma(s) := \gamma_*(s_H) + (7.5 - \gamma_*(s_H))(s - s_H)$ on $[s_H, s_H + 1]$, and $\gamma(s) := 7.5$ for $s \ge s_H + 1$. Define

$$\log c(z) := \max\big(4\ell^2 - \log 4,\ \gamma(s)\Lambda\big), \qquad \textit{that is,} \qquad c = \max\big(\tfrac{1}{4}H_{\mathrm{std}}^{1/2},\ X_0^{\gamma(s)}\big).$$

Then, for every $\Lambda \ge 10^8$, parts (a) and (b) hold at every dyadic height $z \ge X_0$, and part (c) holds for all heights $z \ge X_0$:

(a) *(Counting.) The probability, on $\mathrm{Good}(< t_z)$, that some grid window J at height z has $\operatorname{rank} S(J) < r_0$ and $|S(J)| \ge r_0 + \varphi(r_0)$ is at most L^{-3}. For such J, let V be a heavy subspace of least dimension for $S = S(J)$ (Lemma 6.13); then $S' := S(J) \cap V$ satisfies $\mathrm{Cap}(S') \ge c(z)$.*

(b) *(Absorption compatibility.) If $s \le s_H$, then $y(z) < X_0^- = X_0 - H(X_0)/2$, so $\mathcal{A}$ has no element at height z. If $s > s_H$, condition (77) of Lemma 6.25 holds at z.*

(c) *(Regularity.)*
- *$c(z)$, $\bar{M}_1(z)$ and $H(z) = \max(H_{\mathrm{std}}(z), \bar{M}_1(z)^2)$ are continuous, non-decreasing and piecewise C^1 in z, with locally finitely many break points; in particular H is absolutely continuous on compact intervals. The per-scale parameters M_1, L_0, y and B are non-decreasing across scales.*
- *At every $z \ge X_0$, both one-sided derivatives of $\log H$ with respect to $\log z$ are at most $16\ell/L + 10^{-7} \le 4 \cdot 10^{-6}$. (At a break point, $d\log H/d\log z$ denotes the larger one-sided derivative.) Here $16\ell/L$ comes from H_{std}, and the raise contributes at most 10^{-7}. Hence, H being continuous, $H(vw)/H(v) \le w^{4\cdot 10^{-6}}$ for $v \ge X_0$ and $w \ge 1$ (the regularity (18)), $\rho_H \le 1 + 1.2 \cdot 10^{-5}$, and $H(v)/v$ is strictly decreasing.*
- *$\log H(z) - \log H_{\mathrm{std}}(z) \le 10^{-6} \log z$; so the raise adds nothing to the lag.*
- *$c = \frac{1}{4}H_{\mathrm{std}}^{1/2}$ for $\ell \ge \ell_{\mathrm{std}} := \sqrt{(7.5\Lambda + \log 4)/4}$, which is $1.37 \cdot 10^4$ at $\Lambda = 10^8$.*

Parts (b) and (c) are deterministic statements about the parameters. Their proofs (Steps 5 and 6 below) use only the definitions of γ, c, M_1, $\bar{r}_0$, $\bar{M}_1$, L_0, y, H and B, and not part (a). In particular (F3′) and ρ_H, which use (c), do not depend on (a).

Proof. Step 0 (reduction). Let J and $S' = S(J) \cap V$ be as in (a), so $\mathrm{Cap}(S') \ge c$ by Lemma 6.13(i). By Lemma 6.20(i) and Theorem 6.16, S' yields a configuration with $|R| \ge 1 + j$ rows outside a minimum cover, where $j \ge (\log c - 1)/(2a) - 3/2$ and $a := \log(r_0(D+1))$. By Lemma 6.20(iii) it suffices to show $(1+j)\log(1/w) \ge L + 10\ell + 10$ for a common bound w on the row weights. We use:

- $r_0 \le L/(2\ell)$ (as $L \ge \Lambda \ge 10^8$) and $L/\Lambda \le D \le 2L/\Lambda$;
- $m := r_0 D \le L^2/(\ell\Lambda)$, so $\log 2m \le A$. Also $a \le A - \log 2 + \log(1 + 1/D) \le A$, since $D \ge 1$, and $a \le A - \log 2 + 10^{-9}$ when $D \ge 10^9$ (as in Step 3);
- $A \le 2\ell$, and $\log c \ge 4\ell^2 - \log 4 \ge 3.9\ell^2$ (since $\ell \ge \log\Lambda \ge 18.42$).

Step 1 (G is decreasing in Λ). At fixed s, $\partial A/\partial\Lambda = (3 - 2/\ell)/\Lambda \in (0, 3/\Lambda)$. So

$$\partial_\Lambda \log G^2 \le \frac{3}{\Lambda}\Big(\frac{1}{A} + \frac{1}{A+1}\Big) - \frac{1 - 3/\Lambda}{\Lambda - A - 1} < 0,$$

because $A \ge \log(2\Lambda/\ell) \ge 16 > 6$. Every admissible pair has $\Lambda \ge \Lambda_s$, since $z \ge X_0$ means $s \ge -\log\Lambda$. Hence $\log c \ge \gamma_*(s)\Lambda \ge 1.3\,G(\Lambda, s)\Lambda$ for every $s \le s_H$. Moreover γ_* is non-decreasing in s, since at fixed Λ both $e^{s/2}$ and A increase with s ($\partial_s A = 2 - 1/\ell > 0$, and G increases with A), while Λ_s is non-increasing in s and $\Lambda \mapsto G$ decreases.

Step 2 (rows of high degree: $s \le s_H + 1$). Recall $Q = c^{1/2}$.

- The vectors of degree at most k in m coordinates number at most $\binom{m+k}{k} \le \exp(k(1+\log 2m))$. So the elements of degree at most $D^* := \lfloor \log Q/(1+\log 2m)\rfloor$ number at most Q.
- If $D^* \ge D$, then Cap $\ge c$ is impossible, by Lemma 6.13(iii) and (V1). Otherwise Lemma 6.20(i) completes the cover with rows of degree at least $k := D^*+1 \ge \log c/(2(A+1))$. If no such row exists, Lemma 6.13(iii) gives a contradiction; this includes the case $D^* = D-1$ with D an integer.
- Lemma 6.21(b) applies, since $\pi \le m \le \Lambda^3 e^{14} \ll X_0^-$. It gives $\log(1/w) \ge k(\log X_0^- - A - 1) - \log 3 \ge k(\Lambda - A - 1) - 1.1$, since $(\pi+k)/k \le 2m$, $\log 2m \le A$ and $k(\Lambda - \log X_0^-) \le De^{-0.98\Lambda} \le 10^{-3}$ (here $D \le 2\Lambda e^{s_H+1}$).
- Therefore, using $(A+1)/\log c \le (2\ell+1)/(3.9\ell^2) \le 0.03$,

$$(1+j)\log(1/w) \ge \frac{(\log c)^2(\Lambda - A - 1)}{4A(A+1)}(1-0.031) \ge 1.69\cdot 0.969\cdot \Lambda^2 e^s \ge 1.63L \ge L + 10\ell + 10,$$

  as soon as $\log c \ge 1.3\, G(\Lambda, s)\Lambda$.
- This holds for $s \le s_H$ by Step 1. On $[s_H, s_H+1]$ it holds because $\gamma(s) \ge \gamma_*(s)$ there: $\Lambda_s = 10^8$ on this range, and $G(10^8, \cdot)$ grows at logarithmic rate at most $\frac12 + 2/A + 10^{-7} \le 0.532$ (as $A \ge 64.1$ there), so $\gamma_*(s_H+t) \le \gamma_*(s_H)e^{0.532t} \le \gamma_*(s_H) + 0.24t \le \gamma(s_H+t)$ for $t \in [0,1]$.

Step 3 (Dickman rows: $s \ge s_H+1$). Here $\log c \ge 7.5\Lambda$. We have $D-1 \ge \Lambda e^s - 1 \ge 2u_1$, and for $\ell \le 0.3\Lambda$ the other hypotheses of Lemma 6.21(d) hold: $D \le 2L/\Lambda \le X_0^{0.3}$ and $\pi(X_0^-)^{-1}(eD)^{0.9} \le e^{0.87\Lambda - \Lambda + 1}$. So $\log(1/w) \ge g \ge 0.9\Lambda e^s(\log\Lambda + s)(1-10^{-9})$; the term $-e^{-0.4\Lambda}$ in g is absorbed by the factor $1-10^{-9}$, and $\log X_0^- - \log(3\pi) \ge 0$. Also $1+j \ge (\log c - 1 - a)/(2a) \ge (7.5\Lambda - 1 - a)/(2a)$, with $a \le a_+ := 3\log\Lambda + 2s - 2.9 \le 0.6\Lambda$ (as $\ell \le 0.3\Lambda$). Hence, since the bound decreases in a,

$$\frac{(1+j)g}{L} \ge 0.45\,(1-10^{-9})\Big(7.5 - \frac{1+a_+}{\Lambda}\Big)\frac{\log\Lambda + s}{a_+}, \qquad \frac{\log\Lambda + s}{a_+} = \frac13 + \frac{s+2.9}{3a_+}.$$

If $a_+ \le 10^{-3}\Lambda$, this is at least $0.45\cdot 7.498/3 \ge 1.12$. Otherwise $s + 2.9 \ge (a_+ - 3\log\Lambda)/2$, so $(\log\Lambda + s)/a_+ \ge \frac12 - 10^{-4}$, and the bound is at least $0.45\cdot 6.89\cdot 0.4999 \ge 1.5$. In both cases it exceeds $1 + (10\ell+10)/L$. Where the standard term dominates $\log c$, j is only larger.

Step 4 (tails).

- (T1) $\ell_{\rm std} \le \ell \le 0.3\Lambda$. Here $\log c = 4\ell^2 - \log 4$ and $j \ge \ell - 2$. The hypotheses of Lemma 6.21(d) hold as in Step 3, and it gives $(1+j)g/L \ge 0.9(\ell-1)(\ell - \log\Lambda - 0.01)(1-\Lambda/L)/\Lambda \ge 0.9\cdot 1.87 > 1.6$.
- (T2) $\ell \ge 0.3\Lambda$. Lemma 6.21(e) gives $\log(1/w) \ge L/4 - 5.3X_0^{1/4}/\Lambda - 5.6(X_0+L^2)^{1/4} - \log\Lambda - 3 \ge 0.24L$, since $L \ge e^{0.3\Lambda}$. Then $j \ge 4$ suffices.

So (a) holds at every scale, including the scales $L \ge X_0^2$ of the zeros.

Step 5 (proof of (b)).

- $s \le s_H$. The bounds $\log M_1, \log\bar M_1 \le \log c + \log 2$, $\log H = \max(8\ell^2, 2\log\bar M_1)$ and $\log L_0 \le \log 48 - 2\log c' + \log M_1 + 3\ell = 7.67 + \log M_1 + 3\ell$ (with $c' = 0.15$) give $\log y \le 3\log c + 9\ell + 11.14$. The constant is 11.14 when $H_{\rm std}$ dominates ($8\ell^2 \le 2\log c + 2\log 4$), and 9.8 when $\bar M_1^2$ does. If $\log c = \gamma\Lambda \le (\frac13 - 10^{-5})\Lambda$, this is at most $\Lambda - 2597 < \log X_0^-$, since $9\ell + 11.14 \le 3\cdot 10^{-5}\Lambda - 2597$ for $\ell \le 2\log\Lambda + 6$, and $\log X_0^- \ge \Lambda - e^{-0.98\Lambda}$. If $\log c = 4\ell^2 - \log 4$, it is at most $12\ell^2 + 9\ell + 7 \le \Lambda$.
- $s \ge s_H$ *with* $\log c = \gamma\Lambda \le 7.5\Lambda$. Here $\log 2y \le 3.01\gamma\Lambda$ and $\log C_d + 4\ell \le 1.01\gamma\Lambda$. With $u_0 \ge \theta_b\Lambda e^s/(3.01\gamma)$, condition (77) follows from $e^s\log(\theta_b\Lambda e^s/(3.01\gamma)) \ge 107.6\gamma^2$. The left side is increasing in Λ and s. At $\Lambda = 10^8$ and $s = s_H + t$ it is at least $393e^t(19.2+t)$, against $107.6(0.334 + 7.17t)^2 \le 6053$ on the ramp and $= 6053$ after it. Also $u_0 \ge 100$.

- *Standard regime, $\ell \geq \ell_{\mathrm{std}}$.* Here $\log 2y \leq 12.1\ell^2$, the right side is at most $16.3\ell^2$, and $u_0 \geq e^\ell/(97\ell^2)$. So (77) holds with an enormous margin.

Step 6 (proof of (c)).

- *Continuity and monotonicity.* $\Lambda_s = \max(10^8, e^{-s})$ and γ_* are continuous; γ is continuous at s_H and at $s_H + 1$ by construction; and γ is non-decreasing. So $c(z) = \exp(\max(4\ell^2 - \log 4, \gamma(s)\Lambda))$ is continuous and non-decreasing in z, and so are $\bar{r}_0$, $\bar{M}_1$ and H. Each is built from analytic functions of $\log\log z$ by finitely many maxima and the three break points $s = -\log 10^8, s_H, s_H + 1$ of γ, so each is piecewise C^1 with locally finitely many break points. The per-scale $M_1 = r_0 + \varphi(r_0) + 1$, L_0, y and B are non-decreasing across scales, since r_0 and $c(x)$ are.
- *Derivative.* Where $H = H_{\mathrm{std}}$, $d\log H/dL = 16\ell/L$. Where $H = \bar{M}_1^2 > H_{\mathrm{std}}$, we have $c \geq \bar{M}_1/2 > \frac{1}{4}H_{\mathrm{std}}^{1/2}$, so we are on the raised range and $\log c = \gamma(s)\Lambda$. There $\bar{M}_1 \geq c$, $d\bar{r}_0/dL \leq 1/(3\ell)$, $\partial\bar{M}_1/\partial\bar{r}_0 = \frac{3}{2} + c^{1/2} + \bar{r}_0 \leq 1.01c^{1/2}$ and $\partial\bar{M}_1/\partial c = 1 + \bar{r}_0/(2c^{1/2})$, so
$$\frac{d\log H}{dL} = 2\,\frac{d\log \bar{M}_1}{dL} \leq 2\big(1 + \bar{r}_0 c^{-1/2}\big)\frac{\Lambda\gamma'(s)}{L} + \frac{0.7}{\ell c^{1/2}}.$$
Here $\bar{r}_0 c^{-1/2} \leq 2Le^{-2\ell^2} \leq e^{-600}$ and $0.7/(\ell c^{1/2}) \leq e^{-600}$, because $c \geq \frac{1}{4}e^{4\ell^2}$ and $\ell \geq 18.42$. On the ramp $2\Lambda\gamma'(s)/L \leq 2 \cdot 7.2/(\Lambda e^{s_H}) \leq 4 \cdot 10^{-10}$; for $s \geq s_H + 1$, $\gamma' = 0$; for $s \leq s_H$ the next item applies.
- For $s \leq s_H$, where $\gamma_*\Lambda \geq 4\ell^2 - \log 4$ (so that γ_* dominates the standard term), the raise contributes $2\Lambda\gamma_*'(s)/L = 2\kappa\gamma_*\Lambda/L$, with $\kappa := d\log\gamma_*/ds$. Put $\ell_s := 2\log\Lambda_s + s$ and $A_s := A(\Lambda_s, s)$; then $18.42 \leq \ell_s \leq \ell$, since $\Lambda_s \leq \Lambda$ and $s \geq -\log\Lambda$. Dominance gives $1/\Lambda \leq \gamma_*/(4\ell^2 - \log 4)$, so
$$\frac{\gamma_*\Lambda}{L} = \frac{\gamma_*}{\Lambda e^s} \leq \frac{\gamma_*^2}{(4\ell^2 - \log 4)e^s} = \frac{6.76A_s(A_s+1)}{(4\ell^2 - \log 4)(\Lambda_s - A_s - 1)} \leq \frac{1.7A_s(A_s+1)}{\ell_s^2\Lambda_s}.$$
  - If $s \geq -\log 10^8$, then $\Lambda_s = 10^8$ and $\ell_s \leq 42.82$. Here $\kappa = \frac{1}{2} + \frac{1}{2}A'(\frac{1}{A} + \frac{1}{A+1}) + \frac{1}{2}A'/(\Lambda_s - A - 1)$ with $A' = 2 - 1/\ell_s$, so $\kappa \leq \frac{1}{2} + 2/A_s + 10^{-7} \leq 0.63$, as $A_s \geq 16.2$. Also $A_s \leq 2\ell_s - 20.6$, so $A_s(A_s+1)/\ell_s^2 \leq (2 - 19.6/42.82)^2 \leq 2.38$. The contribution is at most $2 \cdot 0.63 \cdot 1.7 \cdot 2.38 \cdot 10^{-8} \leq 5.1 \cdot 10^{-8}$.
  - If $s < -\log 10^8$, then $\Lambda_s = e^{-s} = e^{\ell_s}$ and $A_s = \log 2 + \ell_s - \log\ell_s$, so $A_s + 1 \leq \ell_s$. The variation of Λ_s adds at most $\frac{1}{2}(1 + 10^{-6})$ to the logarithmic rate, while the A-terms are now non-positive, so $\kappa \leq 1 + 10^{-6}$. The contribution is at most $2(1 + 10^{-6}) \cdot 1.7 \cdot e^{-\ell_s} \leq 3.5 \cdot 10^{-8}$.

  So $2\Lambda\gamma_*'(s)/L \leq 5.1 \cdot 10^{-8}$ for $s \leq s_H$.
- *Conclusion.* The pieces are C^1 up to their break points, so the bounds above hold for both one-sided derivatives at every point: $d\log H/dL \leq \max\big(16\ell/L,\ (1 + e^{-600})\,5.1 \cdot 10^{-8} + e^{-600}\big) \leq 16\ell/L + 10^{-7}$, and $16\ell/L \leq 16\log\Lambda/\Lambda \leq 3 \cdot 10^{-6}$. Since H is continuous and piecewise C^1, integration gives $\log H(vw) - \log H(v) \leq 4 \cdot 10^{-6}\log w$ for $v \geq X_0$ and $w \geq 1$, which is (18); in particular $\rho_H \leq 16^{4\cdot 10^{-6}} \leq 1 + 1.2 \cdot 10^{-5}$. Also $d\log(H(v)/v)/d\log v \leq 4 \cdot 10^{-6} - 1 < 0$, so $H(v)/v$ is strictly decreasing.
- $(\log H - \log H_{\mathrm{std}})/L \leq \max(0, 2\log\bar{M}_1 - 8\ell^2)/L \leq (2\gamma\Lambda + 2)/L \leq 10^{-6}$ at every height, using $\bar{M}_1 \leq 2c$. □

Remark 6.23 (Numerical cross-check). The proof of Theorem 6.22 is analytic. As an independent check, the precise criteria behind Steps 2–5 were evaluated numerically with c as above: D^* from $\binom{m+k}{k}$, the exact j, Lemma 6.21(b) and (d), and (77) with the exact y. On the grid $\Lambda \in \{10^8, 10^9, 10^{10}, 10^{12}, 10^{16}, 10^{20}\}$, $s \in [-\log\Lambda, 30]$ with step 0.01 (about 35 000 points), there

are no failures. An earlier cell-by-cell check at $\Lambda = 10^8$ alone, with a knot-based ramp, also passes: 419 591 cells, each checked at its worst corners with the degree cap D evaluated at the top of the cell, and no failures. The maximum of $(\log H - \log H_{\mathrm{std}})/L$ over $\Lambda \in [10^8, 10^{30}]$ is $1.3 \cdot 10^{-8}$, attained near $s \approx -9.9$ at $\Lambda = 10^8$. (Numerical; see Appendix B.5.)

Remark 6.24. At $\Lambda = 10^8$, s_H corresponds to $\ell = 42.82$ and the ramp ends at $\ell = 43.82$. On the raised range, $c \le X_0^{7.5}$ and the absorption threshold satisfies $y \le X_0^{22.6} L^9$; elsewhere $y = H_{\mathrm{std}} L_0 L^6 = e^{O(\ell^2)}$. In both cases $y \le n^{o(1)}$, as Lemma 4.4 requires.

6.8. **Absorption.** Recall from Section 3 the absorption set

$$\mathcal{A} = \{n \in \mathcal{N} : \ \log n \ge L_*, \ \text{the latest prime } r \text{ of } n \text{ is jittered, and } r < y(n)\},$$

with $L_* = 10^7$ and $y(n) := y(x) = H(x)L_0(x)L(x)^6$, where x is the dyadic scale with $n \in [x, 2x)$; that is, y is frozen per scale, as in Section 6.1. Put $\theta_b := \theta/2$ (so $\theta_b = \frac{1}{8}$ for $\theta = \frac{1}{4}$), $h(u) := u^{\theta_b}$ and $\rho_{\mathcal{A}}(u) := \#(\mathcal{A} \cap [u, u+h])/h$. The target T gets a unit jump at each $n \in \mathcal{A}$, and T' gets $-\rho_{\mathcal{A}}$ (see (5)). Membership is decided when r is decided, at slot approximately $r < y(n) \ll n$, so T is adapted.

Every jittered prime is at least $X_0^- = X_0 - H(X_0)/2$ (centred placement), so $\mathcal{A}$ *has no element at heights where* $y < X_0^-$. By Theorem 6.22(b) these are all heights with $s = \log\log n - 2\log\Lambda \le s_H = 5.9745$ (that is, $\log\log n \le 42.82$ at $\Lambda = 10^8$). So below that height Lemma 6.25 is vacuous, and, since $\Lambda \ge 10^8 > L_*$, the condition $\log n \ge L_*$ plays no role.

Lemma 6.25 (Pointwise sparsity of $\mathcal{A}$)**.** *Let $u \ge X_0$ and $L := \log u$, and put $y := \sup_{v \in [u,2u]} y(v)$ and $C_d := M_1(u)L^6/2$. Suppose*

$$0.9 u_0 \log u_0 \ge \log(2y) + \log C_d + 4\log L, \qquad u_0 := \frac{\theta_b L}{\log(2y)}, \qquad u_0 \ge 100. \tag{77}$$

Then $\#(\mathcal{A} \cap [u, u+h]) \le h\, L^{-4}$ on $\mathrm{Good}(< t)$ for every $t \ge t_u := 9u/L^{11} + H(u)$. In particular this holds on $\mathrm{Good}(< t)$ at heights up to $t + 1.01H(t)$, since $9(t+1.01H)/L^{11} + H < t$. Condition (77) holds at every u at which $\mathcal{A}$ meets $[u, 2u]$, by Theorem 6.22(b).

Proof. (i) *Smoothness.* Every prime of $n \in \mathcal{A} \cap [u, u+h]$ is decided no later than its latest prime $r < y$, so it has value less than $r + H(r) < 2y$.

(ii) *Divisor trick.* Multiply the primes of n in any order and stop at the first partial product $d > h$. Then $d \in (h, 2yh]$ is a $2y$-smooth element of $\mathcal{N}$, and $n = d \cdot m$ with $m \in \mathcal{N}$.

(iii) Hence $\#(\mathcal{A} \cap [u, u+h]) \le \sum_{d \in \mathcal{G}_{2y} \cap (h, 2yh]} N(I_d)$, where $\mathcal{G}_{2y}$ is the set of $2y$-smooth elements of $\mathcal{N}$ and $I_d := [u/d, (u+h)/d]$ has length less than 1.

(iv) *Window count.* I_d lies at height $u/d \le u^{1-\theta_b}$, and it is covered by at most $L^6/4 + 1$ grid windows there. On $\mathrm{Good}(< u) \subset E_{\mathrm{cl}}^{(u)}$ each of them holds at most $M_1(u/d) \le M_1(u)$ elements, so $N(I_d) \le C_d$. (Alternatively, use Corollary 6.3 at height u/d.)

(v) *Rankin's bound for the realised system.* The primes below $2y$ are the rational primes below X_0, together with jittered primes, which satisfy $\pi_{\mathcal{N}}(s) \le 2s/\log s$ (Lemma 5.1(i); the local prime number theorem itself is not needed below X_{LP}, (84)). For $Y \ge h$ put $u_* := \log Y/\log(2y)$ and $\sigma := 1 - \log u_*/\log(2y)$, and suppose $u_* \ge 2\Lambda$ and $\log\log 2y \le 1.79 u_*$. (In use $u_* \ge u_0$, and $u_0 \ge 2\Lambda$ wherever $\mathcal{A} \ne \emptyset$: by Step 5 of the proof of Theorem 6.22, $u_0 \ge \theta_b \Lambda e^s/(3.01\gamma) \ge 5\Lambda$ on the raised range, and $u_0 \ge e^{\ell}/(97\ell^2)$ beyond it. The same step gives $\log 2y \le \max(22.6\Lambda, 12.1\ell^2)$, so $\log\log 2y \le 1.79 u_*$ as well.) Rankin's bound and partial summation as in Lemma 6.21(c), with $2s/\log s$ in place of $1.26s/\log s$, give

$$\Psi_{\mathcal{N}}(Y, 2y)/Y \le \exp\big(-u_*\log u_* + 2.3u_*/\log u_* + \log\log 2y\big) \le \exp(-0.9 u_* \log u_*);$$

here 2.3 bounds $2(1 + 2/\log u_*)$ plus the absolute constants, since $\log u_* \geq 19$. For $Y \geq h$ we have $u_* \geq u_0$.

(vi) So $\#(\mathcal{A} \cap [u, u+h]) \leq C_d \Psi_{\mathcal{N}}(2yh, 2y) \leq C_d \cdot 2yh \cdot \exp(-0.9 u_0 \log u_0) \leq hL^{-4}$, by (77). □

No Diophantine input is used: rigid runs are tamed in step (iv), where every count is moved into windows of length less than 1.

Lemma 6.26 (Consequences for T')**.** (a) $\sup \rho_{\mathcal{A}} \leq L^{-4}$ *at every height.*

(b) *The source perturbation* $\sigma_{\mathcal{A}} := K_{\mathcal{A}}\tilde{\rho} - \rho_{\mathcal{A}}$ *of Section 5 satisfies* $|\sigma_{\mathcal{A}}| \leq 2L^{-4}$ *pointwise. Its compensator form uses* $u = L/\log 2y \geq 10^8$ *wherever* $\mathcal{A} \neq \emptyset$*, the raised range included.*

(c) *(Current-height form.) On* Good$(< u)$ *the* $\mathcal{A}$*-part of* T' *is* $\sigma_{\mathcal{A}}$*, since* $\bar{\rho}_{\mathcal{A}} = \rho_{\mathcal{A}}$*. So the* $\mathcal{A}$*-part of the oscillation term* o_u *of Section 5 satisfies*

$$o_{\mathcal{A}}(u) \leq 2 \sup |\sigma_{\mathcal{A}}| \leq 4L^{-4}$$

(or $\leq 2 \sup \rho_{\mathcal{A}} + 2 \sup |K_{\mathcal{A}}\tilde{\rho}| \leq 8L^{-4}$ *if the two parts are taken separately). This uses only (a) and (b), that is, Lemma 6.25, which reads* E_{cl} *at heights below* u*. So it involves no queue bound and no event* Land *at heights* $\geq u$*. Then* $\sup_v \log v \cdot o_{\mathcal{A}}(v) \leq 8 \min L^{-3} \leq e^{-120}$ *wherever* $\mathcal{A} \neq \emptyset$*, since there* $L \geq \Lambda^2 e^{s_H}$*.*

Part (c) is the form used where o_u *enters* $R1'$ *at the current height (see (63) and Proposition 5.38).*

Proof. (a) is Lemma 6.25. (b) $K_{\mathcal{A}}$ is the part of K_{pt} whose landing is absorbed, that is, whose latest prime is less than $y(\cdot)$. So its cofactors w are y-smooth with $w \geq y/y(y)$. With the *sure* bound $\tilde{\rho} \leq 1.01$ (at most one prime per continuous slot), $K_{\mathcal{A}}\tilde{\rho} \leq 1.01 \sum 1/w$ over such w. This is at most L^{-4} by Rankin's bound, with $u = L/\log 2y \geq 10^8$ and Lemma 5.1(i) for the realised primes below $y(y)$, which are decided long before. Together with (a) this gives $|\sigma_{\mathcal{A}}| \leq 2L^{-4}$. No local count at the current height is used. (c) An oscillation is at most twice the supremum. □

Proposition 6.27 (Small latest primes are absorbed)**.** (S1) *(The small-q truncation in the noise bounds of Lemma 5.22.) The case of latest primes* $q < y$ *with jittered cofactors needs* $\log n < L_*$*. Since* $L_* = 10^7 < \Lambda$*, it is vacuous. Latest primes* $q \in [X_0^-, y)$ *are absorbed.*

(S2) *(The full blocks of Theorem 6.6 at large heights.) The arrivals that would break (72) are rigid runs* $r \cdot r' \cdot d$ *driven by small jittered latest primes. These have latest prime less than* y*, so they lie in* $\mathcal{A}$*. Non-absorbed arrivals have latest prime* $q \geq y$*, so Corollary 6.11 applies to them, and (72) holds as shown in Theorem 6.6.*

In particular no small-latest-prime event appears in the budget (Definition 6.8).

Proof. This is contained in the statement: (S2) combines the definition of $\mathcal{A}$ with Corollary 6.11 and the proof of Theorem 6.6(ii); (S1) is the comparison $L_* < \Lambda$. □

Lemma 6.28 (Raised threshold)**.** *On the raised range of Theorem 6.22, where* $\gamma(s)\Lambda > 4\ell^2 - \log 4$*, we have* $y(n) \geq y_{\mathrm{std}}(n) := H_{\mathrm{std}}(x) L_{0,\mathrm{std}}(x) L(x)^6$ *(with* x *the dyadic scale of* n*), since* H*,* M_1 *and* L_0 *are all at least their standard values. So* $\mathcal{A}_{\mathrm{std}} \subset \mathcal{A}$*: raising* y *only* adds *to* $\mathcal{A}$ *the composites whose latest jittered prime* q *lies in* $[y_{\mathrm{std}}(n), y(n))$*. The consumers of* $\mathcal{A}$ *are unaffected:*

- *Theorem 6.6 and the noise bounds (Lemma 5.22) see only latest primes* $q \geq y$*. Corollary 6.11 needs* $|I/q| \leq H/y = 1/(L_0 L^6)$*, which holds because* y *uses the same raised* H *and* L_0*. No consumer assumes* $y = y_{\mathrm{std}}$*. Latest primes in* $[y_{\mathrm{std}}, y)$ *are absorbed, not repaired.*
- *Lemma 5.21: its "y" is the start of the operator, not the absorption threshold. Its bound uses Mertens' bound over all of* $\mathcal{N}$*, so it is independent of* y*. Its threshold* $H(v) \geq (\log v)^{8.5+\delta}$ *only gets easier as* H *grows.*

- *Proposition 5.38: its only $\mathcal{A}$-input is Lemma 6.26(c), which does not involve y and reads only heights below u.*
- *Lemma 6.25: (77) holds by Theorem 6.22(b). It is read with $y := \sup_{[u,2u]} y$. Since y is frozen per scale, this sup involves at most two scales, and along the ramp $y(2x)/y(x) \le 2^{10^{-5}}(1+2c^{-1/2})$ by Theorem 6.22(c) and $M_1 \le \bar{M}_1 \le M_1(1+2c^{-1/2})$. So this changes nothing numerically.*
- *Lemma 4.4: the bound on the absorbed part holds for $y = X_0^{O(1)} \le n^{o(1)}$.*
- *Queue size and (12): $B \le 5M_1L^2/c' \le X_0^{7.6}L^2$ enters additively or as x^η, and is $\ll x^\theta$ at these heights.*
- *Causality: $y(n) = y(x)$ is a deterministic function of the dyadic scale x of n, so membership in $\mathcal{A}$ is decided at the decision of the latest prime.*

Proof. Each item is a direct check against the cited statement, as indicated. □

6.9. **Dependencies and non-circularity.** Table 2 lists the inputs of the results of this section and of Appendix A.
Non-circularity.

- Theorem 6.6 at slot s_j reads only events with times at most s_j: E_{cl} at cofactor heights at most $4u/L^{11}$, and Corollary 5.40 and Proposition 5.38.
- E_{cl} reads no queue bound at its own heights. Its inputs are (F1)–(F4) at heights at most $s-H$, carried by progressive events.
- Corollary 6.3 is used only at strictly smaller heights.
- Lemma 6.25 uses E_{cl} only at heights $u/d < u$.
- The first-failure argument (Lemma 6.9) orders all of these by event time.

7. The base range: the linear-window target

This section supplies the source term of the margin in Section 5: the rigid part of the target, its short-interval prime-number input, and the decomposition of the sharp reference density. The margin itself and its noise bound are proved in Section 5.

The design. This is the hard greedy rule and the target of Section 3.2, with the rigid set averaged over the *linear* window $[u,(1+\kappa)u]$ and normalised by $1/(\kappa u)$, where $\kappa = X_0^{-5/12+2\varepsilon_\kappa}$. For a measure h on $[1,\infty)$ and $w>0$ write

$$\mathrm{MA}_w[h](x) := \frac{h([x,x+w])}{w}, \tag{78}$$

so that $\rho_{\mathrm{det}}(u) = \mathrm{MA}_{\kappa u}[\mathcal{D}](u)$, and write $h_{(m)}$ for the dilate of h by m, i.e. $h_{(m)}(A) = h(A/m)$.

Lemma 7.1 (Properties of the linear-window target)**.**

(1) Excess identity. *Rigid integers sit on unit jumps of T, so Lemma 3.9 holds verbatim. Also $\rho_{\mathrm{det}} \le 1 + 1/(\kappa u)$, and $1-\rho_{\mathrm{det}} \ge 0.99/\log u - 1/(\kappa u)$ by (80).*

(2) Dilation commutation. *For any point measure h and $m \ge 1$,*

$$\mathrm{MA}_{\kappa x}[h_{(m)}](x) = m^{-1}\,\mathrm{MA}_{\kappa q}[h](q), \qquad q = x/m,$$

because the window is linear. Write the non-rigid integers as $\bigsqcup_{m'} m' \cdot \mathbb{P}_{\ge \max(P^+(m'),X_0)}$, prime powers included. The m'-component of the target at x equals $1/m'$ times the target's fresh-prime density at $q = x/m'$. So the mismatch between nested windows vanishes identically, for every m'.

(3) Partial windows. *When $P^+(m') \in (q,(1+\kappa)q]$, the component is a partial window. Those $m' = pm$ have $p \in (\sqrt{x/m}, \sqrt{(1+\kappa)x/m}]$ and total weight $\ll \kappa/\log x$, so the correction is a relative $O(\kappa)$.*

(4) Edges near X_0. *The ramps of integers in $[X_0, X_0(1+\kappa))$ are truncated at X_0. This gives E a constant offset $|E_0| \le \kappa X_0 + 1$, which changes only C_{X_0}. In the margin it is part of μ_0, through Δ (Lemma 7.3(b), Step 1).*

(5) Drift. *With the linear normalisation the ramp mass is $1 - \kappa/2 + O(\kappa^2)$, so E drifts by $+(\kappa/2)\Psi(x, X_0)$. This is contained in (7), and every consumer uses only $|drift| \le \kappa x^{0.011}$ at the scales of the zeros. In the margin this smooth source is part of μ_0, with $|\mu_0(u)| \le 2\kappa(\log u)^3$ (Lemma 7.3(b)).*

Proof. (2) $\mathrm{MA}_{\kappa x}[h_{(m)}](x) = h([x,(1+\kappa)x]/m)/(\kappa x) = m^{-1}h([q,(1+\kappa)q])/(\kappa q)$. The other items are immediate from the definitions, as indicated. □

7.1. **The short-interval prime number theorem.** Every window of the construction lies in the range of the short-interval prime number theorem. Indeed, for every $q \ge X_0$,

$$\kappa q \ge q^{7/12+\varepsilon_\kappa} \iff q^{5/12-\varepsilon_\kappa} \ge X_0^{5/12-2\varepsilon_\kappa} \impliedby q \ge X_0,$$

and the window has length $\le q$. The input used is the asymptotic count of primes in intervals of length between $q^{7/12+\varepsilon_\kappa}$ and q, uniformly.

Short windows. Heath-Brown proved [9, Theorem, (1.2), p. 23]: for any function $0 \le \varepsilon_{\mathrm{HB}}(x) \le \frac{1}{12}$, $\pi(x) - \pi(x-y) = (y/\log x)\{1 + O(\varepsilon_{\mathrm{HB}}(x)^4) + O((\log\log x/\log x)^4)\}$ uniformly for $x^{7/12-\varepsilon_{\mathrm{HB}}(x)} \le y \le x/(\log x)^4$. We use $\varepsilon_{\mathrm{HB}} \equiv 0$:

$$\pi(x) - \pi(x-y) = \frac{y}{\log x}\Big\{1 + O\Big(\Big(\frac{\log\log x}{\log x}\Big)^4\Big)\Big\} \qquad \text{uniformly for } x^{7/12} \le y \le x/(\log x)^4. \tag{79}$$

The asymptotic for $y \ge x^{7/12+\varepsilon_0}$ with fixed $\varepsilon_0 > 0$ is Huxley's [11], as Heath-Brown credits [9, p. 22]; Heath-Brown removes the ε_0.

Long windows. For $y > x/(\log x)^4$, which the windows κq reach once $\log q > \kappa^{-1/4}$, we use the prime number theorem with the de la Vallée Poussin error, $\pi(t) = \mathrm{li}(t) + O(t\exp(-c_{\mathrm{dVP}}\sqrt{\log t}))$. Applied at both ends of an interval, it gives relative error $\le C(\log q)^5\exp(-c_{\mathrm{dVP}}\sqrt{\log q})$.

The threshold X_{Hux}. For $q \ge 2$ put

$$\eta_{\mathrm{H}}(q) := \sup\Big\{\Big|\frac{\pi(I)}{\int_I dt/\log t} - 1\Big| :\ I \subset [q,2q] \text{ an interval, } |I| \ge (2q)^{7/12}\Big\},$$
$$X_{\mathrm{Hux}} := \min\big\{X \ge e^{10^8} :\ \eta_{\mathrm{H}}(q) \le (\log q)^{-3} \text{ for all } q \ge X\big\}. \tag{80}$$

Every window $[q,(1+\kappa)q]$ with $q \ge X_0$ is such an I. Let $x \le 2q$ be the right end of I. If $|I| \le x/(\log x)^4$, then (79) applies, since $|I| \ge (2q)^{7/12} \ge x^{7/12}$ and $\int_I dt/\log t = (|I|/\log x)(1 + O((\log x)^{-5}))$; otherwise the long-window bound applies. So

$$\eta_{\mathrm{H}}(q) \le \max\big(C_{\mathrm{HB}}(\log\log q/\log q)^4,\ C(\log q)^5 e^{-c_{\mathrm{dVP}}\sqrt{\log q}}\big) = o((\log q)^{-3}),$$

and X_{Hux} is finite. It depends on no parameter of the construction (in particular not on κ). It is effective in principle, since every input is effective, but it has not been computed: we do not know of an explicit form of the Huxley or Heath-Brown asymptotic. Theorem 2.1 is an existence statement, so $X_0 \ge X_{\mathrm{Hux}}$ is enough. The exponent 3 is what Lemma 7.3(c) needs.

Consequence used throughout. For every $q \ge X_0 \ge X_{\mathrm{Hux}}$, $\mathrm{MA}[\mathbb{P}](q)\log q \ge (1-(\log q)^{-3})(1 - \kappa/\log q) \ge 0.99$ (Lemma 7.3(a)). This feeds (6), the bound $1 - \rho_{\mathrm{det}} \ge 0.99/\log u - 1/(\kappa u)$, and the base values of Proposition 5.35 (0.82 for $x \le X_0^{1.01}$ and 0.97 for $x \ge X_0^{1.01}/2$; Lemma 7.3(d)).

Remark 7.2 (A fully explicit route, not used). Johnston and Yang [12, Thm. 1.1 and Table 1, row $X = 10^8$] prove $|\psi(x) - x| \le 2.18 \cdot 10^{-4694}x$ for $\log x \ge 10^8$; for $\theta(x)$ their Corollary 1.2 gives the same form with $A_1 = A + 0.01$. With $\kappa \approx 200 \cdot e^{-10807.6}$ in place of the short-interval input above, this would remove the threshold X_{Hux}. It would not by itself make X_0 explicit:

the constants of the local prime number theorem are not explicit, and with this κ the range $\log x \le (0.0048/\kappa)^{1/5} \approx e^{2159}$ of the μ_0 bound in Lemma 7.3(b) would no longer cover range (A) of Proposition 5.35.

7.2. **The margin decomposition.** Write the non-rigid rational integers as $\bigsqcup_{m'} m' \cdot \mathbb{P}_{\ge \max(P^+(m'),X_0)}$, and put

$$\mathrm{MA}[\mathbb{P}](q) := \mathrm{MA}_{\kappa q}[\text{rational primes} \ge X_0](q),$$

the window density of the new primes. It vanishes for $(1+\kappa)q \le X_0$, it is a partial window for $X_0/(1+\kappa) < q < X_0$, and for $q \ge X_0$ it is the density of all rational primes in $[q,(1+\kappa)q]$. Let $K_{\mathbb{Q}}$ be the latest-prime renewal operator with *rational* cofactors. It acts on a window density $h(q) = \mathrm{MA}_{\kappa q}[\nu](q)$ of a signed measure ν on $[X_0,\infty)$ by components,

$$(K_{\mathbb{Q}}h)(x) := \sum_{m'\in\mathbb{N},\ m'\ge 2} \frac{1}{m'}\,\mathrm{MA}_{\kappa q}\big[\nu|_{[P^+(m'),\infty)}\big](q)\Big|_{q=x/m'},$$

and $K_{\mathbb{Q}}(h\mathbf{1}_{[X_0,\infty)})$ keeps only the components with $x/m' \ge X_0$. For $h = \mathrm{MA}[\mathbb{P}]$ the component of m' is $m'^{-1}\mathrm{MA}_{\kappa q}[\text{rational primes} \ge \max(P^+(m'),X_0)](q)$ at $q = x/m'$. Let K_{pt} be the sharp operator of Section 5, with realised cofactors and latest prime in decision order (Section 3.4). By dilation commutation (Lemma 7.1(2)) we have, for $x \ge X_0$, the exact rational identity

$$T'_{\mathrm{cont}} - R' + \bar\rho_{\mathcal{A}} = (1+K_{\mathbb{Q}})\,\mathrm{MA}[\mathbb{P}] + O(1/(\kappa x)). \tag{81}$$

The identity is stated only for $x \ge X_0$, where T'_{cont} is defined by (5); below X_0 the target has $T'_{\mathrm{cont}} = 0$, although $\mathrm{MA}[\mathbb{P}] > 0$ on $(X_0/(1+\kappa), X_0)$. That window enters (81) only through $K_{\mathbb{Q}}$: a cofactor $N \ge 2$ with $x/N \in (X_0/(1+\kappa), X_0)$ contributes the partial window of the integers Np, $p \ge X_0$, in $[x,(1+\kappa)x]$.

Put $\rho^{\mathrm{sh}} := (1+K_{\mathrm{pt}})^{-1}(T'_{\mathrm{cont}} - R' + \bar\rho_{\mathcal{A}})$, a function on $[X_0^-,\infty)$ that vanishes on $[X_0^-, X_0)$. Near the cut-off the two operators include the following components.

- $K_{\mathbb{Q}}$ (rational system): every cofactor N whose new prime $p \ge X_0$ lies in the window at $q = u/N$; for $q \in (X_0/(1+\kappa), X_0)$ this is a partial window.
- K_{pt} (cut-off $u/w \ge X_0^-$): every realised cofactor w with $u/w \ge X_0^-$. On $[X_0^-, X_0)$ it reads only functions that vanish there (ρ^{sh} and the functions in (82) below), so its cut-off may be taken to be X_0; and it never reads $(X_0/(1+\kappa), X_0^-)$.

So on $(X_0/(1+\kappa), X_0)$ only $K_{\mathbb{Q}}$ contributes, and its contribution there is exactly μ_{edge} below. Writing $T'_{\mathrm{cont}} - R' + \bar\rho_{\mathcal{A}} = (1+K_{\mathrm{pt}})(\mathrm{MA}[\mathbb{P}]\mathbf{1}_{[X_0,\infty)}) + (K_{\mathbb{Q}} - K_{\mathrm{pt}})(\mathrm{MA}[\mathbb{P}]\mathbf{1}_{[X_0,\infty)}) + \mu_{\mathrm{edge}} + O(1/(\kappa x))$ on $[X_0,\infty)$, and applying $(1+K_{\mathrm{pt}})^{-1}$, gives, for $x \ge X_0$,

$$\rho^{\mathrm{sh}} = \mathrm{MA}[\mathbb{P}] + \rho_{\mathrm{osc}} + \rho_{\mathrm{edge}} + (1+K_{\mathrm{pt}})^{-1}(\mu_0 + \mu_1), \qquad \rho_{\mathrm{edge}} := (1+K_{\mathrm{pt}})^{-1}\mu_{\mathrm{edge}}, \tag{82}$$

where ρ_{osc} is the response to R' and the absorbed source is carried in Section 5. Here we split $\mathrm{MA}[\mathbb{P}] = f_0 + f_1$ with $f_0(q) := \mathrm{MA}_{\kappa q}[\mathbf{1}_{t\ge X_0}\,dt/\log t](q)$ and $f_1 := \mathrm{MA}[\mathbb{P}] - f_0$, so that $|f_1(q)| \le \eta_{\mathrm{H}}(q) f_0(q) \le \eta_{\mathrm{H}}(q)/\log q$ for $q \ge X_0$, and we set

$$\mu_{\mathrm{edge}}(u) := \sum_{N\ge 2} \frac{1}{N}\,\mathrm{MA}_{\kappa q}[\text{rational primes} \ge \max(P^+(N),X_0)](q)\Big|_{q=u/N} \mathbf{1}\Big[\frac{u}{N} \in \Big(\frac{X_0}{1+\kappa}, X_0\Big)\Big],$$

$$\mu_1 := (K_{\mathbb{Q}} - K_{\mathrm{pt}})(f_1\mathbf{1}_{[X_0,\infty)}),$$

$$\mu_0 := (K_{\mathbb{Q}} - K_{\mathrm{pt}})(f_0\mathbf{1}_{[X_0,\infty)}) + (\text{decision-order mismatches}) + O(1/(\kappa x)).$$

So μ_{edge} is the part of $K_{\mathbb{Q}}\mathrm{MA}[\mathbb{P}]$ that lies below X_0; for $N < X_0$ its summand is $\mathrm{MA}[\mathbb{P}](u/N)/N$. For $u < 2X_0$ every cofactor $w \ge 2$ has $u/w < X_0$, so $\mu_0 = O(1/(\kappa u))$ and $\mu_1 = 0$ there; and $\mu_{\mathrm{edge}}(u) = 0$ for $u \le 2X_0/(1+\kappa)$, while $\mu_{\mathrm{edge}}(u) = \frac{1}{2}\mathrm{MA}[\mathbb{P}](u/2)$ on $(2X_0/(1+\kappa), 2X_0)$.

Lemma 7.3 (The margin decomposition)**.**

(a) (Huxley part.) *For every $q \ge X_0 \ge X_{\mathrm{Hux}}$,* $\mathrm{MA}[\mathbb{P}](q)\log q \ge 0.99$*, and the left side tends to* 1.

(b) (Smooth part.) *On* $\mathrm{Good}(<u)$*, for every $u \ge X_0$ with* $\log\log u \le \Lambda/8$*,*

$$|\mu_0(u)| \le 2\kappa(\log u)^3.$$

(c) (Huxley-error part.) *Put $v := \log x/\Lambda$. On* $\mathrm{Good}(<x)$*, for every $x \ge X_0$,*

$$|\mu_1(x)| \le 2\Lambda^{-3}(\log v + 1.01),$$

and, for $v \ge 4$ with $U := 1 + 2\log v$,

$$|\mu_1(x)| \le 2\Big[\Big(\frac{U}{\log x}\Big)^3(\log U + 1.01) + 2e^{\gamma}e^{\mathrm{Ein}(2)}\frac{v}{U}v^{-4}\Lambda^{-3}\Big], \qquad \mathrm{Ein}(2) < 3.69.$$

Consequently

$$\sum_{u \ \mathrm{dyadic} \ge X_0} \log(2u) \sup_{[u,2u]} |\mu_1| \le \frac{690}{\log X_0} \le 7\cdot 10^{-6}. \tag{83}$$

(d) (Edge at X_0.) *On* $\mathrm{Good}(<x)$*:*

(i) for $X_0 \le x \le X_0^{1.9}$, $\rho_{\mathrm{edge}}(x) = -\sum_{N\ge 2:\ x/N \in (X_0/(1+\kappa),\, X_0)} \mu(N)\,\dfrac{\mathrm{MA}[\mathbb{P}](x/N)}{N}$*;*

(ii) for $X_0 \le x \le X_0^{1.01}$, at most one N occurs in (i), and $\rho_{\mathrm{edge}}(x)\log x \ge -0.167$*;*

(iii) for $x \ge X_0^{1.01}/2$ with $\log x \le X_0^{1/100}$*,* $|\rho_{\mathrm{edge}}(x)|\log x \le 3X_0^{-0.01} + X_0^{-0.34}$.

Proof of (a). The window $I := [q,(1+\kappa)q]$ lies in $[q,2q]$ and has $|I| = \kappa q \ge q^{7/12+\varepsilon_\kappa} \ge (2q)^{7/12}$, so it is admissible in (80), and $\eta_{\mathrm{H}}(q) \le (\log q)^{-3}$ there (by (79) if $|I| \le x/(\log x)^4$ with $x := (1+\kappa)q$, and by the de la Vallée Poussin bound if $|I|$ is longer). Hence $\pi(I) \ge (1-\eta_{\mathrm{H}}(q))\int_I dt/\log t \ge (1-(\log q)^{-3})\kappa q/\log((1+\kappa)q)$. Dividing by κq, $\mathrm{MA}[\mathbb{P}](q)\log q \ge (1-(\log q)^{-3})(1-\kappa/\log q) \ge 1 - 10^{-20}$, since $\log q \ge 10^8$ and $\kappa = X_0^{-0.3967}$. □

Proof of (b). Step 1: the element count. Put $\Delta := N_{\mathcal{N}} - \lfloor\cdot\rfloor$. Then $\Delta = 0$ on $[1, X_0^-)$, and on $\mathrm{Good}(<u)$, for $t \le u$,

$$|\Delta(t)| \le \delta(t) := 2\kappa t + 2t^\theta \log^* t + 2t^{\theta/2} + B(t) + 3H(t) + 6.$$

The pieces are: $|D| \le B + 3H + 3$ ((11) and the count in the proof of Proposition 4.1); $|E| \le |E_0| + \max\big((\kappa/2)\Psi(t,X_0),\, \Psi([t,(1+\kappa)t],X_0)\big) \le 2\kappa t + 2$ ((8); Lemma 7.1(4),(5): the drift and the mass still in flight in the top window have opposite signs, and $\Psi([t,(1+\kappa)t],X_0) \le \kappa t + 1$); $|R| \le 2t^\theta \log^* t$; and the absorbed part of (7).

Step 2: the prime count. The Chebyshev identity $\log\cdot dN = d\psi * dN$ holds in both systems. Write $\Delta_\psi := \psi_{\mathcal{N}} - \psi$, which is supported on $[X_0^-,\infty)$ because the base primes coincide. Then $d\Delta_\psi * dN_{\mathcal{N}} = dh$ with $dh := \log\cdot d\Delta - d\psi * d\Delta$, and

$$h(t) = \Delta(t)\log t - \int_1^t \Delta(v)\frac{dv}{v} - \int \Delta(t/a)\,d\psi(a).$$

So $|h(t)| \le 5\kappa t\log t$: the κ-part of δ gives $4\kappa t\log t + 2\kappa t$, and the rest of δ gives $\le \kappa t\, X_0^{-0.1}$, using $(B+3H+3)(s) \le s^{3\cdot 10^{-5}}$ (proof of Proposition 4.1) and $\theta \le \frac12 - 10^{-4}$. Invert with the Möbius measure $\mu_{\mathcal{N}}$ of $\mathcal{N}$, using $|\mu_{\mathcal{N}}| \le dN_{\mathcal{N}}$ and $\sum_{v\in\mathcal{N},\, v\le s} 1/v \le \log s + 1$ on Good. This gives

$$|\psi_{\mathcal{N}}(t) - \psi(t)| \le 5\kappa t\log^2 t \qquad (t \le u).$$

Step 3: telescoping the cofactors. We have $(K_{\mathrm{pt}} - K_{\mathbb{Q}})f_0(u) = \sum_{w\in\mathcal{N}} F(w)\chi_w - \sum_{m\in\mathbb{N}} F(m)\chi_m$, where $F(v) := f_0(u/v)/v$ and χ is the latest-prime filter ($P^+ \le u/v$, in value order; the decision-order mismatch is treated below). Write each cofactor as $d\cdot$(a multiset of $r = k-1$ large primes), with $d \in \mathcal{D}$ (the same set in both systems) and the large primes jittered and $\ge X_0^-$, respectively rational and $\ge X_0$.

- *Symmetric telescoping.* The sum over multisets of r large primes is $1/r!$ times the sum over ordered r-tuples (the product measure), plus a diagonal term from the multisets with a repeated large prime. The diagonal term has total weight $\le 1.01e^{\gamma}\Lambda\, e^{\ell_u}/(X_0^-\Lambda)$ in each system, where ℓ_u is defined below. Telescope the product measure: the j-th intermediate sum has its first $j-1$ variables realised and its last $r-j$ rational, with no ordering between them.
- *One replacement.* Fix the other variables, and put $m_j := d\prod_{i\ne j} p_i$ and M' for the largest of them (or $P^+(d)$). The admissible set of the j-th variable p is an interval $[X_0^-, b]$. The filter $\max(p, M')\, m_j p \le u$ reads $p \le \sqrt{u/m_j}$ if $M'^2 m_j \le u$, and $p \le u/(M'm_j)$ otherwise; the support condition $u/(m_jp) \ge X_0$ of $f_0\mathbf{1}_{[X_0,\infty)}$ reads $p \le u/(X_0 m_j)$; and b is the smaller of the two bounds. So f_0 is evaluated only on $[X_0,\infty)$; the partial windows below X_0 are in μ_{edge}, not here. The summand is $G(p) = f_0(u/(m_jp))/(m_jp)$, and on $[X_0,\infty)$, $|qf_0'(q)| \le f_0(q)/\Lambda$, so $|(G/\log)'| \le 1.1G/(p\log p)$. Abel summation against $d(\psi_{\mathcal{N}} - \psi)/\log$ (prime powers cost $O(\sqrt{b}\log b)$) shows that one replacement changes the sum by at most $10\kappa f_0 \log b/m_j + 2.8\kappa f_0 \log^2 b/m_j \le 4\kappa \log^2 u/(m_j\Lambda)$.
- *Summing over the other variables.* They range over a product set, so $\sum 1/m_j \le 1.01e^{\gamma}\Lambda\cdot \ell_u^{r-1}$, with $\ell_u := \sum_{X_0^-\le p\le u} 1/p \le \log(\log u/\Lambda) + 2.2/\Lambda$ in each system (Lemma 5.1(iv) and Mertens' theorem). The r replacements against the factor $1/r!$ give $4\kappa(\log^2 u/\Lambda)\cdot 1.01e^{\gamma}\Lambda\cdot \ell_u^{k-2}/(k-2)!$ for each k; with the diagonal terms, the total is

$$\le 4.1e^{\gamma}\kappa\log^2 u\, e^{\ell_u} \le \frac{7.4\,\kappa(\log u)^3}{\Lambda} \le \kappa(\log u)^3.$$

The rest. Each of the following is $\le 10^{-3}\kappa(\log u)^3$: the decision-order mismatch, bounded directly as follows. Value order and decision order of the new prime and the top large prime P of the cofactor $w = Pw'$ differ only when u/w lies within $H/2$ of P, so $|P - \sqrt{u/w'}| \le H/4$: P lies in an interval of length $\le H/2$ around $\sqrt{u/w'}$, which holds at most $1.5H + 2 \le 2(H+1)$ realised primes. Summing over w', with $\sum 1/w' \le \log u + 1$ and $H(v)/v$ decreasing, bounds the mismatch by $2.02(H(X_0)+1)(\log u + 1)X_0^{-1}\sup|h|$ for the function h the operators act on ($h = f_0$ here, and $h = g$ in (d)). The other items are the partial windows of Lemma 7.1(3) (relative $O(\kappa)$), $f_0 - 1/\log q = O(\kappa/\log^2 q)$, and $O(1/(\kappa u))$. □

Proof of (c). $|\mu_1| \le |K_{\mathbb{Q}} f_1| + |K_{\mathrm{pt}} f_1|$, and both are bounded in the same way; the restrictions $P^+(m') \le x/m'$ are simply dropped.

- Components with $q = x/m' \ge y := x^{1/U}$ have $\eta_{\mathrm{H}}(q) \le (\log x/U)^{-3}$, and $\sum_{m'\le x/y} 1/(m'\log(x/m')) \le \log U + 0.01$. This uses $\sum_{m\le t} 1/m = \log t + \gamma + O(2\kappa)$ on Good($< x$) for $t \le x/X_0$, from Step 1 of (b).
- Components with $q \in [X_0, y)$ have $\eta_{\mathrm{H}} \le \Lambda^{-3}$, $1/\log q \le 1/\Lambda$, and $m' \ge x/y$ with $P^+(m') \le y$. Rankin's method with exponent $1 - 2/\log y$ bounds $\sum 1/m'$ by $e^{-2(U-1)}\prod_{p\le y}(1 - p^{2/\log y - 1})^{-1} \le e^{-2(U-1)}\cdot 2e^{\gamma}e^{\mathrm{Ein}(2)}\log y$. Mertens' theorem for the realised primes is Lemma 5.1(iii),(iv), and $\mathcal{N}$ is free on its primes.
- The first bound is the case $U = 1$, with $\sum_{m'\le x/X_0} 1/(m'\log(x/m')) \le \log v + 0.01$.

Summing over dyadic u, with $\log(2u) \le 1.01 \log u$, gives $(2.02/\log 2)\Lambda^{-1}(I_1 + I_2 + I_3)$. Here $I_1 = \int_1^4 v(\log v + 1.01)\,dv = 14.92$ ($U = 1$ on $v \le 4$), and $I_2 = 214.8$ and $I_3 = 6.7$ are the integrals over $v \ge 4$ of the two terms of the second bound. The integrands are elementary, and the total is $\le 690/\Lambda$ (with Ein(2) and the integrals certified in Appendix B.4). The bound is proportional to $1/\Lambda$, so it holds uniformly in $\Lambda \ge 10^8$. □

Proof of (d). (i) If $\mu_{\mathrm{edge}}(x/n) \ne 0$ then $x/n > 2X_0/(1+\kappa)$, so $n < X_0^{0.9} < X_0^-$ for $x \le X_0^{1.9}$. Such n are rational integers whose primes are decided before X_0; each product in the condition $\chi_x^t(n) = 1$ of Lemma 5.5 is at most $n^2 < x$, so the coefficient $c_n(x)$ of that lemma is $\mu(n)$; and the N with $x/N \in (X_0/(1+\kappa), X_0)$ satisfy $N < 1.01 X_0^{0.9} < X_0$. Writing $N = dM$ with $M \ge 2$,

$$\rho_{\mathrm{edge}}(x) = \sum_n \frac{\mu(n)}{n}\,\mu_{\mathrm{edge}}\Big(\frac{x}{n}\Big) = \sum_{N \ge 2} \frac{\mathrm{MA}[\mathbb{P}](x/N)}{N}\,\mathbf{1}\Big[\frac{x}{N} \in \Big(\frac{X_0}{1+\kappa}, X_0\Big)\Big] \sum_{d \mid N,\, d < N} \mu(d),$$

since every proper divisor d of such an N has $x/d \ge 2x/N > X_0$; and $\sum_{d \mid N,\, d<N} \mu(d) = -\mu(N)$.

(ii) The N in (i) lie in $(x/X_0, (1+\kappa)x/X_0)$, an interval of length $\kappa x/X_0 \le X_0^{-0.38} < 1$. A negative term needs $\mu(N) = 1$, so $N \ge 6$. For $q \in (X_0/(1+\kappa), X_0)$ we have $\mathrm{MA}[\mathbb{P}](q) \le 1.0001/\Lambda$: the window $[X_0, (1+\kappa)q]$ lies in $[X_0, 2X_0]$, so if its length is $\ge (2X_0)^{7/12}$, (80) at X_0 bounds its primes by $(1+\Lambda^{-3})\kappa q/\Lambda$, and otherwise it holds at most $(2X_0)^{7/12} + 1 \le X_0^{-0.01}\kappa q$ primes. As $x < NX_0$,

$$\rho_{\mathrm{edge}}(x) \log x \ge -\frac{1.0001}{N}\Big(1 + \frac{\log N}{\Lambda}\Big) \ge -\frac{1.0001}{6}\Big(1 + \frac{2}{\Lambda}\Big) \ge -0.167,$$

the middle expression being decreasing in $N \ge 6$.

(iii) Let $K'_{\mathbb{Q}}$ be the sharp latest-prime operator of the rational system, with cut-off X_0:

$$(K'_{\mathbb{Q}} h)(y) := \sum_{m \in \mathbb{N},\ m \ge 2} \frac{1}{m}\,\mathbf{1}[P^+(m) < y/m]\,\mathbf{1}[y/m \ge X_0]\,h(y/m).$$

Its filter is strict, as in K_{pt} and Lemma 5.5. Put $g := (1 + K'_{\mathbb{Q}})^{-1}\mu_{\mathrm{edge}}$, and expand $g(x)$ as in (i), now in the rational system. Fix $N \ge 2$ with $x/N \in (X_0/(1+\kappa), X_0)$ and a window prime $p \ge X_0$, and write $N = N_{\le} N_{>}$, where $N_{>}$ collects the prime factors of N larger than p. The filter admits a factorisation $N = nM$ only if $N_{>} \mid n$. For such $n < N$ we have $n \le N/2$, so $x/n \ge X_0$; and the condition $\chi_x^t(n) = 1$ of Lemma 5.5 depends only on $N_{>}$, because $q\,n \le pN/2 < x$ for every prime $q \le p$. Writing $\chi_{N_>}(x) \in \{0, 1\}$ for its value, the coefficient of p in the N-term is

$$\chi_{N_>}(x) \sum_{s \mid N_{\le},\ s \ne N_{\le}} \mu(N_> s) = \begin{cases} -\mu(N)\,\chi_{N_>}(x), & N_{\le} > 1, \\ 0, & N_{\le} = 1. \end{cases}$$

So each N-term is at most $\mathrm{MA}[\mathbb{P}](x/N)/N$ in absolute value. As x increases, its numerator changes monotonically: primes enter the window and coefficients switch on, all with the sign $-\mu(N)$. Hence $|g(x)| \le (1.0001/\Lambda)\sum_{x/X_0 < N < (1+\kappa)x/X_0} 1/N \le (1.0001/\Lambda)(\kappa + X_0/x)$, and $|g| \le 0.51/\Lambda$ everywhere. For $x \ge X_0^{1.01}/2$ and $\log x \le X_0^{1/100}$ this gives $|g(x)| \log x \le 1.0001(\log x/\Lambda)(\kappa + X_0/x) \le 3X_0^{-0.01}$. By the resolvent identity,

$$\rho_{\mathrm{edge}} - g = (1 + K_{\mathrm{pt}})^{-1}(K'_{\mathbb{Q}} - K_{\mathrm{pt}})g.$$

The operators $K'_{\mathbb{Q}}$ and K_{pt} agree on cofactors $< X_0^-$ (their primes are decided at their values, and both filters are strict), and $g(u/w) \ne 0$ needs $u/w > 2X_0/(1+\kappa)$; so $(K'_{\mathbb{Q}} - K_{\mathrm{pt}})g(u) = 0$ for $u < 2X_0X_0^-/(1+\kappa)$. For larger u the telescoping of Step 3 of (b) applies with f_0 replaced by g; it uses only $|g| \le 0.51/\Lambda$ and the variation of g, not a formula for g. By the monotonicity just shown, each N-term varies by at most $3.1/(N\Lambda)$ on its window (at most $1.0001(1+3\kappa)/(N\Lambda)$

while it grows, and the drop to 0 at the end), so g has total variation $\le (3.1/\Lambda)(\log(\beta/\alpha)+1)$ on $[\alpha,\beta]$. As g is not smooth, the Abel summation of one replacement is done discretely, over the merged atoms of $\psi_{\mathcal{N}}$ and ψ in $[X_0^-, b]$. With $\tilde g(t) := g(u/(m_j t))$, the product rule splits each difference of $\tilde g(t)/(m_j t\log t)$ into a difference of $1/(m_j t\log t)$ and a difference of $\tilde g$, and $|\Delta_\psi(t)|/(m_j t\log t) \le 5\kappa\log b/m_j$ is taken out as a supremum. The boundary terms give $5.1\kappa\log b/(m_j\Lambda)$, the differences of $1/(t\log t)$ give $1.40\kappa(\log^2 b+2\log b)/(m_j\Lambda)$, and the variation of $\tilde g$, which is that of g on $[u/(m_j b), u/(m_j X_0^-)]$, gives $15.5\kappa\log^2 b/(m_j\Lambda)$, as $\log X_0^- \ge 1$. Since $79 \le \log X_0^- \le \log b \le \log u$, one replacement costs at most

$$\big(5.1\log b + 1.40(\log^2 b + 2\log b) + 15.5\log^2 b\big)\frac{\kappa}{m_j\Lambda} \le \frac{17\kappa\log^2 u}{m_j\Lambda},$$

in place of $4\kappa\log^2 u/(m_j\Lambda)$. Summing over the other variables as in Step 3 gives at most $17.2e^\gamma\kappa\log^2 u\, e^{\ell_u} \le 31\kappa(\log u)^3/\Lambda$. The remaining items of "the rest" are smaller: the decision-order count holds with $\sup|g| \le 0.51/\Lambda$, and there are no partial windows, as both operators are sharp. So $|(K'_{\mathbb{Q}} - K_{\mathrm{pt}})g(u)| \le 4\kappa(\log u)^3$ for $\log\log u \le \Lambda/8$. By Lemma 5.5, $\log x\,|\rho_{\mathrm{edge}} - g|(x) \le 4.1\kappa(\log x)^5 \le X_0^{-0.34}$ for $\log x \le X_0^{1/100}$. □

Use. Proposition 5.35 uses (a)–(d) on its range (A), $\log x \le X_0^{1/100}$. For (b), Lemma 5.5 gives $\log x\,|(1+K_{\mathrm{pt}})^{-1}\mu_0(x)| \le \log x(\log x + B_1)\sup_{u\le x}|\mu_0(u)| \le 2.1\kappa(\log x)^5 \le X_0^{-0.34}$ there. For (c), the Saias-weighted response (Corollary 5.9) needs $\sum\log(2u)\sup|\mu_1|$ below the budget $c'/(8C_S) \approx 2.4\cdot 10^{-4}$ ($c' = 0.15$, $C_S \le 77$), and (83) has a factor ≥ 30 to spare. For (d), ρ_{edge} enters (82) directly. It lowers $\rho^{\mathrm{sh}}\log x$ by at most 0.167 for $X_0 + H \le x \le X_0^{1.01}$, and only on the windows $N\cdot(X_0/(1+\kappa), X_0)$ with $\mu(N) = 1$; for $x \ge X_0^{1.01}/2$ it changes it by at most $3X_0^{-0.01} + X_0^{-0.34}$. So the lower bound $0.99 - 6\cdot 10^{-4} - X_0^{-0.34} - e^{-80}$ of range (A) becomes ≥ 0.822 on $[X_0 + H, X_0^{1.01}]$ and stays ≥ 0.97 on $[X_0^{1.01}/2, X_{\mathrm{sw}}]$; the two ranges overlap.

7.3. **Where the local prime number theorem is used.** The local prime number theorem (Theorem 5.24) is used only above a threshold $X_{\mathrm{LP}}(X_0)$. Landau's argument against T uses discs of radius $\asymp 1/\log X_0$, because the rigid series is huge at $\sigma = (1+\theta')/2$ (θ' the exponent of Theorem 5.24); so the zero-free constant is $c_{\mathrm{zf}} \propto 1/\Lambda$. Landau's step also needs the zero-free width to beat $17\log\log y/\log y$, which is a binding constraint here. Hence

$$\log X_{\mathrm{LP}} = O(\Lambda^2\log^2\Lambda) \tag{84}$$

(Theorem 5.24), still far below $\log x_1 \ge X_0^2$. Below X_{LP} the fresh-prime density $\mathrm{MA}_{\kappa q}[\mathbb{P}]$ comes from (80), with no local prime number theorem. Above it, Theorem 5.24 applies. The two ranges overlap: range (A) of Proposition 5.35 runs to $\log x = X_0^{1/100}$, inside the limit $\log x \le (0.0048/\kappa)^{1/5} \approx X_0^{0.079}$ of the μ_0 bound, and range (B) starts there, far above $\log X_{\mathrm{LP}}$.

Appendix A. Rank-full configurations

This appendix proves the moment bound for rank-full clusters of generalized integers in a short window (Theorem A.12). It is used in the proof of the cluster event (Proposition 6.10). Its machinery (Lemmas A.2–A.8 and Theorem A.10) is also used in the Cover Bound (Proposition 6.18). Probabilistically, the argument rests only on the standing facts (F1)–(F4) of Section 6.1:

- the slot count (F1);
- the jitter density (F2);
- the harmonic sums (F3′);
- the sure Mertens bound (F4).

These are carried by progressive good events.

The idea is to integrate out the jitter of k suitably chosen primes (the *pivots*), one for each element of the cluster. Each pivot is integrated against its conditional density. Every other prime (a *free* prime) is summed over its realised values by a sure bound.

Two difficulties have to be overcome:

- a prime that occurs in several elements constrains several of them at once;
- a free prime may be decided after some pivot, so its realisation is not measurable at that pivot's decision time.

The *Volume Lemma* (Lemma A.2) handles the first. The pivots are integrated one at a time, each against its conditional density given the past (Lemma A.5). The resulting iterated Lebesgue integral is then evaluated by a single change of variables, which replaces per-constraint widths by an exact Jacobian; no independence between primes is used. The *Pivot-Integration Lemma* (Lemma A.5) and the *Staged Volume Lemma* (Theorem A.10) handle the second: integrate the pivots in stages, and bound the free primes decided between stages by sure sums, each against a weight that is harmonic in some row.

A.1. **Setting.** Throughout, $L = \log x$, $\ell = \log\log x$ and $\Lambda = \log X_0 \geq 10^8$, as in Section 6.1. Fix a dyadic height $z \geq X_0/2$ and a window $J \subset [z, 2z]$ of length $\delta \leq 1$, and put $\varepsilon := \delta/z$.

Definition A.1 (Configuration types)**.** A *configuration type* τ fixes:

(a) k rows $n_1, \dots, n_k \in J$, which are elements of $\mathcal{N}$;
(b) the jittered primes occurring in them (the *columns*), listed in decision order;
(c) the integer exponent matrix $E = (E_{ij})$, where E_{ij} is the exponent of the j-th column in n_i.

The *tail* d_i of a row, the product of its rational primes $< X_0$, is not part of the type; tails are summed in the end step of Lemma A.5. The type τ is *rank-full* if $\operatorname{rank} E = k$. For a column p, write $e_p \in \mathbb{Z}_{\geq 0}^k$ for its column of E.

Greedy pivots. Run through the columns latest-first, and declare p a *pivot* if and only if

$$e_p \notin \operatorname{span}\{e_q : \ q \text{ a pivot decided later than } p\}.$$

Let P be the set of pivots. For a rank-full type, $|P| = k = \operatorname{rank} E$ and the square matrix E_P (the columns of E indexed by P) is invertible. The remaining columns are *free*. For $i \leq k$ write

$$n_i = c_i \prod_{j \in P} q_j^{E_{ij}},$$

where q_j is the value of the j-th pivot and $c_i \in \mathcal{N}$ collects the free jittered primes of n_i together with its tail d_i.

Runs and blocks. Split P into maximal runs $P_1, \dots, P_T$ of consecutive pivots in decision order, latest first. For $t < T$, the *late free block* F_t consists of the free columns decided between P_t and P_{t+1}. The *early block* F_T consists of the free columns decided before all pivots. Write $P_{\leq t} = P_1 \cup \dots \cup P_t$, and similarly $F_{\leq t}$. By construction of the greedy pivots,

$$e_f \in \operatorname{span}\{e_p : \ p \in P_{\leq t}\} \qquad \text{for every } f \in F_{\leq t}. \tag{85}$$

Indeed, a free column f decided after P_{t+1} lies in the span of the pivots decided later than f, and these belong to $P_{\leq t}$.

Skeletons. The *skeleton* $\mathfrak{s}$ of a rank-full type is its pivot pattern E_P, with the pivot columns in decision order, together with its run structure $(|P_1|, \dots, |P_T|)$, a composition of k. Types with the same skeleton differ only in their free columns: their number, their exponents, and the blocks in which they lie. The runs, the blocks F_t (as slot ranges) and the stage rows of Lemma A.8 depend only on the skeleton. For $t < T$ the block F_t is non-empty, since the runs are maximal; the early block may be empty. There are 2^{k-1} run structures.

Slot positions. Pivots are distinct primes, one per slot. Slot spacing is ≥ 0.99, and the value q of a jittered prime differs from its slot s_q by at most $H(s_q)/2$ (Section 6.1). Jittered primes are decided at slots $\geq X_0$. Put $\sigma_j := X_0 + 0.99(j-1)$. The j-th earliest pivot has slot $\geq \sigma_j$, and $v \mapsto v - H(v)/2$ is increasing. So its value is at least $\sigma_j - H(\sigma_j)/2$, and at least $X_0 - H(X_0)/2 = X_0^-$. Hence, whatever the labels are, the j-th earliest pivot has value at least

$$\tau_j := \max\big(X_0^-,\ \sigma_j - 2H(\sigma_j)\big). \tag{86}$$

Recall that H is continuous and non-decreasing, and that $H(v)/v$ is decreasing (Theorem 6.22(c)). This makes $v \mapsto v - H(v)/2$ increasing above: for $v < v'$, since $H(v')/v' \leq H(v)/v < 2$, $H(v') - H(v) \leq (v' - v)H(v)/v < 2(v' - v)$. By (9) and Theorem 6.22(c), $\log H(X_0) \leq 8(\log \Lambda)^2 + 10^{-6}\Lambda$, so $H(X_0)/X_0 \leq e^{-0.99\Lambda} \leq e^{-10^7}$. Values of jittered primes are only bounded below by $X_0^- = X_0 - H(X_0)/2$, as in Section 6.1; every lower bound on such a value below uses X_0^-, while slots are $\geq X_0$. Since $H(X_0)/X_0 \leq e^{-10^7}$, replacing X_0 by X_0^- changes each constant by a relative amount at most e^{-10^7}.

Probabilistic set-up. $\mathcal{F}_{s-}$ is the information available before the decision at slot s. Good$(< s)$ is the good event of Definition 6.8, and $G_{<s}$ abbreviates it. For a window J, t_J denotes the time at which the cluster event of J is examined; it satisfies $t_J \leq t(E1)$ (Definition 6.8). Every bound below is proved on Good$(< t_J)$, which is carried progressively along the pivots. No queue bound at heights inside the window is used quantitatively.

A.2. **The Volume Lemma and the pattern sum.**

Lemma A.2 (Volume Lemma)**.** *Fix the cofactors $c_1, \dots, c_k$. The map $q_P = (q_j)_{j \in P} \mapsto (n_i)_{i \leq k}$, $n_i = c_i \prod_j q_j^{E_{ij}}$, is injective on $\mathbb{R}_+^k$, and*

$$\operatorname{vol}\big\{q_P \in \mathbb{R}_+^k :\ n_i \in J\ \forall i\big\} = \int_{J^k} \frac{\prod_j q_j}{|\det E_P| \prod_i n_i}\, dn, \qquad \prod_j q_j = \frac{\prod_i (n_i/c_i)}{\prod_j q_j^{s_j - 1}}, \tag{87}$$

where $s_j := \sum_i E_{ij} \geq 1$ is the j-th column sum. In particular, if all $q_j \geq X_0^-$, then the volume is at most $\delta^k / \big(|\det E_P| \prod_i c_i\, (X_0^-)^{M-k}\big)$, with $M := \sum_{i,j} E_{ij}$.

Proof. Pass to logarithmic coordinates $x_j = \log q_j$. Then $\log n_i = \log c_i + \sum_j E_{ij} x_j$, an affine map with invertible linear part E_P, so the map is injective. Its Jacobian matrix is $\partial n_i / \partial q_j = E_{ij} n_i / q_j$, whose determinant is $\det(E_P) \prod_i n_i / \prod_j q_j$. The change of variables $q_P \mapsto n$ therefore gives the integral in (87).

For the second identity, multiply the relations $n_i/c_i = \prod_j q_j^{E_{ij}}$ over i. This gives $\prod_j q_j^{s_j} = \prod_i (n_i/c_i)$, hence $\prod_j q_j = \prod_i (n_i/c_i) / \prod_j q_j^{s_j - 1}$. No column of the invertible matrix E_P vanishes, so $s_j \geq 1$. If all $q_j \geq X_0^-$, then $\prod_j q_j^{s_j - 1} \geq (X_0^-)^{M-k}$. Hence the integrand is at most $1/\big(|\det E_P| \prod_i c_i (X_0^-)^{M-k}\big)$, and J^k has volume δ^k. □

Remark A.3. Lemma A.2 involves no per-constraint widths. Bounding the constraints one at a time by Gaussian elimination would combine them into integer combinations whose ℓ^1-norms are not bounded by k in general (this already happens for $k = 4$). The Jacobian avoids this entirely.

Lemma A.4 (Pattern sum)**.** *Summing over the invertible nonnegative integer matrices E_P with columns in decision order,*

$$\frac{1}{k!}\sum_{E_P}\prod_j \tau_j^{-(s_j-1)} \;\le\; \prod_j \bigl(1-1/\tau_j\bigr)^{-k} \;\le\; e^{(3H(X_0)+2)k/X_0}\bigl(1+k/X_0\bigr)^{1.02k}.$$

Since $H(X_0)/X_0 \le e^{-10^7}$ (Section A.1), the right side is at most $e^{k\,e^{-10^6}}\bigl(1+k/X_0\bigr)^{1.02k}$.

Proof. By the Leibniz expansion, an invertible E_P has a nonzero term $\prod_i E_{i,\sigma(i)}$ for some permutation σ. So its support contains the permutation matrix of σ. Subtract that permutation matrix. The rest is a nonnegative integer matrix X, and the column sums satisfy $s_j = 1+\sum_i X_{ij}$. Choosing one such σ for each E_P makes the map $E_P \mapsto (\sigma, X)$ injective. Hence

$$\sum_{E_P}\prod_j \tau_j^{-(s_j-1)} \le k!\prod_j\sum_{m\ge 0}\binom{k+m-1}{m}\tau_j^{-m} = k!\prod_j(1-1/\tau_j)^{-k},$$

since a column of extra mass m can be distributed over the k rows in $\binom{k+m-1}{m}$ ways. Next, $(1-1/\tau)^{-k} \le \exp\bigl(k/(\tau-1)\bigr)$. Since $H(v)/v$ is decreasing, $H(\sigma+0.99)-H(\sigma) \le 0.99\,H(X_0)/X_0$ for $\sigma \ge X_0$. So $\sigma_j - 2H(\sigma_j)$ increases by at least 0.989 from one index to the next. Let N_0 be the number of indices with $\tau_j = X_0^-$. These satisfy $0.99(j-1) \le 2H(\sigma_j) - H(X_0)/2 \le 2H(X_0)(1+10^{-8})$, so $N_0 \le 2.03H(X_0)+1$. The remaining indices satisfy $\tau_{N_0+i} \ge X_0^- + 0.989(i-1)$ for $i \ge 1$. Hence

$$\sum_{j\le k}\frac{1}{\tau_j-1} \le \frac{N_0+1}{X_0^- - 1} + \sum_{i=1}^{k}\frac{1}{X_0^- - 1 + 0.989i} \le \frac{3H(X_0)+2}{X_0} + 1.02\log\Bigl(1+\frac{k}{X_0}\Bigr).$$

Here we used $X_0^- - 1 \ge X_0(1-H(X_0)/X_0)$, which gives $(2.03H(X_0)+2)/(X_0^-1) \le (3H(X_0)+2)/X_0$. We also used $0.989/(X_0^- - 1) \le 1/X_0$ and $1/0.989 \le 1.02$. Exponentiating gives the same bound as before, $\prod_j(1-1/\tau_j)^{-k} \le e^{(3H(X_0)+2)k/X_0}(1+k/X_0)^{1.02k}$. □

A.3. **The Pivot-Integration Lemma.** Fix a skeleton $\mathfrak{s}$. Let $X_\mathfrak{s}$ be the number of realised configurations whose type has skeleton $\mathfrak{s}$ and which satisfy all window and feasibility constraints. These are k-tuples of distinct elements of $\mathcal{N}\cap J$ with the pivot pattern and run structure of $\mathfrak{s}$; their free columns and tails are arbitrary. Let $p_1, p_2, \dots$ be the pivots, latest first. Further data may be fixed in advance, as in Proposition 6.18; the lemma holds with the same proof. Where such data restrict the block contents, the sums in (iii) may still run over all contents, since this only enlarges them.

Lemma A.5 (Pivot-Integration Lemma)**.** *For each run P_t, let s_t be the slot of its last (earliest decided) pivot, let f_t be the latest-decided column of the block F_t below it, and let u_t be the first slot after the slot of f_t. For $t < T$ the block F_t is non-empty, since the runs are maximal. If the early block F_T is empty, put $u_T := X_0$, the first slot. Every pivot value is at most $2z$, since it divides some $n_i \in J$, and $|q-s| \le H(s)/2$; so every slot label is at most $s_{\max} := 2z + 0.51H(2z)$. The decision order gives $u_t \le s_t$. Put*

$$\chi_t := \mathbf{1}(u_t \le s_{\max})\,\mathbf{1}(G_{<u_t}),$$

a function of the block data and of ω only. For a pending integrand Ψ, write $\Psi|_{s_t=u_t}$ for Ψ with the label s_t replaced by u_t. Define $S_\mathfrak{s}$ by the following recursion, run by run, latest first, starting from

$$\Psi := \prod_i \mathbf{1}(n_i \in J)\prod_{j\in P}\mathbf{1}(q_j \ge \tau_j),$$

with τ_j as in (86) for the j-th earliest pivot. The caps hold for every realised configuration. For $t = 1, \dots, T$:

- *(i)* $\Psi \leftarrow \chi_t \, \Psi|_{s_t = u_t}$*;*
- *(ii) at each pivot p of P_t, latest first:* $\Psi \leftarrow \int \Psi \, dq_p$ *(Lebesgue measure; the window restriction of p is dropped);*
- *(iii) for $t < T$, at the late block F_t:* $\Psi \leftarrow$ *any bound for the realised sum $\sum_{F_t} \Psi$ over the block data (every non-empty content of F_t: the number of its free columns, their primes and their exponents) that holds for every ω and is a function of the label s_{t+1} (and of no other label), of the variables not yet processed, and of the primes decided before s_{t+1}, non-increasing in s_{t+1}. An example is Lemma A.6 (S3)(a). For $t = T$, at the early block and the tails:* $\Psi \leftarrow$ *a deterministic bound for the realised sum over them (the early block may be empty). For the harmonic Euler chains an example is Lemma A.6 (S3)(c); tails of rows that carry no chain may instead be fixed in advance.*

Then

$$\mathbb{E}\big[X_{\mathfrak{s}} \, \mathbf{1}(\mathrm{Good}(< t_J))\big] \le \theta^k \, \bar{S}_{\mathfrak{s}}, \qquad \theta := 1 + 1/\Lambda.$$

Step (i) records the label sum at the last pivot of P_t (see the proof). It comes before the integrals of the run, and the block bound in (iii) comes after them. In (iii) the indicator χ_t is part of each summand, and the bound must hold for every ω. After the end step no label survives, so $S_{\mathfrak{s}}$ is a deterministic number; we call it the envelope *and also write $\bar{S}_{\mathfrak{s}} := S_{\mathfrak{s}}$.*

Proof. Write $X_{\mathfrak{s}}$ as a sum, over the slot labels of the pivots and over the realised free data, of $\prod_i \mathbf{1}(n_i \in J)$; the caps may be inserted, since they hold for every realised configuration. Every pivot is decided before t_J. Along the pivots, latest first, the good events are nested:

$$\mathbf{1}(G_{<t_J}) \le \mathbf{1}(G_{<s_{p_1}}) \le \mathbf{1}(G_{<s_{p_2}}) \le \cdots .$$

Each $\mathbf{1}(G_{<s_p})$ is $\mathcal{F}_{s_p-}$-measurable (Definition 6.8). We induct along the decision order, latest first, keeping the invariant that after each step the pending integrand, apart from the good-event indicator carried along the pivots, is a nonnegative function of the variables not yet processed. As a function of the labels of the run P_t being processed, it depends only on s_t, through factors that are non-increasing in s_t and through the order constraint that s_t is later than the slot of f_t. The pivots of a run are processed one after another, and the block directly below a run right after the last pivot of the run. The factors changed in step (i) do not involve the pivots of P_t, so step (i) commutes with their integrals; it is justified below, at the last pivot of P_t.

Pivot step. Let p be the current pivot, with slot s, and condition on $\mathcal{F}_{s-}$. Everything still pending is then a function of q_p, of the label s, and of data decided before s. These data include the realised primes of the block directly below p, if p is the last pivot of its run. By (F2), the position of the prime decided at slot s has conditional density $\le 1/H(s)$ on its window W_s. Hence

$$\mathbb{E}\big[\Psi \, \mathbf{1}(G_{<s}) \,\big|\, \mathcal{F}_{s-}\big] \le \mathbf{1}(G_{<s}) \, \frac{1}{H(s)} \int_{W_s} \Psi \, dq_p.$$

Only continuous slots receive primes, so s runs over continuous slots; the existence of a prime at slot s is bounded by 1 and dropped. So are all decision-order constraints involving the label s, between pivots and between a pivot and free data, except the constraint that s is later than the slot of f_t when p is the last pivot of P_t. Let s' be the slot of the next pivot to be processed. If p is not the last pivot of its run, Ψ does not depend on s, and we use $\mathbf{1}(G_{<s}) \le \mathbf{1}(G_{<s'})$. If p is the last pivot of the run P_t, so $s = s_t$, then Ψ depends on s only through factors that are non-increasing in s, and through the decision order, which requires s to be later than the slot of f_t, i.e. $s \ge u_t$. Also $s \le s_{\max}$. So block data with $u_t > s_{\max}$ have no admissible s, and

contribute nothing. The good events are nested, so $\mathbf{1}(G_{<s}) \le \mathbf{1}(G_{<u_t})\,\mathbf{1}(G_{<s'})$. Hence, for every admissible s,

$$\mathbf{1}(G_{<s})\,\Psi \le \mathbf{1}(G_{<s'})\,\chi_t\,\Psi|_{s_t=u_t},$$

and the right side does not depend on s. (If p is the earliest pivot, there is no s' and its factor is omitted.) This bound holds pointwise in q_p, in the block data and in ω; in particular, no property of the value at which the window constraints pin q_p is used. Write Ψ for the right side. All terms are nonnegative, so the sum over s may be exchanged with the integral and with the sum over the block data, and

$$\sum_s \frac{1}{H(s)} \int_{W_s} \Psi\,dq = \int \Psi(q) \sum_s \frac{\mathbf{1}(q \in W_s)}{H(s)}\,dq \le \theta \int \Psi\,dq.$$

This uses (F1): at most $H(q)(1 + \kappa_+/\log q) + 3$ windows of continuous slots contain the point q, and H varies by a factor $1 + o(1/\Lambda)$ over a window. Since $q \ge X_0^-$ and $\kappa_+ = 0.8$, the inner sum is at most $1 + 0.8/\Lambda + 3/H(X_0) + o(1/\Lambda) \le 1 + 1/\Lambda = \theta$.

Block step. After the label sum at s_t and the integral over q_p, the pending integrand is

$$\theta\,\mathbf{1}(G_{<s'}) \int \chi_t\,\Psi|_{s_t=u_t}\,dq_p,$$

where s' is the slot of the first pivot of P_{t+1}. The factor χ_t is a function of the block data and of ω, so it is carried through the integrals unchanged. The factor $\mathbf{1}(G_{<s'})$ does not involve the block F_t, whose primes are decided after s'. Sum over the realised block data, and replace the sum by the bound of (iii). The slot u_t is a function of the block data: it is known inside the conditional expectation at s_t, but not at the next pivot. It is consumed by the pathwise bound of (iii), whose output does not contain it, so no measurability of u_t is needed. The result is a function of the label s_{t+1}, of the variables not yet processed and of the primes decided before s_{t+1}, non-increasing in s_{t+1}. These primes are decided before every pivot of P_{t+1}. So the invariant holds for the run P_{t+1}, and at its pivots other than the last one the label sums are unaffected.

For the Euler chains, the bound is that of Lemma A.6 (S3)(a). It sums over every content of the block, and its output is the carried function $\Phi_t(s_{t+1})$ of Lemma A.6 (S3)(b), which has the properties just listed. It uses the sure bound of Lemma A.6 (S3)(d) at slots $u \in [X_0, s_{\max}]$, valid on $G_{<u}$; the factor χ_t provides $u_t \le s_{\max}$ and $G_{<u_t}$.

Order of decision versus order of value. Slots and values are compared in three places:

- in the slot bounds τ_j of (86);
- in Proj_t (Theorem A.10), which use slot spacing directly;
- along the chains, which are indexed by slot, only in the sure bound for the top of a chain (Lemma A.6 (S3), from the proof of (F3′)).

In the last case the top is at a slot $u \in [X_0, s_{\max}]$, and the primes involved have values $\ge u - H(u) \ge u/2$. So the cost is at most $e^{3(H(u)+3)/u} \le e^{4H(X_0)/X_0}$, since $H(v)/v$ is decreasing; it is the factor $e^{5H(u)/u}$ already contained in that bound, and it is absorbed into $A_{\mathrm{top}} \le A_R\,e^{2.2/\Lambda}$.

End step. At the earliest pivot, with slot s_T, the pivot step leaves $\theta \int \chi_T\,\Psi|_{s_T=u_T}\,dq_p$. The early block and the tails are then summed over their realised values, and the sum is replaced by the deterministic bound of (iii). For the Euler chains this is Lemma A.6 (S3)(c). Taking expectations completes the induction. □

A.4. **Factorisation and sure sums.**

Lemma A.6 (Factorisation; sure sums; EAC) *Let f be a free column and P_{late} the set of pivots decided later than f, so $e_f = E_{P_{\mathrm{late}}} v_f$ for a unique v_f by (85). After the pivots in P_{late} have*

been integrated,

$$\Psi(q_f, \mathit{earlier}) \le q_f^{-\beta_f}\, \mathbf{1}\big(q_f \in C_\varepsilon(\mathit{earlier})\big)\, \Psi_0(\mathit{earlier}), \qquad \beta_f := \mathbf{1}^{\mathsf{T}} v_f.$$

Here C_ε is the projection of the feasibility caps (later pivots $\ge \tau_j$, and decision order) onto the q_f-axis, over the later-pivot values compatible with the window constraints. When the later pivots are pinned to relative width $O(\varepsilon)$, this widens the caps by relative $O(\varepsilon)$; the main line uses only the projection. The thin window constraints are invariant under the shift $x_f \mapsto x_f + s$, $x_{P_{\mathrm{late}}} \mapsto x_{P_{\mathrm{late}}} - s\, v_f$ (with $x = \log q$). So no late free prime ever enters a thin indicator.

(S1′) *Let c range over cofactors all of whose primes are $\le P := s_{p_t} + H(s_{p_t})$. Then for $\beta \ge 1$,*

$$\sum_c c^{-\beta} \le \mathrm{Eu}(P), \qquad \mathrm{Eu}(x) := \prod_{\mathit{realised}\ p \le x} (1 - 1/p)^{-1}.$$

(S2′) *If $\beta < 1$ and $c \le Y$, then $\sum_c c^{-\beta} \le Y^{1-\beta}\, \mathrm{Eu}(P)$. In particular $\#\{c \le Y :\ P^+(c) \le P\} \le Y\, \mathrm{Eu}(P)$.*

(S3) (Euler chains, indexed by slot.) *For a slot s put*

$$\mathrm{Eu}_s(s) := \prod_{\mathit{realised}\ p\ \mathit{decided\ before}\ s} (1 - 1/p)^{-1},$$

the product over the realised primes of slot $< s$, rational primes $< X_0$ included. So $\mathrm{Eu}_s(X_0)$ is the product $\mathrm{Eu}_{\mathrm{rat}}$ over the rational primes $< X_0$. Let s_t, f_t, u_t, $s_{\max}$ and χ_t be as in Lemma A.5, and put $s_{T+1} := X_0$. Some rows are chain rows*, each with a starting run. Let K_t be the number of chain rows starting at one of $P_1, \dots, P_t$ (so $K_0 = 0$), and $j_t := K_t - K_{t-1} \ge 0$ the number starting at P_t; $j_t = 0$ is allowed. Suppose that every free column of F_t occurs in a chain row starting at one of $P_1, \dots, P_t$, and give a prime g of F_t the weight g^{-a_g}, where $a_g \ge 1$ is its total exponent in those chain rows. Each chain row y has its tail d_y, with weight $1/d_y$. Put*

$$A_{\mathrm{top}} := e^\gamma \log\big(2z + H(2z)\big) \cdot \exp\big(2.21/\Lambda + 5H(X_0)/X_0\big),$$

and define functions $G_t \ge 0$ on $(0, \infty)$ by $G_0 :\equiv 1$ and, for $t \ge 1$,

$$G_t(E) := \int_E^{A_{\mathrm{top}}} K_t\, x^{j_t - 1}\, G_{t-1}(x)\, dx \quad (E < A_{\mathrm{top}}), \qquad G_t(E) := 0 \quad (E \ge A_{\mathrm{top}}). \tag{88}$$

Each G_t is non-increasing. Put $\Phi_t(u) := G_t(\mathrm{Eu}_s(u))/\mathrm{Eu}_s(u)^{K_t}$. Then:

(a) (block bound with variable content) *on every ω,*

$$\sum_{F_t} \chi_t \prod_{g \in F_t} g^{-a_g} \cdot \Phi_{t-1}(u_t) \ \le\ \Phi_t(s_{t+1}),$$

where the sum runs over every non-empty content of F_t: every finite set of realised primes with slots in $[s_{t+1}, s_{\max}]$, f_t being the latest, and every choice of their exponents in the chain rows with all $a_g \ge 1$;

(b) the carried function Φ_t is non-increasing. The factor $\Phi_t(s_{t+1})$ involves only the label s_{t+1} and primes decided before s_{t+1}. Step (i) of Lemma A.5 for the run P_{t+1} turns it into $\Phi_t(u_{t+1})$, the carried factor in (a) for the next block;

(c) at the end, for $t = T$: the sum of the left side of (a) over all contents of the early block, the empty content included, multiplied by $\prod_y \sum_{d_y} 1/d_y$ over the tails of the K_T chain rows, is at most $G_T(\mathrm{Eu}_{\mathrm{rat}}) + \mathrm{Eu}_{\mathrm{rat}}^{j_T}\, G_{T-1}(\mathrm{Eu}_{\mathrm{rat}})$;

(d) (sure top bound) *for a slot* $u \in [X_0, s_{\max}]$, *on* $G_{<u}$,

$$\mathrm{Eu}_s(u) \le e^\gamma \log u \cdot \exp\bigl(2.21/\Lambda + 5H(u)/u\bigr) \le A_{\mathrm{top}} \le A_R\, e^{2.2/\Lambda},$$

with $A_R = 1.8(\log z + 4)$ *as in Theorem A.12. The case* $u = X_0$ *involves only the rational primes, and gives* $\mathrm{Eu}_{\mathrm{rat}} \le A_{\mathrm{top}}$.

The recursion (88) is evaluated in Theorem A.12 and in Proposition 6.18. It does not factorise into one bound per row; its output, summed over the run structures, is at most $(k^k/k!)\, A_{\mathrm{top}}^k$ *when every row is a chain row.*

Proof. (FAC). The window constraints read $\log n_i = \log c_i + \sum_p E_{ip} x_p \in \log J$. Since $e_f = E_{P_{\mathrm{late}}} v_f$, the shift $x_f \mapsto x_f + s$, $x_{P_{\mathrm{late}}} \mapsto x_{P_{\mathrm{late}}} - s v_f$ leaves every $\sum_p E_{ip} x_p$ unchanged. In the integral over the later pivots, substitute $y := x_{P_{\mathrm{late}}} + v_f x_f$. The window indicators then no longer involve x_f. The Jacobian $\prod_{p \in P_{\mathrm{late}}} q_p = e^{\mathbf{1}^\top x_{P_{\mathrm{late}}}} = e^{\mathbf{1}^\top y}\, q_f^{-\beta_f}$ produces the factor $q_f^{-\beta_f}$. The remaining dependence on x_f is through the feasibility caps, $y_p - v_{f,p} x_f \ge \log \tau_p$ and the decision order. These are half-spaces in x_f for each y. Bounding them by their projection over the y-values compatible with the window constraints gives $\mathbf{1}(q_f \in C_\varepsilon)$.

(S1′)–(S2′). For $\beta \ge 1$, $\sum_c c^{-\beta} \le \sum_c c^{-1} \le \prod_{p \le P}(1 - 1/p)^{-1} = \mathrm{Eu}(P)$, since every c is a product of realised primes $\le P$ (rational primes $< X_0$ included). For $\beta < 1$ and $c \le Y$, $c^{-\beta} = c^{-1} c^{1-\beta} \le Y^{1-\beta} c^{-1}$, and the same Euler product bounds $\sum c^{-1}$.

(S3). (a) Fix the latest prime f of the content, with slot σ, and put $u := u_t$ (the first slot after σ), $x := 1/f$, $E_1 := \mathrm{Eu}_s(\sigma)$ and $E_2 := \mathrm{Eu}_s(u)$. There is one prime per slot, so $E_2 = E_1/(1-x)$. Write $K := K_t$, $j := j_t$ and $m := K_{t-1}$, so $K = m + j$, and $a := s_{t+1}$. The other primes of the content have slots in $[a, \sigma)$. For a realised prime g with slot in $[a, \sigma)$, the sum of g^{-a_g} over its exponent vectors in the K chain rows, the zero vector (meaning g is absent) included, is $(1 - 1/g)^{-K}$; for f itself the zero vector is excluded, and the sum is $(1-x)^{-K} - 1$. Dropping all other constraints, the contents with latest prime f contribute at most

$$\chi_t \Bigl(\frac{E_1}{\mathrm{Eu}_s(a)}\Bigr)^K \bigl((1-x)^{-K} - 1\bigr) \frac{G_{t-1}(E_2)}{E_2^m} = \chi_t\, E_1^j \bigl((1-x)^{-j} - (1-x)^m\bigr) \frac{G_{t-1}(E_2)}{\mathrm{Eu}_s(a)^K}.$$

Put $v := \log(1/(1-x)) > 0$. If $j \ge 1$, then, as $1 - e^{-mv} \le mv \le (m/j)(e^{jv} - 1)$,

$$(1-x)^{-j} - (1-x)^m = (e^{jv} - 1) + (1 - e^{-mv}) \le \frac{K}{j}\,(e^{jv} - 1),$$

so $E_1^j((1-x)^{-j} - (1-x)^m) \le (K/j)(E_2^j - E_1^j) = \int_{E_1}^{E_2} K y^{j-1}\, dy$. If $j = 0$, then $1 - (1-x)^m \le mv = \int_{E_1}^{E_2} K y^{-1}\, dy$. In both cases, since G_{t-1} is non-increasing, the contribution is at most

$$\chi_t\, \mathrm{Eu}_s(a)^{-K} \int_{E_1}^{E_2} K y^{j-1} G_{t-1}(y)\, dy.$$

The intervals $[\mathrm{Eu}_s(\sigma), \mathrm{Eu}_s(u)]$ of distinct realised primes are disjoint, since Eu_s is non-decreasing along the slots, and they lie in $[\mathrm{Eu}_s(a), \infty)$. If $\chi_t = 1$, then $u \le s_{\max}$ and $G_{<u}$ holds, so $E_2 \le A_{\mathrm{top}}$ by (d). Summing over f gives at most $\mathrm{Eu}_s(a)^{-K} \int_{\mathrm{Eu}_s(a)}^{A_{\mathrm{top}}} K y^{j-1} G_{t-1}(y)\, dy = \Phi_t(a)$, which is (a); if $\mathrm{Eu}_s(a) \ge A_{\mathrm{top}}$, no term has $\chi_t = 1$ and both sides vanish. Both elementary inequalities are sharp as $x \to 0$, so (a) loses almost nothing when the primes are large; the factor $k^k/k!$ of Theorem A.12 comes from the order in which the labels are replaced, not from these inequalities.

(b) G_t is non-increasing and Eu_s is non-decreasing. The primes of slot $< s_{t+1}$ are decided before every pivot of P_{t+1}.

(c) For $t = T$, every prime of the early block has slot $\ge X_0 = s_{T+1}$, so (a) bounds the non-empty contents by $\Phi_T(X_0) = G_T(\mathrm{Eu}_{\mathrm{rat}})/\mathrm{Eu}_{\mathrm{rat}}^{K_T}$. The empty content has $u_T = X_0$ and

contributes $\chi_T \Phi_{T-1}(X_0) \le G_{T-1}(\mathrm{Eu}_{\mathrm{rat}})/\mathrm{Eu}_{\mathrm{rat}}^{K_{T-1}}$. The tails are the X_0-smooth numbers, so $\sum_d 1/d = \mathrm{Eu}_{\mathrm{rat}}$ for each of the K_T chain rows. Multiplying gives (c).

(d) The second case of the proof of (F3′) bounds exactly the product over the realised primes decided before u, on Good$(<u)$, by $e^\gamma \log u \cdot \exp(2.21/\Lambda + 5H(u)/u)$. That bound uses the rational primes [16, Thm. 8, (3.29)] and the sure local Mertens bound of Lemma 5.1 (iv) for the realised primes with values in $[X_0^-, u - H(u)]$, applied once per chain (Remark 5.2).

Finally, for $u \le s_{\max} \le 2z + H(2z)$ we have $\log u \le \log(2z + H(2z))$ and $5H(u)/u \le 5H(X_0)/X_0$, since $H(v)/v$ is decreasing; this gives A_{top}. Moreover $\log(2z + H(2z)) \le \log z + 0.7$ and $5H(X_0)/X_0 \le e^{-10^6}$, so $A_{\mathrm{top}} \le e^\gamma(\log z + 0.7)e^{2.21/\Lambda}(1 + 10^{-8}) \le 1.8(\log z + 4)\, e^{2.2/\Lambda}$. □

Remark A.7 (Widened caps; an optional route). Widen the caps of (FAC) by the pinned range ρ_t of the later pivots, instead of by $O(\varepsilon)$. Then they become functions of outer variables only, the block sums commute with the later integrals, and the recursion of Lemma A.5 becomes a monotone functional of the final summand. The widened box costs a factor $e^{O(k\rho_t)}$. Let $m := \log 2z/\Lambda$, the maximal exponent sum of a row. Then $\|E_P^{-1}\|_\infty \le k\, m^{k-1}$, by Cramer's rule and Hadamard's inequality. The order is attained: rows $n_i = q_i q_{i+1}^{m-1}$ give $E_P = I + (m-1)N$ with N the nilpotent shift, whose inverse has a row of ℓ^1-norm $\sum_{l<k}(m-1)^l$. So the route is harmless when

$$(k-1)\log m + O(\log k) \;\ll\; \log(z/\delta).$$

The main line does not use this route. It uses the stagewise form (Theorem A.10 with Lemma A.8), which needs no conditioning hypothesis.

A.5. **Nested stage rows.** Let B' be a rank-full set of r rows, with exponent matrix E on its jittered primes, and with pivots, runs and blocks as in Section A.1. A *cover* is a set $C \subseteq B'$ of rows such that every prime occurring in B' occurs in some row of C. For each t, let M_t be the row matroid of $E_{\cdot,P_{\le t}}$, i.e. the matroid on the rows $y \in B'$ in which a set of rows is independent iff their restrictions to the columns $P_{\le t}$ are linearly independent.

Lemma A.8 (Nested stage rows)**.** *Order the rows so that those of C come before those of $B' \setminus C$, and let $\mathcal{R}_{\le t}$ be the lexicographically first basis of M_t for this order. Then:*

(1) $\mathcal{R}_{\le 1} \subseteq \mathcal{R}_{\le 2} \subseteq \cdots \subseteq \mathcal{R}_{\le T} = B'$;
(2) $|\mathcal{R}_{\le t}| = |P_{\le t}|$, and $E_{\mathcal{R}_{\le t}, P_{\le t}}$ is invertible;
(3) every column $j \in P_{\le t} \cup F_{\le t}$ has a nonzero entry in some row of $C \cap \mathcal{R}_{\le t}$.

Proof. (2). Since E_P is invertible, the columns $P_{\le t}$ are linearly independent, so $\operatorname{rank} M_t = |P_{\le t}|$. A basis of M_t is therefore exactly a set $\mathcal{R}$ of $|P_{\le t}|$ rows with $E_{\mathcal{R},P_{\le t}}$ invertible.

(1). Suppose the restriction of a row y to $P_{\le t+1}$ lies in the span of the restrictions of a set X of rows. Then so does its restriction to $P_{\le t}$. Hence $\mathrm{cl}_{M_{t+1}}(X) \subseteq \mathrm{cl}_{M_t}(X)$. A row y lies in the lexicographically first basis of a matroid N iff $y \notin \mathrm{cl}_N(\text{rows listed before } y)$. Therefore

$$y \in \mathcal{R}_{\le t} \;\Rightarrow\; y \notin \mathrm{cl}_{M_t}(\text{earlier rows}) \;\Rightarrow\; y \notin \mathrm{cl}_{M_{t+1}}(\text{earlier rows}) \;\Rightarrow\; y \in \mathcal{R}_{\le t+1}.$$

Finally $\mathcal{R}_{\le T} = B'$, because M_T has rank $r = |B'|$.

(3). Because the rows of C are listed first, the lexicographically first basis contains a maximal independent subset of C. So it maximises $|C \cap \mathcal{R}|$ among the bases $\mathcal{R}$ of M_t. Let $j \in P_{\le t} \cup F_{\le t}$, and let $y \in C$ have $E_{y,j} \neq 0$; such a y exists because C is a cover. If $y \in \mathcal{R}_{\le t}$, we are done. Otherwise

$$E_{y,P_{\le t}} = \sum_{w \in \mathcal{R}_{\le t}} \alpha_w E_{w,P_{\le t}}.$$

Column j lies in the span of the columns $P_{\le t}$ over all rows. This is trivial for $j \in P_{\le t}$, and holds by (85) for $j \in F_{\le t}$. Hence $E_{y,j} = \sum_w \alpha_w E_{w,j}$, so some $w \in \mathcal{R}_{\le t}$ has $\alpha_w E_{w,j} \neq 0$. If $w \in C$,

column j is covered. If $w \notin C$, then $\alpha_w \neq 0$, so $\mathcal{R}_{\le t} - w + y$ is a basis of M_t with one more row of C. This contradicts maximality. □

Remark A.9. Neither minimality of C nor any bound on $|C|$ is used. The order matters: if the rows of $B' \setminus C$ are listed first, conclusion (3) can fail. As an independent check, not a step of the proof, Lemma A.8 was also confirmed by exact rational rank computations: the nested sets exist on 32 133 random multi-stage configurations, and the explicit construction works on 13 433 (configuration, cover) pairs (computer check; see Appendix B).

A.6. **The Staged Volume Lemma.** Let B' be a rank-full set of k rows, $C \subseteq B'$ a cover, and $\mathcal{R}_{\le t}$ the nested bases of Lemma A.8. The *stage rows* are $C_t := \mathcal{R}_{\le t} \setminus \mathcal{R}_{<t}$. For a row y and a stage t put

$$c_y^{(\le t)} := n_y \Big/ \prod_{j \in P_{\le t}} q_j^{E_{yj}},$$

the product of all primes of y outside $P_{\le t}$, free primes and tail included. The entries of the free columns in the rows outside C are fixed in advance (skeleton (s4) of Proposition 6.18).

Theorem A.10 (Staged Volume Lemma)**.** *In the recursion of Lemma A.5, integrate the pivots stage by stage, the stage-t pivots P_t by the change of variables $q_{P_t} \mapsto (n_y)_{y \in C_t}$. Then the following hold.*

(i) (Stage identity with the cover split.) *After stage t, the Jacobian integrand is $\prod_{P_{\le t}} q / \big(|\det| \prod_{y \in \mathcal{R}_{\le t}} n_y\big)$, and*

$$\frac{\prod_{j \in P_{\le t}} q_j}{\prod_{y \in \mathcal{R}_{\le t}} n_y} = W_t^{\mathrm{cov}} \prod_{y \in \mathcal{R}_{\le t} \setminus C} \frac{1}{n_y} \prod_{j \in P_{\le t}} q_j^{\,1 - s_j^{\mathrm{cov}}}. \tag{89}$$

Here $W_t^{\mathrm{cov}} := \prod_{y \in C \cap \mathcal{R}_{\le t}} 1/c_y^{(\le t)}$, and $s_j^{\mathrm{cov}} := \sum_{y \in C \cap \mathcal{R}_{\le t}} E_{yj} \ge 1$ by Lemma A.8 (3). Further, $|\det| = |\det E_{\mathcal{R}_{\le t}, P_{\le t}}| = \prod_{s \le t} |\det S_s|$, the product of the Schur complements. This is a nonzero integer, so $|\det| \ge 1$, and the integrand is at most the right side of (89). Individual $|\det S_s|$ may be < 1. The rows outside the cover keep the exact factor $1/n_y$, where $n_y \in J$ is their stage integration variable.

(ii) (Late primes are harmonic.) *For $f \in F_t$, Lemma A.8 (3) puts f in a cover row of $\mathcal{R}_{\le t}$, so W_t^{cov} carries q_f^{-a} with $a \ge 1$. The whole column of f is determined by its $\mathcal{R}_{\le t}$-entries:*

$$e_f = E_{\cdot, P_{\le t}}\, E^{-1}_{\mathcal{R}_{\le t}, P_{\le t}}\, e_{f, \mathcal{R}_{\le t}}.$$

Hence the block data enter only through the weight $\prod_{g \in F_t} g^{-a_g}$, with $a_g \ge 1$ the total exponent of g in the cover rows of $\mathcal{R}_{\le t}$; the entries in the other rows of $\mathcal{R}_{\le t}$ carry no weight. The indicators of the remaining rows do not involve f. The sum over all contents of F_t (the number, the primes and the exponents of its free columns) is bounded by Lemma A.6 (S3)(a), with the rows of C as chain rows, each starting at the stage where it joins; no count of free structures is needed.

(iii) (Later stages; projected feasibility.) *The caps $\mathbf{1}(q_j \ge \tau_j)$ of Lemma A.5 hold at every point of the integration domain. After the block bound of each block, multiply by $\mathbf{1}(\mathrm{Proj}_t)$. Here Proj_t is the projection onto the remaining variables of the real completions: summed primes $\ge X_0^-$, and integrated pivots $\ge \tau_j$ in slot order. It is deterministic, and $\mathcal{F}_{p-}$-measurable at the next pivot p. On Proj_t, a row y joining at a stage $t' > t$ satisfies*

$$n_y \;\ge\; c_y^{\mathrm{noF}} \prod_{j \in P_{t'}} q_j^{E_{yj}} \prod_{j \in P_{<t'}} \tau_j^{E_{yj}}\, (X_0^-)^{e_y(F_{\le t})}. \tag{90}$$

Here c_y^{noF} is $c_y^{(\le t')}$ with the primes of $F_{\le t}$ removed, and $e_y(F_{\le t})$ is the total exponent of those primes in y. This is the only inequality the later stages use. In particular, the pivot mass of later rows carries τ_j^{-E}, not X_0^{-E}.

(iv) (Slot-indexed chains.) *The Euler chains are indexed by slot, as in Lemma A.6 (S3). They are carried from block to block by the functions Φ_t of Lemma A.6 (S3)(b), and the end step (S3)(c) closes them. The block sums do not factorise over the rows.*

(v) (Joint caps.) *A row's free pieces across blocks, together with its tail, take one joint cap: $\mathbf{1}\big(\prod_t c_t \le Y\big) \le Y/\prod_t c_t$. Capping block by block would give Y^T. For rows outside the cover the joint cap is not needed: their free part is a monomial in primes of the cover rows, and it is fixed in advance together with their X_0-smooth tail (skeleton (s4) of Proposition 6.18).*

Proof. (i). We integrate the runs $P_1, P_2, \dots$ in this order, as Lemma A.5 requires. Suppose stages $1, \dots, t-1$ are done. The rows of $\mathcal{R}_{<t}$ have become integration variables $n_y \in J$. The later pivots $P_{<t}$ are then functions of these variables and of the variables not yet processed; this uses the invertibility of $E_{\mathcal{R}_{<t},P_{<t}}$ (Lemma A.8 (2)). In logarithmic coordinates, for $y \in C_t$,

$$\log n_y = S_t\, x_{P_t} + (\text{terms in } n_{\mathcal{R}_{<t}} \text{ and in variables decided before } P_t),$$

$$S_t := E_{C_t,P_t} - E_{C_t,P_{<t}} E^{-1}_{\mathcal{R}_{<t},P_{<t}} E_{\mathcal{R}_{<t},P_t}.$$

Here S_t is the Schur complement of $E_{\mathcal{R}_{<t},P_{<t}}$ in $E_{\mathcal{R}_{\le t},P_{\le t}}$, and it is invertible because both matrices are. The change of variables $q_{P_t} \mapsto (n_y)_{y\in C_t}$ has Jacobian $|\det S_t| \prod_{y\in C_t} n_y / \prod_{j\in P_t} q_j$. Multiplying over $s \le t$, and using $\det E_{\mathcal{R}_{\le t},P_{\le t}} = \prod_{s\le t} \det S_s$ (block LU factorisation), gives the stated integrand.

For the identity (89), write $n_y = c_y^{(\le t)} \prod_{j\in P_{\le t}} q_j^{E_{yj}}$ for each $y \in \mathcal{R}_{\le t}$. For the cover rows, the product is

$$\prod_{y\in C\cap\mathcal{R}_{\le t}} n_y = \prod_{y\in C\cap\mathcal{R}_{\le t}} c_y^{(\le t)} \cdot \prod_j q_j^{s_j^{\mathrm{cov}}}.$$

Dividing $\prod_{P_{\le t}} q_j$ by this, and by the factors n_y of the remaining rows of $\mathcal{R}_{\le t}$, gives (89). The integer matrix $E_{\mathcal{R}_{\le t},P_{\le t}}$ is invertible, so $|\det E_{\mathcal{R}_{\le t},P_{\le t}}| \ge 1$.

(ii). By (85), $e_f = E_{\cdot,P_{\le t}} v_f$ for some v_f. Restricting to the rows of $\mathcal{R}_{\le t}$ gives $e_{f,\mathcal{R}_{\le t}} = E_{\mathcal{R}_{\le t},P_{\le t}} v_f$, hence $v_f = E^{-1}_{\mathcal{R}_{\le t},P_{\le t}} e_{f,\mathcal{R}_{\le t}}$ and the displayed formula. In particular, once the entries in the rows outside C are fixed in advance (in Theorem A.12 there are no such rows; in Proposition 6.18 they are part of the skeleton), the block data are determined by the cover-row pieces $(c_{y,F_t})_{y\in C\cap\mathcal{R}_{\le t}}$, where c_{y,F_t} is the product of the primes of F_t in y with their exponents. This map is injective.

By Lemma A.8 (3), f has a nonzero entry in a cover row of $\mathcal{R}_{\le t}$. So $W_t^{\mathrm{cov}} = \prod_{y\in C\cap\mathcal{R}_{\le t}} c_{y,F_t}^{-1} \cdot$ (factors free of F_t), with every prime of F_t appearing to a power ≥ 1. Before the block bound, replace $q_j^{1-s_j^{\mathrm{cov}}}$ by $\tau_j^{1-s_j^{\mathrm{cov}}}$; this is valid at every point of the domain, by the caps of Lemma A.5, since $s_j^{\mathrm{cov}} \ge 1$. After that, the only dependence on F_t is through the cover-row pieces. By injectivity, the block sum is a sum over the realised primes of the columns of F_t and their cover-row exponents, with weight $\prod_{y\in C\cap\mathcal{R}_{\le t}} c_{y,F_t}^{-1} = \prod_{g\in F_t} g^{-a_g}$, $a_g \ge 1$. Its sum over all contents of F_t is bounded by Lemma A.6 (S3)(a), with the rows of C as chain rows, each starting at the stage where it joins.

For the remaining rows $y \notin \mathcal{R}_{\le t}$, the coefficient of x_f in their constraints, after substituting the pivots of $P_{\le t}$, is

$$e_{f,y} - E_{y,P_{\le t}} E^{-1}_{\mathcal{R}_{\le t},P_{\le t}} e_{f,\mathcal{R}_{\le t}} = e_{f,y} - E_{y,P_{\le t}} v_f = 0.$$

So their indicators do not involve f. The non-cover rows of $\mathcal{R}_{\le t}$ carry $1/n_y$ with n_y an integration variable, which does not involve f either.

(iii). After a block bound, the summed primes are no longer variables. The remaining integrand is a function of the variables not yet processed, and every actual configuration lies in Proj_t. So inserting $\mathbf{1}(\mathrm{Proj}_t)$ costs nothing; with the caps of Lemma A.5 it holds automatically for the integrated pivots. Proj_t is defined by real inequalities among deterministic quantities and variables decided before the next pivot, so it has the stated measurability.

For a row y joining at stage $t' > t$, write

$$n_y = c_y^{\mathrm{noF}} \cdot (\text{primes of } F_{\le t} \text{ in } y) \cdot \prod_{j \in P_{t'}} q_j^{E_{yj}} \prod_{j \in P_{<t'}} q_j^{E_{yj}}.$$

The left side, as a function of the remaining variables, does not depend on the completion. On Proj_t some completion has all summed primes $\ge X_0^-$ and all integrated pivots $\ge \tau_j$. This gives (90). In the stage-t' identity, this inequality replaces the exact factorisation of the rows joining at t'; it drops the summed primes from their cofactors and keeps $\tau_j^{-E_{yj}}$ for their earlier-integrated pivot mass.

(iv) is Lemma A.6 (S3). *(v)*: if $\prod_t c_t \le Y$ then $Y/\prod_t c_t \ge 1$. For a row outside the cover, every prime lies in some cover row, and its exponents and its tail are fixed in advance (skeleton (s4) of Proposition 6.18). □

Remark A.11. As a numerical corroboration only, and not part of the proof, some structural facts used in the proof of Theorem A.10 were checked by exact rational arithmetic in the case $C = B'$ (computer check; see Appendix B). There was no violation on random and adversarial families with 8–12 rows and up to nine stages. The facts checked are:

- the stage rows exist;
- column masses are ≥ 1 on the stage prefix;
- late free columns are covered and are determined by their stage-prefix entries, with vanishing coefficient in the remaining rows;
- the cumulative determinants are nonzero integers.

The identity (89) itself is not part of that check.

A.7. **Theorem R.**

Theorem A.12 (Rank-full clusters)**.** *For every window $J \subset [z, 2z]$ of length $\delta \le 1$ and every $k \ge 1$,*

$$\begin{aligned}
&\mathbb{E}\Big[\#\{\textit{rank-}k\ k\textit{-subsets of } \mathcal{N} \cap J\} \cdot \mathbf{1}\big(\mathrm{Good}(< t_J)\big)\Big] \\
&\quad \le \frac{k^k}{k!} (\theta A_{\mathrm{top}} \delta)^k\, e^{(3H(X_0)+2)k/X_0} \big(1 + k/X_0\big)^{1.02k} \\
&\quad \le (e\,\theta A_R \delta)^k\, e^{(3H(X_0)+2)k/X_0} \big(1 + k/X_0\big)^{1.02k}\, e^{2.2k/\Lambda}, \qquad A_R := 1.8(\log z + 4),
\end{aligned}$$

with A_{top} as in Lemma A.6 (S3). Since $H(X_0)/X_0 \le e^{-10^7}$, the factor $e^{(3H(X_0)+2)k/X_0}$ is at most $e^{k\,e^{-10^6}}$.

Proof. Reduction to skeletons. A k-subset of rank k has $k!$ orderings. Fix one ordering. The type of the ordered configuration has a skeleton $\mathfrak{s}$: its pattern E_P, with the pivot columns in decision order, and its run structure. The expectation is at most $1/k!$ times the sum, over patterns E_P and run structures, of $\mathbb{E}[X_{\mathfrak{s}} \mathbf{1}(\mathrm{Good}(< t_J))]$, where $X_{\mathfrak{s}}$ counts the ordered configurations with skeleton $\mathfrak{s}$, with all their free columns and tails.

One skeleton. Apply Lemma A.5 to $X_{\mathfrak{s}}$, evaluating the recursion by Theorem A.10 with the cover $C := B'$. Every row is then a chain row, starting at the stage where it joins, so $K_t = |P_{\le t}|$ and $j_t = |P_t| \ge 1$. Condition (3) of Lemma A.8 holds trivially: pivot columns are nonzero on $\mathcal{R}_{\le t}$ by (2), and late free columns by (85) and (2). Collect the factors:

- every row's n_y is integrated over J in exactly one stage, which gives δ^k;
- $1/|\det E_P| \le 1$;
- the pivot label sums give θ^k;
- the caps give $\prod_j q_j^{1-s_j} \le \prod_j \tau_j^{-(s_j-1)}$, with s_j the full column sums ((89) and (90));
- the free columns, in all blocks, and the tails: by Lemma A.6 (S3)(a)–(c), at most

$$\mathrm{Fin}(\mathfrak{s}) := G_T(\mathrm{Eu}_{\mathrm{rat}}) + \mathrm{Eu}_{\mathrm{rat}}^{j_T}\, G_{T-1}(\mathrm{Eu}_{\mathrm{rat}}),$$

  with the recursion (88) for the run structure $(j_1, \dots, j_T)$.

The factor $e^{2.2/\Lambda}$ in $A_{\mathrm{top}} \le A_R e^{2.2/\Lambda}$ is the error term of the sure Mertens bound (F4); the rational primes (the tails) enter through [16, Thm. 8, (3.29)], inside the proof of (F3′); and the slot–value fuzz of Lemma A.5 is contained in A_{top}, as its factor $e^{5H(X_0)/X_0}$.

The sum over run structures. Put $A := A_{\mathrm{top}}$ and $\rho := \mathrm{Eu}_{\mathrm{rat}}/A \in (0,1]$ (Lemma A.6 (S3)(d)). Substituting $x = Ay$ in (88) shows $G_t(E) = A^{K_t} g_t(E/A)$, where g_t is the same recursion with $A = 1$. Define polynomials H_K on $[0,1]$ by $H_0 := 1$ and

$$H_K(E) := K \sum_{j=1}^{K} \int_E^1 x^{j-1} H_{K-j}(x)\, dx \qquad (K \ge 1). \tag{91}$$

Splitting off the last run, the sum of g_T over the compositions $(j_1, \dots, j_T)$ of K is H_K. Hence

$$\sum_{\text{run structures}} \mathrm{Fin}(\mathfrak{s}) = A^k\, F_k(\rho), \qquad F_k(\rho) := H_k(\rho) + \sum_{j=1}^{k} \rho^j H_{k-j}(\rho).$$

Each H_K is nonnegative and non-increasing, with $H_K'(E) = -K \sum_j E^{j-1} H_{K-j}(E)$. So

$$F_k'(\rho) = \sum_{j=1}^{k} (j-k)\rho^{j-1} H_{k-j}(\rho) + \sum_{j=1}^{k} \rho^j H_{k-j}'(\rho) \le 0,$$

and $F_k(\rho) \le F_k(0) = H_k(0)$. To evaluate $H_k(0)$, let $T(w) := \sum_{n\ge1} n^{n-1} w^n/n!$ be the tree function, the inverse series of $w \mapsto we^{-w}$, so $T(w)e^{-T(w)} = w$. Consider the power series $P(E,z) := \sum_{K\ge0} H_K(E) z^K$. Comparing coefficients, (91) is equivalent to

$$\partial_E P = -z\, \partial_z\Big(\frac{zP}{1-Ez}\Big), \qquad P(1,z) = 1,$$

which determine P. Put $\zeta := T(ze^{-Ez})$, so $\log\zeta - \zeta = \log z - Ez$. Then $\partial_E \zeta = -z\zeta/(1-\zeta)$ and $\partial_z \zeta = \zeta(1-Ez)/(z(1-\zeta))$, and a direct computation shows that $P = (1-Ez)/(1-\zeta)$ satisfies the equation: both sides equal $-z/(1-\zeta) - z\zeta(1-Ez)/(1-\zeta)^3$. Also $T(ze^{-z}) = z$ gives $P(1,z) = 1$. At $E = 0$, $P(0,z) = 1/(1-T(z)) = \sum_K K^K z^K/K!$, since $zT'(z) = T/(1-T)$. Hence

$$\sum_{\text{run structures}} \mathrm{Fin}(\mathfrak{s}) \le A_{\mathrm{top}}^k \frac{k^k}{k!} \le e^k A_{\mathrm{top}}^k.$$

(More precisely, $F_k(\rho) = [z^k]\, 1/(1 - T(ze^{-\rho z}))$, which equals 1 at $\rho = 1$ and $k^k/k!$ at $\rho = 0$.)

The sum over patterns. Summing $\prod_j \tau_j^{-(s_j-1)}$ over patterns with the factor $1/k!$ gives $e^{(3H(X_0)+2)k/X_0}(1+k/X_0)^{1.02k}$, by Lemma A.4. Finally $A_{\mathrm{top}} \le A_R e^{2.2/\Lambda}$ by Lemma A.6 (S3)(d). □

Remark A.13. Theorem A.12 contains two simpler cases:

- tree-like types;
- types in which every free prime is decided before every pivot. For these, Lemma A.2 alone gives the bound, with the free data summed at the end.

Corollary A.14 (The case of rank at least r_0)**.** *Let $k = r_0 = \lceil L/(3\ell)\rceil + 3$ and $\delta = 4L^{-6}$, and let $L \ge 465$. Then the right side of Theorem A.12, multiplied by the number $2z/\delta$ of (overlapping) grid windows at height $z \le x$, is at most L^{-4}. This is the rank-$\ge r_0$ case of Proposition 6.10, and it is valid for every $L \ge 465$.*

Proof. Per element, the factor in Theorem A.12 is $e\,\theta A_R \delta\, e^{(3H(X_0)+2)/X_0}(1 + r_0/X_0)^{1.02} e^{2.2/\Lambda}$. Here:

- $\theta\, e^{(3H(X_0)+2)/X_0} e^{2.2/\Lambda} \le 1.01$, since $H(X_0)/X_0 \le e^{-10^7}$ (Section A.1), $\theta = 1 + 1/\Lambda$ and $\Lambda \ge 10^8$;
- $A_R \le 1.8(L+4)$, since $z \le x$;
- $1 + r_0/X_0 \le 1 + L \le 2L$.

Hence the factor is at most

$$e \cdot 1.8(L+4) \cdot 4L^{-6} \cdot (2L)^{1.02} \cdot 1.01 \le 15e\, L^{-3.98} \qquad (L \ge 465).$$

Since $2z/\delta \le e^L L^6/2$ and $\log(15e) - 3.98\ell < 0$, and $r_0 \ge L/(3\ell) + 3$,

$$\log\Big(\frac{2z}{\delta} \cdot \mathrm{RHS}\Big) \le L + 6\ell + r_0\big(\log(15e) - 3.98\ell\big) \le L + 6\ell + \Big(\frac{L}{3\ell} + 3\Big)\big(\log(15e) - 3.98\ell\big)$$
$$\le L\Big(-0.32666 + \frac{1.2361}{\ell}\Big) - 5.94\,\ell + 11.13 \;\le\; -4\ell$$

for $L \ge 465$. Here $1 - 3.98/3 = -0.32666\ldots$, $\log(15e)/3 \le 1.2361$, $3\log(15e) \le 11.13$, and the term $-11.94\ell = 3 \cdot (-3.98\ell)$ is kept. For the last step, put $f(L) := L(-0.32666 + 1.2361/\ell) - 1.94\ell + 11.13$; then $f(465) \le -59$, and $f'(L) \le -0.32666 + 1.2361/\ell < 0$ for $L \ge 465$. □

Appendix B. The computer-assisted parts

The scripts are in the directory `certificates/` of the repository accompanying this paper. They are written in Python 3; the interval computations use ball arithmetic from `python-flint` (Arb). Each subsection states what is checked, the method, the command, and the expected output. Commands are run from the repository root unless stated otherwise.

Five of them are steps of the proof: the Beurling–Saias certificate (Section B.1), the finite computations on the integers (Section B.2), the small range of the X_0-band bound (Section B.3), the numerical constants of Sections 5 and 7 (Section B.4), and the Rankin constants of Lemma 6.21 (`rankin_R.py`, Section B.5). The others are independent numerical cross-checks of statements proved in the text (Theorem 6.22, Lemma A.8, Theorem A.10) or of the bookkeeping of Section 2.3.

B.1. **The Beurling–Saias certificate (`bs-cert`).** *What it certifies.* The proof of Theorem 5.6 (and of Lemma 5.8) reduces the induction step to an explicit inequality $\Sigma(\ell, u) \ge 0$, where $\ell = \log Y$, $u = \ell/\log z$ and $\tau = \log t \ge 10^8$. Two parts of this reduction are computer-assisted.

- *The region* $30\tau \le \ell < 10^{18}$, $u \ge 2$. A closed-form lower bound $S(\ell,u) \le \Sigma(\ell,u)$ is built from the majorants of the proof (it is written out in Step 7(iii) of the proof of Theorem 5.6) and from bounds $r_{\mathrm{low}} \le r \le r_{\mathrm{up}}$ for the Dickman ratio $r = -\rho'/\rho$. The script checks $S > 0$ on cells covering the whole region, uniformly in $\tau \ge 10^8$.
- *One-variable inequalities:* the positivity $c_0(u) := c(u) - \ell\Phi_{\mathrm{ben}} \ge K'u^2\log(1+u)$ for $27 \le u \le 4\cdot 10^5$, used for $\ell \ge 10^{18}$; the one-step inequality of Lemma 5.3 for $1 \le u \le 10^6$; and that one-step inequality on $u \in [1,2]$ including both extra terms of the proof: the excess $X(u)$ of the small heights $1 \le Y' < 72$ (present for $u < 1 + \log 72/L_z$), and a conservative allowance $2K_M r(u)(e^{0.141}-1)$. Beyond these ranges they are proved analytically.

Method.

- All arithmetic is ball arithmetic (Arb, through `python-flint` 0.9.0); the bound S is in `bs_cert_arb.py`, the interval layer in `ivarb.py`. Every elementary operation and transcendental function is enclosed. The bounds r_{low} (from the recursion $r(v) \ge r(v-1)/(1-e^{-r(v-1)})$, $r(2) = 1/(2-2\log 2)$) and r_{up} (from $r(x) \le \log((x+1)/\delta)/(1-\delta)$, minimised over $\delta \in \{0.05, 0.1, 0.15, 0.2, 0.3, 0.4, 0.5\}$) are computed in balls. Decimal constants (such as $K' = 2.2$, $K = 3.2$, $C_0 = 0.70$ and 0.4) enter as exact balls. Floating-point numbers enter only as exactly representable cell endpoints and parameters (the float complement $1-\delta$ in r_{up} is compensated by a factor $1 + 2^{-50}$), and as outward-rounded (`nextafter`) images of ball endpoints: in the table lookups for r_{low} and r_{up}, in a few intermediate upper bounds that are fed back into the ball computation, in the final sign test, and in the printed output. Each such conversion moves a lower bound down or an upper bound up, so it can only weaken the certified inequality.
- Each cell is the product of an ℓ-interval and a u-interval (or an interval in $v := u - \ell^{1/4}$). The bound S is evaluated on the whole box, so no Lipschitz or interpolation argument is needed.
- The range $\ell \in [3\cdot 10^9, 10^{18}]$ is split into 403 bands of ratio 1.05. In each band, u-cells cover $[2, \max(40, u_0/2)]$ and v-cells cover the rest up to the largest junction $\bar{U}(\ell) = U(\ell;\tau = 10^8)$. The endpoints of the v-cells are rounded outward, so the seams between u-cells and v-cells, and the top edge, are covered exactly for every ℓ in the band. Cells that fail are bisected adaptively.
- Every cell also checks $r(u-1) > \gamma_1$ and $r(u-1) > \gamma_{\max}$, which is the condition for the integrals of the second majorant to converge.

How to run (with `python-flint` 0.9.0 installed for the interpreter used, e.g. in a virtual environment; `-I` ignores the user site directory):

```
python3 -I -B certificates/bs-cert/cert_run_arb_patched.py 1.05 10
    # main region, 10 workers, ~20 s
python3 -I -B certificates/bs-cert/aux_cert_arb.py
    # the one-variable checks, ~1 s
```

Expected output (abridged: long lines are wrapped, with the continuation indented, and the run time of the first line is omitted):

```
[patched seams] q=1.05 bands=403 cells=302013 failed=0 min certified l*S=2.491e+06
    at (82792994654.45967, 86932644387.18266, 200.1275416813517, 210.13391876541928, False)
c_0 >= K'u^2 log(1+u): [27,400000] cells=9609 failed=0 min certified margin=0.189
Lemma V one-step (L*Q_V, L = max(100,u)): [1,1e+06] cells=13823 failed=0
    min certified margin=1.724
Lemma V one-step on [1,2] with both extra terms: [1,2] cells=1387 failed=0
    min certified margin=0.829
```

That is: 302 013 cells (adaptive bisections included), 0 failures, minimum certified $\ell \cdot S = 2.491 \cdot 10^6$, at $\ell \approx 8.3 \cdot 10^{10}$, $u \in [200, 210]$; for $c_0(u)$, 9 609 cells on $[27, 4 \cdot 10^5]$, 0 failures, minimum margin 0.189; for the one-step inequality of Lemma 5.3, 13 823 cells on $u \in [1, 10^6]$, 0 failures, minimum margin 1.724; and for that inequality on $u \in [1, 2]$ with both extra terms, 1 387 cells, 0 failures, minimum margin 0.829. The script `cert_run_arb.py 1.05 10` gives the same result. The double-precision twins `bs_cert.py`, `cert_run.py` and `aux_cert.py` (minimum $2.598 \cdot 10^6$ for the main region) are a cross-check only and are not part of the certificate. In `bs_cert_arb.py`, the first majorant is `betaI`, `alpha`; the second is `betaII`, `gam1`, `gmax`, `tailII`; the error terms are `E1`, `E2`, `E3` with `eta`; the tail is `ET`, `T`; and the v-cells are `vmode`.

B.2. **Finite computations (`finite-n`).** *What it certifies.* The finite computations on $\mathbb{N}$ quoted in Section 5. Each is a supremum or a sum over a finite range; the numbers refer to the lines of the output.

1. Lemma 5.1(i): the maximum of $(\psi(x) - \psi(x/2))/x$ over $1 \le x < 1500$, and the elementary bound for $x \ge 1500$.
2. Lemma 5.1(ii): the range of $S(x) - \log x$ on $1 \le x \le 10^7$, where $S(x) = \sum_{p^k \le x} \log p / p^k$.
3. Lemma 5.3, small heights: the maximum of $\sum_{Y < n \le 2Y} 1/n$ over $72 \le Y < 10^4$ and the value $\log 2 + 10^{-4}$ for $Y \ge 10^4$ (line 3a), and the maximum over $1 \le Y < 72$ (line 3b).
4. The maximum of $y^{-1/2} \sum_{n \le y} \mu^2(n) n^{-1/2}$ over $1 \le y \le 10^6$, that is, $F_{1/2}$ in the squarefree average of Section 5.8.
5. Lemma 5.23: the maximum over primes $p < 400$ of $\prod_{q<p}(1 - 1/q)^{-1} - 1.8 \log p$.
6. Lemma 5.1(iii): $\sum_p \log p/(p(p-1))$, as a partial sum to 10^7 plus an explicit tail bound.

Method. Each supremum over a real variable is reduced to finitely many points by the step structure of the function; the reduction is explained in a comment above the corresponding check. The products in lines 3b and 5 are exact rationals; every other quantity is computed in ball arithmetic (`python-flint` 0.9.0), and each comparison with the bound used in the paper is decided on the ball. Each line prints the ball, or the exact value, and the result of that comparison.

How to run (about 3 seconds, with `python-flint`):

```
python3 -I -B certificates/finite-n/finite_checks.py
```

Expected output (long lines are wrapped, with the continuation indented; the value on the runtime line varies):

```
1  CM(i), N, 1<=x<1500: max (psi(x)-psi(x/2))/x = [0.68023948 +/- 3.67e-9] at x = 5
    (paper: 0.6803 <= 0.71): True; x>=1500: 0.71x - (x log2 + 3log x + 3) =
    [0.339568 +/- 1.21e-9] at x=1500, derivative >= [0.01485 +/- 2.82e-6] for x>=1500: True
2  CM(ii), N, 1<=x<=1e7: S(x)-log x in [[-0.75203870 +/- 1.62e-9], 0];
    inf approached as x -> 3-, sup at x = 1 (paper: [-0.753, 0]): True; 665134 prime powers
3a Lemma V, N, 72<=Y<1e4: max sum_{Y<n<=2Y} 1/n = [0.69658357 +/- 3.91e-9]
    for Y in [72.5, 73.0) (paper: 0.69659 <= C0 = 0.70): True;
    Y>=1e4: log2 + 1e-4 = [0.69324718 +/- 5.61e-10] (paper: 0.6933): True
3b Lemma V, N, 1<=Y<72: max sum_{Y<n<=2Y} 1/n = 5/6 for Y in [1.5, 2.0), exact
    (paper: 0.8334): True
4  F_{1/2}, N, 1<=y<=1e6: max y^{-1/2} sum_{n<=y} mu^2(n) n^{-1/2} = [1.3296349 +/- 4.90e-8]
    at y = 7 (paper: 1.32964 <= 1.33): True
5  Lemma threshold-mass, primes p<400: max prod_{q<p}(1-1/q)^{-1} - 1.8 log p =
    [0.24736173 +/- 1.71e-9] at p = 7 (paper: <= 0.25): True
6  CM(iii), rational primes: sum log p/(p(p-1)) <= [0.75536651 +/- 8.30e-10] +
    [1.71e-6 +/- 1.81e-9] = [0.75536822 +/- 2.64e-9] (paper: 0.75537): True
runtime 2.8 s
```

Each line states the bound used in the paper as `(paper: ...)`.

B.3. **The jump at X_0, small range (`x0-band`).** *What it certifies.* The small range of Lemma 5.27, i.e. Proposition 5.31(i): the bound $B(n) \le 5/12$ for every integer $2 \le n \le 10^6$, where

$$B(n) = \frac{1}{n}\Big[\tfrac{1}{2}\mu^2(n) + \tfrac{1}{2}\sum_{d|n,\,1<d<n}\Big(2\mu^2(d) + \tfrac{1}{2}\lambda\frac{X(d)}{d}\Big)\Big],$$
$$X(d) = \sum_{e|d,\,1<e<d} \mu^2(e)\, e\, H_{\mathrm{ord}}(d/e), \qquad \lambda = 1.0001.$$

The maximum is attained only at $n = 6$, where $B(6) = 5/12$ exactly; the script checks this exactly, comparing every other (inflated) $B(n)$ with $5/12$ as a rational and recomputing $B(6)$ in rationals. It also prints the suprema over $n \ge 10^a$ for $a = 1, \dots, 5$, and the analytic tail bound $(\frac{1}{2} + 1.64\sqrt{n})/n + \frac{1}{4} \cdot 7.4915 \cdot 2^{0.2} n^{-0.2}$ of the proof of Proposition 5.31(i) at $n = 10^6$, which is $0.137383 \le 0.14$ (the constants 7.4915 and 0.14 are certified in Section B.4). For $n > 10^6$ the lemma uses that analytic bound, not the computation.

Method. Sieves compute, exactly in integers: μ^2; $H_{\mathrm{ord}}(m)$ by $H_{\mathrm{ord}}(1) = 1$, $H_{\mathrm{ord}}(m) = \sum_{d|m,\,d<m} H_{\mathrm{ord}}(d)$; $X(d)$; and the divisor sums. The integer arrays are checked against 64-bit overflow (the maximum of X is about $2.9 \cdot 10^8$). The only floating-point steps are the divisions $X(d)/d$ and sums of at most $\tau(n) \le 240$ nonnegative terms, whose relative rounding error is $< 10^{-12}$; the script multiplies every $B(n)$ by $1 + 10^{-9}$ before taking maxima, so every printed value is an upper bound. All terms are nonnegative and every integer quantity is exact, so no interval arithmetic is needed.

How to run (standard library only, about 5 seconds):

```
cd certificates/x0-band && python3 -I -B x0band_exact.py 1000000
```

Expected output (abridged):

```
N= 1000000  Hmax= 83934208  Xmax= 288745984  (int64 ok: True )
top: [(6, 0.41667), (12, 0.27778), (10, 0.25), (4, 0.25), (2, 0.25), (30, 0.23389), ...]
sup B(n), 10<=n<=N: 0.277781
sup B(n), 100<=n<=N: 0.088059
sup B(n), 1000<=n<=N: 0.022826
sup B(n), 10000<=n<=N: 0.007607
sup B(n), 100000<=n<=N: 0.002121
B(n) <= 5/12 for 2<=n<=N, equality only at n=6: True
analytic tail bound for n>=N: 0.137383
```

The companion script `x0band.py` is an earlier floating-point exploration up to $2 \cdot 10^4$; it is not part of the certificate.

B.4. **Numerical constants (`constants`).** *What it certifies.* Constants that the text quotes, in ball arithmetic, without another certificate. Here $A(s) := 1/(2-\zeta(s)) - 1 = \sum_{m \ge 2} H_{\mathrm{ord}}(m) m^{-s}$ for $\mathcal{N} = \mathbb{N}$.

- The ordered factorisations of Section 5.5.2: the root $\rho_0 \in (1.7286, 1.7287)$ of $\zeta(\rho_0) = 2$; $A_0(0.03) = A(1.97) = 2.06733 \le 2.07$; $A_1(0.03) = -\zeta'(1.97)/(2-\zeta(1.97))^2 = 9.40888 \le 9.42$; $\zeta(1.8) = 1.88223$, so $\zeta(1.8) + X_0^{-0.7} \le 1.883$; and $1/(2 - 1.883) \le 8.6$.
- Proposition 5.31: $A(1.8 - 3 \cdot 10^{-6}) = 7.49142 \le 7.4915$; $\frac{1}{4} \cdot 7.4915 \cdot 2^{0.2} \le 2.152$; $4/\sqrt{6} \le 1.64$; the tail $(\frac{1}{2} + 1.64\sqrt{n})/n + 2.152 n^{-0.2} = 0.13742 \le 0.14$ at $n = 10^6$; $A(1.77) = 13.0822 \le 13.1$; $\zeta(1.97) - 1 = 0.67398 \le 0.674$; and $-\zeta'(1.97) = 1.00004 \le 1.0001$.
- Lemma 7.3(c) and (83): $\mathrm{Ein}(2) = \int_0^2 (e^w - 1)\, dw/w = 3.68387 < 3.69$; $I_1 = 14.9154$, $I_2 = 214.7714$ and $I_3 = 6.6962$; $(2.02/\log 2)(I_1 + I_2 + I_3) = 688.88 \le 690$; and $690/\Lambda \le 7 \cdot 10^{-6}$.

Method. Ball arithmetic (`python-flint` 0.9.0) throughout: ζ, and ζ' from the Taylor series of ζ; $\mathrm{Ein}(2) = \mathrm{Ei}(2) - \gamma - \log 2$; ρ_0 from the signs of $\zeta - 2$ at 1.7286 and 1.7287 (ζ is decreasing). I_1 is evaluated from its antiderivative, and I_3 from the closed form $2e^{\gamma}e^{\mathrm{Ein}(2)} \cdot \frac{1}{2}e^{1/2}E_1(\log 4 + \frac{1}{2})$. For I_2, after the substitution $v = e^t$, the range $[\log 4, 60]$ is integrated by Arb's certified integration (`acb.integral`), and the rest is bounded explicitly by $e^{-60}p(121)(\log 121 + 1.01)/121$ with $p(y) = y^4 + 8y^3 + 48y^2 + 192y + 384$ (a comment in the script gives the derivation). Each comparison with the bound used in the text is decided on the ball.

How to run (under a second, with `python-flint`):

```
python3 -I -B certificates/constants/consts.py
```

Expected output (long lines are wrapped, with the continuation indented):

```
k1  rho_0 (root of zeta(s) = 2): zeta(1.7286) - 2, zeta(1.7287) - 2 =
    [8.5893e-5 +/- 2.83e-10], [-9.5920e-5 +/- 1.69e-10]: True
k2  A_0(0.03) = A(1.97) (paper: 2.0673, used <= 2.07) = [2.0673297 +/- 3.79e-8]: True
k3  A_1(0.03) = -zeta'(1.97)/(2-zeta(1.97))^2 (paper: 9.409, used <= 9.42) =
    [9.4088776 +/- 2.57e-8]: True
k4  zeta(1.8) (paper: zeta(1.8) + X_0^-0.7 <= 1.883) = [1.8822296 +/- 1.82e-8]: True
k5  1/(2 - 1.883) (paper: sum H_ord(m) m^-1.8 <= 8.6) = [8.5470085 +/- 4.71e-8]: True
x1  A(1.8 - 3e-6) (paper: <= 7.4915) = [7.49142324 +/- 3.98e-9]: True
x2  (1/4) 7.4915 2^0.2 (paper: <= 2.152) = [2.1513684 +/- 3.17e-8]: True
x3  2^omega(n)/sqrt(n) <= prod_{p|n} 2p^-1/2 <= 4/sqrt(6) (paper: 1.64) =
    [1.6329932 +/- 3.82e-8]: True
x4  (1/2 + 1.64 sqrt n)/n + 2.152 n^-0.2 at n = 1e6 (paper: <= 0.14 for n > 1e6) =
    [0.13742252 +/- 5.33e-10]: True
x5  A(1.77) (paper: 13.08, used <= 13.1; abscissa 1.7286 < 1.77) =
    [13.082206 +/- 2.21e-7]: True
x6  sum_{w>=2} w^-1.97 = zeta(1.97) - 1 (paper: 0.674) = [0.67398353 +/- 3.66e-9]: True
x7  sum_{w>=2} w^-1.97 log w = -zeta'(1.97) (paper: <= 1.0001) = [1.0000389 +/- 3.48e-8]: True
m1  Ein(2) = int_0^2 (e^w-1)/w dw = Ei(2) - gamma - log 2 (paper: < 3.69) =
    [3.68387151 +/- 5.41e-10]: True
m2  I_1 = int_1^4 v(log v + 1.01) dv (paper: 14.92) = [14.915355 +/- 1.12e-7]: True
m3  I_2 = int_4^oo U^3 (log U + 1.01) v^-2 dv, U = 1 + 2 log v (paper: 214.8) =
    [214.771420 +/- 2.64e-7]: True
m4  I_3 = 2 e^gamma e^Ein(2) int_4^oo dv/(U v^2) (paper: 6.7) = [6.6961946 +/- 4.30e-9]: True
m5  (2.02/log 2)(I_1 + I_2 + I_3) (paper: <= 690) = [688.87765 +/- 2.83e-6]: True
m6  690/Lambda at Lambda = 1e8 (paper eq:mu1sum: <= 7e-6) = [6.900e-6 +/- 3e-14]: True
```

B.5. **The capacity theorem: numerical cross-check and Rankin constants (`capl-cert`).** Theorem 6.22 is proved analytically. Apart from `rankin_R.py` (see below), the scripts in `certificates/capl-cert/` re-evaluate, in double-precision floating point, the precise criteria that Steps 2–5 of that proof bound analytically: the degree cap D^* computed from $\binom{m+k}{k}$; the exact $j = \max\{j : e(r_0(D+1))^{2j+1} < c\}$; the row-weight bounds of Lemma 6.21(b),(d); and condition (77) of Lemma 6.25 with the exact absorption threshold y. They are not interval-certified, and nothing in the proof depends on them. They use only the Python standard library and each takes a few seconds.

- `capl_model.py`: the shared model (high-degree-row criterion of Step 2, Dickman-row criterion, and the criterion (77) of Lemma 6.25 including $u_0 \geq 100$), in log space. It prints nothing.
- `analytic_capl.py`: computes $s_H = 5.9745$ (the root of $\gamma_*(s_H) = \frac{1}{3} - 10^{-5}$); checks that $G(\Lambda, s)$ decreases in Λ for Λ from 10^8 to 10^{40} at sample values of s; and evaluates the precise criteria with $c = \max(\frac{1}{4}H_{\mathrm{std}}^{1/2}, X_0^{\gamma(s)})$ on the grid $\Lambda \in \{10^8, 10^9, 10^{10}, 10^{12}, 10^{16}, 10^{20}\}$, $s \in [-\log\Lambda, 30]$ at step 0.01. Expected output: `s_H = 5.9745`, "decreasing" on every sample line, and 4842, 5072, 5302, 5763, 6684, 7605 points with 0 failures (35 268 in all).

- `caplw_check_independent.py`: an independent re-implementation of Steps 1, 2, 3 and 5 at 200 000 random pairs with $\log_{10}\Lambda$ uniform in $[8, 30]$ and $s \in [-\log\Lambda, 60]$. Expected output (the second line is wrapped here):

```
s_H = 5.974538031693934
Step1 violations: 0  Step2 violations: 0 (min lhs/rhs 1.6938)
    Dickman violations: 0 (min ratio 1.1897)  Step5 violations: 0
```

- `cert_capl.py`: an earlier cell-by-cell check at $\Lambda = 10^8$ only, with a piecewise-linear ramp of maximum 5.95 in place of γ, covering $\ell \in [\log\Lambda, 2 \cdot 10^4]$ with cells of width 0.002 up to $\ell = 60$ and 0.05 beyond; each cell is checked at its worst corners. It is superseded by the analytic, Λ-uniform proof. Expected output (first two lines; 419 591 cells in all):

```
cells 20790 on [18.421, 60]: failures 0; max slope dlogH/dL = 2.94e-06
cells 398801 on [60, 2e4]: failures 0; max slope = 0.00e+00
```

- `needs.py`: a diagnostic that prints, at sample ℓ, the least $\log c$ needed by the precise criteria and the largest allowed by Lemma 6.25. It checks nothing.
- `raise_max.py`: the value $1.3 \cdot 10^{-8}$ quoted in Remark 6.23. It computes the raise $\max(0, 2\log\bar{M}_1 - 8\ell^2)/L$ of H over H_{std}, with c, $\gamma(s)$ and the smooth majorant $\bar{M}_1 = \bar{r}_0 + \varphi(\bar{r}_0) + 1$, $\bar{r}_0 = L/(3\ell) + 4$, that defines H (Section 2.2), exactly as in that theorem (in log space), on the grid $\log_{10}\Lambda \in [8, 30]$ at step 0.05 and $s \in [-\log\Lambda, 30]$ at step 0.01 (3 252 514 points), and compares the maximum with the analytic bound 10^{-6}, which is all the proof uses. Expected output (about 10 seconds; the second line is wrapped here):

```
s_H = 5.9745; points = 3252514
max (log H - log H_std)/L = 1.292e-08 at log10 Lambda = 8.00, s = -9.92;
    analytic bound 1e-6: OK
```

A proof step in this directory: `rankin_R.py`. Unlike the scripts above, this one is a step of the proof, of Lemma 6.21(c),(e). In ball arithmetic it checks:

- $R(\sigma) := \sum_p \big(-\log(1 - p^{-\sigma}) - p^{-\sigma}\big)$ satisfies $R(0.69) \le 0.7618 \le 0.77$ and $R(0.75) \le 0.6212 \le 0.63$. The sum over $p \le 10^7$ is computed in balls, and the rest is bounded by $\sum_{n>N} n^{-2\sigma}/(2(1 - n^{-\sigma})) \le N^{1-2\sigma}/(2(2\sigma - 1)(1 - N^{-\sigma}))$ with $N = 10^7$.
- $\mathrm{Ei}(11) - e^{11}/11 - e^{11}/121 \le 134$, and $2.75/x + 134x^2e^{-x} < 1$ for $x \ge 11$. Together these give $\mathrm{Ei}(x) \le (e^x/x)(1 + 2/x)$ for $x \ge 11$. The second function has derivative $-2.75/x^2 + 134x(2 - x)e^{-x} < 0$ for $x > 2$, so it suffices that it is < 1 at $x = 11$.
- $(3/2)^{3/4} \le 1.3555$ and $\mathrm{Ei}(\frac{1}{4}\log 2) \ge -1$.

Expected output (about 2 seconds, with `python-flint`):

```
R(0.69) <= [0.76178247 +/- 3.88e-9]  (claim <= 0.77): OK
R(0.75) <= [0.62117313 +/- 3.99e-9]  (claim <= 0.63): OK
Ei(11) - e^11/11 - e^11/121 = [133.47496 +/- 4.33e-6]  (claim <= 134): OK
2.75/x + 134 x^2 e^-x < 1 at x = 11 (decreasing for x > 2): OK
(3/2)^(3/4) = [1.355403005 +/- 4.15e-10]  (claim <= 1.3555): OK
Ei(log(2)/4) = [-0.9944989999 +/- 1.03e-11]  (claim >= -1): OK
ALL OK
```

How to run (from the repository root; all but `rankin_R.py` use only the standard library):

```
python3 -I -B certificates/capl-cert/analytic_capl.py
python3 -I -B certificates/capl-cert/caplw_check_independent.py
python3 -I -B certificates/capl-cert/cert_capl.py
python3 -I -B certificates/capl-cert/needs.py
python3 -I -B certificates/capl-cert/raise_max.py
python3 -I -B certificates/capl-cert/rankin_R.py    # needs python-flint 0.9.0
```

B.6. **Lemma A.8: numerical check (`lemma-sc`).** Lemma A.8 is proved in full in Appendix A; these scripts are an independent corroboration, not a step of the proof. They work in exact rational arithmetic (Python's `fractions`).

- *What is tested.* Random rank-full integer exponent matrices E ($r \le 8$ rows, up to $r+6$ columns, entries in $\{0,1,2,3\}$ at random densities; columns that vanish identically are removed, since primes absent from every row must not be counted) whose greedy latest-first pivots form at least two stages. For every *minimum* cover C of the row supports, `sc_check.py` searches by brute force for nested row sets $R_{\le 1} \subset R_{\le 2} \subset \cdots$ with $|R_{\le t}| = |P_{\le t}|$, $E_{R_{\le t}, P_{\le t}}$ invertible, and every pivot of $P_{\le t}$ and every late free column of F_t covered by a row of $C \cap R_{\le t}$. `sc_greedy.py` tests the explicit construction of the proof: the lexicographically first bases of the row matroids M_t, with the rows of C first, checked for nesting, invertibility and coverage. `sc_check_independent.py` is an independent re-implementation over all *minimal* covers; it also shows that the order matters, since putting the rows of C last fails coverage in 1238 of 2342 cases.
- *Commands and expected output* (each run prints one dictionary and takes seconds to a minute):

```
python3 -I -B certificates/lemma-sc/sc_check.py 1
  {'tested': 10615, 'single': 20820, 'all_ok': 10615, 'some_fail': 0, 'all_fail': 0}
python3 -I -B certificates/lemma-sc/sc_check.py 7 20000 7
  {'tested': 5585, 'single': 10441, 'all_ok': 5585, 'some_fail': 0, 'all_fail': 0}
python3 -I -B certificates/lemma-sc/sc_check.py 11 60000 5
  {'tested': 15933, 'single': 31583, 'all_ok': 15933, 'some_fail': 0, 'all_fail': 0}
python3 -I -B certificates/lemma-sc/sc_greedy.py 5 30000 8
  {'ok': 13433}
python3 -I -B certificates/lemma-sc/sc_check_independent.py
  configs 842 {'ok': 2342, 'fail': 0, 'advfail': 1238, 'nest': 0}
```

  Here `tested` counts the multi-stage configurations (single-stage ones are skipped and counted as `single`), and `all_ok` those for which the nested sets exist for every minimum cover; `ok` counts the (configuration, cover) pairs for which the explicit construction works, and `advfail` the coverage failures of the reversed order.

B.7. **The staged volume lemma: structural check (`svl-stress`).** Theorem A.10 is proved in Appendix A; this script is an independent sanity check of the structural facts its proof uses, not a step of the proof. In exact rational arithmetic, on random and adversarial rank-full configurations with 8–12 rows and up to nine stages, it checks that the stage rows exist; that column masses are ≥ 1 on the stage prefix; that late free primes are covered by the stage rows; that a late free column is determined by its stage-prefix entries, with vanishing coefficient in the remaining rows; and that the cumulative determinants are nonzero integers. The checks take the cover to be all stage rows (the case $C = B'$ of Appendix A); the identity of Theorem A.10 itself is not checked.

```
python3 -I -B certificates/svl-stress/svl_stress.py        # 2-3 s, standard library only
  rand configs 60 max stages T 1 violations 0
  late configs 60 max stages T 9 violations 0
  shared configs 60 max stages T 2 violations 0
  nest configs 34 max stages T 4 violations 0
```

B.8. **Non-circularity of the dependency graph (`depcheck`).** Section 2.3 asserts that the dependencies between the results of this paper used at the *same* time form an acyclic graph, every loop closing only through an input used at a strictly earlier time or a strictly smaller height. The script `certificates/depcheck/depcheck.py` encodes the full dependency graph, one edge per internal input (external results enter only as leaves, and only some are listed), each edge marked "same time" or "strictly earlier" as in the table of Section 2.3, of which that table is an abridgement. It runs a depth-first search for a cycle among the same-time edges.

```
python3 -I -B certificates/depcheck/depcheck.py
  same-time subgraph acyclic: True | nodes 79 | edges 193 | strictly-earlier edges 27
```

The time-ordering of the strictly-earlier edges is argued in Section 2.4; the script checks only the bookkeeping. Theorem 6.22 is split into two nodes, part (a) and parts (b),(c): the latter

are deterministic facts about the parameters and have no inputs, while (F3′) and the landing lemma use part (c). With a single node the graph would contain the spurious same-time cycle (F3′) → Theorem 6.22 → Lemma 6.20 → Proposition 6.18 → Lemma A.6 → (F3′). As a sanity check of the script, adding the edge (F3′) → Theorem 6.22(a) makes it report exactly this cycle.

Node names. The nodes are named by the labels of the LaTeX source of this paper. A name `a..b` stands for the consecutive results from `a` to `b`, `a+b` for two items treated together, `lab(a)(c)` for parts (a) and (c) of a result, and `cite:key` for a cited external result; `Rankin` and `Paley-Zygmund` are the standard inequalities. The internal nodes are: `cor:increment` 6.11; `cor:kr-weights` 5.17; `cor:lag` 3.8; `cor:localcount` 6.3; `cor:r1` 5.40; `cor:saias` 5.9; `eq:HBS` (21); `eq:RH` (18); `eq:Rhatbound` (17); `eq:ampl` (2); `eq:huxley` (80); `eq:osc` (63); `eq:rhodet` (3); `eq:sF1+eq:sF2` (F1), (F2); `eq:sF3p+eq:sF4` (F3′), (F4); `lem:C2` 3.5; `lem:G1` 3.3; `lem:S` 5.21; `lem:SC` A.8; `lem:T1` 3.4; `lem:ZA` 4.4; `lem:ZB` 4.2; `lem:ZBlow` 4.3; `lem:adapted` 3.2; `lem:blin` 7.1; `lem:bs-profile` 5.8; `lem:budget` 6.9; `lem:cm` 5.1; `lem:count..lem:excess` 3.6–3.9; `lem:covering..thm:tcl` 6.14–6.17; `lem:designedzero..lem:goodheights` 4.6–4.8; `lem:fac` A.6; `lem:heavy` 6.13; `lem:landing` 6.1; `lem:lemmaM` 5.5; `lem:lemmaP` 6.25; `lem:margin` 7.3; `lem:noise` 5.22; `lem:pil` A.5; `lem:predictable` 5.34; `lem:raisedy` 6.28; `lem:reduction` 6.20; `lem:rhoA` 6.26; `lem:rhoA(a)(b)` 6.26(a,b); `lem:rhoA(c)` 6.26(c); `lem:rowweights` 6.21; `lem:si` 5.32; `lem:smooth-v` 5.3; `lem:tau-near` 5.20; `lem:threshold-mass` 5.23; `lem:volume` A.2; `lem:xb` 5.27; `prop:La` 4.1; `prop:S1S2` 6.27; `prop:br` 5.26; `prop:cover` 6.18; `prop:ecl` 6.10; `prop:g1` 5.35; `prop:pointwise` 5.38; `prop:true-kernels` 5.18; `sec:greedy` §3.4; `thm:A` 2.1; `thm:R` A.12; `thm:bs` 5.6; `thm:caplw(a)` 6.22(a); `thm:caplw(b)(c)` 6.22(b,c); `thm:g2` 5.37; `thm:kr` 5.14; `thm:kr-random` 5.15; `thm:lp` 5.24; `thm:queue` 6.6; `thm:svl` A.10; `thm:zeros` 4.9.

Appendix C. Statement on the use of AI

How this work was produced. The construction, the proofs and the certificate code in this paper were developed with extensive use of AI agents (Anthropic's Claude models), working as a team under the direction of the author. The agents proposed constructions, wrote and revised proofs, wrote the computer-assisted certificates, and drafted this manuscript.

How it has been checked. Every review so far has been carried out by AI agents. Two component referees checked the individual parts of the argument, in several rounds, and an independent end-to-end referee then read the complete proof. That referee found no blocking error, found a number of repairable gaps, all of which were repaired, and finally judged that the argument proves Theorem 2.1 for $\theta \in [\frac{1}{4}, \frac{1}{2} - 10^{-4}]$. The reports are included in the `review/` directory of the repository that accompanies this paper. The certificates of Appendix B were re-run independently during that review.

Corrections found while writing. Turning the refereed working draft into this paper, a final AI consistency check of the paper against that draft (report `review/paper-check.md`), and reports received after that check surfaced slips in the refereed draft (in constants, missing terms, statements broader than their proofs, and individual proof steps). The paper corrects sixty of them; each is logged, with its justification, in the file `CORRECTIONS-vs-v3.md` of the repository, which lists every deviation of this paper from the refereed draft. None of them changes any conclusion. Entries 31–60 of the log were reported after the AI checks were complete (eighteen of them by a reader of the first public version) or were found while repairing or re-checking those; their repairs have been checked only by targeted AI re-checks, whose reports `R4.md`, `R5-pil-repair.md`, `R7-margin-var.md`, `R8-cert-repro.md`, `R9-cond-check-55.md`, `R9b-pil-o3b-55.md` and `R10-lp-transfer.md` are in the directory `review/recheck/`. Further fixes were made, during the same checks, to proofs that were newly written out for this paper.

A gap found and closed after publication. Correction 48 concerns the label sum of the Pivot-Integration Lemma (Lemma A.5). Its first repair proved the lemma for one configuration type. Two later independent AI checks (`review/recheck/R6-cond-check.md` and

`review/recheck/R6b-pil-o3b.md`) found that Theorem A.12 and the Cover Bound (Proposition 6.18) use the bound summed over all free-prime structures, and that the per-type repair does not cover that sum. For a time, therefore, the proof of Theorem 2.1 was incomplete. Correction 55 replaces the block bound by one that sums over every block content (Lemma A.6 (S3)). Theorem A.12 then carries an extra factor $k^k/k! \le e^k$, and the Cover Bound carries factors absorbed by the rows outside the cover; no conclusion changes. Correction 55 was checked by two further independent adversarial AI checks (`review/recheck/R9-cond-check-55.md` and `review/recheck/R9b-pil-o3b-55.md`), one of them by the agent that found the gap. Like every other repair here, it has not been checked by a human.

What has not been done. No human expert has verified the proof. The author takes responsibility for the manuscript, and welcomes scrutiny, corrections and counterexamples.

Components reviewed only once. The end-to-end referee did not re-derive the following components; each was checked by one component referee only.

- The closed form of the cell bound behind the computer-assisted part of Theorem 5.6 (Appendix B).
- The analytic regions of the proof of Theorem 5.6.
- Lemma 5.5 for non-squarefree n.
- The interior of the middle and large ranges of Lemma 5.27.
- The staged volume lemma, Theorem A.10.
- The Landau–Perron argument behind Theorem 5.24.

Material checked only by AI checks made for this paper. The proofs below were written out in full for this paper. In the form given here they have not been seen by the earlier referees. They have been checked only by AI checks made for this paper: the consistency check `review/paper-check.md`, and, for text rewritten after that check reported, the targeted rechecks in `review/recheck/`.

- in the proof of Theorem 5.6: the induction slack Σ of Step 5 ((36)), region Z1, and the remainder lemma for $\ell \ge 10^{18}$ (Lemma 5.7);
- the local prime number theorem, Theorem 5.24, including Landau's lemma and the explicit bound $G(L)$ and its range in part (iii);
- the Kantorovich–Rubinstein bounds, Theorems 5.14 and 5.15;
- Proposition 5.18 and Lemmas 5.20 and 5.21;
- the three ranges of the X_0-band bound (Proposition 5.31) and Lemma 5.32;
- the base case of the induction in the proof of Lemma 5.3;
- the proof of the direction (b)⇒(a) in Remark 1.2, and the proof of Lemma 6.20(iii);
- the bookkeeping of the Pivot-Integration Lemma, Lemma A.5, with the Euler chains (S3) of Lemma A.6;
- the analytic proof of Theorem 6.22, Steps 0–6;
- the explicit constants introduced by corrections #20–#22 of the log: the landing windows and K_{loc} of Corollary 6.3, the constants in Step 5 of the proof of Theorem 6.22, in step (v) of the proof of Lemma 6.25 and in Lemma 6.21(e), and C_0 in Lemma 4.3.

The same holds for the other corrections logged in `CORRECTIONS-vs-v3.md`.

A reader who wishes to check the proof with limited time might start with these two lists.

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result	inputs	used in
(F1)–(F2)	Section 3 (box jitter, slots), Lemmas 3.2 and 3.5	throughout
(F3′), (F4)	Lemma 5.1(iv); Theorem 6.22(c); [16]	Lemma 6.1, Lemma A.5, Lemma A.6
Lemma 6.1	(F1)–(F2); Theorem 6.22(c) (for ρ_H); [8]	Corollary 6.3; event E4
Corollary 6.3	Lemma 6.1; Theorem 6.6 at slots $\le \max J + H$; Lemma 3.6	Lemma 5.23, Lemma 5.20, Proposition 5.26
Theorem 6.6	Corollary 5.40, Proposition 5.38, Proposition 6.10, Corollary 6.11, (F1)–(F3′), Lemma 3.7, Lemma 3.9; [8]	(12), Section 4, Section 5, Lemma 6.9
Definition 6.8, Lemma 6.9	Theorem 6.6, Proposition 6.10, Lemma 6.1, Lemma 5.22	Theorem 2.1
Proposition 6.10	Theorem A.12, Lemma 6.13, Theorem 6.22	Corollary 6.11, Theorem 6.6, Lemma 6.25
Corollary 6.11	Proposition 6.10	Theorem 6.6, Lemma 5.22, Proposition 5.38
Lemma 6.13	none	Proposition 6.10, Lemma 6.20, Theorem 6.22
Lemma 6.14, Lemma 6.15, Theorem 6.16, Theorem 6.17	none (deterministic)	Lemma 6.20, Theorem 6.22
Proposition 6.18	Lemma A.2, Lemma A.6, Theorem A.10, Lemma A.8, (V1)	Lemma 6.20
Lemma 6.20	Lemma 6.13, Lemma 6.14, Theorem 6.16, Proposition 6.18, (F4)	Theorem 6.22
Lemma 6.21	Rankin's bound; [16]; certificate (Appendix B.5)	Theorem 6.22
Theorem 6.22(a)	Lemma 6.13, Lemma 6.20, Lemma 6.21, Theorem 6.16	Proposition 6.10
Theorem 6.22(b), (c)	none (definitions of the parameters)	the parameters of Section 6.1, (F3′), Lemma 6.1, Theorem 6.6, Lemma 6.25
Lemma 6.25	Proposition 6.10 (heights $< u$), Lemma 5.1, Theorem 6.22(b)	Lemma 6.26, (5), Lemma 4.4, Section 5 ($\sigma_{\mathcal{A}}$)
Lemma 6.26	Lemma 6.25 only (heights $< u$)	(c): (63), Proposition 5.38 (current height)
Proposition 6.27	Lemma 6.25, Corollary 6.11	Theorem 6.6, Lemma 5.22
Lemma 6.28	Theorem 6.22, Lemma 6.25, Lemma 6.26	Lemma 5.21, Proposition 5.38, Lemma 4.4
Lemmas A.2–A.6, Lemma A.8, Theorem A.10	(F1)–(F4)	Theorem A.12, Proposition 6.18
Theorem A.12	Lemmas A.2–A.6, Lemma A.8, Theorem A.10	Proposition 6.10

TABLE 2. Dependencies of the results of Section 6 and Appendix A.